

## Chapter 26

# Enhanced Pulse Width Modulator (ePWM)



The enhanced pulse width modulator (ePWM) peripheral is a key element in controlling many of the power electronic systems found in both commercial and industrial equipment. These systems include digital motor control, switch mode power supply control, uninterruptible power supplies (UPS), and other forms of power conversion. The ePWM peripheral can also perform a digital-to-analog (DAC) function, where the duty cycle is equivalent to a DAC analog value; it is sometimes referred to as a power DAC.

This chapter is applicable for ePWM type 4 with added register protection capability. See the [C2000 Real-Time Control Peripheral Reference Guide](#) for a list of all devices with an ePWM module of the same type, to determine the differences between the types, and for a list of device-specific differences within a type.

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## 26.1 Introduction

This chapter includes an overview and information about each submodule:

- [Time Base \(TB\) Submodule](#)
- [Counter Compare \(CC\) Submodule](#)
- [Action Qualifier \(AQ\) Submodule](#)
- [Dead-Band Generator \(DB\) Submodule](#)
- [PWM Chopper \(PC\) Submodule](#)
- [Trip Zone \(TZ\) Submodule](#)
- [Event Trigger \(ET\) Submodule](#)
- [Digital Compare \(DC\) Submodule](#)

The ePWM Type 4 is functionally compatible to Type 2 (a Type 3 does not exist). Type 4 has the following enhancements in addition to the Type 2 features:

- **Register Address Map:** Additional registers are required for new features on ePWM Type 4. The ePWM register address space has been remapped for better alignment and easy usage.
- **Delayed Trip Functionality:** Changes have been added to achieve deadband insertion capabilities to support, for example, delayed trip functionality needed for peak current mode control type application scenarios. This has been accomplished by allowing comparator events to go into the Action Qualifier as a trigger event (Events T1 and T2). If comparator T1 / T2 events are used to edit the PWM, changes to the PWM waveform do not take place immediately. Instead, the waveform synchronizes to the next TBCLK.
- **Dead-Band Generator Submodule Enhancements:** Shadowing of the DBCTL register to allow dynamic configuration changes.
- **One Shot and Global Load of Registers:** The ePWM Type 4 allows one shot and global load capability from shadow to active registers to avoid partial loads in, for example, multiphase applications. ePWM Type 4 also allows a programmable prescale of shadow to active load events. ePWM Type 4 Global Load can simplify ePWM software by removing interrupts and making sure that all registers are loaded at the same time.
- **Trip-Zone Submodule Enhancements:** Independent flags have been added to reflect the trip status for each of the TZ sources. Changes have been made to the trip-zone submodule to support certain power converter switching techniques like valley switching.
- **Digital Compare Submodule Enhancements:** Blanking window filter register width has been increased from 8 to 16 bits. DCCAP functionality has been enhanced to provide more programmability.
- **PWM SYNC Related Enhancements:** The ePWM Type 4 allows PWM SYNCOUT generation based on CMPC and CMPD events. These events can also be used for PWMSYNC pulse selection.

The ePWM Type 2 is fully compatible to Type 1. Type 2 has the following enhancements in addition to the Type 1 features:

- **High-Resolution Dead-Band Capability:** High-resolution capability is added to dead-band RED and FED in half-cycle clocking mode.
- **Dead-Band Generator Submodule Enhancements:** The ePWM Type 2 has features to enable both RED and FED on either PWM outputs. Provides increased dead band with 14-bit counters and dead-band / dead-band high-resolution registers are shadowed
- **High-Resolution Extension available on ePWMxB outputs:** Provides the ability to enable high-resolution period and duty cycle control on ePWMxB outputs. This is discussed in more detail in [Section 26.15](#).
- **Counter Compare Submodule Enhancements:** The ePWM Type 2 allows interrupts and SOC events to be generated by additional counter compares CMPC and CMPD.
- **Event Trigger Submodule Enhancements:** Prescaling logic to issue interrupt requests and ADC start of conversion expanded up to every 15 events. This submodule allows software initialization of event counters on SYNC event.
- **Digital Compare Submodule Enhancements:** Digital Compare Trip Select logic [DCTRIPSEL] has up to 12 external trip sources selected by the Input X-BAR logic in addition to an ability to OR all of them (up to 14 [external and internal sources]) to create the respective DCxEVTs.

- **Simultaneous Writes to TBPRD and CMPx Registers:** This feature allows writes to TBPRD, CMPA:CMPAHR, CMPB:CMPBHR, CMPC and CMPD of any ePWM module to be tied to any other ePWM module, and also allows all ePWM modules to be tied to a particular ePWM module if desired.
- **Shadow to Active Load on SYNC of TBPRD and CMP Registers:** This feature supports simultaneous writes of TBPRD and CMPA/B/C/D registers.

An effective PWM peripheral must be able to generate complex pulse width waveforms with minimal CPU overhead or intervention and must be highly programmable and very flexible while being easy to understand and use. The ePWM unit described here addresses these requirements by allocating all needed timing and control resources on a per PWM channel basis. Cross coupling or sharing of resources has been avoided; instead, the ePWM is built up from smaller single channel submodules with separate resources that can operate together as required to form a system. This modular approach results in an orthogonal architecture and provides a more transparent view of the peripheral structure, helping users to understand the operation quickly.

In this document, the letter x within a signal or submodule name is used to indicate a generic ePWM instance on a device. For example, output signals EPWMxA and EPWMxB refer to the output signals from the ePWMx instance. Thus, EPWM1A and EPWM1B belong to ePWM1 and likewise EPWM4A and EPWM4B belong to ePWM4.

### Type0 to Type1 Enhancements

- **Increased Dead-Band Resolution:** Dead-band clocking has been enhanced to allow half-cycle clocking to double resolution.
- **Enhanced Interrupt and SOC Generation:** Interrupts and ADC start-of-conversion can now be generated on both the TBCTR == zero and TBCTR == period events. This feature enables dual edge PWM control. Additionally, the ADC start-of-conversion can be generated from an event defined in the digital compare submodule.
- **High-Resolution Period Capability:** Provides the ability to enable high-resolution period. This is discussed in more detail in [Section 26.15](#).
- **Digital Compare Submodule:** The digital compare submodule enhances the event triggering and trip zone submodules by providing filtering, blanking and improved trip functionality to digital compare signals. Such features are essential for peak current mode control and for support of analog comparators.

#### Note

The name of the sync signal that goes to the CMPSS and GPDAC has been updated from PWMSYNC to EPWMSYNCPER (SYNCPER/PWMSYNCPER/EPWMxSYNCPER) to avoid confusion with the other EPWM sync signals EPWMSYNCI and EPWMSYNCO. For a description of these signals, see [Table 26-2](#).

## 26.1.1 EPWM Related Collateral

### Foundational Materials

- [C2000 Academy - EPWM](#)
- [Real-Time Control Reference Guide](#)
  - Refer to the EPWM section

### Getting Started Materials

- [C2000 ePWM Developer's Guide Application Report](#)
- [Enhanced Pulse Width Modulator \(ePWM\) Training for C2000 MCUs \(Video\)](#)
- [Flexible PWMs Enable Multi-Axis Drives, Multi-Level Inverters Application Report](#)
- [Getting Started with the C2000 ePWM Module \(Video\)](#)
- [Using PWM Output as a Digital-to-Analog Converter on a TMS320F280x Digital Signal Control Application Report](#)

- Chapters 1 to 6 are Fundamental material, derivations, and explanations that are useful for learning about how PWM can be used to implement a DAC. Subsequent chapters are Getting Started and Expert material for implementing in a system.
- [Using the Enhanced Pulse Width Modulator \(ePWM\) Module Application Report](#)

## Expert Materials

- [C2000 real-time microcontrollers - Reference designs](#)
  - See TI designs related to specific end applications used.
- [CRM/ZVS PFC Implementation Based on C2000 Type-4 PWM Module Application Report](#)
- [Leverage New Type ePWM Features for Multiple Phase Control Application Report](#)

### 26.1.2 Submodule Overview

The ePWM module represents one complete PWM channel composed of two PWM outputs: EPWMxA and EPWMxB. Multiple ePWM modules are instantiated within a device as shown in [Figure 26-1](#). Each ePWM instance is identical with one exception. Some instances include a hardware extension that allows more precise control of the PWM outputs. This extension is the high-resolution pulse width modulator (HRPWM) and is described in [Section 26.15](#). See the device data sheet to determine which ePWM instances include this feature. Each ePWM module is indicated by a numerical value starting with 1. For example, ePWM1 is the first instance and ePWM3 is the third instance in the system and ePWMx indicates any instance.

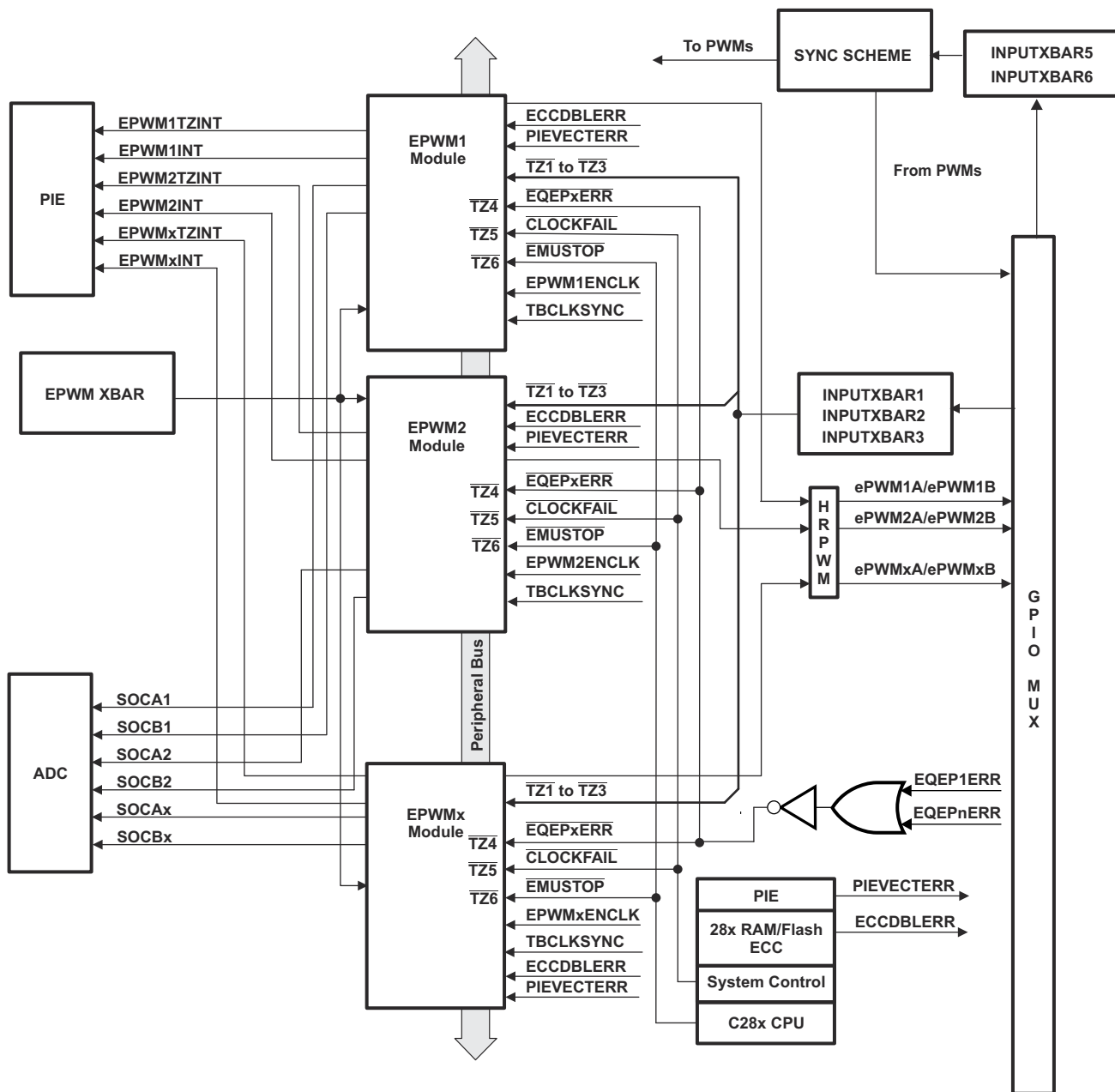
The ePWM modules are chained together by way of a clock synchronization scheme that allows them to operate as a single system when required. Additionally, this synchronization scheme can be extended to the capture peripheral submodules (eCAP). The number of submodules is device-dependent and based on target application needs. Submodules can also operate standalone.

Each ePWM module supports the following features:

- Dedicated 16-bit time-base counter with period and frequency control
- Two PWM outputs (EPWMxA and EPWMxB) that can be used in the following configurations:
  - Two independent PWM outputs with single-edge operation
  - Two independent PWM outputs with dual-edge symmetric operation
  - One independent PWM output with dual-edge asymmetric operation
- Asynchronous override control of PWM signals through software.
- Programmable phase-control support for lag or lead operation relative to other ePWM modules.
- Hardware-locked (synchronized) phase relationship on a cycle-by-cycle basis.
- Dead-band generation with independent rising and falling edge delay control.
- Programmable trip zone allocation of both cycle-by-cycle trip and one-shot trip on fault conditions.
- A trip condition can force either high, low, or high-impedance state logic levels at PWM outputs.
- All events can trigger both CPU interrupts and ADC start of conversion (SOC)
- Programmable event prescaling minimizes CPU overhead on interrupts.
- PWM chopping by high-frequency carrier signal, useful for pulse transformer gate drives.

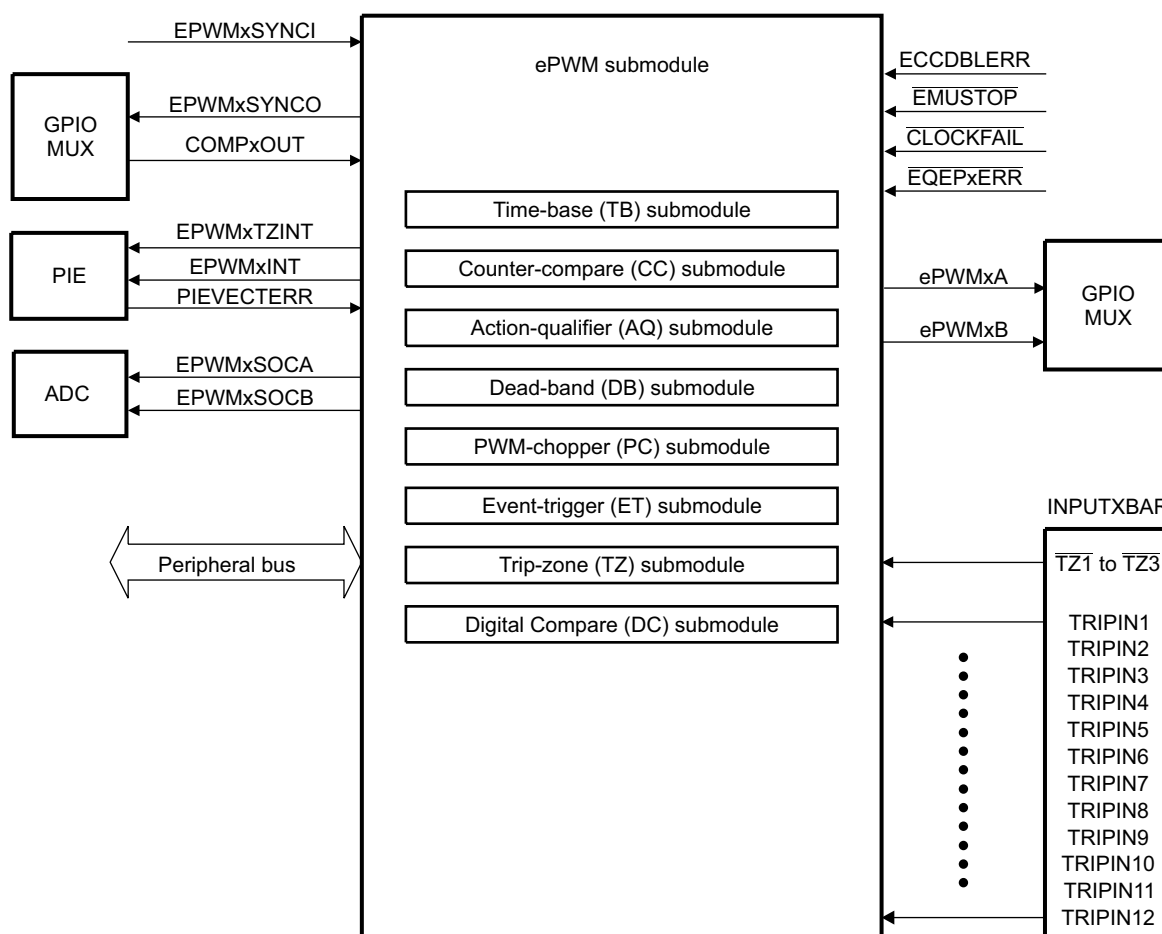
Each ePWM module is connected to the input/output signals shown in [Figure 26-1](#). The signals are described in detail in subsequent sections.

The order in which the ePWM modules are connected can differ from what is shown in [Figure 26-1](#). See [Section 26.4.3.3](#) for the synchronization scheme for a particular device. Each ePWM module consists of eight submodules and is connected within a system by way of the signals shown in [Figure 26-2](#).



A. This signal exists only on devices with an eQEP submodule.

**Figure 26-1. Multiple ePWM Modules**



**Figure 26-2. Submodules and Signal Connections for an ePWM Module**

Figure 26-3 shows more internal details of a single ePWM module. The main signals used by the ePWM module are:

- **PWM output signals (EPWMxA and EPWMxB)**

The PWM output signals are made available external to the device.

- **Trip-zone signals ( $\overline{TZ1}$  to  $\overline{TZ6}$ )**

These input signals alert the ePWM module of fault conditions external to the ePWM module. Each submodule on a device can be configured to either use or ignore any of the trip-zone signals. The  $\overline{TZ1}$  to  $\overline{TZ3}$  trip-zone signals can be configured as asynchronous inputs through the GPIO peripheral using the Input X-BAR logic, refer to Figure 26-51.  $\overline{TZ4}$  is connected to an inverted EQEPx error signal (EQEPxERR), which can be generated from any one of the EQEP submodule (for those devices with an EQEP module).  $\overline{TZ5}$  is connected to the system clock fail logic, and  $\overline{TZ6}$  is connected to the EMUSTOP output from the CPU. This allows configuring a trip action when the clock fails or the CPU halts.

- **Time-base synchronization input (EPWMxSYNCl), output (EPWMxSYNCO), and peripheral (EPWMxSYNCPER) signals**

Each ePWM module can be synchronized with other ePWM modules or other peripherals, using EPWMSYNClNSEL. Each ePWM module can also generate a synchronization output signal. The source of the EPWMxSYNCO can be selected and enabled by EPWMSYNCOEN and TBCTL2.OSHTSYNCMODE. For more information, see Section 26.4.3.3.

Each ePWM module also generates another PWMSYNCl signal called EPWMxSYNCPER. EPWMxSYNCPER goes to the GPDAC and CMPSS for synchronization purposes. Functionality is configured using the HRPCTL register, but has no relation with the HRPWM. For more information on how EPWMxSYNCPER is used by the GPDAC and CMPSS, see the respective chapters.

- **ADC start-of-conversion signals (EPWMxSOCA and EPWMxSOCB)**

Each ePWM module has two ADC start of conversion signals. Any ePWM module can trigger a start of conversion. Whichever event triggers the start of conversion is configured in the event-trigger submodule of the ePWM.

- **Comparator output signals (COMPxOUT)**

Output signals from the comparator module can be fed through the Input X-BAR and EPWM X-BAR to one or all of the 12 trip inputs [TRIPIN1 - TRIPIN12] and in conjunction with the trip zone signals can generate digital compare events.

- **Peripheral bus**

The peripheral bus is 32-bits wide and allows both 16-bit and 32-bit writes to the ePWM register file.



**Figure 26-3. ePWM Modules and Critical Internal Signal Interconnects**



## 26.2 Configuring Device Pins

To connect the device input pins to the module, the Input X-BAR and EPWM X-BAR must be used. Some examples of when an external signal can be needed are TZx, TRIPx, and EXTSYNCIN. Any GPIO on the device can be configured as an input. The GPIO input qualification can be set to asynchronous mode by setting the appropriate GPxQSEL register bits to 11b. The internal pullups can be configured in the GPyPUD register. Since the GPIO mode is used, the GPyINV register can invert the signals. Additionally, some TRIPx (TRIP4-12 excluding TRIP6) signals must be routed through the ePWM X-Bar in addition to the Input X-Bar.

The GPIO mux registers must be configured for this peripheral. To avoid glitches on the pins, the GPyGMUX bits must be configured first (while keeping the corresponding GPyMUX bits at the default of zero), followed by writing the GPyMUX register to the desired value.

See the *General-Purpose Input/Output (GPIO)* chapter for more details on GPIO mux, GPIO settings, and XBAR configuration.

## 26.3 ePWM Modules Overview

Eight submodules are included in every ePWM peripheral. Each of these submodules performs specific tasks that can be configured by software.

[Table 26-1](#) lists the key submodules together with a list of the main configuration parameters. For example, if you need to adjust or control the duty cycle of a PWM waveform, see the counter-compare submodule in [Section 26.5](#) for relevant details.

**Table 26-1. Submodule Configuration Parameters**

| Submodule                            | Configuration Parameter or Option                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Time Base (TB)</a>       | <ul style="list-style-type: none"> <li>Scale the time-base clock (TBCLK) relative to the ePWM clock (EPWMCLK).</li> <li>Configure the PWM time-base counter (TBCTR) frequency or period.</li> <li>Set the mode for the time-base counter: <ul style="list-style-type: none"> <li>count-up mode: used for asymmetric PWM</li> <li>count-down mode: used for asymmetric PWM</li> <li>count-up-and-down mode: used for symmetric PWM</li> </ul> </li> <li>Configure the time-base phase relative to another ePWM module.</li> <li>Synchronize the time-base counter between modules through hardware or software.</li> <li>Configure the direction (up or down) of the time-base counter after a synchronization event.</li> <li>Simultaneous writes to the TBPRD registers on all PWM's corresponding to the configuration on EPWMXLINK.</li> <li>Configure how the time-base counter behaves when the device is halted by an emulator.</li> <li>Specify the source for the synchronization output of the ePWM module</li> <li>Configure one shot and global load of registers in this module.</li> </ul> |
| <a href="#">Counter Compare (CC)</a> | <ul style="list-style-type: none"> <li>Specify the PWM duty cycle for output EPWMxA and output EPWMxB</li> <li>Specify the time at which switching events occur on the EPWMxA or EPWMxB output</li> <li>Specify the programmable delay for interrupt and SOC generation with additional comparators</li> <li>Simultaneous writes to the CMPA, CMPB, CMPC, CMPD registers on all PWM's corresponding to the configuration on EPWMXLINK.</li> <li>Configure one shot and global load of registers in this module.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

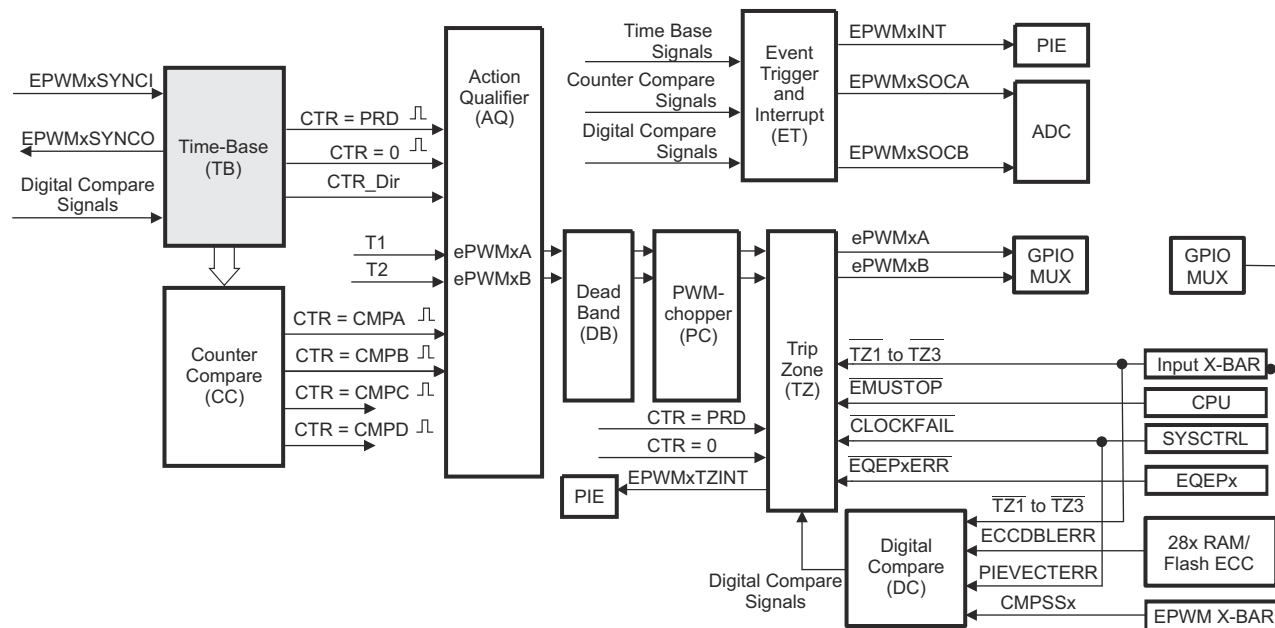
**Table 26-1. Submodule Configuration Parameters (continued)**

| Submodule                | Configuration Parameter or Option                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Action Qualifier (AQ)    | <ul style="list-style-type: none"> <li>Specify the type of action taken when a time-base counter-compare, trip-zone submodule, or comparator event occurs: <ul style="list-style-type: none"> <li>No action taken</li> <li>Output EPWMxA and EPWMxB switched high</li> <li>Output EPWMxA and EPWMxB switched low</li> <li>Output EPWMxA and EPWMxB toggled</li> </ul> </li> <li>Force the PWM output state through software control</li> <li>Configure and control the PWM dead band through software</li> <li>Configure one shot and global load of registers in this module.</li> </ul>                                                                                                                                                                                                                                                                                                                          |
| Dead-Band Generator (DB) | <ul style="list-style-type: none"> <li>Control of traditional complementary dead-band relationship between upper and lower switches</li> <li>Specify the output rising-edge-delay value</li> <li>Specify the output falling-edge delay value</li> <li>Bypass the dead-band module entirely. In this case the PWM waveform is passed through without modification.</li> <li>Option to enable half-cycle clocking for double resolution.</li> <li>Allow ePWMxB phase shifting with respect to the ePWMxA output.</li> <li>Configure one shot and global load of registers in this module.</li> </ul>                                                                                                                                                                                                                                                                                                                 |
| PWM Chopper (PC)         | <ul style="list-style-type: none"> <li>Create a chopping (carrier) frequency.</li> <li>Pulse width of the first pulse in the chopped pulse train.</li> <li>Duty cycle of the second and subsequent pulses.</li> <li>Bypass the PWM chopper module entirely. In this case the PWM waveform is passed through without modification.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Trip Zone (TZ)           | <ul style="list-style-type: none"> <li>Configure the ePWM module to react to one, all, or none of the trip-zone signals or digital compare events.</li> <li>Specify the trip action taken when a fault occurs: <ul style="list-style-type: none"> <li>Force EPWMxA and EPWMxB high</li> <li>Force EPWMxA and EPWMxB low</li> <li>Force EPWMxA and EPWMxB to a high-impedance state</li> <li>Configure EPWMxA and EPWMxB to ignore any trip condition.</li> </ul> </li> <li>Configure how often the ePWM reacts to each trip-zone signal: <ul style="list-style-type: none"> <li>One-shot</li> <li>Cycle-by-cycle</li> </ul> </li> <li>Enable the trip-zone to initiate an interrupt.</li> <li>Bypass the trip-zone module entirely.</li> <li>Programmable option for cycle-by-cycle trip clear</li> <li>If desired, independently configure trip actions taken when time-base counter is counting down.</li> </ul> |
| Event Trigger (ET)       | <ul style="list-style-type: none"> <li>Enable the ePWM events that trigger an interrupt.</li> <li>Enable ePWM events that trigger an ADC start-of-conversion event.</li> <li>Specify the rate at which events cause triggers (every occurrence or every 2nd or up to 15th occurrence)</li> <li>Poll, set, or clear event flags</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Digital Compare (DC)     | <ul style="list-style-type: none"> <li>Enables comparator (COMP) module outputs and trip zone signals which are configured using the Input X-BAR to create events and filtered events</li> <li>Specify event-filtering options to capture TBCTR counter, generate blanking window, or insert delay in PWM output or time-base counter based on captured value.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## 26.4 Time-Base (TB) Submodule

Each ePWM module has a time-base submodule that determines all of the event timing for the ePWM module. Built-in synchronization logic allows the time-base of multiple ePWM modules to work together as a single system.

Figure 26-4 illustrates the time-base submodule within the ePWM.



**Figure 26-4. Time-Base Submodule**

### 26.4.1 Purpose of the Time-Base Submodule

The time-base submodule can be configured for the following:

- Specify the ePWM time-base counter (TBCTR) frequency or period to control how often events occur.
- Manage time-base synchronization with other ePWM modules.
- Maintain a phase relationship with other ePWM modules.
- Set the time-base counter to count-up, count-down, or count-up-and-down mode.
- Generate the following events:
  - CTR = PRD: Time-base counter equal to the specified period (TBCTR = TBPRD).
  - CTR = Zero: Time-base counter equal to zero (TBCTR = 0x00).
- Configure the rate of the time-base clock; a prescaled version of the ePWM clock (EPWMCLK). This allows the time-base counter to increment/decrement at a slower rate.

#### Note

If required by the application code to update the TBCTR value through software while the TBCTR is counting, note that the time-base module needs at least 1 TBCLK cycle for the time-base related events to be realized. Hence, the TBCTR can be written with TBCTR = PRD-1 instead of TBCTR = PRD (in case the counter is counting up) and can be written as TBCTR = 1 instead of TBCTR = 0 (in case the counter is counting down) for the events to be realized.



**Table 26-2. Key Time-Base Signals**

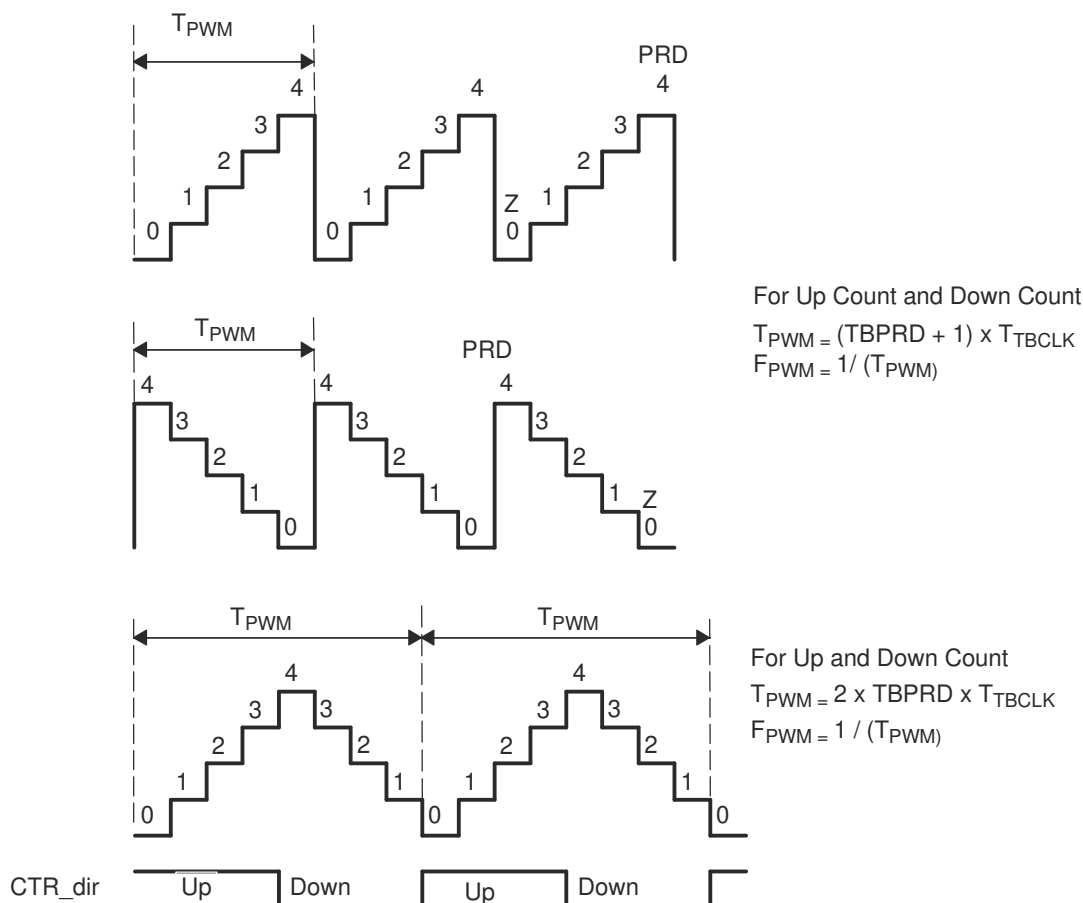
| Signal       | Description                                                                                                                                                                                                                                                                                                                                    |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EPWMxSYNCl   | Time-base synchronization input.<br><br>Input pulse used to synchronize the time-base counter with the counter of other ePWM modules. For more information on all of the signals available for synchronization, see EPWMSYNClNSEL. For information on the synchronization order of a particular device, see <a href="#">Section 26.4.3.3</a> . |
| EPWMxSYNCO   | Time-base synchronization output.<br><br>This output pulse is used to synchronize the counter of other ePWM modules. Using EPWMSYNCOuTEN, TBCTL2, TBCTL3 and GLDCTL2, the source of the output pulse is selected.                                                                                                                              |
| EPWMxSYNCPER | Time-base peripheral synchronization output.<br><br>This output signal is used to synchronize the GPDAC and CMPSS to the EPWM. The output signal can be configured using the HRPCTL register. Note that this signal has no relation with the HRPWM.                                                                                            |
| CTR = PRD    | Time-base counter equal to the specified period.<br><br>This signal is generated whenever the counter value is equal to the active period register value. That is when TBCTR = TBPRD.                                                                                                                                                          |
| CTR = Zero   | Time-base counter equal to zero<br><br>This signal is generated whenever the counter value is zero. That is when TBCTR equals 0x00.                                                                                                                                                                                                            |
| CTR = CMPB   | Time-base counter equal to active counter-compare B register (TBCTR = CMPB).<br><br>This event is generated by the counter-compare submodule and used by the synchronization out logic                                                                                                                                                         |
| CTR_dir      | Time-base counter direction.<br><br>Indicates the current direction of the ePWM's time-base counter. The signal is high when the counter is increasing and the signal is low when the counter is decreasing.                                                                                                                                   |
| CTR_max      | Time-base counter equal max value. (TBCTR = 0xFFFF)<br><br>Generated event when the TBCTR value reaches the maximum value. This signal is only used only as a status bit                                                                                                                                                                       |
| TBCLK        | Time-base clock.<br><br>This is a prescaled version of the ePWM clock (EPWMCLK) and is used by all submodules within the ePWM. This clock determines the rate at which time-base counter increments or decrements.                                                                                                                             |

### 26.4.3 Calculating PWM Period and Frequency

The frequency of PWM events is controlled by the time-base period (TBPRD) register and the mode of the time-base counter. Figure 26-6 shows the period ( $T_{PWM}$ ) and frequency ( $F_{PWM}$ ) relationships for the up-count, down-count, and up-down-count time-base counter modes when the period is set to 4 (TBPRD = 4). The time increment for each step is defined by the time-base clock (TBCLK) which is a prescaled version of the ePWM clock (EPWMCLK).

The time-base counter has three modes of operation selected by the time-base control register (TBCTL):

- **Up-Down Count Mode:** In up-down count mode, the time-base counter starts from zero and increments until the period (TBPRD) value is reached. When the period value is reached, the time-base counter then decrements until the counter reaches zero. At this point, the counter repeats the pattern and begins to increment.
- **Up-Count Mode:** In up-count mode, the time-base counter starts from zero and increments until the counter reaches the value in the period register (TBPRD). When the period value is reached, the time-base counter resets to zero and begins to increment once again.
- **Down-Count Mode:** In down-count mode, the time-base counter starts from the period (TBPRD) value and decrements until the counter reaches zero. When the counter reaches zero, the time-base counter is reset to the period value and begins to decrement once again.



**Figure 26-6. Time-Base Frequency and Period**

### 26.4.3.1 Time-Base Period Shadow Register

The time-base period register (TBPRD) has a shadow register. Shadowing allows the register update to be synchronized with the hardware. The following definitions are used to describe all shadow registers in the ePWM module:

- **Active Register:** The active register controls the hardware and is responsible for actions that the hardware causes or invokes.
- **Shadow Register:** The shadow register buffers provide a temporary holding location for the active register and have no direct effect on any control hardware. At a strategic point in time, the shadow register content is transferred to the active register. This prevents corruption or spurious operation due to the register being asynchronously modified by software.

The memory address of the shadow period register is the same as the active register. Which register is written to or read from is determined by the TBCTL[PRDL] bit. This bit enables and disables the TBPRD shadow register as follows:

- **Time-Base Period Shadow Mode:** The TBPRD shadow register is enabled when TBCTL[PRDL] = 0. Reads from and writes to the TBPRD memory address go to the shadow register. The shadow register contents are transferred to the active register (TBPRD (Active) ← TBPRD (shadow)) when the time-base counter equals zero (TBCTR = 0x00) and/or a sync event as determined by the TBCTL2[PRDLDSYNC] bit. The PRDLDSYNC bit is valid only if TBCTL[PRDL] = 0. By default the TBPRD shadow register is enabled. The sources for the SYNC input is explained in [Section 26.4.3.3](#).

The global load control mechanism can also be used with the time-base period register by configuring the appropriate bits in the global load configuration register (GLDCFG). When global load mode is selected the transfer of contents from shadow register to active register, for all registers that have this mode enabled, occurs at the same event as defined by the configuration bits in Global Shadow to Active Load Control Register (GLDCTL). Global load control mechanism is explained in [Section 26.4.7](#).

- **Time-Base Period Immediate Load Mode:** If immediate load mode is selected (TBCTL[PRDL] = 1), then a read from or a write to the TBPRD memory address goes directly to the active register.

### 26.4.3.2 Time-Base Clock Synchronization

The TBCLKSYNC bit in the peripheral clock enable registers allows all users to globally synchronize all enabled ePWM modules to the time-base clock (TBCLK). When set, all enabled ePWM module clocks are started with the first rising edge of TBCLK aligned. For synchronized TBCLKs, the prescalers for each ePWM module must be set identically.

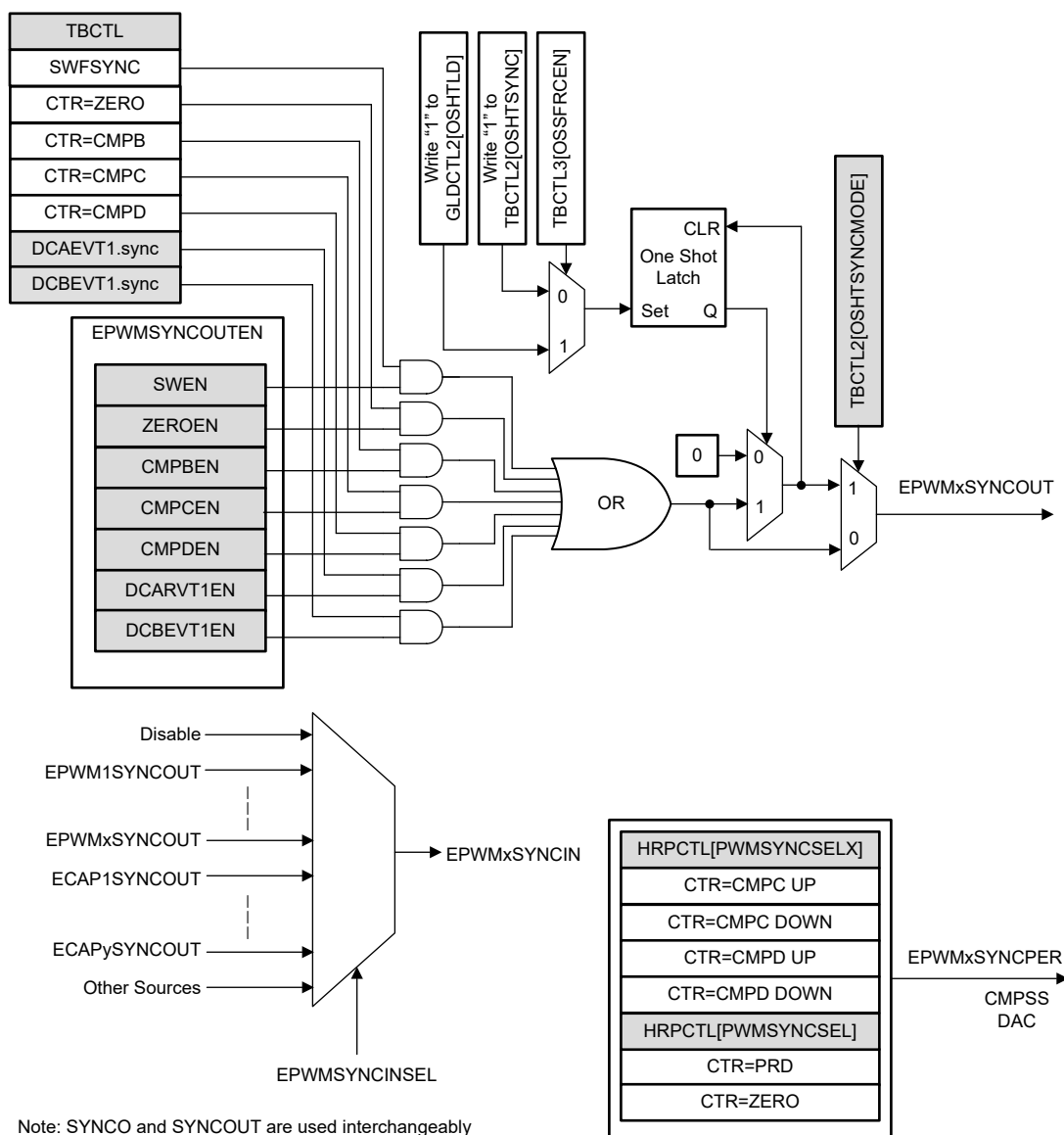
The proper procedure for enabling ePWM clocks is as follows:

1. Enable ePWM module clocks in the PCLKCRx register
2. Set TBCLKSYNC= 0
3. Configure ePWM modules
4. Set TBCLKSYNC= 1

In a multicore environment, GTBCLKSYNC can be used to override the core-specific TBCLKSYNC. When GTBCLKSYNC is set, TBCLKSYNC is ignored in all cores and therefore clearing or setting the TBCLKSYNC has no affect. If this feature is not required, GTBCLKSYNC must be cleared. In a multicore environment where different ePWM modules are assigned to different cores, the GTBCLKSYNC bit can be used to enable and disable the Time-Base clock of all ePWM modules simultaneously.

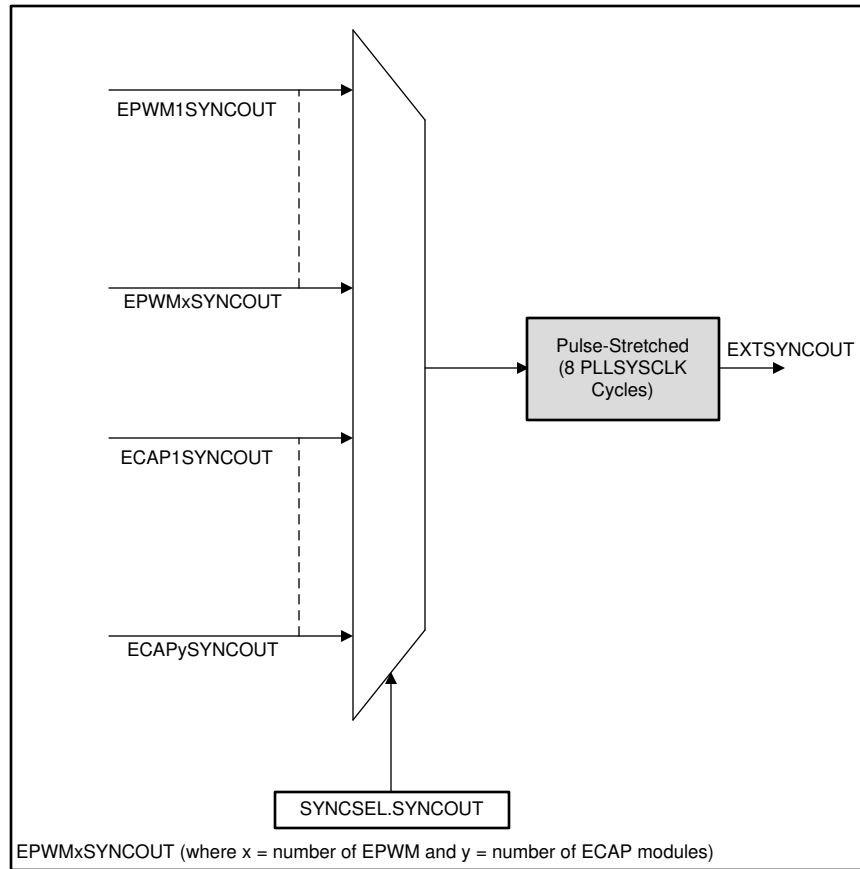
### 26.4.3.3 Time-Base Counter Synchronization

The ePWM synchronization scheme allows for increased flexibility of synchronization of the ePWM modules. Each ePWM module has a synchronization input (SYNCI), a synchronization output (SYNCO) and a peripheral synchronization output (SYNCPER). In Figure 26-7, EXTSYN CIN1 is sourced from INPUTXBAR5 and EXTSYN CIN2 is sourced from INPUTXBAR6, which can be configured to select any GPIO as the synchronization input. Refer to Section 26.4.3.4 for a list of all sync inputs including INPUTXBAR5 and INPUTXBAR6. Figure 26-8 shows the sources that can be used for EXTSYN COUT.



**Figure 26-7. Time-Base Counter Synchronization Scheme**





**Figure 26-8. ePWM External SYNC Output**

**Note**

See the data sheet for the number of ePWM and eCAP modules available on your specific device.

Each ePWM module can be configured to use or ignore the synchronization input. If the TBCTL[PHSEN] bit is set, then the time-base counter (TBCTR) of the ePWM module is automatically loaded with the phase register (TBPHS) contents when one of the following conditions occur:

- **EPWMxSYNCl: Synchronization Input Pulse:** The value of the phase register is loaded into the counter register when an input synchronization pulse is detected (TBPHS → TBCTR). This operation occurs on the next valid time-base clock (TBCLK) edge.

The delay from internal control module to target modules is given by:

- if (TBCLK = EPWMCLK): 2 x EPWMCLK
- if (TBCLK < EPWMCLK): 1 x TBCLK
- **Software Forced Synchronization Pulse:** Writing a 1 to the TBCTL[SWFSYNC] control bit invokes a software forced synchronization. This pulse is ORed with the synchronization input signal, and therefore has the same effect as a pulse on EPWMxSYNCl.
- **Digital Compare Event Synchronization Pulse:** DCAEVT1 and DCBEVT1 digital compare events can be configured to generate synchronization pulses which have the same affect as EPWMxSYNCl.

**Note**

If the EPWMxSYNCl signal is held high, the sync does not continuously occur. The EPWMxSYNCl is rising edge activated.

This feature enables the ePWM module to be automatically synchronized to the time base of another ePWM module. Lead or lag phase control can be added to the waveforms generated by different ePWM modules to synchronize them. In up-down-count mode, the TBCTL[PHSDIR] bit configures the direction of the time-base counter immediately after a synchronization event. The new direction is independent of the direction prior to the synchronization event. The PHSDIR bit is ignored in count-up or count-down modes. See [Figure 26-9](#) through [Figure 26-12](#) for examples.

Clearing the TBCTL[PHSEN] bit configures the ePWM to ignore the synchronization input pulse.

#### 26.4.3.4 ePWM SYNC Selection

[Table 26-3](#) specifies the sources for the ePWM SYNC input and output.

**Table 26-3. ePWM SYNC Selection**

| EPWMSYNCINSEL.SEL, ECAPSYNCINSEL.SEL | SYNC Source    |
|--------------------------------------|----------------|
| 0                                    | Reserved       |
| 1                                    | EPWM1_SYNCOUT  |
| 2                                    | EPWM2_SYNCOUT  |
| 3                                    | EPWM3_SYNCOUT  |
| 4                                    | EPWM4_SYNCOUT  |
| 5                                    | EPWM5_SYNCOUT  |
| 6                                    | EPWM6_SYNCOUT  |
| 7                                    | EPWM7_SYNCOUT  |
| 8                                    | EPWM8_SYNCOUT  |
| 9                                    | EPWM9_SYNCOUT  |
| 10                                   | EPWM10_SYNCOUT |
| 11                                   | EPWM11_SYNCOUT |
| 12                                   | EPWM12_SYNCOUT |
| 13                                   | EPWM13_SYNCOUT |
| 14                                   | EPWM14_SYNCOUT |
| 15                                   | EPWM15_SYNCOUT |
| 16                                   | EPWM16_SYNCOUT |
| 17                                   | ECAP1_SYNCOUT  |
| 18                                   | ECAP2_SYNCOUT  |
| 19                                   | ECAP3_SYNCOUT  |
| 20                                   | ECAP4_SYNCOUT  |
| 21                                   | ECAP5_SYNCOUT  |
| 22                                   | ECAP6_SYNCOUT  |
| 23                                   | ECAP7_SYNCOUT  |
| 24                                   | INPUTXBAR5     |
| 25                                   | INPUTXBAR6     |
| 26                                   | ECAT_SYNC0     |
| 27                                   | ECAT_SYNC1     |
| 28-31                                | Reserved       |

#### 26.4.4 Phase Locking the Time-Base Clocks of Multiple ePWM Modules

The TBCLKSYNC bit can be used to globally synchronize the time-base clocks of all enabled ePWM modules on a device. This bit is part of the device's clock enable registers and is described in the *System Control and Interrupts* section of this manual. When TBCLKSYNC = 0, the time-base clock of all ePWM modules is stopped (default). When TBCLKSYNC = 1, all ePWM time-base clocks are started with the rising edge of TBCLK aligned. For perfectly synchronized TBCLKs, the prescaler bits in the TBCTL register of each ePWM module must be set identically. The proper procedure for enabling the ePWM clocks is:

1. Enable the individual ePWM module clocks. This is described in the *System Control and Interrupts* chapter.
2. Set TBCLKSYNC = 0. This stops the time-base clock within any enabled ePWM module.
3. Configure the prescaler values and desired ePWM modes.
4. Set TBCLKSYNC = 1.

#### 26.4.5 Simultaneous Writes to TBPRD and CMPx Registers Between ePWM Modules

For variable frequency applications, there is a need for simultaneous writes of TBPRD and CMPx registers between ePWM modules. This prevents situations where a CTR = 0 or CTR = PRD pulse forces a shadow to active load of these registers before all registers are updated between ePWM modules (resulting in some registers being loaded from new shadow values while others are loaded from old shadow values). To support this, an ePWM register linking scheme for TBPRD:TBPRDHR, CMPA:CMPAHR, CMPB:CMPBHR, CMPC, and CMPD registers between PWM modules has been added.

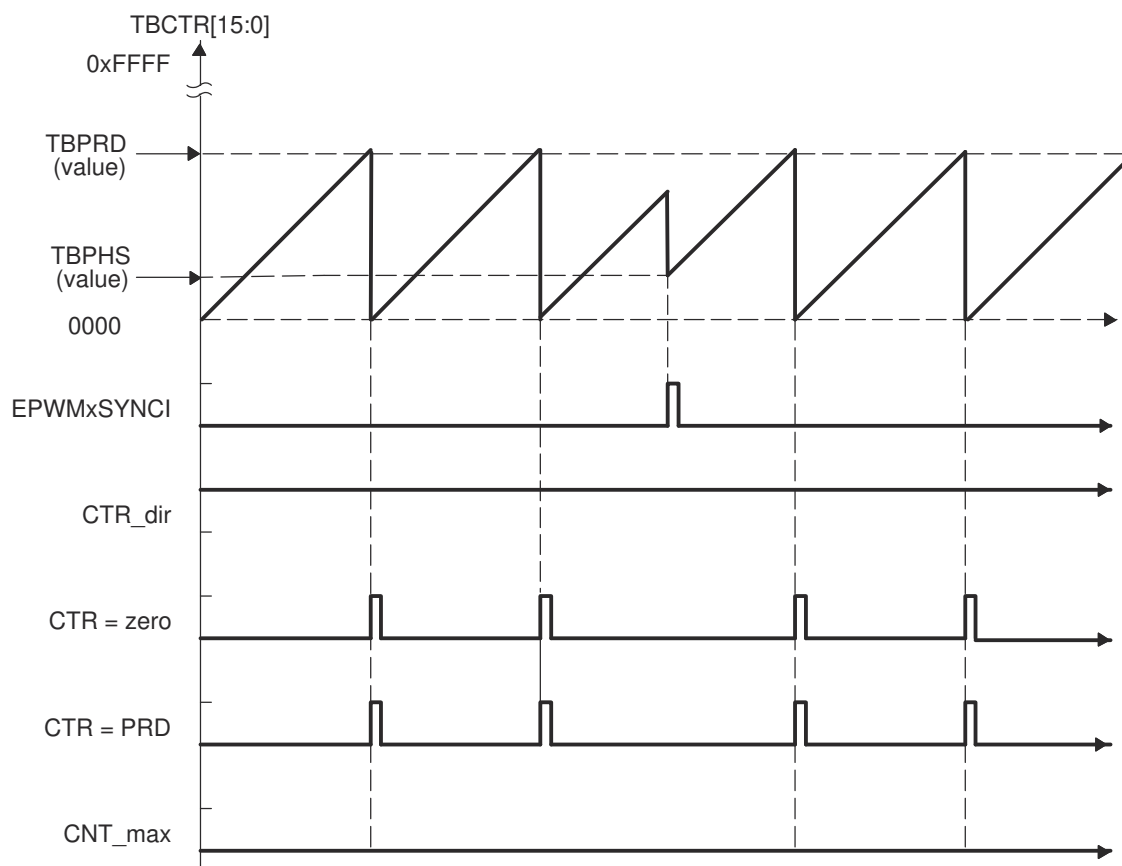
Refer to the register description for EPWMXLINK to see the linked register bit-field values for corresponding ePWM. An example of using the EPWMXLINK is linking ePWM2 CMPA with CMPA of ePWM1 through ePWM2's EPWMXLINK[CMPELINK] register bit-field. In this case, a write to CMPA of ePWM1 also changes the CMPA value for ePWM2.

#### 26.4.6 Time-Base Counter Modes and Timing Waveforms

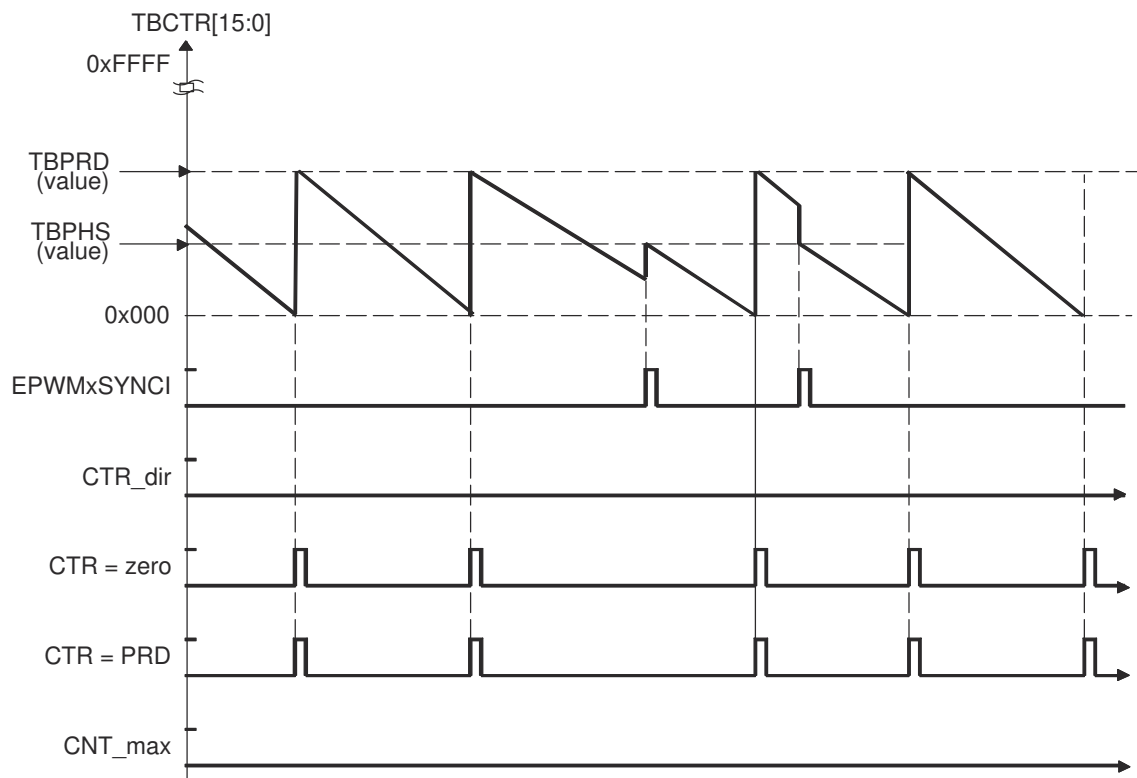
The time-base counter operates in one of four modes:

- Up-count mode that is asymmetrical
- Down-count mode that is asymmetrical
- Up-down-count that is symmetrical
- Frozen where the time-base counter is held constant at the current value

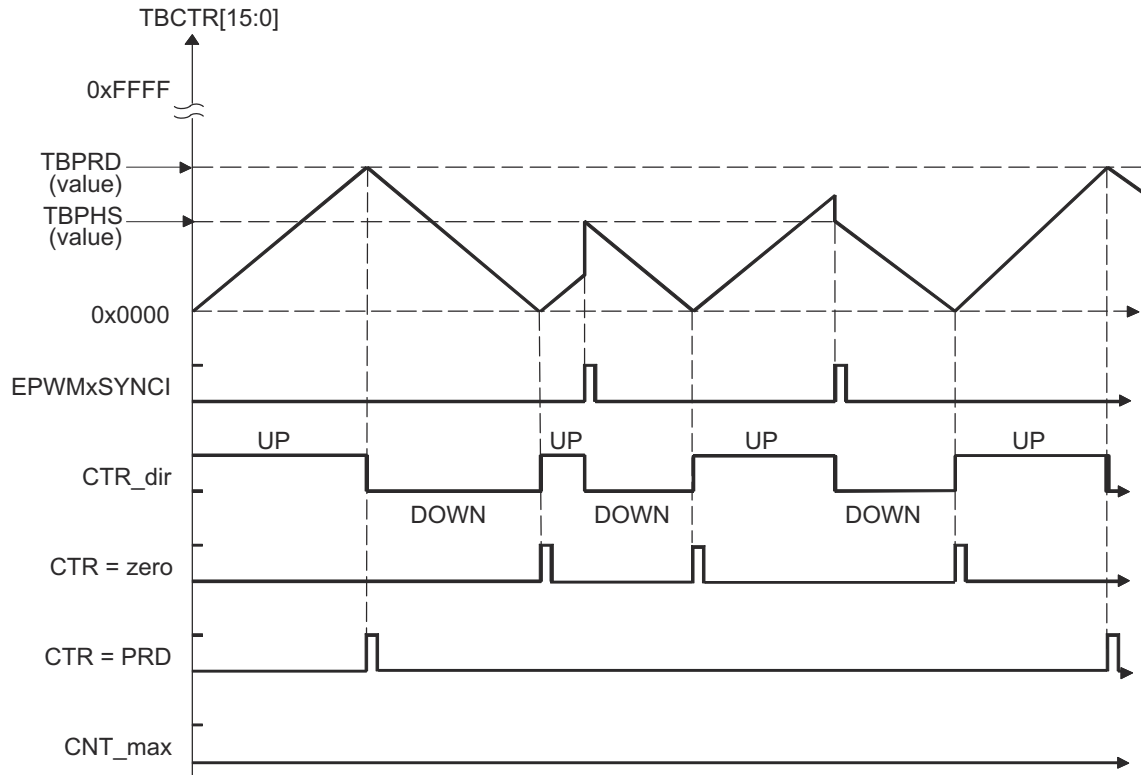
To illustrate the operation of the first three modes, the following timing diagrams show when events are generated and how the time-base responds to an EPWMxSYNCl signal.



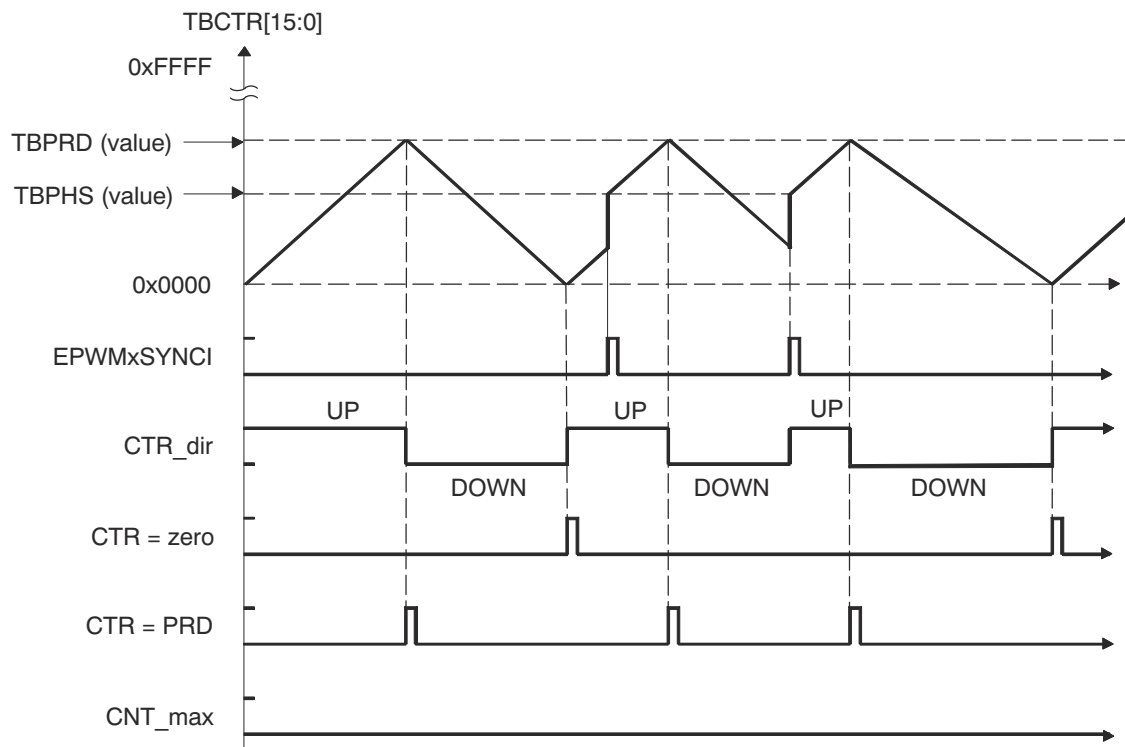
**Figure 26-9. Time-Base Up-Count Mode Waveforms**



**Figure 26-10. Time-Base Down-Count Mode Waveforms**



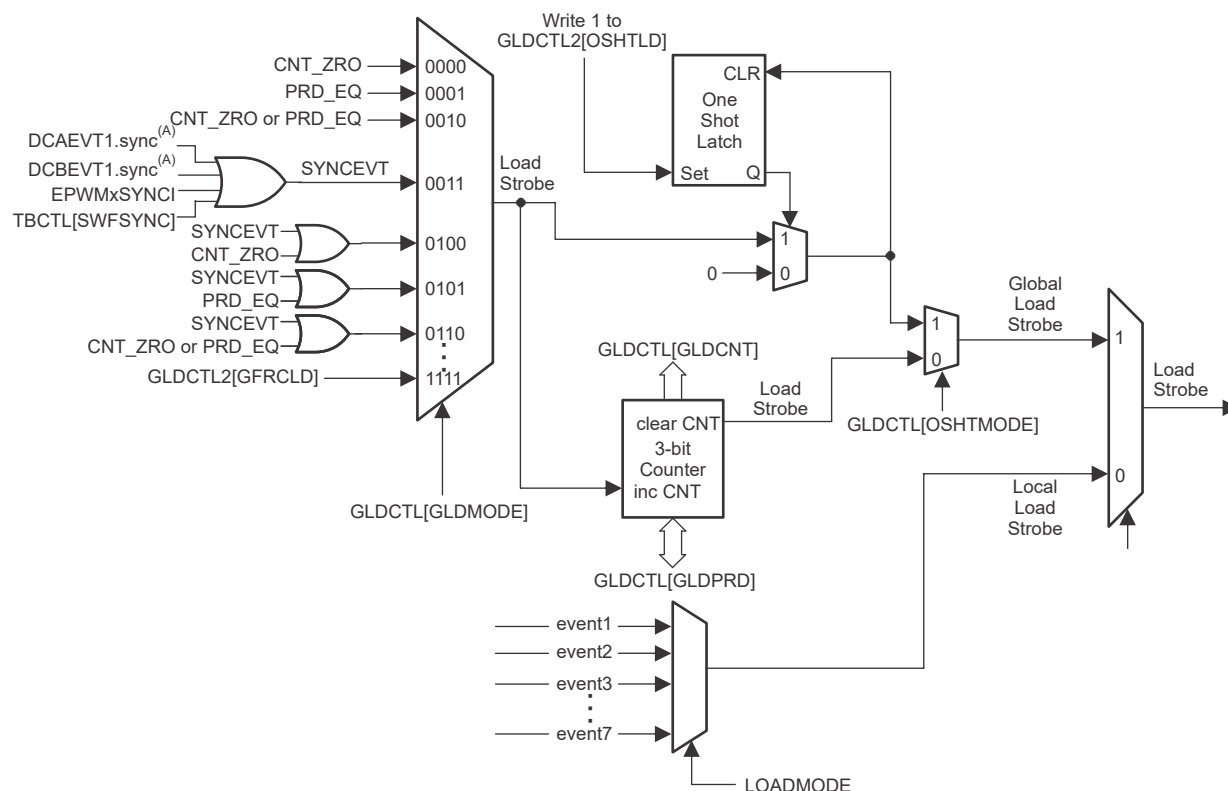
**Figure 26-11. Time-Base Up-Down-Count Waveforms, TBCTL[PHSDIR = 0] Count Down On Synchronization Event**



**Figure 26-12. Time-Base Up-Down Count Waveforms, TBCTL[PHSDIR = 1] Count Up On Synchronization Event**

## 26.4.7 Global Load

Figure 26-13 shows the signals and registers associated with the global load feature.



**Figure 26-13. Global Load: Signals and Registers**

### Note

The SYNCEVT signal is only propagated through when PHSEN is SET.

When this feature is enabled, the transfer of contents from the shadow register to the active register, for all registers that have this mode enabled, occurs at the same event as defined by the configuration bits in Global Shadow to Active Load Control Register (GLDCTL[GLDMODE]). When GLDCTL[GLD] = 1, shadow to active load event selection bits for individual shadowed registers are ignored and global load mode takes effect for the corresponding registers enabled by GLDCFG[REGx].

When GLDCTL[GLD] = 1 and GLDCFG[REGx] = 0, global load mode does not affect the corresponding register (REGx). Shadow to active load event selection bits for individual shadowed registers decide how the transfer of contents from shadow register to active register takes place.

### 26.4.7.1 Global Load Pulse Pre-Scalar

This feature provides the capability to choose shadow to active transfers to happen once in 'N' occurrences of selected global load pulse (GLDCTL[GLDMODE]). This pre-scale functionality is not available for registers that cannot or are not configured to use the global load mechanism (that is, GLDCTL[GLD] = 0 or GLDCFG[REGx] = 0).



### 26.4.7.2 One-Shot Load Mode

This feature allows users to cause the shadow register to active register transfers to occur once. When  $GLDCTL2[OSHTLD] = 1$  the shadow to active register transfer, for registers that are configured to use the global load mechanism, takes place on the event selected by  $GLDCTL[GLDMODE]$ .

Software force loading of contents from shadow register to active register is possible by using  $GLDCTL2[GFRCLD]$ . The  $GLDCTL2$  register can also be linked across multiple PWM modules by using  $EPWMXLINK[GLDCTL2LINK]$ . This, along with the one-shot load mode feature discussed above, provides a method to correctly update multiple PWM registers in one or more PWM modules at certain PWM events or, if desired, in the same clock cycle. This is very useful in variable frequency applications and/or multi-phase interleaved applications.

#### Note

One-shot load mode must not be used when high-resolution mode is enabled.

### 26.4.7.3 One-Shot Sync Mode

To enable the one-shot sync mode to generate a SYNCOUT pulse, configure the  $TBCTL2[OSHTSYNCMODE]$  bit and set the  $TBCTL2[OSHTSYNC]$  bit as shown in Figure 26-14.

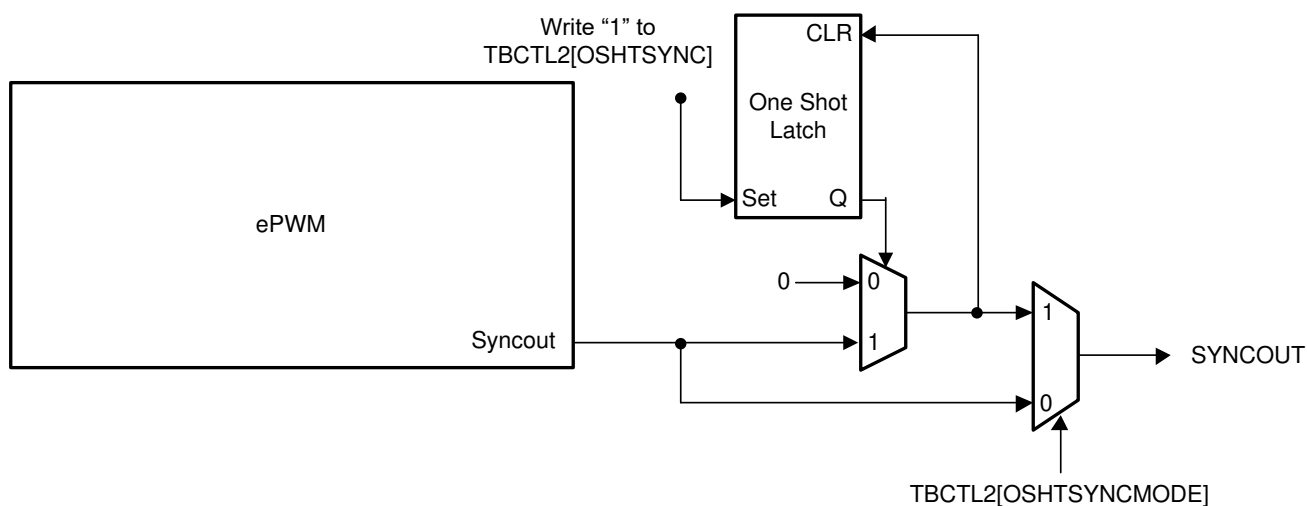
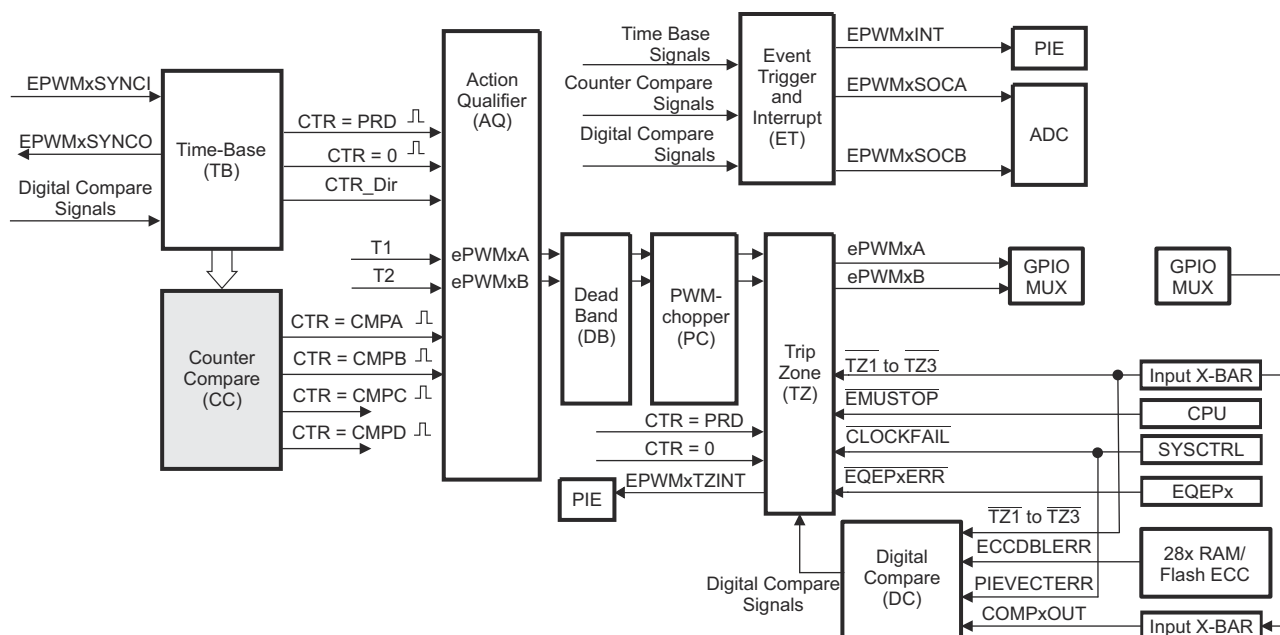


Figure 26-14. One-Shot Sync Mode

## 26.5 Counter-Compare (CC) Submodule

Figure 26-15 illustrates the counter-compare submodule within the ePWM.



**Figure 26-15. Counter-Compare Submodule**

### 26.5.1 Purpose of the Counter-Compare Submodule

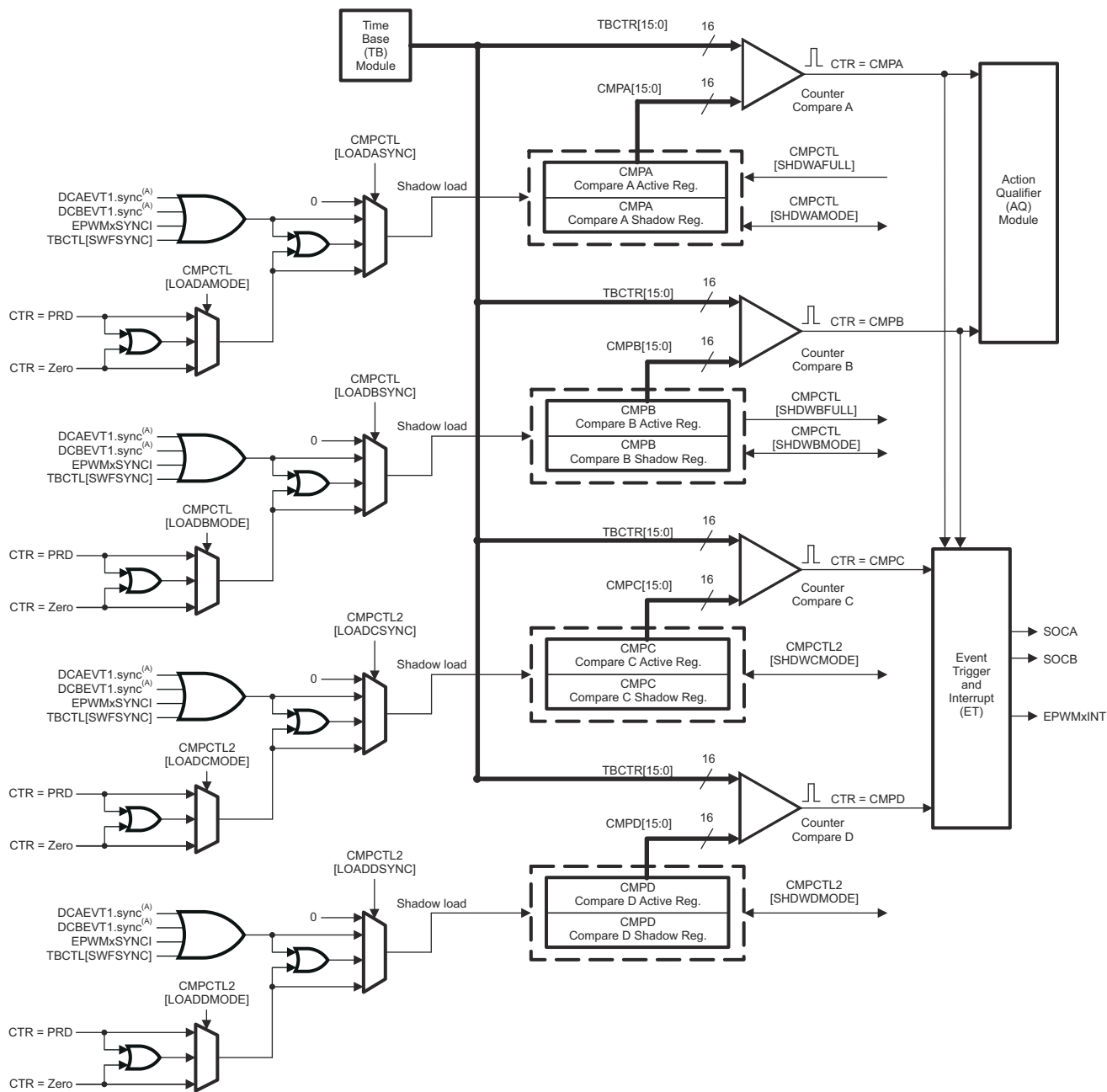
The counter-compare submodule takes as input the time-base counter value. This value is continuously compared to the counter-compare A (CMPA), counter-compare B (CMPB), counter-compare C (CMPC), and counter-compare D (CMPD) registers. When the time-base counter is equal to one of the compare registers, the counter-compare unit generates an appropriate event.

The counter-compare:

- Generates events based on programmable time stamps using the CMPA, CMPB, CMPC, and CMPD registers:
  - CTR = CMPA: Time-base counter equals counter-compare A register (TBCTR = CMPA)
  - CTR = CMPB: Time-base counter equals counter-compare B register (TBCTR = CMPB)
  - CTR = CMPC: Time-base counter equals counter-compare C register (TBCTR = CMPC)
  - CTR = CMPD: Time-base counter equals counter-compare D register (TBCTR = CMPD)
- Controls the PWM duty cycle, if the action-qualifier submodule is configured appropriately using counter-compare A (CMPA) and counter-compare B (CMPB)
- Shadows new compare values to prevent corruption or glitches during the active PWM cycle

## 26.5.2 Controlling and Monitoring the Counter-Compare Submodule

The counter-compare submodule operation is shown in Figure 26-16.



- A. These events are generated by the ePWM digital compare (DC) submodule based on the levels of the TRIPIN inputs (for example, CMPSSx and TZ signals).

**Figure 26-16. Detailed View of the Counter-Compare Submodule**

### 26.5.3 Operational Highlights for the Counter-Compare Submodule

The counter-compare submodule is responsible for generating events that can be used in the action-qualifier and event-trigger submodules. There are four independent compare events:

1. CTR = CMPA: Time-base counter equal to counter-compare A register (TBCTR = CMPA).
2. CTR = CMPB: Time-base counter equal to counter-compare B register (TBCTR = CMPB).
3. CTR = CMPC: Time-base counter equal to counter-compare C register (TBCTR = CMPC). This event can be used to generate an event in the event trigger submodule only.
4. CTR = CMPD: Time-base counter equal to counter-compare D register (TBCTR = CMPD). This event can be used to generate an event in the event trigger submodule only

For up-count or down-count mode, each event occurs only once per cycle. For up-down count mode, each event occurs twice per cycle if the compare value is between 0x00-TBPRD; and once per cycle if the compare value is equal to 0x00 or equal to TBPRD. These events are applied to the action-qualifier submodule where the events are qualified by the counter direction and converted into actions if enabled. Refer to [Section 26.6.1](#) for more details.

The counter-compare registers CMPA and CMPB each have an associated shadow register. Shadowing provides a way to keep updates to the registers synchronized with the hardware. When shadowing is used, updates to the active registers only occur at strategic points. This prevents corruption or spurious operation due to the register being asynchronously modified by software. The memory address of the active register and the shadow register is identical. The register that is written to or read from is determined by the CMPCTL[SHDWAMODE] and CMPCTL[SHDWBMODE] bits. These bits enable and disable the CMPC shadow register and CMPD shadow register, respectively. The behavior of the two load modes is:

#### Shadow Mode:

The shadow mode for the CMPA is enabled by clearing the CMPCTL[SHDWAMODE] bit and the shadow register for CMPB is enabled by clearing the CMPCTL[SHDWBMODE] bit. Shadow mode is enabled by default for both CMPA and CMPB.

If the shadow register is enabled then the content of the shadow register is transferred to the active register on one of the following events as specified by the CMPCTL[LOADAMODE], CMPCTL[LOADBMODE], CMPCTL[LOADASYNC], and CMPCTL[LOADBSYNC] register bits:

- CTR = PRD: Time-base counter equal to the period (TBCTR = TBPRD).
- CTR = Zero: Time-base counter equal to zero (TBCTR = 0x00)
- Both CTR = PRD and CTR = Zero
- SYNC event caused by DCAEVT1 or DCBEVT1 or EPWMxSYNCl or TBCTL[SWFSYNC]
- Both SYNC event or a selection made by LOADAMODE/LOADBMODE

Only the active register contents are used by the counter-compare submodule to generate events to be sent to the action-qualifier.

#### Note

Refer to [Section 26.6.5](#) for valid configurations of CMPA/CMPB and LOADAMODE/LOADBMODE.

#### Immediate Load Mode:

If the immediate load mode is selected (that is, CMPCTL[SHDWAMODE] = 1 or CMPCTL[SHDWBMODE] = 1), then a read from or a write to the register goes directly to the active register.

## Additional Comparators

The counter-compare submodule on ePWMs type 2 and later are responsible for generating two additional independent compare events based on two compare registers, which is fed to Event Trigger submodule:

1. CTR = CMPC: Time-base counter equal to counter-compare C register (TBCTR = CMPC).
2. CTR = CMPD: Time-base counter equal to counter-compare D register (TBCTR = CMPD).

The counter-compare registers CMPC and CMPD each have an associated shadow register. By default this register is shadowed. The memory address of the active register and the shadow register is identical. The value in the active CMPC and CMPD register is compared to the time-base counter (TBCTR). When the values are equal, the counter compare module generates a “time-base counter equal to counter compare C or counter compare D ” event respectively. Shadowing of this register is enabled and disabled by the CMPCTL2[SHDWCMODE] and CMPCTL2[SHDWDMODE] bit. These bits enable and disable the CMPC shadow register and CMPD shadow register respectively. The behavior of the two load modes is described below:

### Shadow Mode:

The shadow mode for the CMPC is enabled by clearing the CMPCTL2[SHDWCMODE] bit and the shadow register for CMPD is enabled by clearing the CMPCTL2[SHDWDMODE] bit. Shadow mode is enabled by default for both CMPC and CMPD.

If the shadow register is enabled then the content of the shadow register is transferred to the active register on one of the following events as specified by the CMPCTL2[LOADCMODE], CMPCTL2[LOADDMODE], CMPCTL2[LOADCSYNC], and CMPCTL2[LOADDSYNC] register bits:

- CTR = PRD: Time-base counter equal to the period (TBCTR = TBPRD).
- CTR = Zero: Time-base counter equal to zero (TBCTR = 0x00)
- Both CTR = PRD and CTR = Zero
- SYNC event caused by DCAEVT1 or DCBEVT1 or EPWMxSYNCl or TBCTL[SWFSYNC]
- Both SYNC event or a selection made by LOADCMODE/LOADDMODE

Only the active register contents are used by the counter-compare submodule to generate events to be sent to the action-qualifier.

### Immediate Load Mode:

If the immediate load mode is selected (that is, CMPCTL2[SHDWCMODE] = 1 or CMPCTL2[SHDWDMODE] = 1), then a read from or a write to the register goes directly to the active register.

## Global Load Support

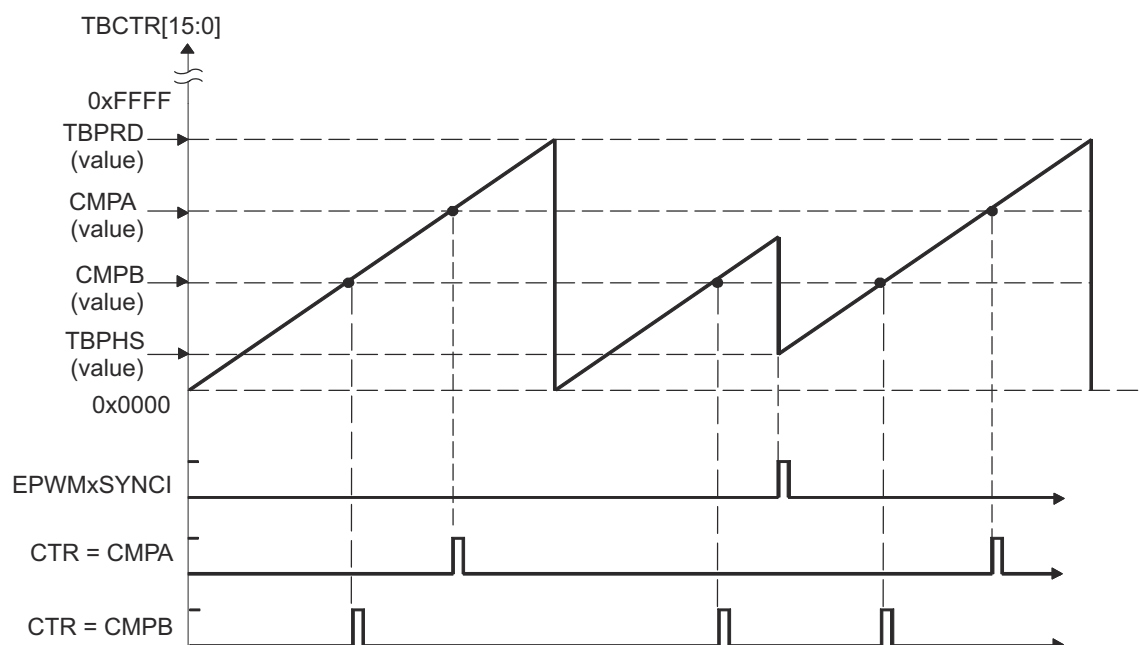
The global load control mechanism can also be used for all counter-compare registers by configuring the appropriate bits in the global load configuration register (GLDCFG). When the global load mode is selected the transfer of contents from shadow register to active register, for all registers that have this mode enabled, occurs at the same event as defined by the configuration bits in the Global Shadow to Active Load Control Register (GLDCTL). The global load control mechanism is explained in [Section 26.4.7](#).

### 26.5.4 Count Mode Timing Waveforms

The counter-compare module can generate compare events in all three count modes:

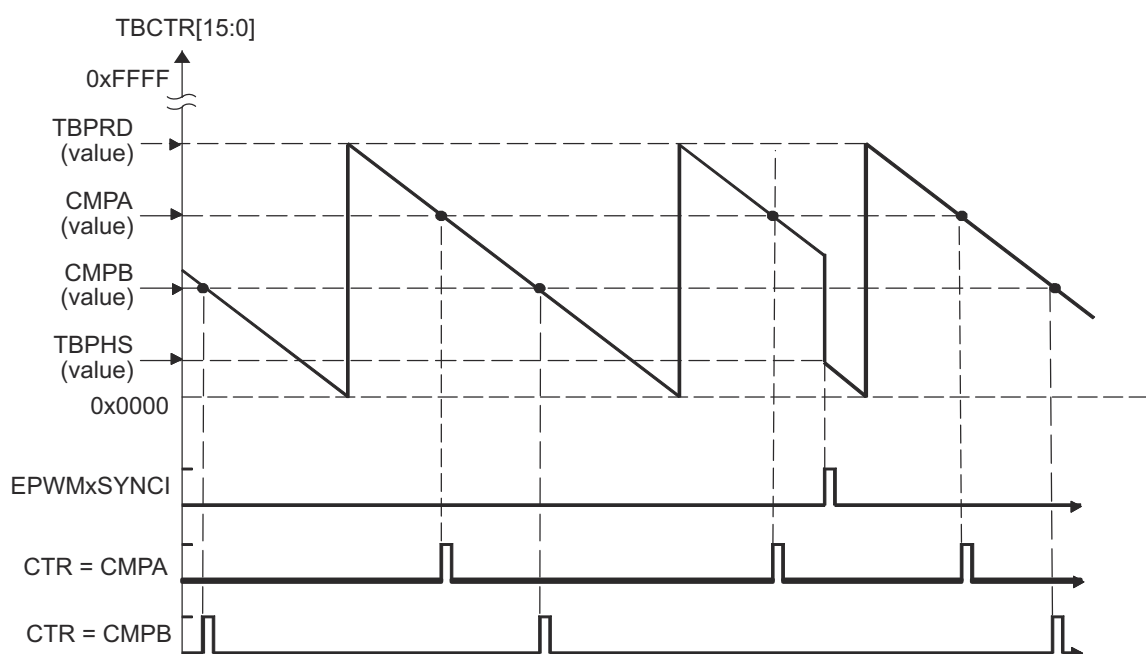
- Up-count mode: used to generate an asymmetrical PWM waveform.
- Down-count mode: used to generate an asymmetrical PWM waveform.
- Up-down-count mode: used to generate a symmetrical PWM waveform.

To best illustrate the operation of the first three modes, the timing diagrams in [Figure 26-17](#) through [Figure 26-20](#) show when events are generated and how the EPWMxSYNCl signal interacts.

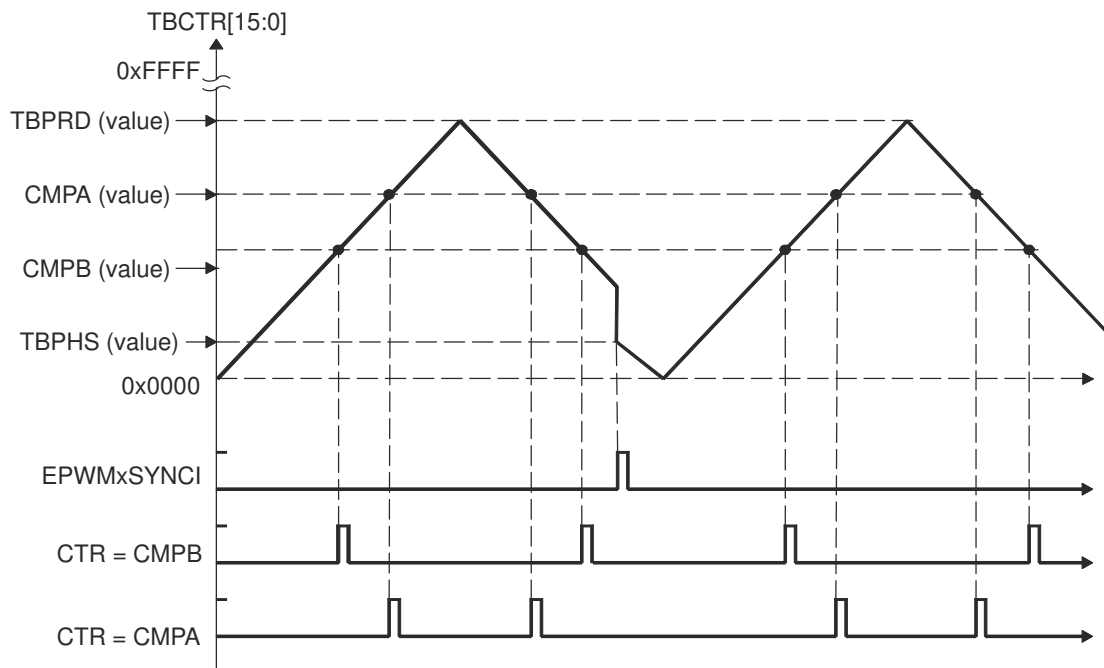


An EPWMxSYNCl external synchronization event can cause a discontinuity in the TBCTR count sequence. This can lead to a compare event being skipped. This skipping is considered normal operation and must be taken into account.

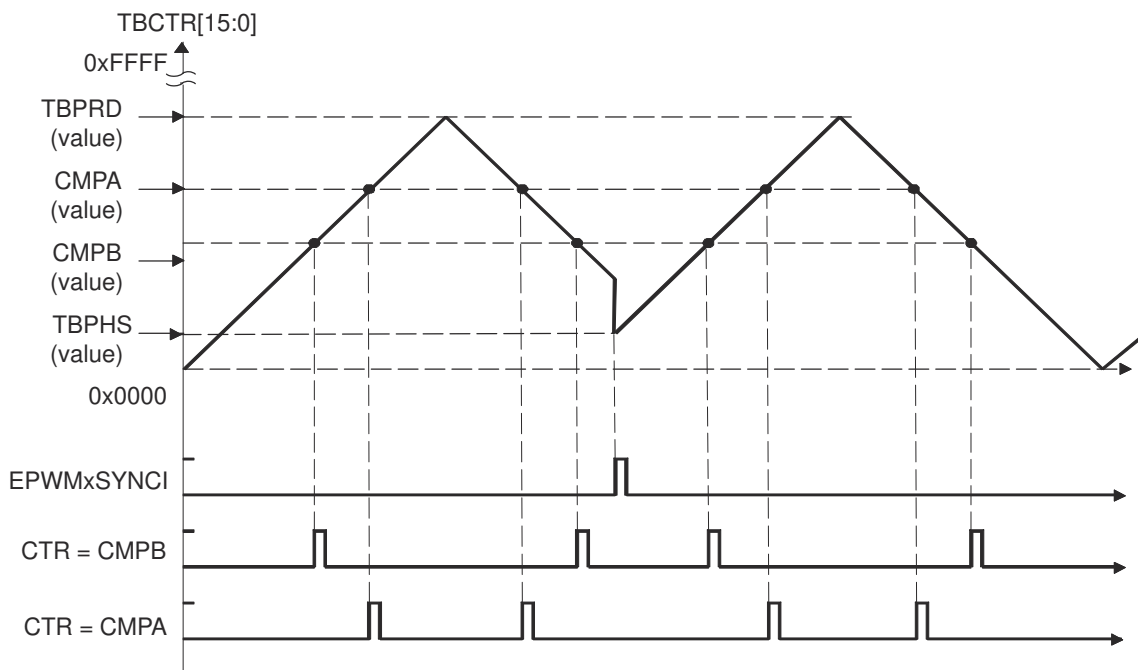
**Figure 26-17. Counter-Compare Event Waveforms in Up-Count Mode**



**Figure 26-18. Counter-Compare Events in Down-Count Mode**



**Figure 26-19. Counter-Compare Events In Up-Down-Count Mode,  $TBCTL[PHSDIR] = 0$  Count Down On Synchronization Event**

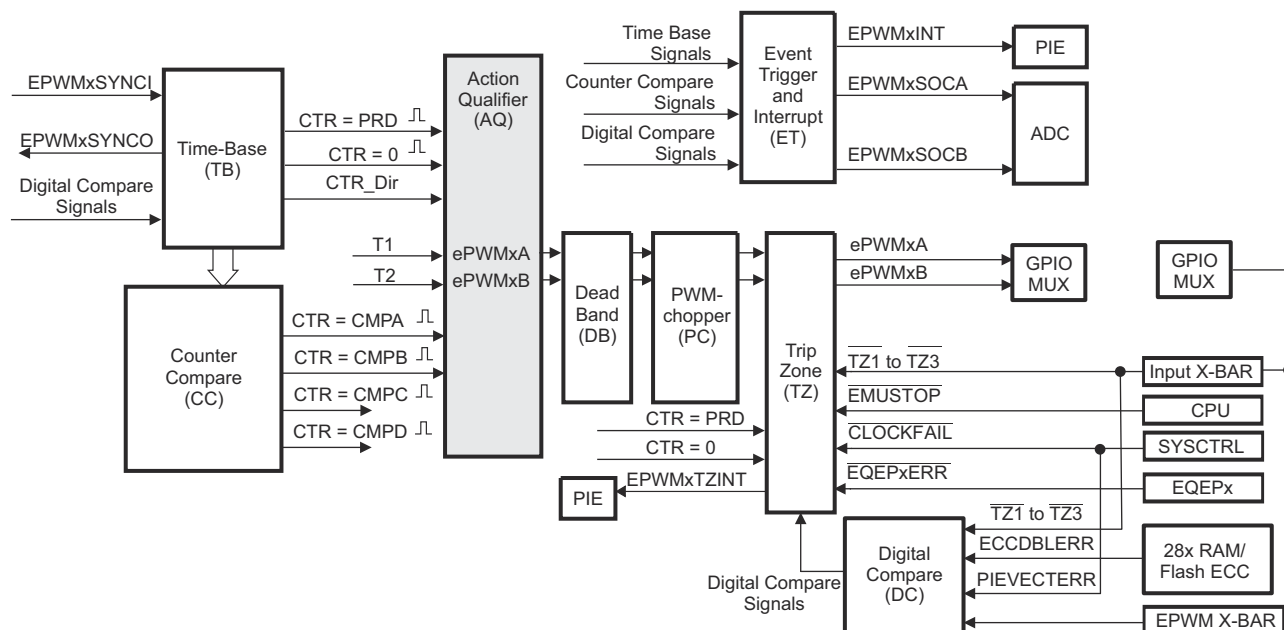


**Figure 26-20. Counter-Compare Events In Up-Down-Count Mode,  $TBCTL[PHSDIR] = 1$  Count Up On Synchronization Event**

## 26.6 Action-Qualifier (AQ) Submodule

The action-qualifier submodule has the most important role in waveform construction and PWM generation. The action-qualifier submodule decides which events are converted into various action types, thereby, producing the required switched waveforms at the EPWMxA and EPWMxB outputs.

Figure 26-21 illustrates the action-qualifier submodule within the ePWM.



**Figure 26-21. Action-Qualifier Submodule**

### 26.6.1 Purpose of the Action-Qualifier Submodule

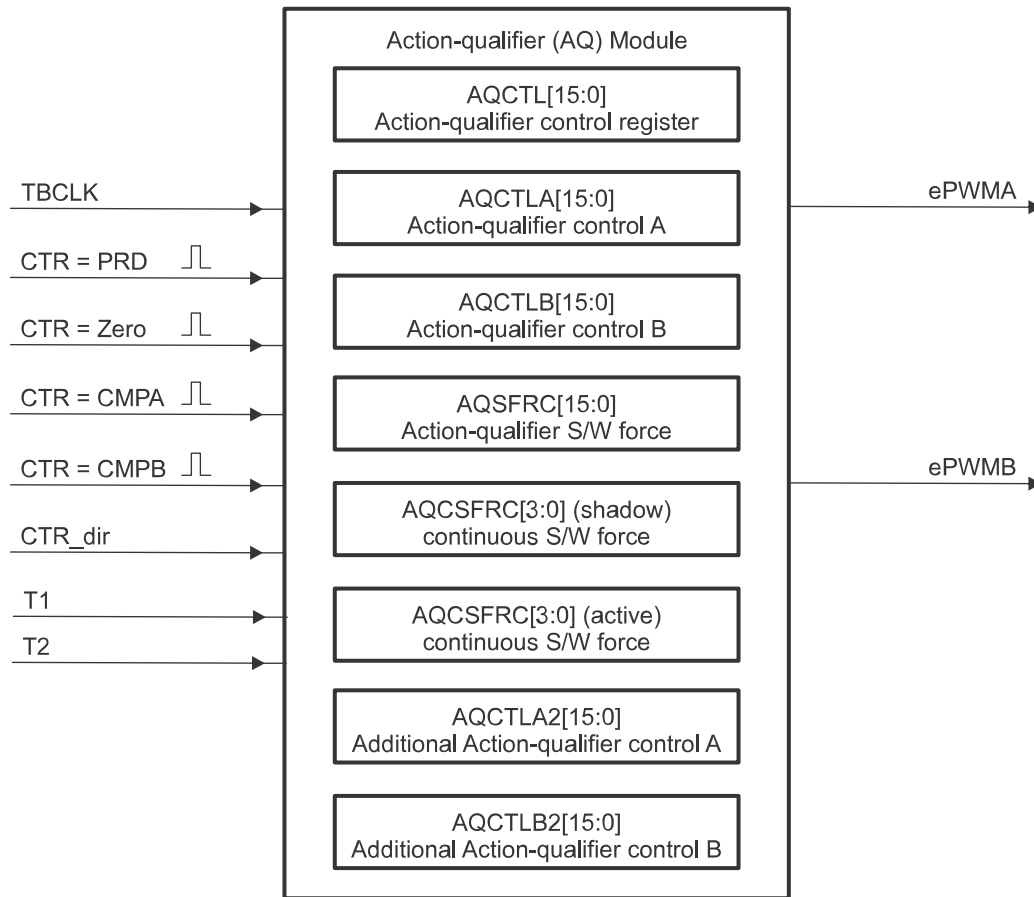
The action-qualifier submodule is responsible for the following:

- Qualifying and generating actions (set, clear, toggle) based on the following events:
  - CTR = PRD: Time-base counter equal to the period (TBCTR = TBPRD).
  - CTR = Zero: Time-base counter equal to zero (TBCTR = 0x00)
  - CTR = CMPA: Time-base counter equal to the counter-compare A register (TBCTR = CMPA)
  - CTR = CMPB: Time-base counter equal to the counter-compare B register (TBCTR = CMPB)
- T1, T2 events: Trigger events based on comparator, trip or syncin events
- Managing priority when these events occur concurrently
- Providing independent control of events when the time-base counter is increasing and when it is decreasing



## 26.6.2 Action-Qualifier Submodule Control and Status Register Definitions

The action-qualifier submodule operation is shown in [Figure 26-22](#) and monitored by way of the registers in [Section 26.17](#).



**Figure 26-22. Action-Qualifier Submodule Inputs and Outputs**

For convenience, the possible input events are summarized again in [Table 26-4](#).

**Table 26-4. Action-Qualifier Submodule Possible Input Events**

| Signal                | Description                                      | Registers Compared |
|-----------------------|--------------------------------------------------|--------------------|
| CTR = PRD             | Time-base counter equal to the period value      | TBCTR = TBPRD      |
| CTR = Zero            | Time-base counter equal to 0                     | TBCTR = 0x00       |
| CTR = CMPA            | Time-base counter equal to the counter-compare A | TBCTR = CMPA       |
| CTR = CMPB            | Time-base counter equal to the counter-compare B | TBCTR = CMPB       |
| T1 event              | Based on comparator, trip, or syncin events      | None               |
| T2 event              | Based on comparator, trip, or syncin events      | None               |
| Software forced event | Asynchronous event initiated by software         |                    |

The software forced action is a useful asynchronous event. This control is handled by the AQSFRC and AQCSFRC registers.

### Note

If the CSFA is not used in shadow mode, the RLDCSF bit must be configured to disable shadow mode.

The action-qualifier submodule controls how the two outputs EPWMxA and EPWMxB behave when a particular event occurs. The event inputs to the action-qualifier submodule are further qualified by the counter direction (up or down). This allows for independent action on outputs on both the count-up and count-down phases.

The possible actions imposed on outputs EPWMxA and EPWMxB are:

- **Set High:** Set output EPWMxA or EPWMxB to a high level.
- **Clear Low:** Set output EPWMxA or EPWMxB to a low level.
- **Toggle:** If EPWMxA or EPWMxB is currently pulled high, then pull the output low. If EPWMxA or EPWMxB is currently pulled low, then pull the output high.
- **Do Nothing:** Keep outputs EPWMxA and EPWMxB at same level as currently set. Although the "Do Nothing" option prevents an event from causing an action on the EPWMxA and EPWMxB outputs, this event can still trigger interrupts and ADC start of conversion. See the description in [Section 26.10](#) for details.

Actions are specified independently for either output (EPWMxA or EPWMxB). Any or all events can be configured to generate actions on a given output. For example, both CTR = CMPA and CTR = CMPB can operate on output EPWMxA. All qualifier actions are configured using the control registers found in the *ePWM Registers* section.

For clarity, the illustrations in this chapter use a set of symbolic actions. These symbols are summarized in [Figure 26-23](#). Each symbol represents an action as a marker in time. Some actions are fixed in time (zero and period) while the CMPA and CMPB actions are moveable and the time positions are programmed by way of the counter-compare A and B registers, respectively. To turn off or disable an action, use the "Do Nothing option"(the default at reset).

| SW<br>force | TB Counter equals |           |           |        | Trigger Events |    | Actions    |
|-------------|-------------------|-----------|-----------|--------|----------------|----|------------|
|             | Zero              | Comp<br>A | Comp<br>B | Period | T1             | T2 |            |
|             |                   |           |           |        |                |    | Do Nothing |
|             |                   |           |           |        |                |    | Clear Lo   |
|             |                   |           |           |        |                |    | Set Hi     |
|             |                   |           |           |        |                |    | Toggle     |

**Figure 26-23. Possible Action-Qualifier Actions for EPWMxA and EPWMxB Outputs**

The Action Qualifier Trigger Event Source Selection register (AQTSRCSEL) is used to select the source for T1 and T2 events. T1/T2 selection and configuration of a trip/digital-compare event in Action Qualifier submodule is independent of the configuration of that event in the Trip-Zone submodule. A particular trip event can or cannot

be configured to cause trip action in the Trip Zone submodule, but the same event can be used by the Action Qualifier to generate T1/T2 for controlling PWM generation.

### 26.6.3 Action-Qualifier Event Priority

It is possible for the ePWM action qualifier to receive more than one event at the same time. In this case, events are assigned a priority by the hardware. The general rule is events occurring later in time have a higher priority and software forced events always have the highest priority. The event priority levels for up-down count mode are shown in [Table 26-5](#). A priority level of 1 is the highest priority and level 10 is the lowest. The priority changes slightly depending on the direction of TBCTR.

**Table 26-5. Action-Qualifier Event Priority for Up-Down-Count Mode**

| Priority Level | Event If TBCTR is Incrementing<br>TBCTR = Zero up to TBCTR = TBPRD | Event If TBCTR is Decrementing<br>TBCTR = TBPRD down to TBCTR = 1 |
|----------------|--------------------------------------------------------------------|-------------------------------------------------------------------|
| 1 (Highest)    | Software forced event                                              | Software forced event                                             |
| 2              | T1 on up-count (T1U)                                               | T1 on down-count (T1D)                                            |
| 3              | T2 on up-count (T2U)                                               | T2 on down-count (T2D)                                            |
| 4              | Counter equals CMPB on up-count (CBU)                              | Counter equals CMPB on down-count (CBD)                           |
| 5              | Counter equals CMPA on up-count (CAU)                              | Counter equals CMPA on down-count (CAD)                           |
| 6 (Lowest)     | Counter equals zero                                                | Counter equals period (TBPRD)                                     |

[Table 26-6](#) shows the action-qualifier priority for up-count mode. In this case, the counter direction is always defined as up; therefore, down-count events never are taken.

**Table 26-6. Action-Qualifier Event Priority for Up-Count Mode**

| Priority Level | Event                                   |
|----------------|-----------------------------------------|
| 1 (Highest)    | Software forced event                   |
| 2              | Counter equal to period (TBPRD)         |
| 3              | T1 on up-count (T1U)                    |
| 4              | T2 on up-count (T2U)                    |
| 5              | Counter equal to CMPB on up-count (CBU) |
| 6              | Counter equal to CMPA on up-count (CAU) |
| 7 (Lowest)     | Counter equal to Zero                   |

[Table 26-7](#) shows the action-qualifier priority for down-count mode. In this case, the counter direction is always defined as down; therefore, up-count events never are taken.

**Table 26-7. Action-Qualifier Event Priority for Down-Count Mode**

| Priority Level | Event                                     |
|----------------|-------------------------------------------|
| 1 (Highest)    | Software forced event                     |
| 2              | Counter equal to Zero                     |
| 3              | T1 on down-count (T1D)                    |
| 4              | T2 on down-count (T2D)                    |
| 5              | Counter equal to CMPB on down-count (CBD) |
| 6              | Counter equal to CMPA on down-count (CAD) |
| 7 (Lowest)     | Counter equal to period (TBPRD)           |

It is possible to set the compare value greater than the period. In this case, the action takes place as shown in [Table 26-8](#).

**Table 26-8. Behavior if CMPA/CMPB is Greater than the Period**

| Counter Mode       | Compare on Up-Count Event<br>CAD/CBD                                                                                                                                                                             | Compare on Down-Count Event<br>CAD/CBD                                                                                                                                                                           |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Up-Count Mode      | If $CMPA/CMPB \leq TBPRD$ period, then the event occurs on a compare match ( $TBCTR = CMPA$ or $CMPB$ ).<br>If $CMPA/CMPB > TBPRD$ , then the event does not occur.                                              | Never occurs.                                                                                                                                                                                                    |
| Down-Count Mode    | Never occurs.                                                                                                                                                                                                    | If $CMPA/CMPB < TBPRD$ , the event occurs on a compare match ( $TBCTR = CMPA$ or $CMPB$ ).<br>If $CMPA/CMPB \geq TBPRD$ , the event occurs on a period match ( $TBCTR = TBPRD$ ).                                |
| Up-Down Count Mode | If $CMPA/CMPB < TBPRD$ and the counter is incrementing, the event occurs on a compare match ( $TBCTR = CMPA$ or $CMPB$ ).<br>If $CMPA/CMPB \geq TBPRD$ , the event occurs on a period match ( $TBCTR = TBPRD$ ). | If $CMPA/CMPB < TBPRD$ and the counter is decrementing, the event occurs on a compare match ( $TBCTR = CMPA$ or $CMPB$ ).<br>If $CMPA/CMPB \geq TBPRD$ , the event occurs on a period match ( $TBCTR = TBPRD$ ). |

### 26.6.4 AQCTLA and AQCTLB Shadow Mode Operations

To enable Action Qualifier mode changes which must occur at the end of a period even when the phase changes, shadowing of the AQCTLA and AQCTLB registers has been added on ePWMs type 2 and later. Additionally, shadow to active load on SYNC of these registers is supported as well. Shadowing of this register is enabled and disabled by the AQCTL[SHDWAQAMODE] and AQCTL[SHDWAQBMODE] bits. These bits enable and disable the AQCTLA shadow register and AQCTLB shadow register, respectively. The behavior of the two load modes is:

#### Shadow Mode:

The shadow mode for the AQCTLA is enabled by setting the AQCTL[SHDWAQAMODE] bit, and the shadow register for AQCTLB is enabled by setting the AQCTL[SHDWAQBMODE] bit. Shadow mode is disabled by default for both AQCTLA and AQCTLB

If the shadow register is enabled, then the content of the shadow register is transferred to the active register on one of the following events as specified by the AQCTL[LDAQAMODE], AQCTL[LDAQBMODE], AQCTL[LDAQASYNC], and AQCTL[LDAQBSYNC] register bits:

- CTR = PRD: Time-base counter equal to the period ( $TBCTR = TBPRD$ ).
- CTR = Zero: Time-base counter equal to zero ( $TBCTR = 0x00$ )
- Both CTR = PRD and CTR = Zero
- SYNC event caused by DCAEVT1 or DCBEVT1 or EPWMxSYNCl or TBCTL[SWFSYNC]
- Both SYNC event or a selection made by LDAQAMODE/LDAQBMODE

#### Global Load Support

Global load control mechanism can also be used for AQCTLA:AQCTLA2, AQCTLB:AQCTLB2, and AQCSFRC registers by configuring the appropriate bits in the global load configuration register (GLDCFG). When global load mode is selected, the transfer of contents from shadow register to active register for all registers that have this mode enabled, occurs at the same event as defined by the configuration bits in the Global Shadow to Active Load Control Register (GLDCTL). The global load control mechanism is explained in [Section 26.4.7](#).

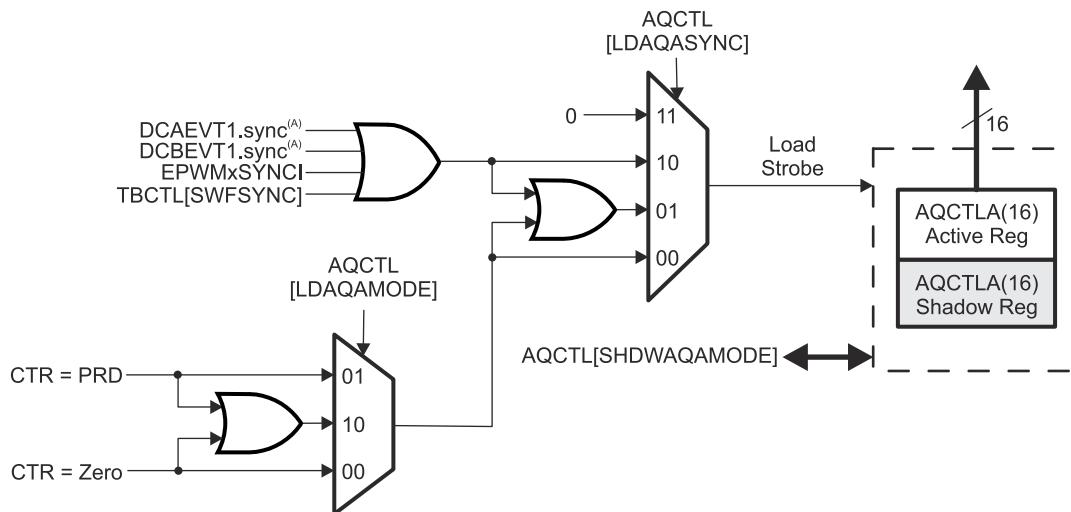
#### Immediate Load Mode:

If the immediate load mode is selected (that is, AQCTL[SHDWAQAMODE] = 0 or AQCTL[SHDWAQBMODE] = 0), then a read from or a write to the register goes directly to the active register. See [Figure 26-24](#) and [Figure 26-25](#).

### Note

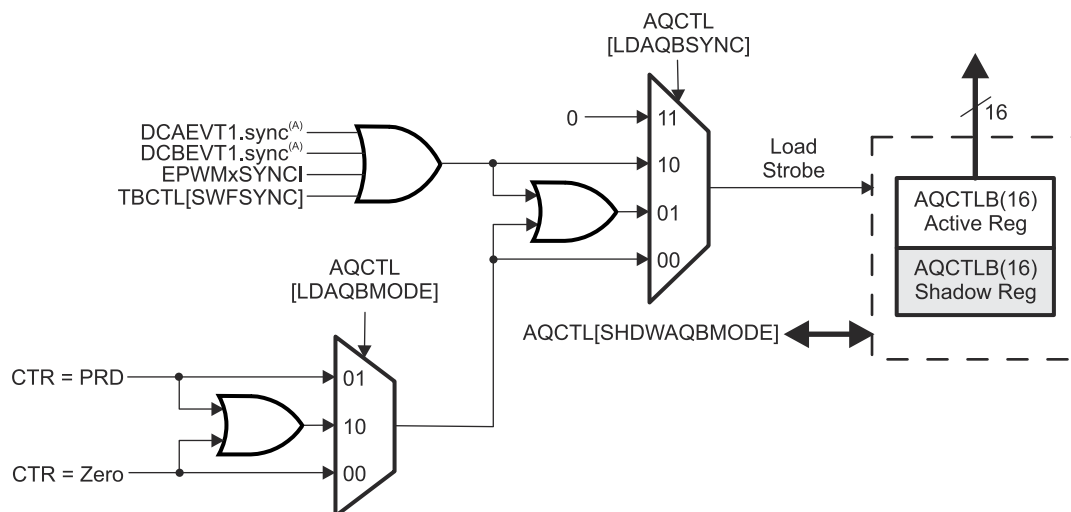
Shadow to Active Load of Action Qualifier Output A/B Control Register [AQCTLA and AQCTLB] on CMPA = 0 or CMPB = 0 boundary

If the Counter-Compare A Register (CMPA) or Counter-Compare B Register (CMPB) is set to a value of 0 and the action qualifier action on AQCTLA and AQCTLB is configured to occur in the same instant as a shadow to active load (that is, CMPA=0 and AQCTLA shadow to active load on TBCTR=0 using AQCTL register LDAQAMODE and LDAQAMODE bits), then both events enter contention. It is recommended to use a Non-Zero Counter-Compare when using Shadow to Active Load of Action Qualifier Output A/B Control Register on TBCTR = 0 boundary.



- A. These events are generated by the ePWM digital compare (DC) submodule based on the levels of the TRIPIN inputs (for example, CMPSSx and TZ signals).

**Figure 26-24. AQCTL[SHDWAQAMODE]**



- A. These events are generated by the ePWM digital compare (DC) submodule based on the levels of the TRIPIN inputs (for example, CMPSSx and TZ signals).

**Figure 26-25. AQCTL[SHDWAQBMODE]**

## 26.6.5 Configuration Requirements for Common Waveforms

### Note

The waveforms in this chapter show the behavior of the ePWMs for a static compare register value. In a running system, the active compare registers (CMPA and CMPB) are typically updated from the respective shadow registers once every period. Specify when the update takes place: either when the time-base counter reaches zero or when the time-base counter reaches the period. There are some cases when the action based on the new value can be delayed by one period or the action based on the old value can take effect for an extra period. Some PWM configurations avoid this situation. These include, but are not limited to, the following:

#### Use up-down count mode to generate a symmetric PWM:

- If loading CMPA/CMPB on zero, then use CMPA/CMPB values greater than or equal to 1.
- If loading CMPA/CMPB on period, then use CMPA/CMPB values less than or equal to TBPRD-1.

This means there is always a pulse of at least one TBCLK cycle in a PWM period which, when very short, tend to be ignored by the system.

#### Use up-down count mode to generate an asymmetric PWM:

- To achieve 50%-0% asymmetric PWM use the following configuration: Load CMPA/CMPB on period and use the period action to clear the PWM and a compare-up action to set the PWM. Modulate the compare value from 0 to TBPRD to achieve 50%-0% PWM duty.

#### When using up-count mode to generate an asymmetric PWM:

- To achieve 0-100% asymmetric PWM, you **must** load CMPA/CMPB on TBPRD. When CMPA/CMPB is not loaded on TBCTR=PRD, boundary conditions can occur depending on the timing of the write and the value written to CMPA/CMPB. Use the Zero action to set the PWM and a compare-up action to clear the PWM. Modulate the compare value from 0 to TBPRD+1 to achieve 0-100% PWM duty.

#### When using up-count mode to generate an asymmetric PWM with deadband enabled:

- To achieve 0%-100% PWM use the following configuration: When the CMPA value is too close to 0 or PRD such that the following conditions are met ( $CMPX < \text{Deadband}$ ) or ( $CMPX > PRD - \text{Deadband}$ ), the actions specified by the AQCTL register for CMPX do not take effect. To avoid this, the AQCTL settings must be altered under these conditions only to generate either high or low pulses for both CAU or CAD events (both set or both clear). Make sure that this software update is occurring synchronous to the PWM carrier cycle, and shadow mode is enabled.

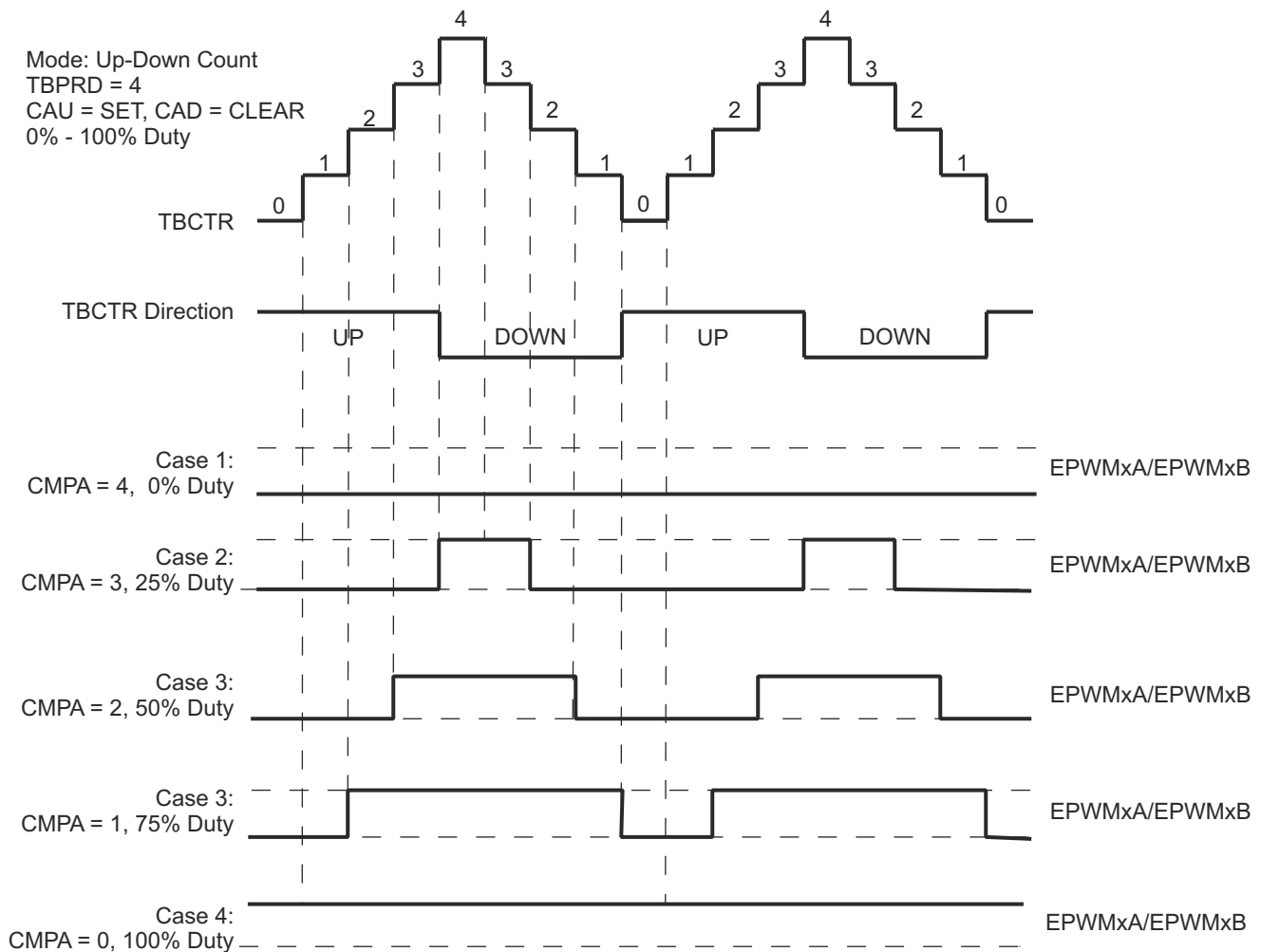
#### When using up-down count mode to generate an asymmetric PWM with deadband enabled:

- To achieve 0%-100% PWM use the following configuration: When the CMPA value is too close to 0 or PRD such that the following conditions are met ( $CMPX < \text{Deadband}/2$ ) or ( $CMPX > PRD - (\text{Deadband})/2$ ), the actions specified by the AQCTL register for CMPX do not take effect. To avoid this, the AQCTL settings must be altered under these conditions only to generate either high or low pulses for both CAU or CAD events (both set or both clear). Make sure that this software update is occurring synchronous to the PWM carrier cycle, and shadow mode is enabled.

See [Using Enhanced Pulse Width Modulator \(ePWM\) Module for 0-100% Duty Cycle Control](#).

Figure 26-26 shows how a symmetric PWM waveform can be generated using the up-down-count mode of the TBCTR. In this mode, 0%-100% DC modulation is achieved by using equal compare matches on the up count and down count portions of the waveform. In the example shown, CMPA is used to make the comparison. When the counter is incrementing, the CMPA match pulls the PWM output high. Likewise when the counter is decrementing, the compare match pulls the PWM signal low. When  $CMPA = 0$ , the PWM signal is high for the entire period giving a 100% duty waveform. When  $CMPA = TBPRD$ , the PWM signal is low achieving 0% duty.

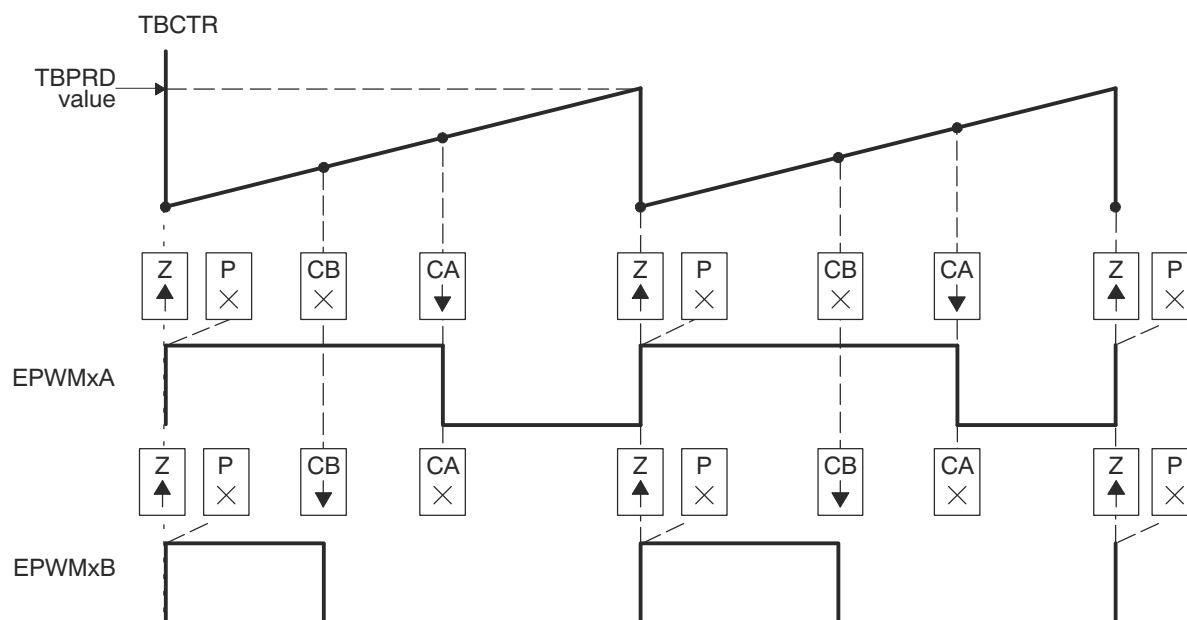
When using this configuration in practice, if loading CMPA/CMPB on zero, then use CMPA/CMPB values greater than or equal to 1. If loading CMPA/CMPB on period, then use CMPA/CMPB values less than or equal to TBPRD-1. This means there is always a pulse of at least one TBCLK cycle in a PWM period which, when very short, tend to be ignored by the system.



**Figure 26-26. Up-Down Count Mode Symmetrical Waveform**

The PWM waveforms in [Figure 26-27](#) through [Figure 26-32](#) show some common action-qualifier configurations. Some conventions used in the figures and examples are as follows:

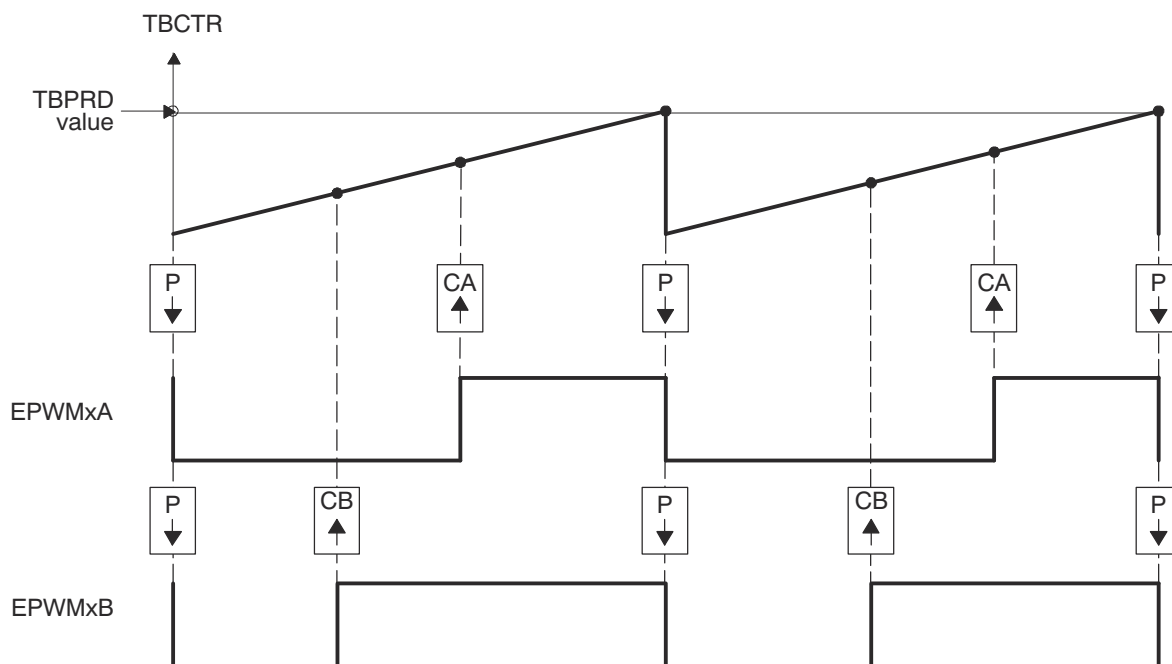
- TBPRD, CMPA, and CMPB refer to the value written in the respective registers. The active register, not the shadow register, is used by the hardware.
- CMPx, refers to either CMPA or CMPB.
- EPWMxA and EPWMxB refer to the output signals from ePWMx
- Up-Down means count-up-and-count-down mode, Up means up-count mode and Down means down-count mode
- Sym = Symmetric, Asym = Asymmetric



- PWM period =  $(TBPRD + 1) \times T_{TBCLK}$
- Duty modulation for EPWMxA is set by CMPA, and is active high (that is, high time duty proportional to CMPA).
- Duty modulation for EPWMxB is set by CMPB and is active high (that is, high time duty proportional to CMPB).
- The "Do Nothing" actions (X) are shown for completeness, but are not shown on subsequent diagrams.
- Actions at zero and period, although appearing to occur concurrently, are actually separated by one TBCLK period. TBCTR wraps from period to 0000.

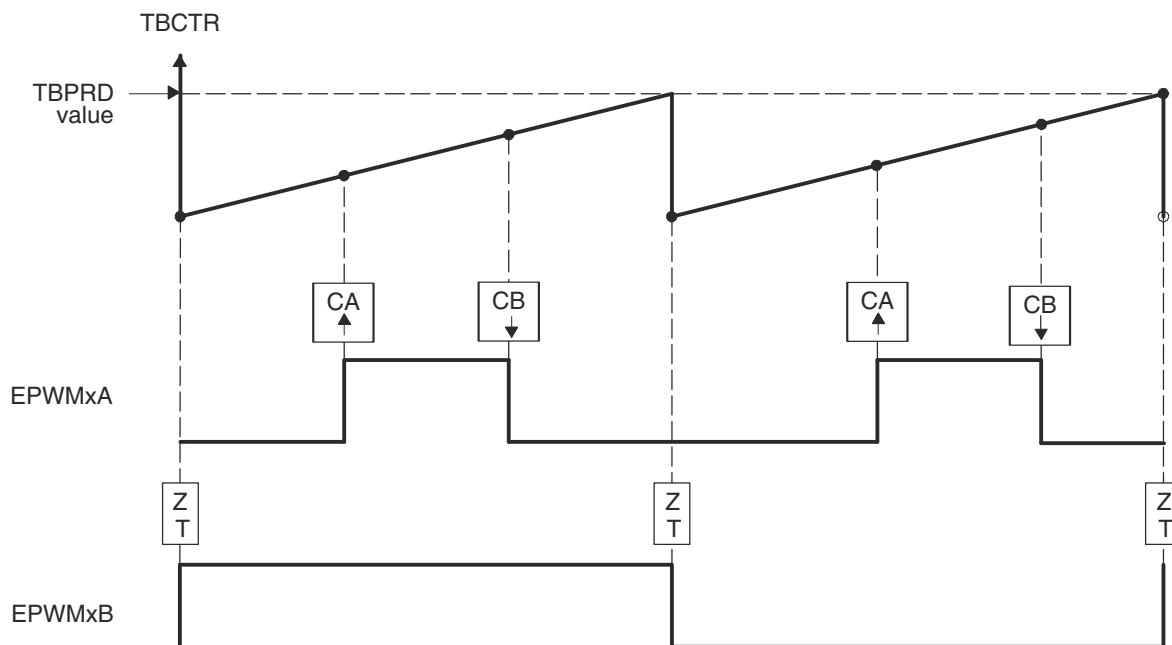
**Figure 26-27. Up, Single Edge Asymmetric Waveform, with Independent Modulation on EPWMxA and EPWMxB—Active High**





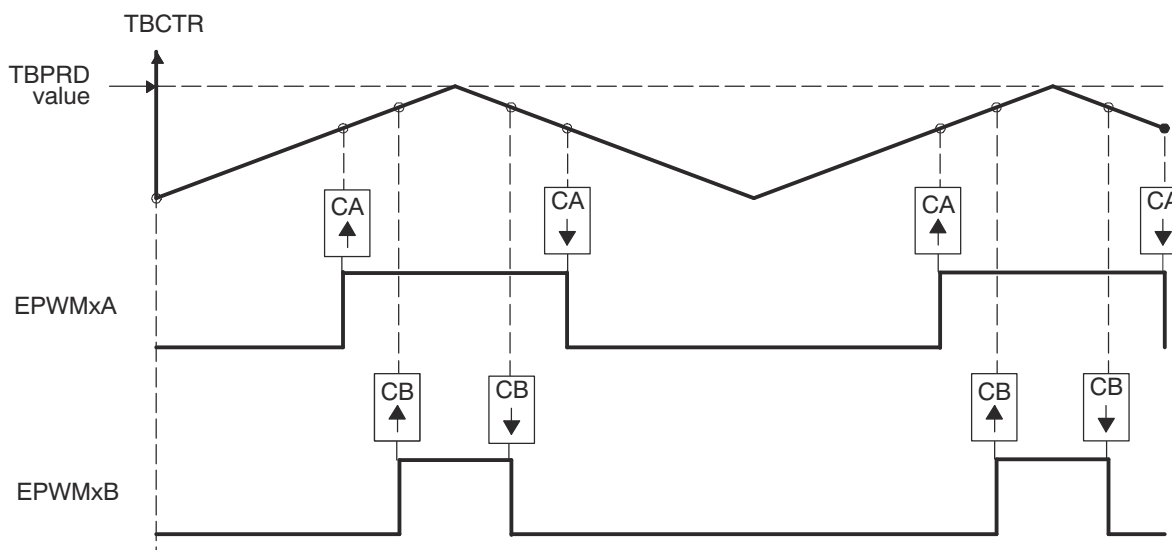
- PWM period =  $(TBPRD + 1) \times T_{TBCLK}$
- Duty modulation for EPWMxA is set by CMPA, and is active low (that is, the low time duty is proportional to CMPA).
- Duty modulation for EPWMxB is set by CMPB and is active low (that is, the low time duty is proportional to CMPB).
- Actions at zero and period, although appearing to occur concurrently, are actually separated by one TBCLK period. TBCTR wraps from period to 0000.

**Figure 26-28. Up, Single Edge Asymmetric Waveform with Independent Modulation on EPWMxA and EPWMxB—Active Low**



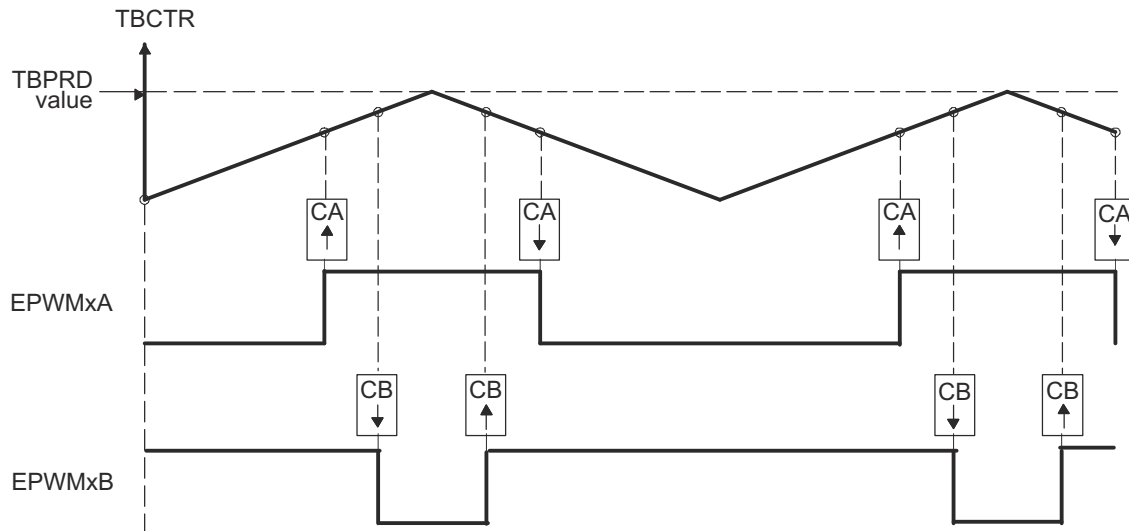
- A.  $\text{PWM frequency} = 1/((\text{TBPRD} + 1) \times T_{\text{TBCLK}})$
- B. Pulse can be placed anywhere within the PWM cycle (0000 - TBPRD)
- C. High time duty proportional to  $(\text{CMPB} - \text{CMPA})$

**Figure 26-29. Up-Count, Pulse Placement Asymmetric Waveform With Independent Modulation on EPWMxA**



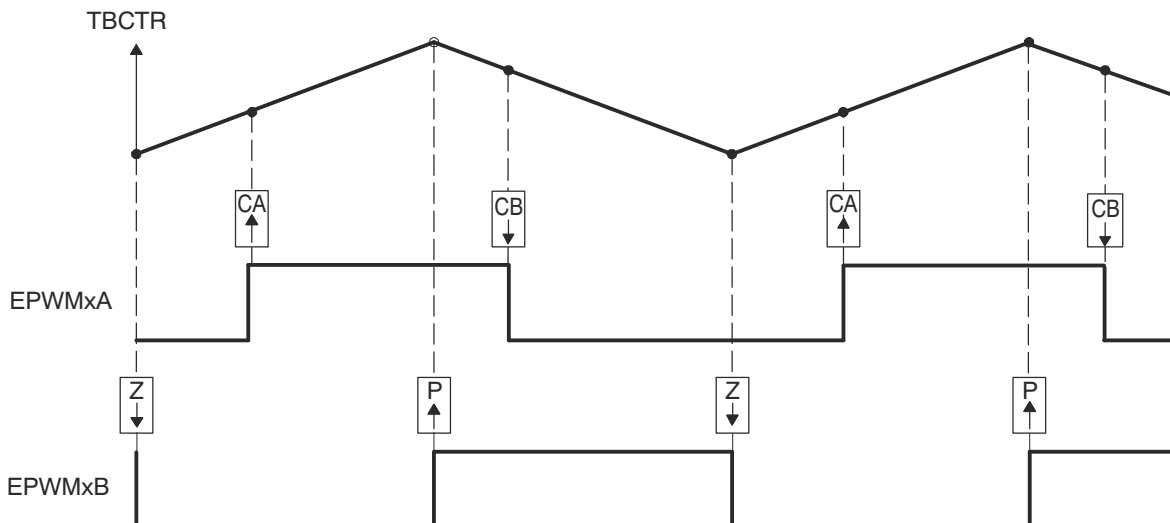
- A.  $\text{PWM period} = 2 \times \text{TBPRD} \times T_{\text{TBCLK}}$
- B. Duty modulation for EPWMxA is set by CMPA, and is active low (that is, the low time duty is proportional to CMPA).
- C. Duty modulation for EPWMxB is set by CMPB and is active low (that is, the low time duty is proportional to CMPB).
- D. Outputs EPWMxA and EPWMxB can drive independent power switches.

**Figure 26-30. Up-Down Count, Dual-Edge Symmetric Waveform, with Independent Modulation on EPWMxA and EPWMxB — Active Low**



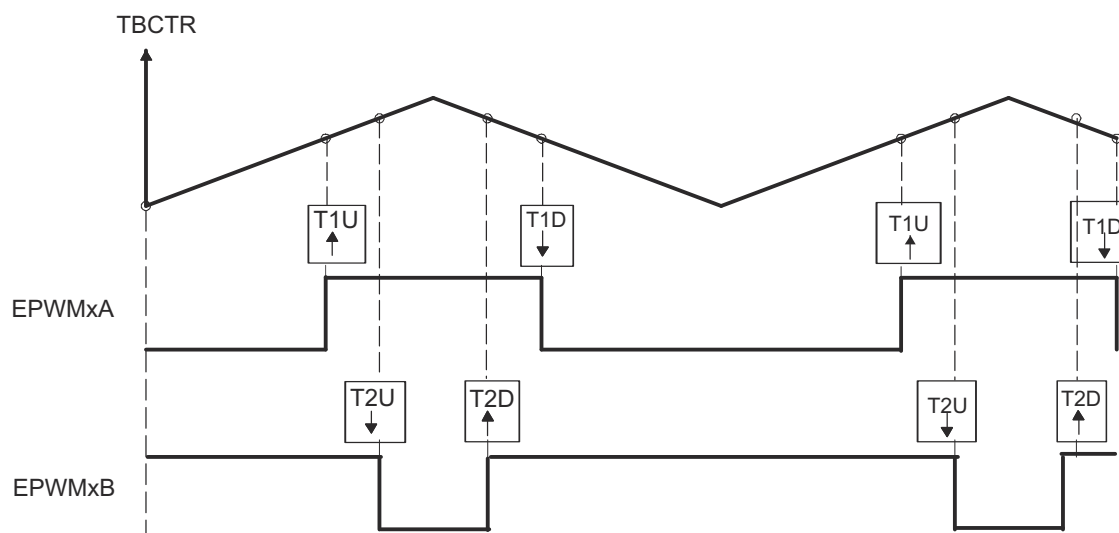
- A.  $\text{PWM period} = 2 \times \text{TBPRD} \times T_{\text{TBCLK}}$
- B. Duty modulation for EPWMxA is set by CMPA, and is active low, that is, low time duty proportional to CMPA.
- C. Duty modulation for EPWMxB is set by CMPB and is active high, that is, high time duty proportional to CMPB.
- D. Outputs EPWMx can drive upper/lower (complementary) power switches.
- E. Dead-band =  $\text{CMPB} - \text{CMPA}$  (fully programmable edge placement by software). Note the dead-band module is also available if the more classical edge delay method is required.

**Figure 26-31. Up-Down Count, Dual-Edge Symmetric Waveform, with Independent Modulation on EPWMxA and EPWMxB — Complementary**



- A.  $\text{PWM period} = 2 \times \text{TBPRD} \times \text{TBCLK}$
- B. Rising edge and falling edge can be asymmetrically positioned within a PWM cycle. This allows for pulse placement techniques.
- C. Duty modulation for EPWMxA is set by CMPA and CMPB.
- D. Low time duty for EPWMxA is proportional to  $(\text{CMPA} + \text{CMPB})$ .
- E. To change this example to active high, CMPA and CMPB actions need to be inverted (that is, Clear on CMPA, Set on CMPB).
- F. Duty modulation for EPWMxB is fixed at 50% (utilizes spare action resources for EPWMxB).

**Figure 26-32. Up-Down Count, Dual-Edge Asymmetric Waveform, with Independent Modulation on EPWMxA—Active Low**

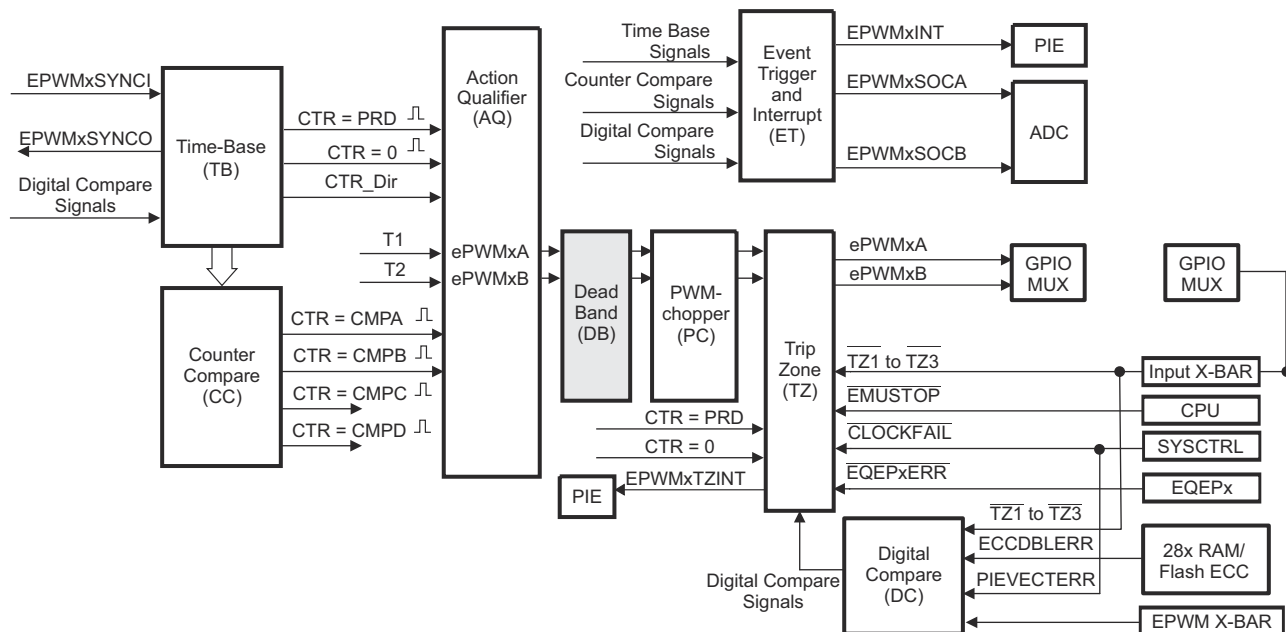


- A.  $\text{PWM period} = 2 \times \text{TBPRD} \times \text{TTBCLK}$
- B. Independent T1 event actions when counter is counting up and when the counter is counting down are used to generate EPWMxA output.
- C. Independent T2 event actions when counter is counting up and when the counter is counting down are used to generate EPWMxB output.
- D. TZ1 is selected as the source for T1.
- E. TZ2 is selected as the source for T2.

**Figure 26-33. Up-Down Count, PWM Waveform Generation Utilizing T1 and T2 Events**

## 26.7 Dead-Band Generator (DB) Submodule

Figure 26-34 illustrates the dead-band submodule within the ePWM.



**Figure 26-34. Dead\_Band Submodule**

### 26.7.1 Purpose of the Dead-Band Submodule

The action-qualifier (AQ) module section discussed how the AQ module can generate the required dead band by having full control over edge placement using both the CMPA and CMPB resources of the ePWM module. However, if the more classical edge delay-based dead band with polarity control is required, then the dead-band submodule described here must be used.

The key functions of the dead-band module are:

- Generating appropriate signal pairs (EPWMxA and EPWMxB) with dead-band relationship from a single EPWMxA input
- Programming signal pairs for:
  - Active high (AH)
  - Active low (AL)
  - Active high complementary (AHC)
  - Active low complementary (ALC)
- Adding programmable delay to rising edges (RED)
- Adding programmable delay to falling edges (FED)
- Can be totally bypassed from the signal path (note dotted lines in diagram)

### 26.7.2 Dead-band Submodule Additional Operating Modes

On type 1 ePWM RED can appear on one channel output and FED can appear on the other channel output.

The following list shows the distinct difference between type 1 and type 4 modules with respect to dead-band operating modes:

- By adding S6, S7, and S8 in [Figure 26-35](#), RED and FED can appear on both the A-channel and B-channel outputs. Additionally, both RED and FED together can be applied to either the A-channel or B-channel outputs to allow B-channel phase shifting with respect to the A-channel.

---

**Note**

Phase shifting B-channel with respect to the A-channel using the dead-band submodule additional operating modes has limitations with respect to the choice of RED and FED delay with respect to the operating duty cycle of the ePWMxA and ePWMxB outputs.

---

- The dead-band counters have also been increased to 14 bits
- Deadband and deadband high-resolution registers are now shadowed
- High-resolution deadband RED and FED have been enabled using the DBREDHR and DBFEDHR registers

---

**Note**

The PWM chopper is not enabled when high-resolution deadband is enabled.

High-resolution deadband RED and FED requires half-cycle clocking mode (DBCTL[HALFCYCLE] = 1).

Cannot have both RED and FED together applied to both ePWMxA and ePWMxB. RED and FED together can be applied only to either OutA OR OutB.

Phase shifting B-channel with respect to the A-channel: When PWMxB is derived from PWMxA using the DEDB\_MODE bit and by delaying rising edge and falling edge by the phase shift amount. When the duty cycle value on PWMxA is less than this phase shift amount, PWMxA's falling edge has precedence over the delayed rising edge for PWMxB. Make sure the duty cycle value of the current waveform applied to the dead-band module is greater than the required phase shift amount.

The Type 4 action qualifier and dead-band outputs of the ePWM module are delayed by one TBCLK cycle in comparison to the Type 2 ePWM module, although the Type 4 behavior is the same as the Type 3 PWM. Both PWMA and PWMB signals are delayed under all circumstances.

---

### Shadow Mode:

The shadow mode for the DBRED is enabled by setting the DBCTL[SHDWDBREDDMODE] bit and the shadow register for DBFED is enabled by setting the DBCTL [SHDWDBFEDMODE] bit. Shadow mode is disabled by default for both DBRED and DBFED

If the shadow register is enabled, then the content of the shadow register is transferred to the active register on one of the following events as specified by the DBCTL [LOADREDDMODE] and DBCTL [LOADFEDMODE] register bits:

- CTR = PRD: Time-base counter equal to the period (TBCTR = TBPRD).
- CTR = Zero: Time-base counter equal to zero (TBCTR = 0x00)
- Both CTR = PRD and CTR = Zero

The DBCTL register can be shadowed. The shadow mode for DBCTL is enabled by setting the DBCTL2[SHDWDBCTLMODE] bit. If the shadow register is enabled then the content of the shadow register is transferred to the active register on one of the following events as specified by the DBCTL2[LOADDBCTLMODE] register bit:

- CTR = PRD: Time-base counter equal to the period (TBCTR = TBPRD)
- CTR = Zero: Time-base counter equal to zero (TBCTR = 0x00)
- Both CTR = PRD and CTR = Zero

---

#### Note

The application software must enable shadow load mode in the DBCTL[SHDWDBREDDMODE] and DBCTL[SHDWDBFEDMODE] **before** programming values for the DBRED and DBFED registers. If the shadow register is enabled **after** programming the DBRED and DBFED registers, the DBRED and DBFED registers are loaded with a value of 0.

---

### Global Load Support

Global load control mechanism can also be used for DBRED:DBREDHR, DBFED:DBFEDHR, and DBCTL registers by configuring the appropriate bits in the global load configuration register (GLDCFG). When global load mode is selected the transfer of contents from shadow register to active register, for all registers that have this mode enabled, occurs at the same event as defined by the configuration bits in the Global Shadow to Active Load Control Register (GLDCTL). The Global load control mechanism is explained in [Section 26.4.7](#).

---

#### Note

When DBRED/DBFED active is loaded with a new shadow value while DB counters are counting, the new DBRED/DBFED value only affects the NEXT PWMx edge and not the current edge.

A Deadband value of zero cannot be used when the Global Shadow to Active Load is set to occur at CTR=ZERO. Similarly, a Deadband value of PRD cannot be used when the Global Shadow to Active Load is set to occur at CTR=PRD.

TBPRDHR cannot be used with Global load. If high-resolution period must be changed in the application, users must write to the individual period registers from an ePWM ISR (The ISR must be synchronous with the PWM switching period), where the Global Load One-Shot bit is also written to.

---

The configuration options for the dead-band submodule are shown in [Figure 26-35](#).



Although all combinations are supported, not all are typical usage modes. [Table 26-9](#) documents some classical dead-band configurations. These modes assume that the DBCTL[IN\_MODE] is configured such that EPWMxA In is the source for both falling-edge and rising-edge delay. Enhanced, or non-traditional modes can be achieved by changing the input signal source. The modes shown in [Table 26-9](#) fall into the following categories:

- **Mode 1: Bypass both falling-edge delay (FED) and rising-edge delay (RED):** Allows the user to fully disable the dead-band submodule from the PWM signal path.
- **Mode 2-5: Classical Dead-Band Polarity Settings:** These represent typical polarity configurations that can address all the active-high and active-low modes required by available industry power switch gate drivers. The waveforms for these typical cases are shown in [Figure 26-36](#). Note that to generate equivalent waveforms to [Figure 26-36](#), configure the action-qualifier submodule to generate the signal as shown for EPWMxA.
- **Mode 6: Bypass rising-edge delay (RED) and Mode 7: Bypass falling-edge delay (FED):** Finally the last two entries in [Table 26-9](#) show combinations where either the falling-edge delay (FED) or rising-edge delay (RED) blocks are bypassed.

Figure 26-36 shows waveforms for typical cases where  $0\% < \text{duty} < 100\%$ .



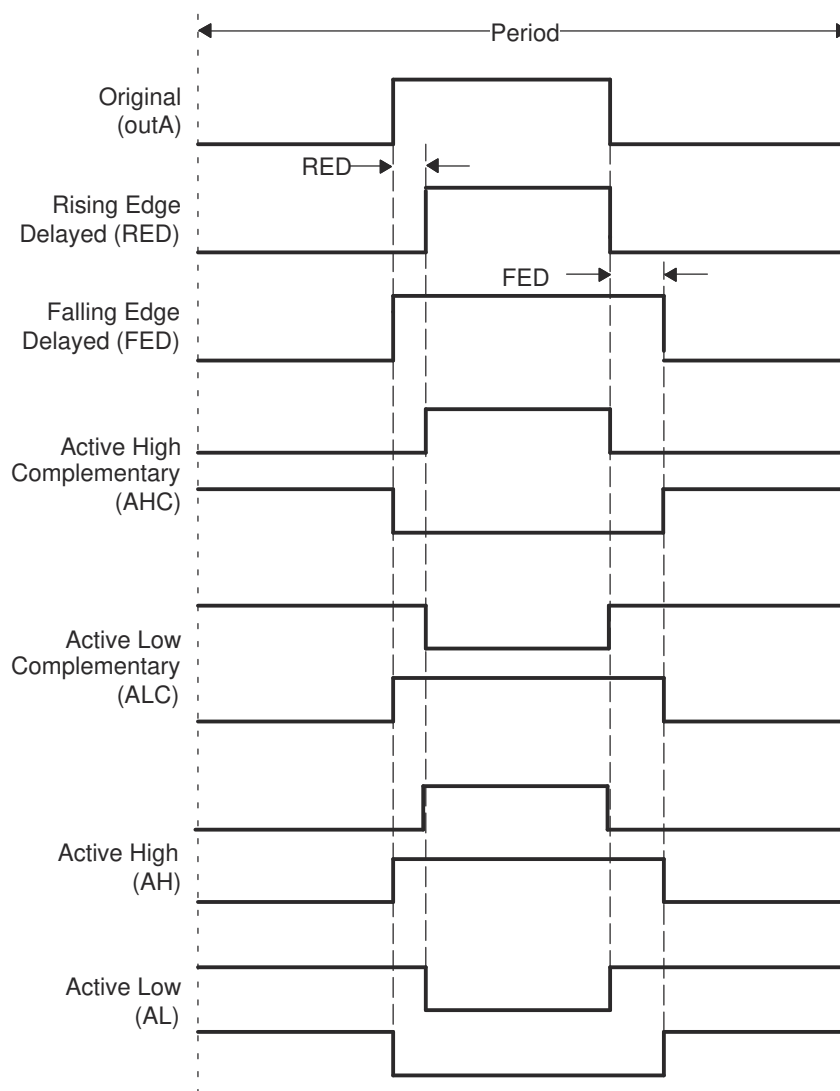
**Table 26-9. Classical Dead-Band Operating Modes**

| Mode | Mode Description                               | DBCTL[POLSEL] |        | DBCTL[OUT_MODE] |    |
|------|------------------------------------------------|---------------|--------|-----------------|----|
|      |                                                | S3            | S2     | S1              | S0 |
| 1    | EPWMxA and EPWMxB Passed Through (No Delay)    | X             | X      | 0               | 0  |
| 2    | Active High Complementary (AHC)                | 1             | 0      | 1               | 1  |
| 3    | Active Low Complementary (ALC)                 | 0             | 1      | 1               | 1  |
| 4    | Active High (AH)                               | 0             | 0      | 1               | 1  |
| 5    | Active Low (AL)                                | 1             | 1      | 1               | 1  |
| 6    | EPWMxA Out = EPWMxA In (No Delay)              | 0 or 1        | 0 or 1 | 0               | 1  |
|      | EPWMxB Out = EPWMxA In with Falling-Edge Delay |               |        |                 |    |
| 7    | EPWMxA Out = EPWMxA In with Rising-Edge Delay  | 0 or 1        | 0 or 1 | 1               | 0  |
|      | EPWMxB Out = EPWMxB In with No Delay           |               |        |                 |    |

**Table 26-10. Additional Dead-Band Operating Modes**

| Mode Description                                                                                                                                 | DBCTL[DEDB-MODE] | DBCTL[OUTSWAP] |    |
|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------|----|
|                                                                                                                                                  | S8               | S6             | S7 |
| EPWMxA and EPWMxB signals are as defined by OUT-MODE bits.                                                                                       | 0                | 0              | 0  |
| EPWMxA = A-path as defined by OUT-MODE bits.                                                                                                     | 0                | 0              | 1  |
| EPWMxB = A-path as defined by OUT-MODE bits (rising-edge delay or delay-bypassed A-signal path)                                                  |                  |                |    |
| EPWMxA = B-path as defined by OUT-MODE bits (falling-edge delay or delay-bypassed B-signal path)                                                 | 0                | 1              | 0  |
| EPWMxB = B-path as defined by OUT-MODE bits                                                                                                      |                  |                |    |
| EPWMxA = B-path as defined by OUT-MODE bits (falling-edge delay or delay-bypassed B-signal path)                                                 | 0                | 1              | 1  |
| EPWMxB = A-path as defined by OUT-MODE bits (rising-edge delay or delay-bypassed A-signal path)                                                  |                  |                |    |
| Rising-edge delay applied to EPWMxA / EPWMxB as selected by S4 switch (IN-MODE bits) on A signal path only.                                      | 0                | X              | X  |
| Falling-edge delay applied to EPWMxA / EPWMxB as selected by S5 switch (IN-MODE bits) on B signal path only.                                     |                  |                |    |
| Rising-edge delay and falling-edge delay applied to source selected by S4 switch (IN-MODE bits) and output to B signal path only. <sup>(1)</sup> | 1                | X              | X  |

- (1) When this bit is set to 1, the user can always either set OUT\_MODE bits such that Apath = InA or set OUTSWAP bits such that EPWMxA=Bpath. Otherwise, EPWMxA is invalid.



**Figure 26-36. Dead-Band Waveforms for Typical Cases (0% < Duty < 100%)**

The dead-band submodule supports independent values for rising-edge (RED) and falling-edge (FED) delays. The amount of delay is programmed using the DBRED and DBFED registers. These are 10-bit registers and the value represents the number of TBCLK (time-base clock) pulses by which a signal edge is delayed. For example, the formula to calculate falling-edge-delay and rising-edge-delay is:

$$FED = DBFED \times T_{TBCLK}$$

$$RED = DBRED \times T_{TBCLK}$$

Where  $T_{TBCLK}$  is the period of TBCLK, the prescaled version of EPWMCLK.

For convenience, delay values for various TBCLK options are shown in [Table 26-11](#). The ePWM input clock frequency that these delay values been computed by is 100MHz.

**Table 26-11. Dead-Band Delay Values in  $\mu$ s as a Function of DBFED and DBRED**

| Dead-Band Value |                   | Dead-Band Delay ( $\mu$ s) |                   |  |
|-----------------|-------------------|----------------------------|-------------------|--|
| DBFED, DBRED    | TBCLK = EPWMCLK/1 | TBCLK = EPWMCLK /2         | TBCLK = EPWMCLK/4 |  |
| 1               | 0.01              | 0.02                       | 0.04              |  |
| 5               | 0.05              | 0.10                       | 0.20              |  |
| 10              | 0.10              | 0.20                       | 0.40              |  |
| 100             | 1.00              | 2.00                       | 4.00              |  |
| 200             | 2.00              | 4.00                       | 8.00              |  |
| 400             | 4.00              | 8.00                       | 16.00             |  |
| 500             | 5.00              | 10.00                      | 20.00             |  |
| 600             | 6.00              | 12.00                      | 24.00             |  |
| 700             | 7.00              | 14.00                      | 28.00             |  |
| 800             | 8.00              | 16.00                      | 32.00             |  |
| 900             | 9.00              | 18.00                      | 36.00             |  |
| 1000            | 10.00             | 20.00                      | 40.00             |  |

When half-cycle clocking is enabled, the formula to calculate the falling-edge-delay and rising-edge-delay becomes:

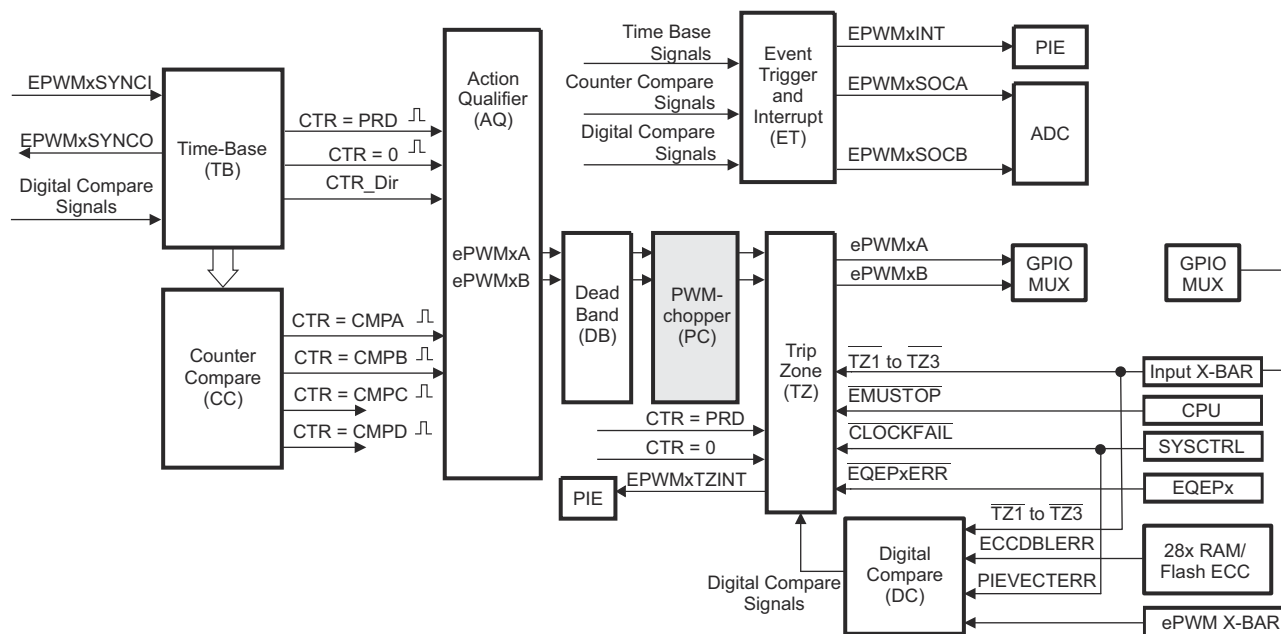
$$FED = DBFED \times T_{TBCLK}/2$$

$$RED = DBRED \times T_{TBCLK}/2$$

## 26.8 PWM Chopper (PC) Submodule

The PWM chopper submodule allows a high-frequency carrier signal to modulate the PWM waveform generated by the action-qualifier and dead-band submodules. This capability is important if pulse transformer-based gate drivers to control the power switching elements are needed.

Figure 26-37 illustrates the PWM chopper submodule within the ePWM.



**Figure 26-37. PWM Chopper Submodule**

### 26.8.1 Purpose of the PWM Chopper Submodule

The key functions of the PWM chopper submodule are:

- Programmable chopping (carrier) frequency
- Programmable pulse width of first pulse
- Programmable duty cycle of second and subsequent pulses
- Can be fully bypassed if not required

### 26.8.2 Operational Highlights for the PWM Chopper Submodule

Figure 26-38 shows the operational details of the PWM chopper submodule. The carrier clock is derived from EPWMCLK. The clock frequency and duty cycle are controlled using the CHPFREQ and CHPDUTY bits in the PCCTL register. The one-shot block is a feature that provides a high energy first pulse to make sure hard and fast power switch turn on, while the subsequent pulses sustain pulses, making sure the power switch remains on. The one-shot width is programmed using the OSHTWTH bits. The PWM chopper submodule can be fully disabled (bypassed) using the CHPEN bit.

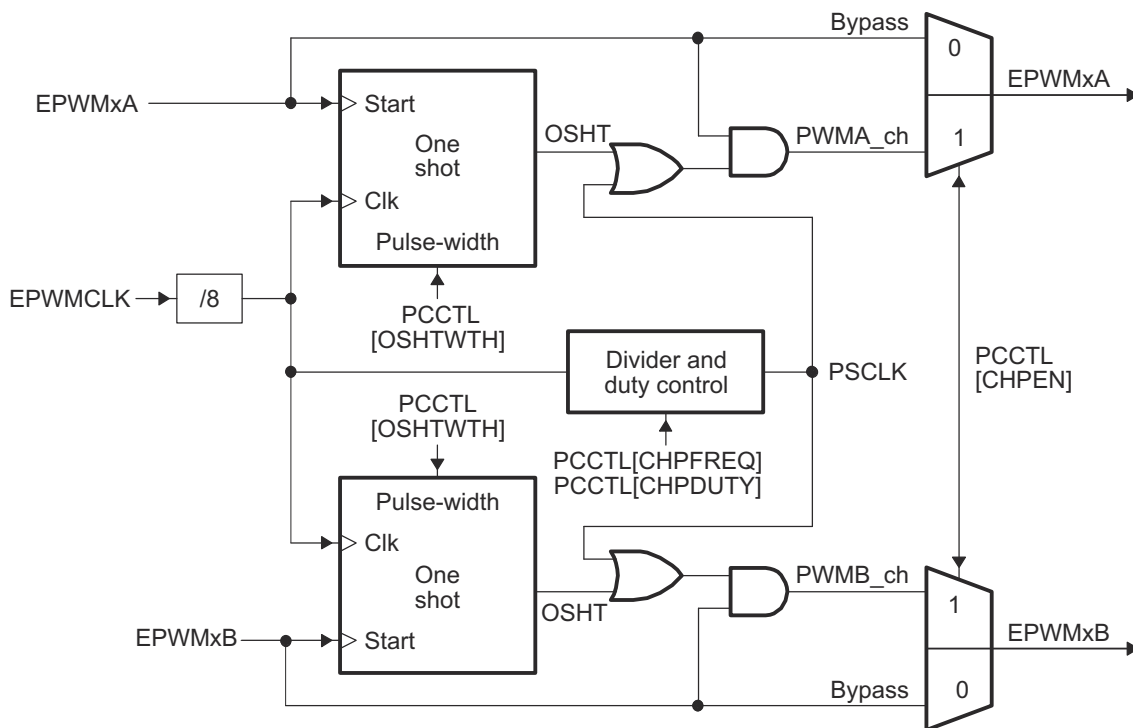


Figure 26-38. PWM Chopper Submodule Operational Details

### 26.8.3 Waveforms

Figure 26-39 shows simplified waveforms of the chopping action only; one-shot and duty-cycle control are not shown. Details of the one-shot and duty-cycle control are discussed in the following sections.

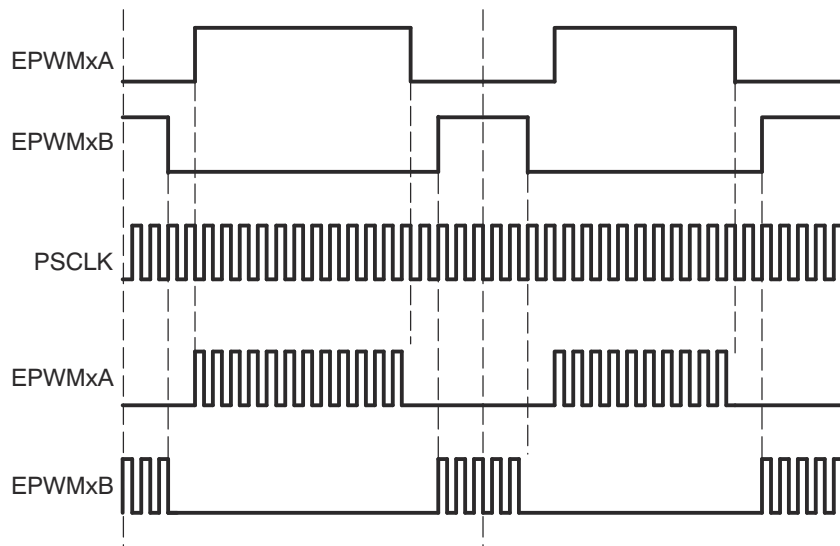


Figure 26-39. Simple PWM Chopper Submodule Waveforms Showing Chopping Action Only

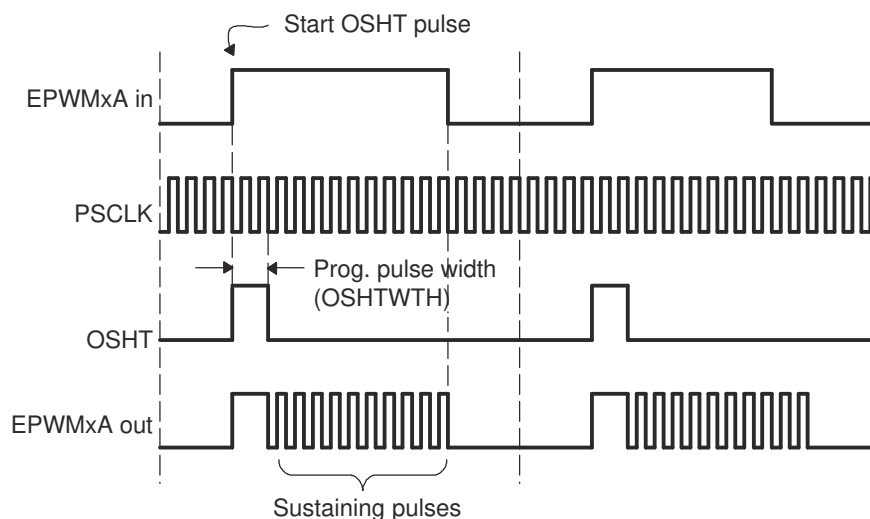
### 26.8.3.1 One-Shot Pulse

The width of the first pulse can be programmed to any of 16 possible pulse width values. The width or period of the first pulse is given by:

$$T_{1stpulse} = T_{EPWMCLK} \times 8 \times OSHTWTH$$

Where  $T_{EPWMCLK}$  is the period of the system clock (EPWMCLK) and OSHTWTH is the four control bits (value from 1 to 16)

Figure 26-40 shows the first and subsequent sustaining pulses and Table 26-12 gives the possible pulse width values for a EPWMCLK = 80MHz.



**Figure 26-40. PWM Chopper Submodule Waveforms Showing the First Pulse and Subsequent Sustaining Pulses**

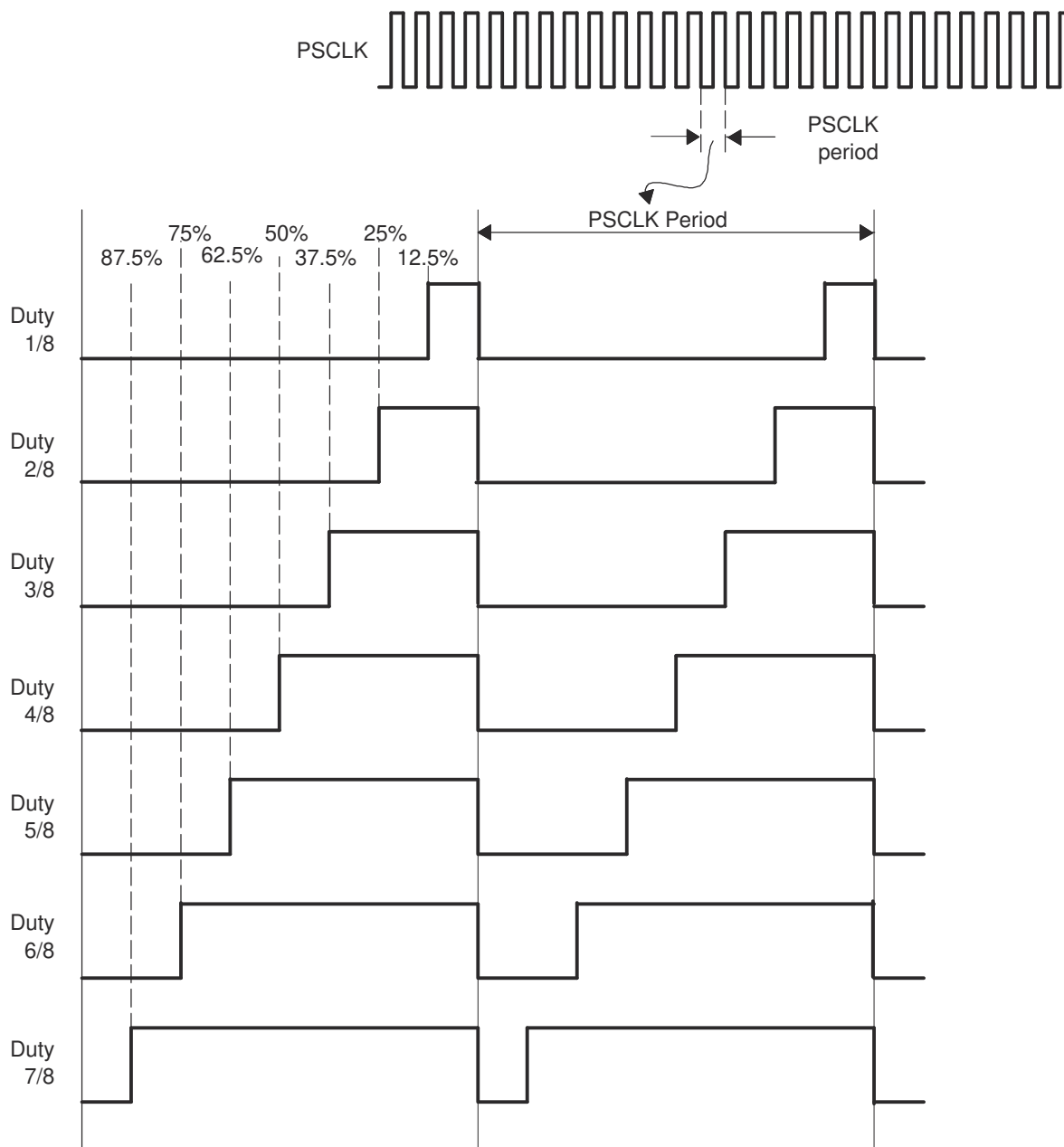
**Table 26-12. Possible Pulse Width Values for  
EPWMCLK = 80MHz**

| OSHTWTH (hex) | Pulse Width (ns) |
|---------------|------------------|
| 0             | 100              |
| 1             | 200              |
| 2             | 300              |
| 3             | 400              |
| 4             | 500              |
| 5             | 600              |
| 6             | 700              |
| 7             | 800              |
| 8             | 900              |
| 9             | 1000             |
| A             | 1100             |
| B             | 1200             |
| C             | 1300             |
| D             | 1400             |
| E             | 1500             |
| F             | 1600             |

### 26.8.3.2 Duty Cycle Control

Pulse transformer-based gate drive designs need to comprehend the magnetic properties or characteristics of the transformer and associated circuitry. Saturation is one such consideration. To assist the gate drive designer, the duty cycles of the second and subsequent pulses have been made programmable. These sustaining pulses make sure the correct drive strength and polarity is maintained on the power switch gate during the on period, and hence a programmable duty cycle allows a design to be tuned or optimized using software control.

Figure 26-41 shows the duty cycle control that is possible by programming the CHPDUTY bits. One of seven possible duty ratios can be selected ranging from 12.5% to 87.5%.



**Figure 26-41. PWM Chopper Submodule Waveforms Showing the Pulse Width (Duty Cycle) Control of Sustaining Pulses**

## 26.9 Trip-Zone (TZ) Submodule

Each ePWM module is connected to six  $\overline{TZn}$  signals ( $\overline{TZ1}$  to  $\overline{TZ6}$ ).  $\overline{TZ1}$  to  $\overline{TZ3}$  are sourced from the GPIO mux.  $\overline{TZ4}$  is sourced from an inverted EQEPxERR signal on those devices with an EQEP module.  $\overline{TZ5}$  is connected to the system clock fail logic, and  $\overline{TZ6}$  is sourced from the EMUSTOP output from the CPU. These signals indicate external fault or trip conditions, and the ePWM outputs can be programmed to respond accordingly when faults occur.

Figure 26-42 illustrates the trip-zone submodule within the ePWM.

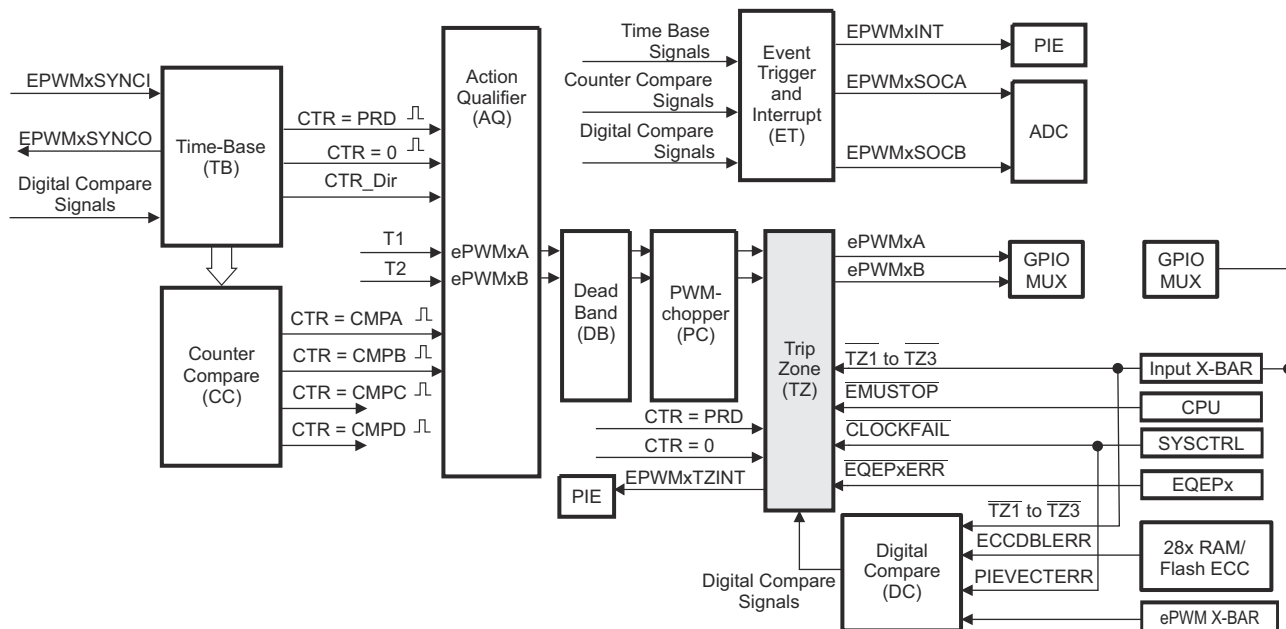


Figure 26-42. Trip-Zone Submodule

### 26.9.1 Purpose of the Trip-Zone Submodule

The key functions of the trip-zone submodule are:

- Trip inputs  $\overline{TZ1}$  to  $\overline{TZ6}$  can be flexibly mapped to any ePWM module.
- Upon a fault condition, outputs EPWMxA and EPWMxB can be forced to one of the following:
  - High
  - Low
  - High-impedance
  - No action taken
- Support for one-shot trip (OSHT) for major short circuits or over-current conditions.
- Support for cycle-by-cycle tripping (CBC) for current limiting operation.
- Support for digital compare tripping (DC) based on state of on-chip analog comparator module outputs and  $\overline{TZ1}$  to  $\overline{TZ3}$  signals.
- Each trip-zone input and digital compare (DC) submodule DCAEVT1/2 or DCBEVT1/2 force event can be allocated to either one-shot or cycle-by-cycle operation.
- Interrupt generation is possible on any trip-zone input.
- Software-forced tripping is also supported.
- The trip-zone submodule can be fully bypassed if the trip-zone submodule is not required.



## 26.9.2 Operational Highlights for the Trip-Zone Submodule

The following sections describe the operational highlights and configuration options for the trip-zone submodule.

The trip-zone signals  $\overline{TZ1}$  to  $\overline{TZ6}$  (also collectively referred to as  $\overline{TZn}$ ) are active-low input signals. When one of these signals goes low, or when a DCAEVT1/2 or DCBEVT1/2 force happens based on the TZDCSEL register event selection, the indication is that a trip event has occurred. Each ePWM module can be individually configured to ignore or use each of the trip-zone signals or DC events. Which trip-zone signals or DC events are used by a particular ePWM module is determined by the TZSEL register for that specific ePWM module. The trip-zone signals can or cannot be synchronized to the ePWMclock (EPWMCLK) and digitally filtered within the GPIO MUX block. A minimum of  $3 \cdot T_{BCLK}$  low pulse width on  $\overline{TZn}$  inputs is sufficient to trigger a fault condition on the ePWM module. If the pulse width is less than this, the trip condition cannot be latched by CBC or OST latches. The asynchronous trip makes sure that if clocks are missing for any reason, the outputs can still be tripped by a valid event present on  $\overline{TZn}$  inputs. The GPIOs or peripherals must be appropriately configured. For more information, see the *System Control and Interrupts* chapter.

Each  $\overline{TZn}$  input can be individually configured to provide either a cycle-by-cycle or one-shot trip event for an ePWM module. DCAEVT1 and DCBEVT1 events can be configured to directly trip an ePWM module or provide a one-shot trip event to the module. Likewise, DCAEVT2 and DCBEVT2 events can also be configured to directly trip an ePWM module or provide a cycle-by-cycle trip event to the module. This configuration is determined by the TZSEL[DCAEVT1/2], TZSEL[DCBEVT1/2], TZSEL[CBCn], and TZSEL[OSHTn] control bits (where n corresponds to the trip input), respectively.

- **Cycle-by-Cycle (CBC):**

When a cycle-by-cycle trip event occurs, the action specified in the TZCTL[TZA] and TZCTL[TZB] bits is carried out immediately on the EPWMxA and EPWMxB outputs. [Table 26-13](#) lists some of the possible actions. Independent actions can be specified based on the occurrence of the event while the counter is counting up or while the counter is counting down by appropriately configuring bits in the TZCTL2 register. Actions specified in the TZCTL2 register take effect only when the ETZE bit in TZCTL2 is set.

Additionally, when a cycle-by-cycle trip event occurs, the cycle-by-cycle trip event flag (TZFLG[CBC]) is set and a EPWMx\_TZINT interrupt is generated when enabled in the TZEINT register and interrupt controller. A corresponding flag for the event that caused the CBC event is also set in register TZCBCFLG.

If the CBC interrupt is enabled using the TZEINT register and DCAEVT2 or DCBEVT2 are selected as CBC trip sources using the TZSEL register, it is not necessary to also enable the DCAEVT2 or DCBEVT2 interrupts in the TZEINT register, as the DC events trigger interrupts through the CBC mechanism.

The specified condition on the inputs is automatically cleared based on the selection made with TZCLR[CBCPULSE] if the trip event is no longer present. Therefore, in this mode, the trip event is cleared or reset every PWM cycle. The TZFLG[CBC] and TZCBCFLG flag bits remain set until the flag bits are manually cleared by writing to the TZCLR[CBC] and TZCBCCLR flag bits. If the cycle-by-cycle trip event is still present when the TZFLG[CBC] and TZCBCFLG register bits are cleared, then these bits are again immediately set.

- **One-Shot (OSHT):**

When a one-shot trip event occurs, the action specified in the TZCTL[TZA] and TZCTL[TZB] bits is carried out immediately on the EPWMxA and EPWMxB output. [Table 26-13](#) lists some of the possible actions. Independent actions can be specified based on the occurrence of the event while the counter is counting up and while the counter is counting down by appropriately configuring bits in TZCTL2 register. Actions specified in TZCTL2 register take effect only when ETZE bit in TZCTL2 is set.

Additionally, when a one-shot trip event occurs, the one-shot trip event flag (TZFLG[OST]) is set and a EPWMx\_TZINT interrupt is generated when enabled in the TZEINT register and interrupt controller. A corresponding flag for the event that caused the OST event is also set in register TZOSTFLG. The one-shot trip condition must be cleared manually by writing to the TZCLR[OST] bit. If desired, the TZOSTFLG register bit can be cleared by manually writing to the corresponding bit in the TZOSTCLR register.

If the one-shot interrupt is enabled using the TZEINT register and DCAEVT1 or DCBEVT1 are selected as OSH trip sources using the TZSEL register, it is not necessary to also enable the DCAEVT1 or DCBEVT1 interrupts in the TZEINT register, as the DC events trigger interrupts through the OSH mechanism.

### Note

Clear the TZFLG and TZOSTFLG flags after making sure that the TRIPIN source of the OST has become inactive. Otherwise, if interrupts are enabled, depending on when the flags are cleared, an OST interrupt can occur and the OST flags are zero.

## • Digital Compare Events (DCAEVT1/2 and DCBEVT1/2):

A digital compare DCAEVT1/2 or DCBEVT1/2 event is generated based on a combination of the DCAH/DCAL and DCBH/DCBL signals as selected by the TZDSEL register. The signals which source the DCAH/DCAL and DCBH/DCBL signals are selected using the DCTRIPSEL register and can be either trip zone input pins or analog comparator CMPSSx signals. For more information on the digital compare submodule signals, see [Section 26.11](#).

When a digital compare event occurs, the action specified in the TZCTL[DCAEVT1/2] and TZCTL[DCBEVT1/2] bits is carried out immediately on the EPWMxA and EPWMxB output. [Table 26-13](#) lists the possible actions. Independent actions can be specified based on the occurrence of the event while the counter is counting up and while the counter is counting down by appropriately configuring bits in TZCTLDCA and TZCTLDCA register. Actions specified in TZCTLDCA and TZCTLDCA registers take effect only when ETZE bit in TZCTL2 is set.

In addition, the relevant DC trip event flag (TZFLG[DCAEVT1/2] / TZFLG[DCBEVT1/2]) is set and a EPWMx\_TZINT interrupt is generated when enabled in the TZEINT register and interrupt controller.

The specified condition on the pins is automatically cleared when the DC trip event is no longer present. The TZFLG[DCAEVT1/2] or TZFLG[DCBEVT1/2] flag bit remains set until the flag is manually cleared by writing to the TZCLR[DCAEVT1/2] or TZCLR[DCBEVT1/2] bit. If the DC trip event is still present when the TZFLG[DCAEVT1/2] or TZFLG[DCBEVT1/2] flag is cleared, then the flag is again immediately set.

The action taken when a trip event occurs can be configured individually for each of the ePWM output pins by way of the TZCTL, TZCTL2, TZCTLDCA, and TZCTLDCA register bit fields. Some of the possible actions, shown in [Table 26-13](#), can be taken on a trip event.

**Table 26-13. Possible Actions On a Trip Event**

| TZCTL Register Bitfield Settings | EPWMxA and EPWMxB   | Comment                                         |
|----------------------------------|---------------------|-------------------------------------------------|
| 0,0                              | High-Impedance      | Tripped                                         |
| 0,1                              | Force to High State | Tripped                                         |
| 1,0                              | Force to Low State  | Tripped                                         |
| 1,1                              | No Change           | Do Nothing.<br>No change is made to the output. |

### Example 26-1. Trip-Zone Configurations

#### Scenario A:

A one-shot trip event on  $\overline{TZ1}$  pulls both EPWM1A, EPWM1B low and also forces EPWM2A and EPWM2B high.

- Configure the ePWM1 registers as follows:
  - TZSEL[OSHT1] = 1: enables  $\overline{TZ1}$  as a one-shot event source for ePWM1
  - TZCTL[TZA] = 2: EPWM1A is forced low on a trip event.
  - TZCTL[TZB] = 2: EPWM1B is forced low on a trip event.
- Configure the ePWM2 registers as follows:
  - TZSEL[OSHT1] = 1: enables  $\overline{TZ1}$  as a one-shot event source for ePWM2
  - TZCTL[TZA] = 1: EPWM2A is forced high on a trip event.
  - TZCTL[TZB] = 1: EPWM2B is forced high on a trip event.

#### Scenario B:

A cycle-by-cycle event on  $\overline{TZ5}$  pulls both EPWM1A, EPWM1B low.

A one-shot event on  $\overline{TZ1}$  or  $\overline{TZ6}$  puts EPWM2A into a high impedance state.

- Configure the ePWM1 registers as follows:
  - TZSEL[CBC5] = 1: enables  $\overline{TZ5}$  as a cycle-by-cycle event source for ePWM1
  - TZCTL[TZA] = 2: EPWM1A is forced low on a trip event.
  - TZCTL[TZB] = 2: EPWM1B is forced low on a trip event.
- Configure the ePWM2 registers as follows:
  - TZSEL[OSHT1] = 1: enables  $\overline{TZ1}$  as a one-shot event source for ePWM2
  - TZSEL[OSHT6] = 1: enables  $\overline{TZ6}$  as a one-shot event source for ePWM2
  - TZCTL[TZA] = 0: EPWM2A is put into a high-impedance state on a trip event.
  - TZCTL[TZB] = 3: EPWM2B ignores the trip event.

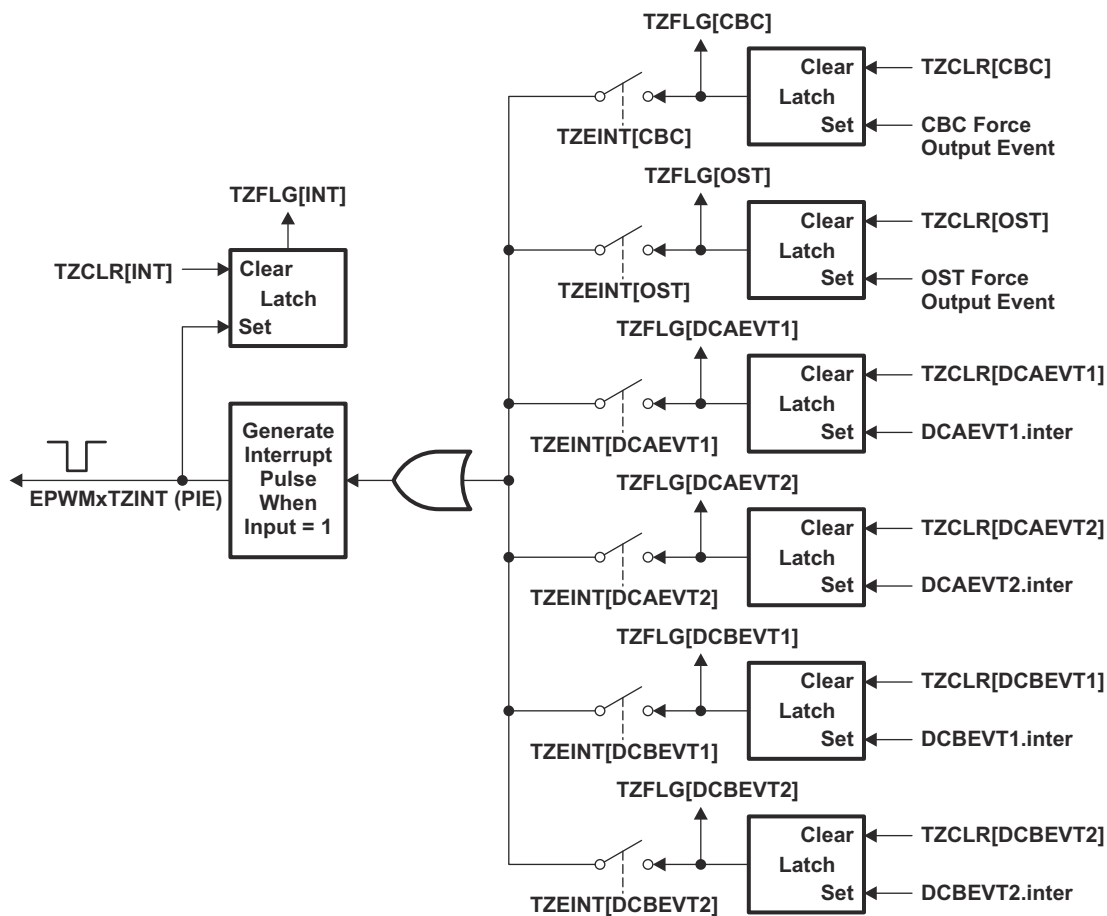
#### Note

When configuring the GPIOs and INPUT X-BAR/EPWM X-BAR options, be aware that a change in the X-BAR input selections can cause an unwanted event. Therefore, set up the GPIO and X-BAR input configurations before enabling the ePWM Trip-Zone. If a requirement is to change the GPIO/X-BAR configurations while the ePWM Trip-Zone is enabled, the user can turn off the TRIPs by clearing the TZSEL register and reconfiguring the TRIP selection (TZSEL) after the INPUT XBAR selection is changed.

### 26.9.3 Generating Trip Event Interrupts

Figure 26-43 and Figure 26-44 illustrate the trip-zone submodule control and interrupt logic, respectively. DCAEVT1/2 and DCBEVT1/2 signals are described in further detail in [Section 26.11](#).





**Figure 26-44. Trip-Zone Submodule Interrupt Logic**

These individual flags for the CBC, OST and DCxEVTy can be used to detect the source of the EPWMxTZINT Interrupt. When multiple sources are used to generate the EPWMxTZINT interrupt, reading and clearing the flags takes different actions based on the specific event.

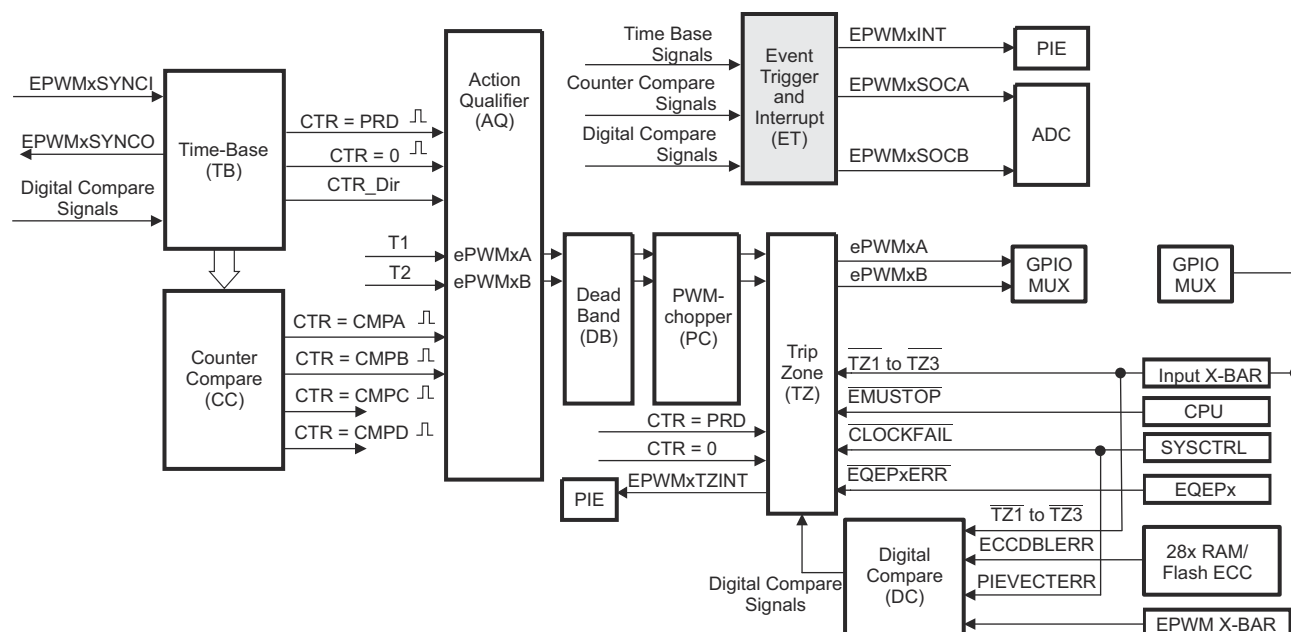
## 26.10 Event-Trigger (ET) Submodule

The key functions of the event-trigger submodule are:

- Receives event inputs generated by the time-base, counter-compare, and digital-compare submodules
- Uses the time-base direction information for up/down event qualification
- Uses prescaling logic to issue interrupt requests and ADC start of conversion at:
  - Every event
  - Every second event
  - Up to every fifteenth event
- Provides full visibility of event generation using event counters and flags
- Allows software forcing of Interrupts and ADC start of conversion

The event-trigger submodule manages the events generated by the time-base submodule, the counter-compare submodule, and the digital-compare submodule to generate an interrupt to the CPU and a start of conversion pulse to the ADC when a selected event occurs.

Figure 26-45 illustrates the event-trigger submodule within the ePWM.

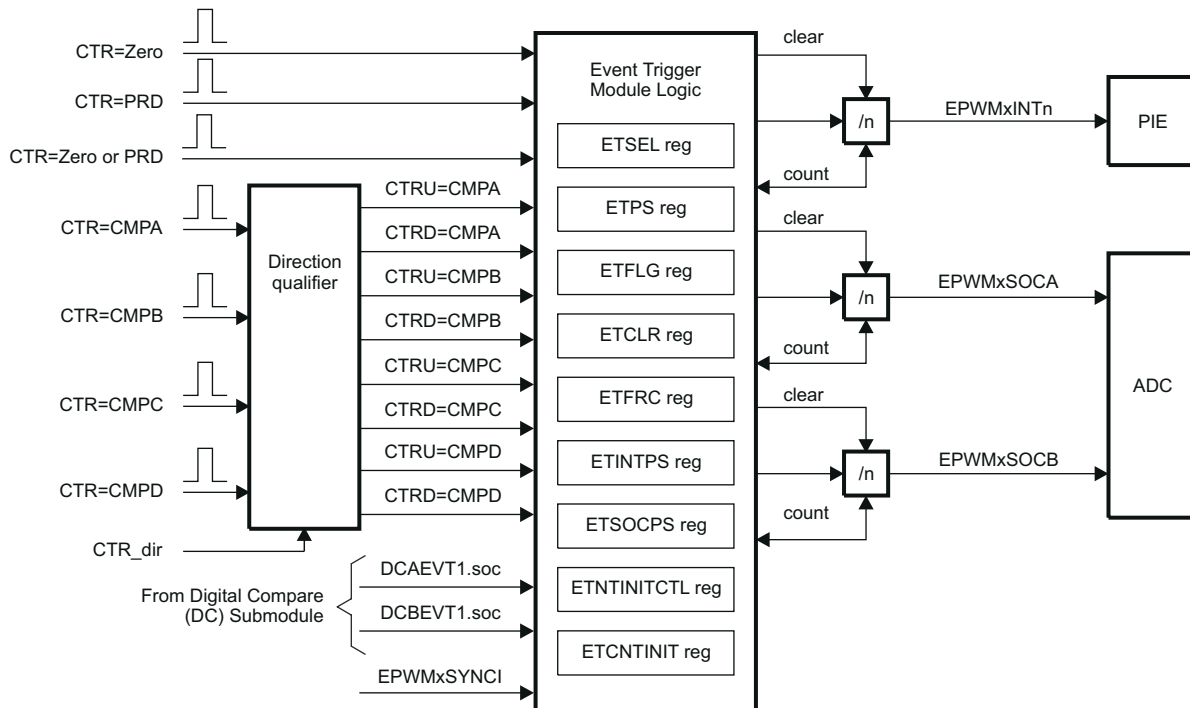


**Figure 26-45. Event-Trigger Submodule**

### 26.10.1 Operational Overview of the ePWM Event-Trigger Submodule

The event-trigger submodule monitors various event conditions (shown as inputs on the left side of Figure 26-46) and can be configured to prescale these events before issuing an Interrupt request or an ADC start of conversion. The event-trigger prescaling logic can issue Interrupt requests and ADC start of conversion at:

- Every event
- Every second event
- Up to every fifteenth event



**Figure 26-46. Event-Trigger Submodule Showing Event Inputs and Prescaled Outputs**

- ETSEL - This selects which of the possible events trigger an interrupt or start an ADC conversion.
- ETPS - This programs the event prescaling options mentioned above.
- ETFLG - These are flag bits indicating status of the selected and prescaled events.
- ETCLR - These bits allow clearing the flag bits in the ETFLG register using software.
- ETFRC - These bits allow software forcing of an event. Useful for debugging or software intervention.
- ETINTPS - This programs the interrupt event prescaling options, supporting count and period up to 15 events.
- ETSOCPS - This programs the SOC event prescaling options, supporting count and period up to 15 events.
- ETCNTINITCTL - These bits enable ETCNTINIT initialization using SYNC event or using software force.
- ETCNTINIT - These bits allow initializing INT/SOCA/SOCB counters on SYNC events (or software force) with user programmed value.

A more detailed look at how the various register bits interact with the Interrupt and ADC start of conversion logic are shown in Figure 26-47, Figure 26-48, and Figure 26-49.

Figure 26-47 shows the event-trigger's interrupt generation logic. The interrupt-period (ETPS[INTPRD]) bits specify the number of events required to cause an interrupt pulse to be generated. The choices available are:

- Do not generate an interrupt.
- Generate an interrupt on every event.
- Generate an interrupt on every second event.
- Generate an interrupt on every third event.



The selection made on ETPS[INTPSSEL] bit determines whether ETINTPS register, INTCNT2 and INTPRD2 bit fields determine frequency of events (interrupt once every 0-15 events).

The event that can cause an interrupt is configured by the interrupt selection (ETSEL[INTSEL]) and (ETSEL[INTSELCMP]) bits. The event can be one of the following:

- Time-base counter equal to zero (TBCTR = 0x00).
- Time-base counter equal to period (TBCTR = TBPRD).
- Time-base counter equal to zero or period (TBCTR = 0x00 || TBCTR = TBPRD).
- Time-base counter equal to the compare A register (CMPA) when the timer is incrementing.
- Time-base counter equal to the compare A register (CMPA) when the timer is decrementing.
- Time-base counter equal to the compare B register (CMPB) when the timer is incrementing.
- Time-base counter equal to the compare B register (CMPB) when the timer is decrementing.
- Time-base counter equal to the compare C register (CMPC) when the timer is incrementing.
- Time-base counter equal to the compare C register (CMPC) when the timer is decrementing.
- Time-base counter equal to the compare D register (CMPD) when the timer is incrementing.
- Time-base counter equal to the compare D register (CMPD) when the timer is decrementing.

The number of events that have occurred can be read from the interrupt event counter ETPS[INTCNT] or ETINTPS[INTCNT2] register bits based off of the selection made using ETPS[INTPSSEL]. That is, when the specified event occurs the ETPS[INTCNT] or ETINTPS[INTCNT2] bits are incremented until the bits reach the value specified by ETPS[INTPRD] or ETINTPS[INTPRD2] determined again by the selection made in ETPS[INTPSSEL]. When ETPS[INTCNT] = ETPS[INTPRD], the counter stops counting and the counter output is set. The counter is only cleared when an interrupt is sent to the interrupt controller.

When ETPS[INTCNT] reaches ETPS[INTPRD], the following behavior occurs. [The following behavior is also applicable to ETINTPS[INTCNT2] and ETINTPS[INTPRD2]:

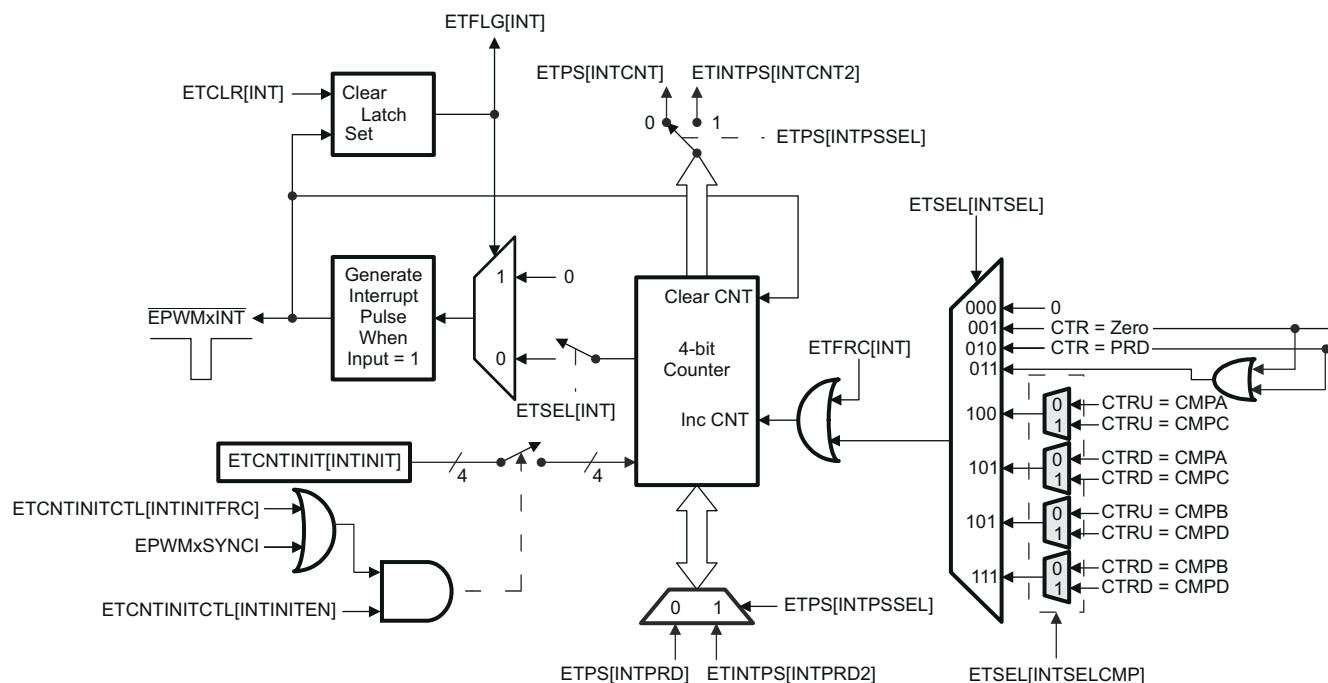
- If interrupts are enabled, ETSEL[INTEN] = 1 and the interrupt flag is clear, ETFLG[INT] = 0, then an interrupt pulse is generated and the interrupt flag is set, ETFLG[INT] = 1, and the event counter is cleared ETPS[INTCNT] = 0. The counter begins counting events again.
- If interrupts are disabled, ETSEL[INTEN] = 0, or the interrupt flag is set, ETFLG[INT] = 1, the counter stops counting events when the counter reaches the period value ETPS[INTCNT] = ETPS[INTPRD].
- If interrupts are enabled, but the interrupt flag is already set, then the counter holds the output high until the ENTFLG[INT] flag is cleared. This allows for one interrupt to be pending while one is serviced.

Writing a 0 to the INTPRD bits automatically clears the counter (INTCNT = 0) and the counter output resets (so no interrupts are generated). For all other writes to INTPRD, INTCNT retains the previous value. INTCNT resets when INTCNT overflows. Writing a 1 to the ETFRC[INT] bit increments the event counter INTCNT. The counter behaves as previously described when INTCNT = INTPRD. When INTPRD = 0, the counter is disabled and hence no events are detected and the ETFRC[INT] bit is also ignored. The same applies to ETINTPS[INTCNT2] and ETINTPS[INTPRD2].

The previous definition means that an interrupt on every event, on every second event, or on every third event if using the INTCNT and INTPRD can be generated. An interrupt on every event up to 15 events if using the INTCNT2 and INTPRD2 can be generated.

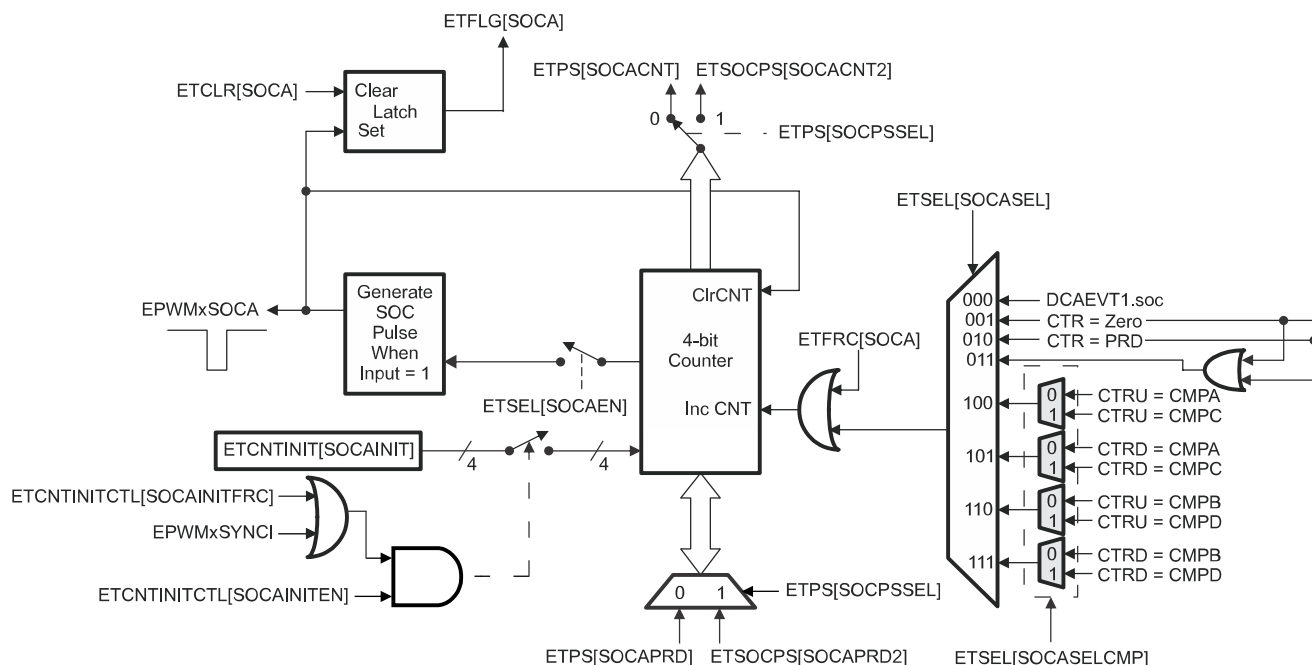
The INTCNT2 value can be initialized with the value from ETCNTINIT[INTINIT] based on the selection made in ETCNTINITCTL[INTINITEN]. When ETCNTINITCTL[INTINITEN] is set, then initialization of INTCNT2 counter with contents of ETCNTINIT[INTINIT] on a SYNC event or software force is determined by ETCNTINITCTL[INTINITFRC].





**Figure 26-47. Event-Trigger Interrupt Generator**

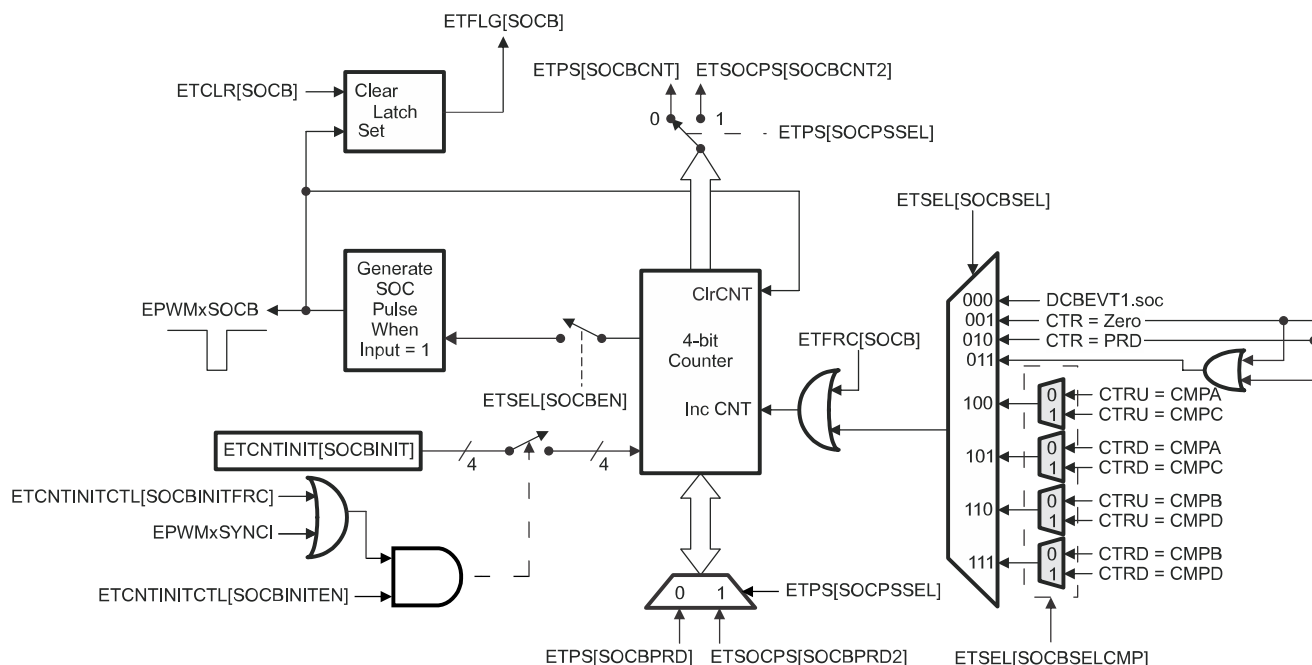
Figure 26-48 shows the operation of the event-trigger's start-of-conversion-A (SOCA) pulse generator. The enhancements include SOCASELCMP and SOCBSELCMP bit fields defined in the ETSEL register enable CMPC and CMPD events respectively to cause a start of conversion. The ETPS[SOCPSSEL] bit field determines whether SOCACNT2 and SOCAPRD2 take control or not. The ETPS[SOCACNT] counter and ETPS[SOCAPRD] period values behave similarly to the interrupt generator except that the pulses are continuously generated. That is, the pulse flag ETFLG[SOCA] is latched when a pulse is generated, but the interrupt generator does not stop further pulse generation. The enable and disable bit ETSEL[SOCASEN] stops pulse generation, but input events can still be counted until the period value is reached as with the interrupt generation logic. The event that triggers an SOCA and SOCB pulse can be configured separately in the ETSEL[SOCASEL] and ETSEL[SOCBSEL] bits. The possible events are the same events that can be specified for the interrupt generation logic with the addition of the DCAEVT1.soc and DCBEVT1.soc event signals from the digital compare (DC) submodule. The SOCACNT2 initialization scheme is very similar to the interrupt generator with respective enable, value initialize and SYNC or software force options.



NOTE: The DCAEVT1.soc signals are generated by the Digital Compare (DC) submodule in [Section 26.11](#).

**Figure 26-48. Event-Trigger SOCA Pulse Generator**

[Figure 26-49](#) shows the operation of the event-trigger's start-of-conversion-B (SOCB) pulse generator. The event-trigger's SOCB pulse generator operates the same way as the SOCA.



NOTE: The DCBEVT1.soc signals are generated by the Digital Compare (DC) submodule in [Section 26.11](#).

**Figure 26-49. Event-Trigger SOCB Pulse Generator**

## 26.11 Digital Compare (DC) Submodule

Figure 26-50 illustrates where the digital compare (DC) submodule signals interface to other submodules in the ePWM system.

The eCAP input signals are sourced from the Input X-BAR signals as shown in Figure 26-51.

On this device, any of the GPIO pins can be flexibly mapped to be the trip-zone input and trip inputs to the trip-zone submodule and digital compare submodule. The Input X-BAR Input Select (INPUTxSELECT) register defines which GPIO pins gets assigned to be the trip-zone inputs / trip inputs.

The digital compare (DC) submodule compares signals external to the ePWM module (for instance, CMPSSx signals from the analog comparators) to directly generate PWM events/actions which then feed to the event-trigger, trip-zone, and time-base submodules. Additionally, blanking window functionality is supported to filter noise or unwanted pulses from the DC event signals.

### Note

The user is responsible for driving the correct state on the selected pin before enabling the clock and configuring the trip input for the respective ePWM peripheral to avoid spurious latch of the TRIP signal.

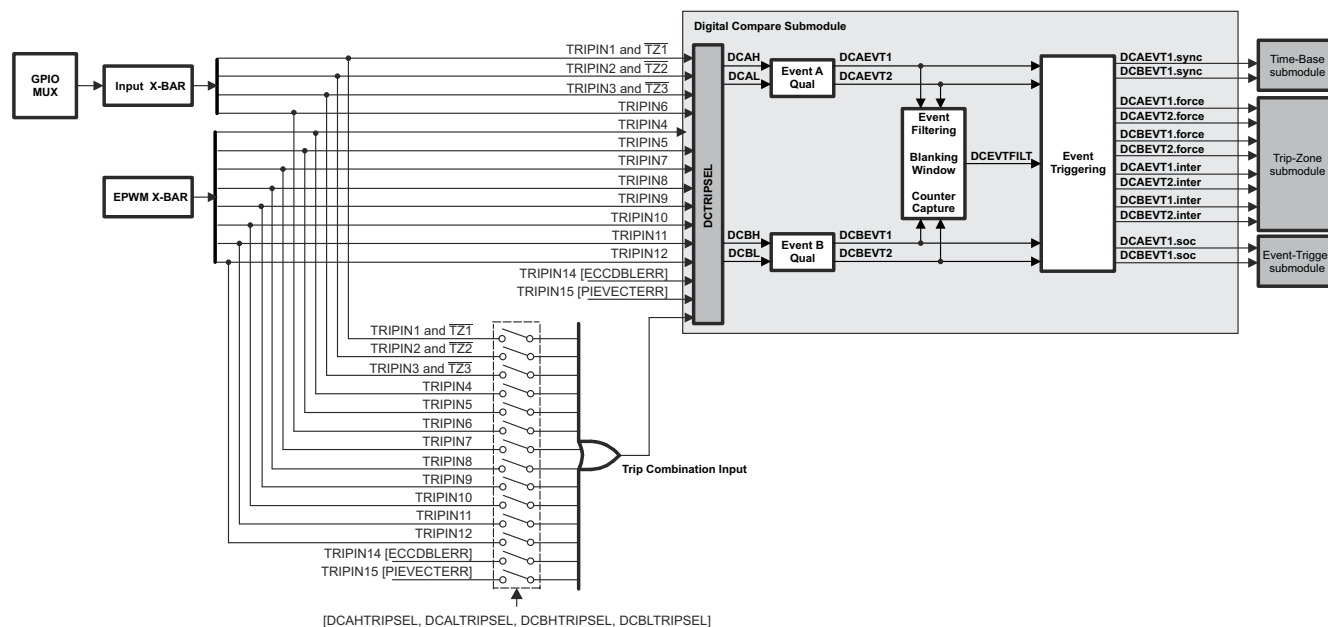
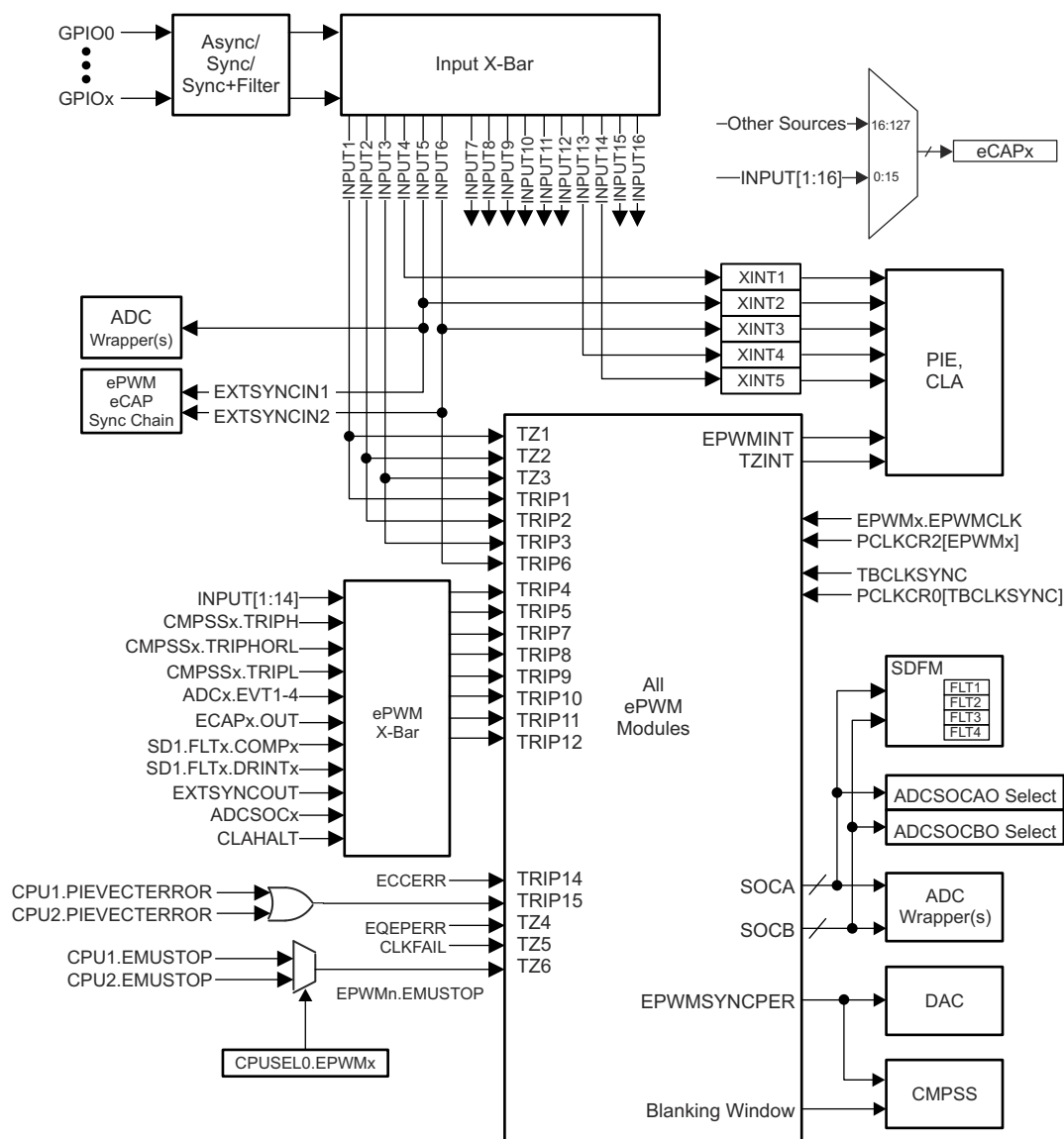


Figure 26-50. Digital-Compare Submodule High-Level Block Diagram



**Figure 26-51. GPIO MUX-to-Trip Input Connectivity**

### 26.11.1 Purpose of the Digital Compare Submodule

The key functions of the digital compare submodule are:

- Analog comparator (COMP) module outputs fed through the Input X-BAR, EPWM X-BAR, externally using the GPIO peripheral, interrupt controller signals, ECC error signals, TZ1, TZ2, and TZ3 inputs generate Digital Compare A High/Low (DCAH, DCAL) and Digital Compare B High/Low (DCBH, DCBL) signals.
- DCAH/L and DCBH/L signals trigger events that can then either be filtered or applied directly to the trip-zone, event-trigger, and time-base submodules to:
  - generate a trip zone interrupt
  - generate an ADC start of conversion
  - force an event
  - generate a synchronization event for synchronizing the ePWM module TBCTR.
- Event filtering (blanking window logic) can optionally blank the input signal to remove noise.

### 26.11.2 Enhanced Trip Action Using CMPSS

To allow multiple CMPSS at a time to affect DCA/BEVTx events and trip actions, there is a OR logic to bring together ALL trip inputs (up to 15) from sources external to the ePWM module and feed into DCAH, DCAL, DCBH, and DCBL as a “combinational input” using the DCTRIPSEL register. This is configured by selecting “Trip combination input” (value of 0xF) in the DCTRIPSEL register.

There is a discrete choice of which trip inputs to put through the combinational logic for generating the DCAH, DCAL, DCBH, and DCBL signals. This is achieved using the DCAHTRIPSEL, DCALTRIPSEL, DCBHTRIPSEL, and DCBLTRIPSEL register selections. Inputs selected for combinational input are passed through to the DCTRIPSEL register.

### 26.11.3 Using CMPSS to Trip the ePWM on a Cycle-by-Cycle Basis

When using the CMPSS to trip the ePWM on a cycle-by-cycle basis, steps can be taken to prevent an asserted comparator trip state in one PWM cycle from extending into the following cycle. The CMPSS can be used to signal a trip condition to the downstream ePWM modules. For applications like peak current mode control, only one trip event per PWM cycle is expected. Under certain conditions, it is possible for a sustained or late trip event (arriving near the end of a PWM cycle) to carry over into the next PWM cycle if precautions are not taken. If either the CMPSS Digital Filter or the ePWM Digital Compare (DC) submodule is configured to qualify the comparator trip signal, “N” number of clock cycles of qualification are introduced before the ePWM trip logic can respond to logic changes of the trip signal. Once an ePWM trip condition is qualified, the trip condition remains active for N clock cycles after the comparator trip signal has de-asserted. If a qualified comparator trip signal remains asserted within N clock cycles prior to the end of a PWM cycle, the trip condition is not cleared until after the following PWM cycle has started. Thus, the new PWM cycle detects a trip condition as soon as the cycle begins.

To avoid this undesired trip condition, the application can take steps to make sure that the qualified trip signal seen by the ePWM trip logic is deasserted prior to the end of each PWM cycle. This can be accomplished through various methods:

- Design the system such that a comparator trip is not asserted within N clock cycles prior to the end of the PWM cycle.
- Activate blanking of the comparator trip signal using the ePWM event filter at least two clock cycles prior to the PWMSYNCPER signal and continue blanking for at least N clock cycles into the next PWM cycle.
- If the CMPSS COMPxLATCH path is used, clear the COMPxLATCH at least N clock cycles prior to the end of the PWM cycle. The latch can be cleared by software (using COMPSTCLR) or by generating an early PWMSYNCPER signal. The ePWM modules on this device include the ability to generate PWMSYNCPER upon a CMPC or CMPD match (using HRPCTL) for arbitrary PWMSYNCPER placement within the PWM cycle.

## 26.11.4 Operation Highlights of the Digital Compare Submodule

The following sections describe the operational highlights and configuration options for the digital compare submodule.

### 26.11.4.1 Digital Compare Events

As described in [Section 26.11.1](#), trip zone inputs ( $\overline{TZ1}$ ,  $\overline{TZ2}$ , and  $\overline{TZ3}$ ) and CMPSSx signals from the analog comparator (COMP) module can be selected using the DCTRIPSEL bits to generate the Digital Compare A High and Low (DCAH/L) and Digital Compare B High and Low (DCBH/L) signals. Then, the configuration of the TZDCSEL register qualifies the actions on the selected DCAH/L and DCBH/L signals, which generate the DCAEVT1/2 and DCBEVT1/2 events (Event Qualification A and B).

#### Note

The  $\overline{TZn}$  signals, when used as a DCEVT tripping functions, are treated as a normal input signal and can be defined to be active-high or active-low inputs. ePWM outputs are asynchronously tripped when either the  $\overline{TZn}$ , DCAEVTx.force, or DCBEVTx.force signals are active. For the condition to remain latched, a minimum of  $3 \times \text{TBCLK}$  sync pulse width is required. If pulse width is  $< 3 \times \text{TBCLK}$  sync pulse width, the trip condition can or can not get latched by CBC or OST latches.

The DCAEVT1/2 and DCBEVT1/2 events can then be filtered to provide a filtered version of the event signals (DCEVTFILT) or the filtering can be bypassed. Filtering is discussed further in [Event Filtering](#). Either the DCAEVT1/2 and DCBEVT1/2 event signals or the filtered DCEVTFILT event signals can generate a force to the trip zone module, a TZ interrupt, an ADC SOC, or a PWM sync signal.

- **force signal:** DCAEVT1/2.force signals force trip zone conditions which either directly influence the output on the EPWMxA pin (using TZCTL, TZCTLDCA, TZCTLDCB register configurations) or, if the DCAEVT1/2 signals are selected as one-shot or cycle-by-cycle trip sources (using the TZSEL register), the DCAEVT1/2.force signals can effect the trip action using the TZCTL or TZCTL2 register configurations. The DCBEVT1/2.force signals behaves similarly, but affect the EPWMxB output pin instead of the EPWMxA output pin.

The priority of conflicting actions on the TZCTL, TZCTL2, TZCTLDCA and TZCTLDCB registers is as follows (highest priority overrides lower priority):

Output EPWMxA:

- TZA (highest) -> DCAEVT1 -> DCAEVT2 (lowest)
- TZAU (highest) -> DCAEVT1U -> DCAEVT2U (lowest)
- TZAD (highest) -> DCAEVT1D -> DCAEVT2D (lowest)

Output EPWMxB:

- TZB (highest) -> DCBEVT1 -> DCBEVT2 (lowest)
- TZBU (highest) -> DCBEVT1U -> DCBEVT2U (lowest)
- TZBD (highest) -> DCBEVT1D -> DCBEVT2D (lowest)

- **interrupt signal:** DCAEVT1/2.interrupt signals generate trip zone interrupts to the interrupt controller. To enable the interrupt, set the DCAEVT1, DCAEVT2, DCBEVT1, or DCBEVT2 bits in the TZEINT register. Once one of these events occurs, an EPWMxTZINT interrupt is triggered, and the corresponding bit in the TZCLR register must be set to clear the interrupt.
- **soc signal:** The DCAEVT1.soc signal interfaces with the event-trigger submodule and can be selected as an event which generates an ADC start-of-conversion-A (SOCA) pulse using the ETSEL[SOCASEL] bit. Likewise, the DCBEVT1.soc signal can be selected as an event which generates an ADC start-of-conversion-B (SOCB) pulse using the ETSEL[SOCBSEL] bit.
- **sync signal:** The DCAEVT1.sync and DCBEVT1.sync events are ORed with the EPWMxSYNCl input signal and the TBCTL[SWFSYNC] signal to generate a synchronization pulse to the time-base counter.

Figure 26-52 and Figure 26-53 show how the DCxEVT1, DCxEVT2, or DCEVTFLT signals are processed to generate the digital compare A and B event force, interrupt, soc and sync signals.

In some of the applications like Phase Shifted Full Bridge (PSFB) Converters, it is required that different actions are taken on a CBC trip event and an OST trip event. This can be achieved using the DCxEVT1LAT.

- This latch can be cleared on CNT=0, CTR=PRD, and CNT=0 OR CTR=PRD events based on the setting of DCxCTL.EVTy.LATCLRSEL setting. This is similar to CBC latch clear mechanism.
- DCxEVTy.force signal can be chosen to be either the latched version or the unlatched version based on DCxCTL.EVTyLATSEL value.
- The status of DCxEVTyLAT signal can be accessed by reading DCxCTL.EVTyLAT field.

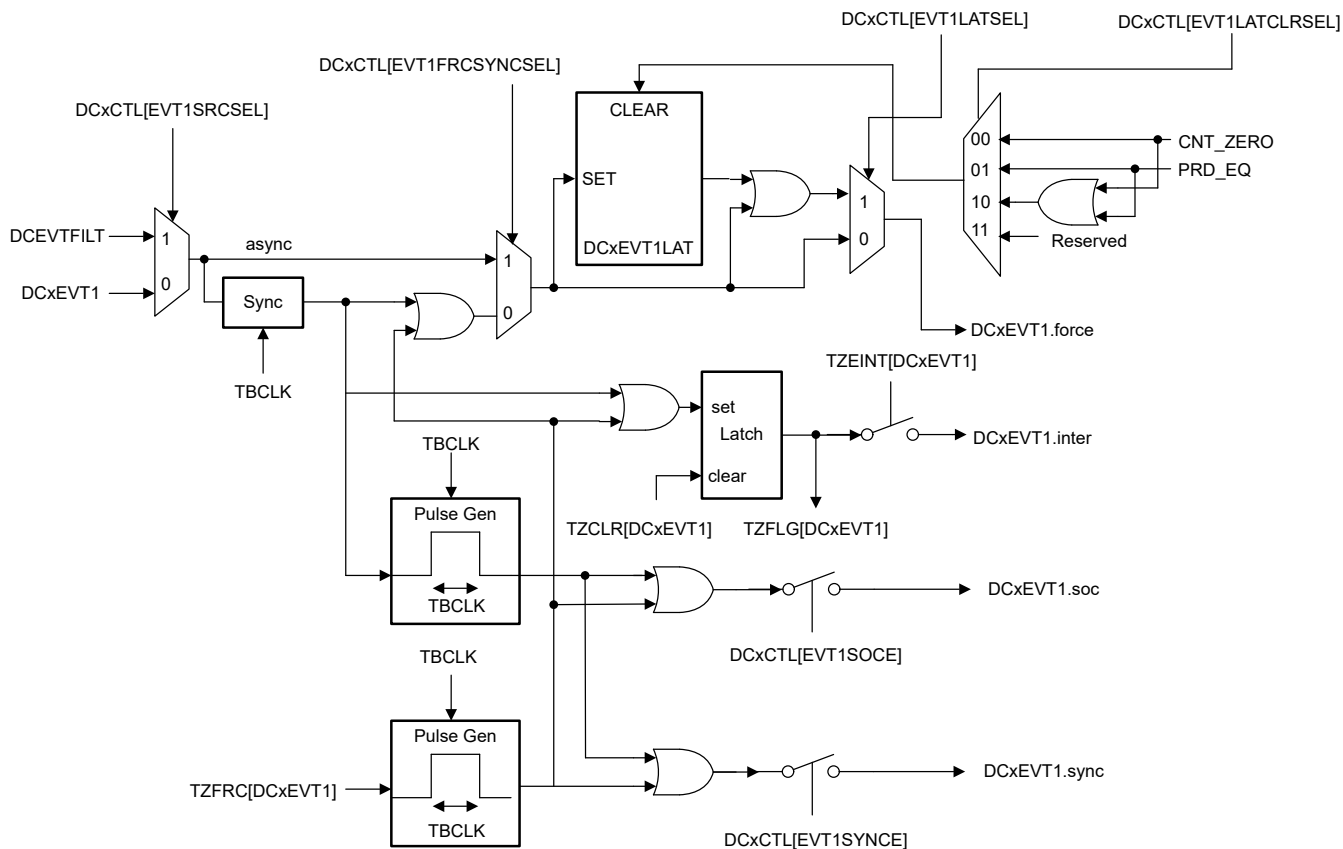
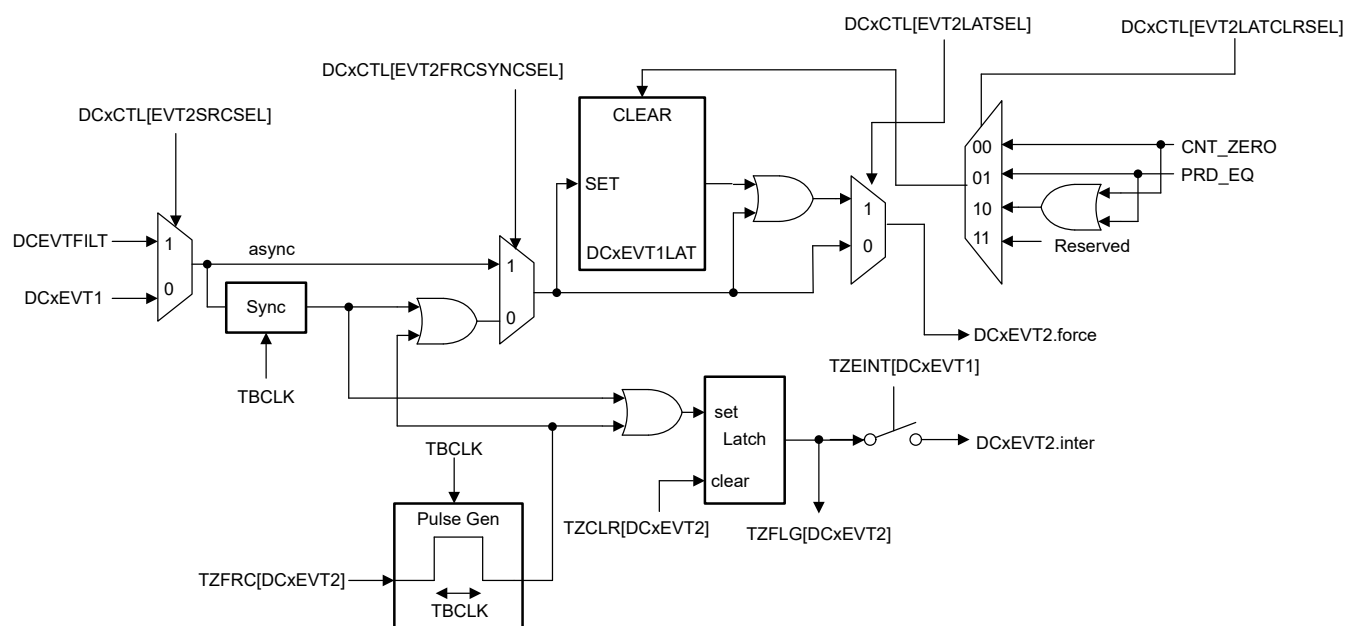


Figure 26-52. DCxEVT1 Event Triggering



**Figure 26-53. DCxEVT2 Event Triggering**



#### 26.11.4.2 Event Filtering

**Blank Control Logic:** The DCAEVT1/2 and DCBEVT1/2 events can be filtered using event filtering logic to remove noise by optionally blanking events for a certain period of time. This is useful for cases where the analog comparator outputs can be selected to trigger DCAEVT1/2 and DCBEVT1/2 events, and the blank control logic is used to filter out potential noise on the signal prior to tripping the PWM outputs or generating an interrupt or ADC start-of-conversion. Blank control logic is used to define a blanking window, which ignores all event occurrences on the signal while the window is active. The blanking window is configured in the DCFCTL, DCFOFFSET, and DCFWINDOW registers. The DCFCTL register enables the blanking window and aligns the blanking window to either a CTR = PRD pulse or a CTR = 0 pulse or both CTR = PRD and CTR = 0 as specified by DCFCTL[PULSESEL]. DCFCTL[SRCSEL] selects the DCxEVType event source for the DCEVTFILT signal. An offset value in TBCLK counts is programmed into the DCFOFFSET register, which determines at what point after the CTR = PRD or CTR = 0 pulse the blanking window starts. The duration of the blanking window, in number of TBCLK counts after the offset counter expires, is written to the DCFWINDOW register. Before and after the blanking window ends, events can generate soc, sync, interrupt, and force signals as before.

Figure 26-54 shows the details of the event filtering logic.

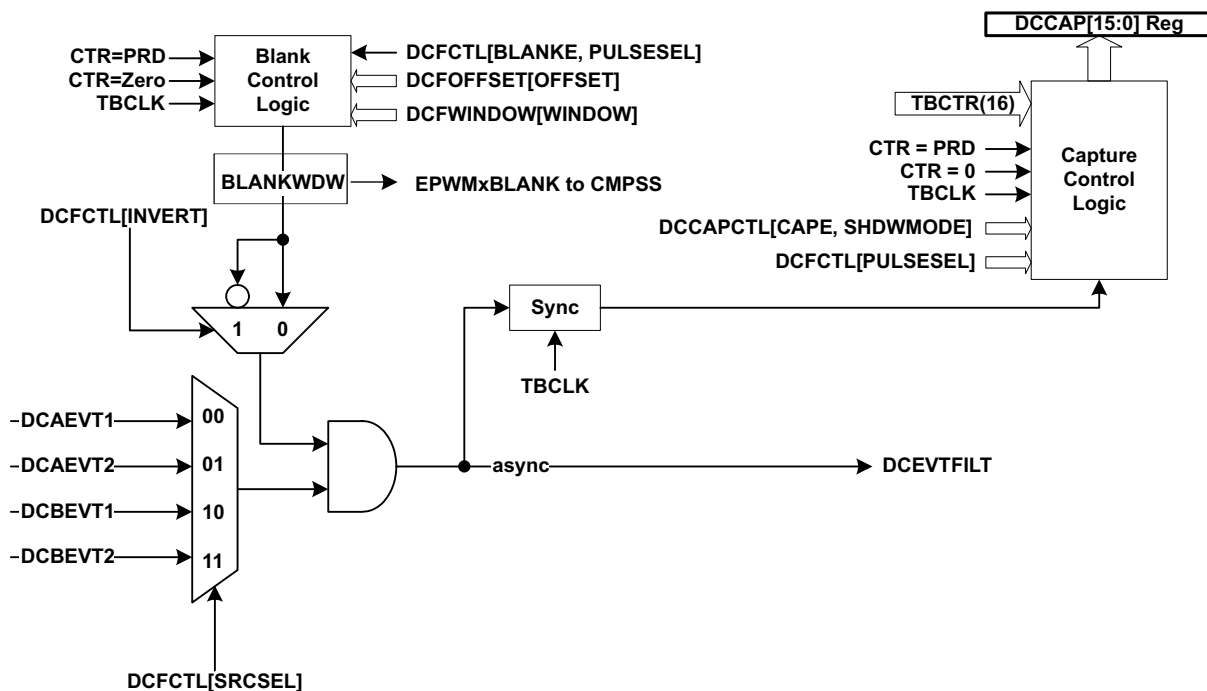


Figure 26-54. Event Filtering

**Capture Control Logic:** The event filtering can also capture the TBCTR value of the selected DCxEVType event as configured in the DCCAPCTL register. When capture control logic is enabled, the selected DCxEVType event triggers capture of the TBCTR to the active register. The CPU reads directly from the active register unless shadow mode is enabled by DCCAPCTL[SHDWMODE]. When shadow mode is enabled, the active register information is copied to shadow register on the event specified by DCFCTL[PULSESEL], and the CPU reads from the shadow register. After the selected DCxEVType event, no further capture events occur until the event specified by DCCAPCTL[CAPMODE]. The CAPMODE can be configured two ways: (1) no further capture events occur until the event defined by DCFCTL[PULSESEL] or (2) no further capture events occur until the compare-event flag at DCCAPCTL[CAPSTS] is cleared by DCCAPCTL[CAPCLR].

#### Note

You must configure the ePWM blanking window appropriately so that the Trip Input stays valid for at least 3 ePWM cycles after the blanking window has expired.

Figure 26-55 illustrates several timing conditions for the offset and blanking window within an ePWM period. Notice that if the blanking window crosses the CTR = 0 or CTR = PRD boundary, the next window still starts at the same offset value after the CTR = 0 or CTR = PRD pulse.

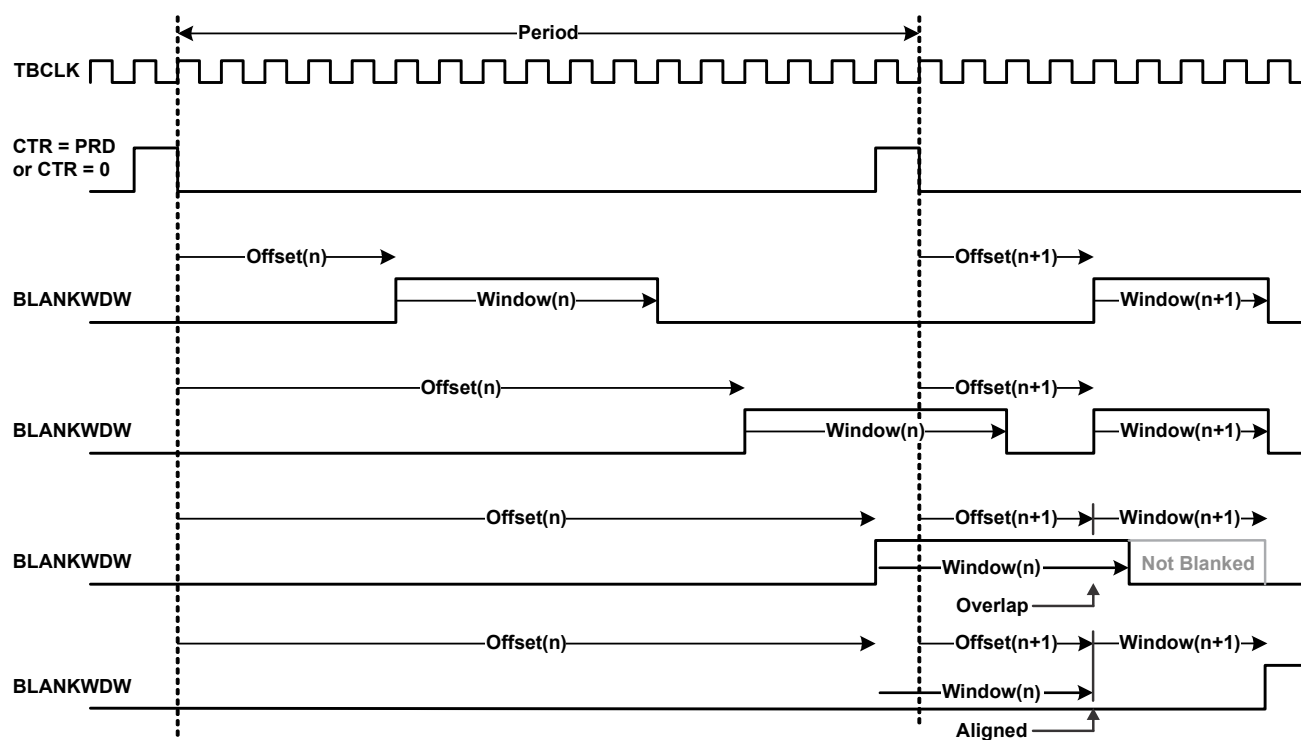


Figure 26-55. Blanking Window Timing Diagram

### 26.11.4.3 Valley Switching

Event filtering depicts the valley switching function along with the event filtering logic described in Event Filtering. This function can be used to achieve programmable valley switching without any additional external circuitry. This module provides an on-chip hardware mechanism that can:

- Capture the oscillation period
- Accurately delay the PWM switching instant
- Allow a programmable number of edges before the delay takes effect
- Provide multiple choices of triggers and events
- Allow easy adaptability for optimum performance under changing system/operating conditions

The DCxEVTy signal needs further processing to support valley switching. Here is a brief description of how valley switching function is enabled:

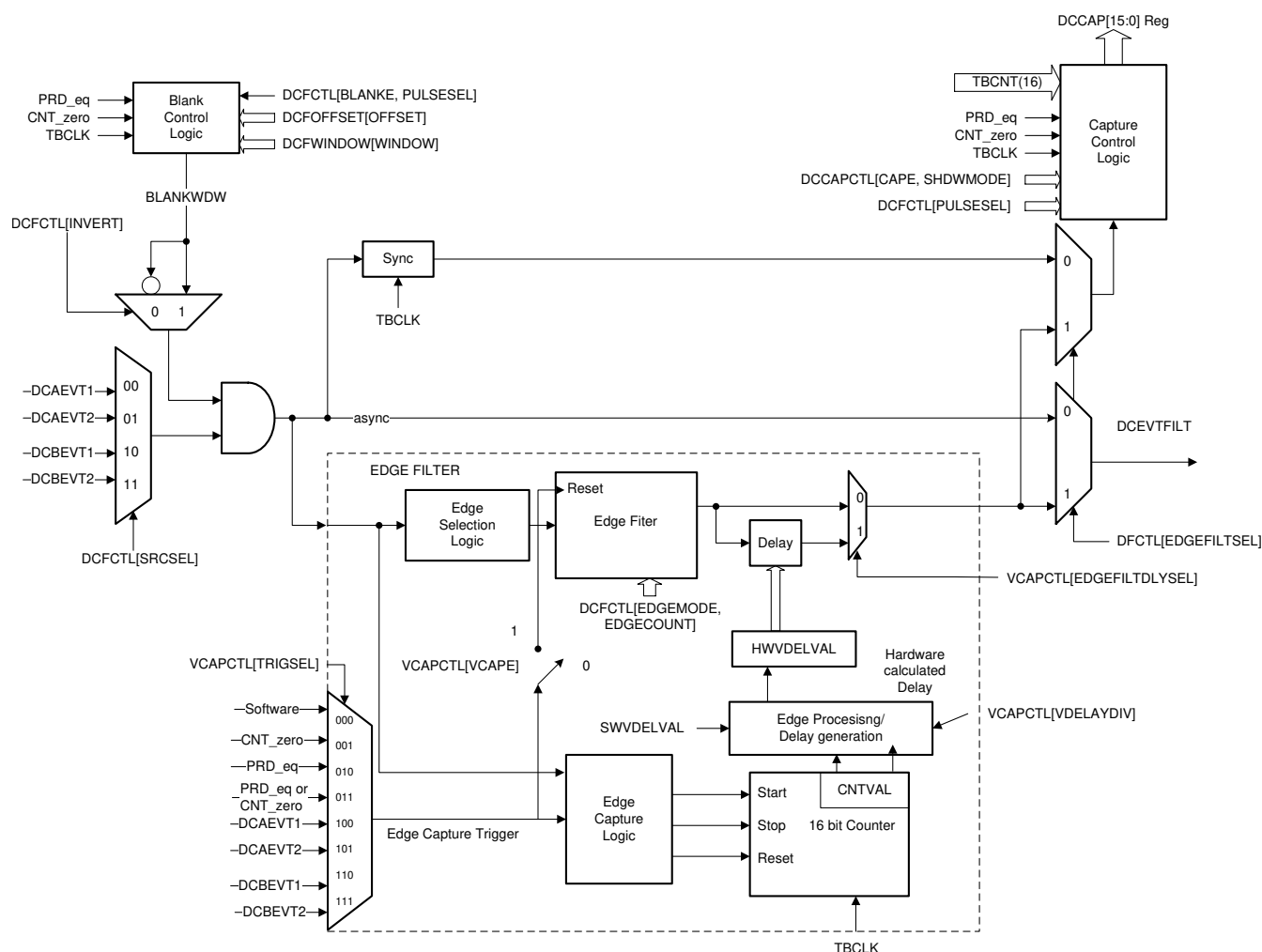
1. Select one of the DCxEVTy events as input to the valley switching block (DCFCTL[SRCSSEL]) with an option to add the blanking window (Blank Control Logic). This is where the comparator output (or external input) above is selected as an input to the valley switching block.
2. Configure the edge filter to capture 'n' rising, falling or both edges through the edge selection logic (DCFCTL[EDGEMODE, EDGECOUNT]).
3. Select the correct event to reset and restart the edge filter (VCAPCTL[TRIGSEL]). Edge capturing event is triggered or armed by this selected edge.
4. Enable valley capture logic (VCAPCTL[VCAPE]).
5. Select the start edge that indicates the start of capture for oscillation period measurement (VCNTCFG[STARTEDGE]). This is where the 16-bit counter starts counting.
6. Select the stop edge (VCNTCFG[STOPEDGE]) that indicates the edge at which the 16-bit counter stops counting. The captured counter value (CNTVAL) provides oscillation period information.
  - The STOPEDGE value must always be greater than STARTEDGE value.
7. Configure and apply the captured delay (CNTVAL) to the edge filtered DCxEVTy signal. The CNTVAL value can be applied as is or applied in conjunction with a software programmed value (useful for offset adjustment) (SWVDELVAL) or only a fraction of the delay can be applied with or without SWVDELVAL. This is useful to correctly apply a delay corresponding to the valley point. (VCAPCTL[VDELAYDIV])
8. Configure VCAPCTL[EDGEFILTDLYSEL] to apply hardware delay based on the captured value above.

Once the counter is stopped, counter value is copied into CNTVAL register and counter is reset to zero. No further captures are done until the logic is triggered again by occurrence of event selected by VCAPCTL[TRIGSEL]. In this implementation, the software trigger is used as the source for VCAPCTL[TRIGSEL]. Upon occurrence of the trigger event, irrespective of the current status of the counter, the counter is reset and starts counting from zero upon occurrence of the STARTEDGE. Similarly, upon occurrence of the trigger event, the edge filter is reset and starts counting from zero upon occurrence of the STARTEDGE.

Output from the valley switching block (DCEVTFILT) is then used to synchronize the PWM time-base. The process is shown in [Figure 26-56](#).

#### Note

A specific application example showcasing the usage of valley switching hardware and software is available in C2000Ware.



**Figure 26-56. Valley Switching**

## 26.12 ePWM Crossbar (X-BAR)

Figure 26-57 shows the architecture of the ePWM Crossbar (X-BAR). This module enables selection of various trigger sources into any of the eight dedicated EPWM trips inputs, namely the TRIP4, TRIP5, TRIP7, TRIP8, TRIP9, TRIP10, TRIP11, and TRIP12.

### Note

Refer to the *Crossbar (X-BAR)* chapter for more information on the X-BAR modules, including X-BAR flags.

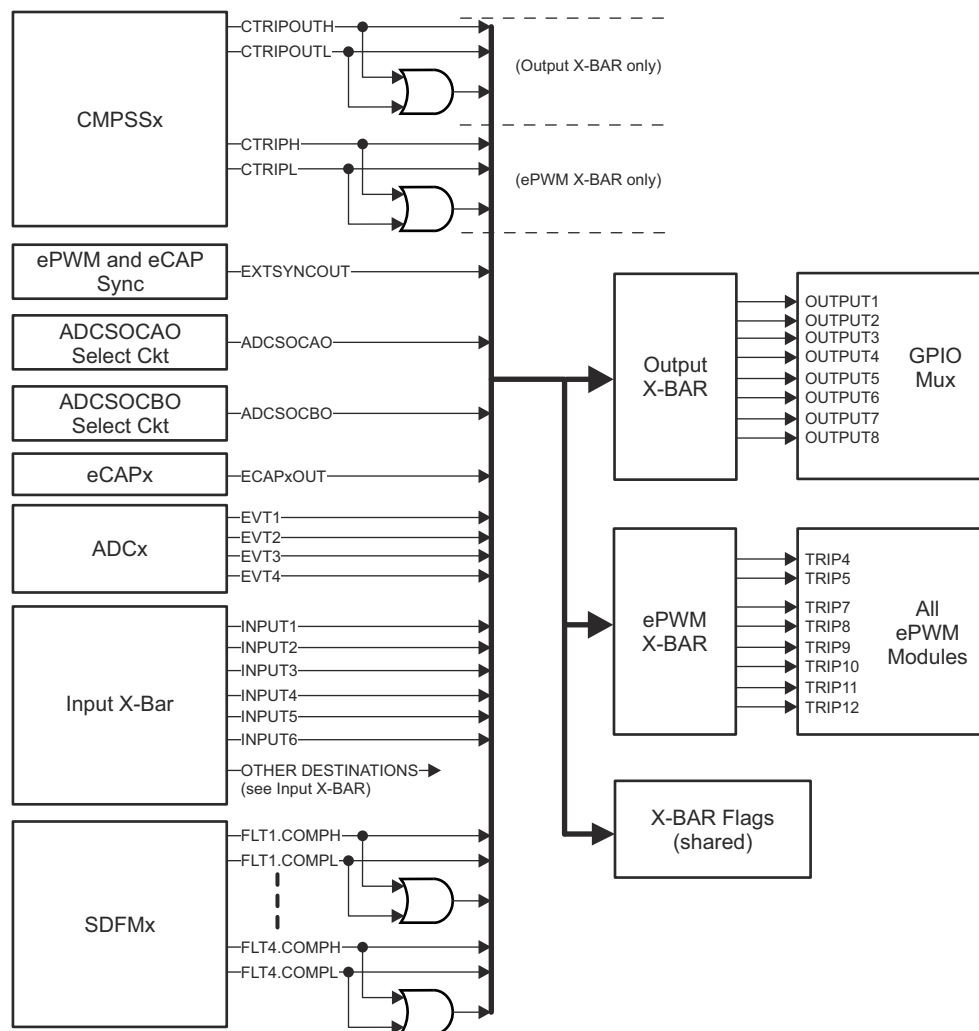


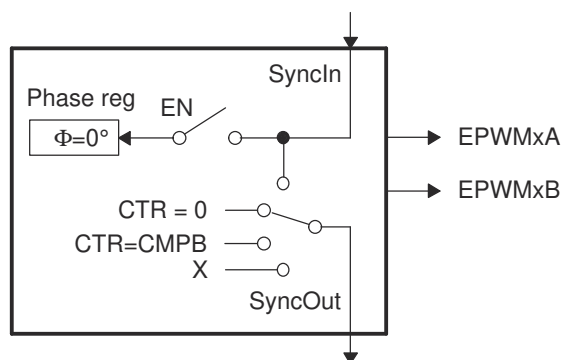
Figure 26-57. ePWM X-BAR

## 26.13 Applications to Power Topologies

An ePWM module has all the local resources necessary to operate completely as a standalone module or to operate in synchronization with other identical ePWM modules.

### 26.13.1 Overview of Multiple Modules

Previously in this chapter, all discussions have described the operation of a single module. To facilitate the understanding of multiple modules working together in a system, the ePWM module described in reference is represented by the more simplified block diagram shown in [Figure 26-58](#). This simplified ePWM block shows only the key resources needed to explain how a multiswitch power topology is controlled with multiple ePWM modules working together.



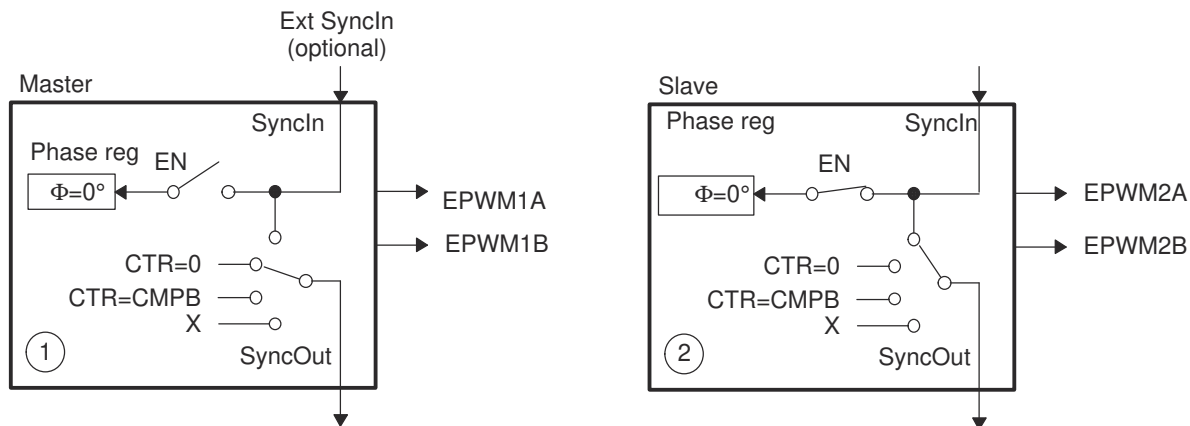
**Figure 26-58. Simplified ePWM Module**

### 26.13.2 Key Configuration Capabilities

The key configuration choices available to each module are as follows:

- Options for SyncIn
  - Load own counter with phase register on an incoming sync strobe—enable (EN) switch closed
  - Do nothing or ignore incoming sync strobe—enable switch open
  - Sync flow-through - SyncOut connected to SyncIn
  - Master mode, provides a sync at PWM boundaries—SyncOut connected to CTR = PRD
  - Master mode, provides a sync at any programmable point in time—SyncOut connected to CTR = CMPB
  - Module is in standalone mode and provides no sync to other modules—SyncOut connected to X (disabled)
- Options for SyncOut
  - Sync flow-through - SyncOut connected to SyncIn
  - Master mode, provides a sync at PWM boundaries—SyncOut connected to CTR = PRD
  - Master mode, provides a sync at any programmable point in time—SyncOut connected to CTR = CMPB
  - Module is in standalone mode and provides no sync to other modules—SyncOut connected to X (disabled)

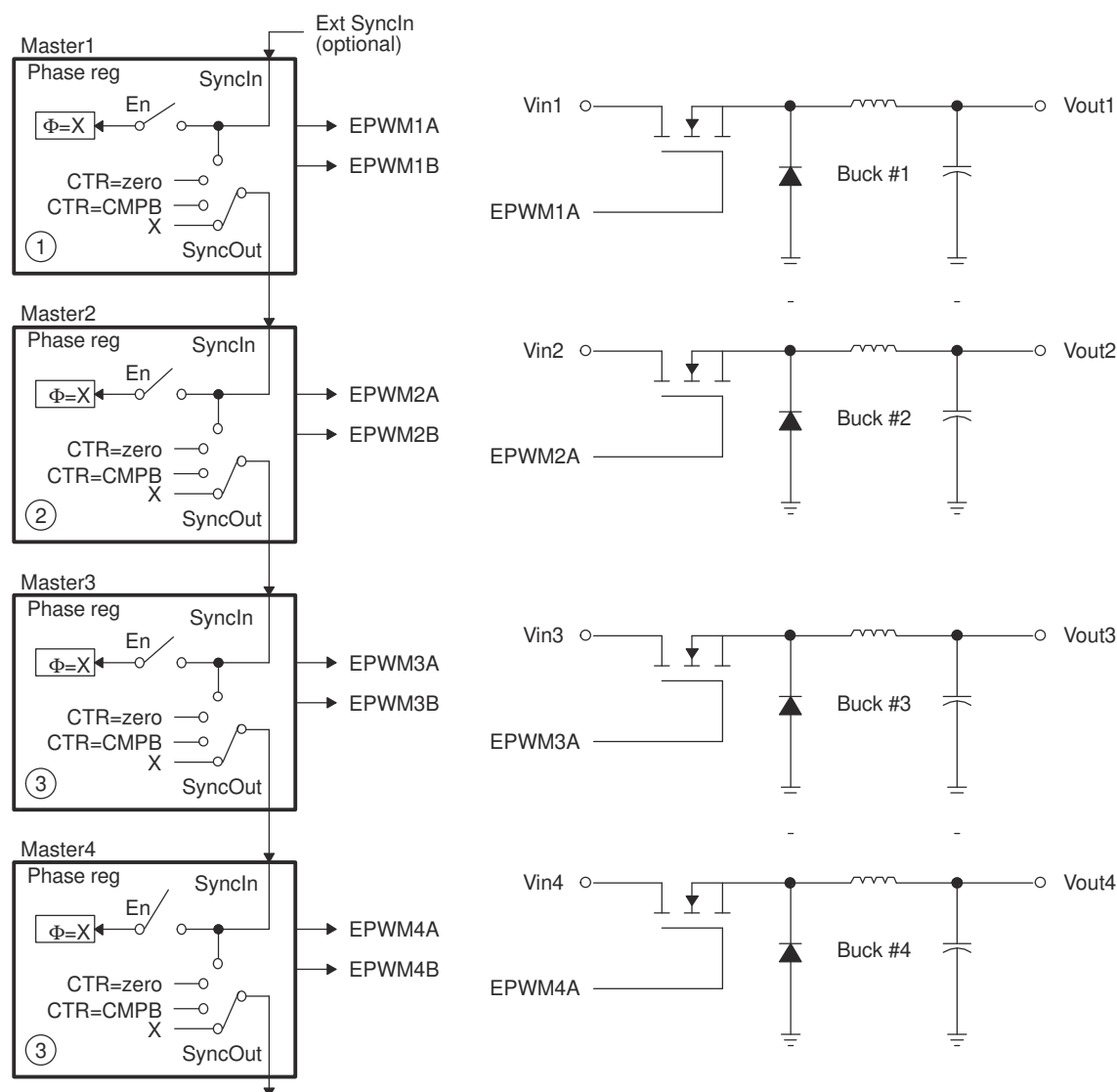
For each choice of SyncOut, a module can also choose to load the counter with a new phase value on a SyncIn strobe input or choose to ignore the value (that is, by the enable switch). Although various combinations are possible, the two most common—Master module and Slave module modes—are shown in [Figure 26-59](#).



**Figure 26-59. EPWM1 Configured as a Typical Master, EPWM2 Configured as a Slave**

### 26.13.3 Controlling Multiple Buck Converters With Independent Frequencies

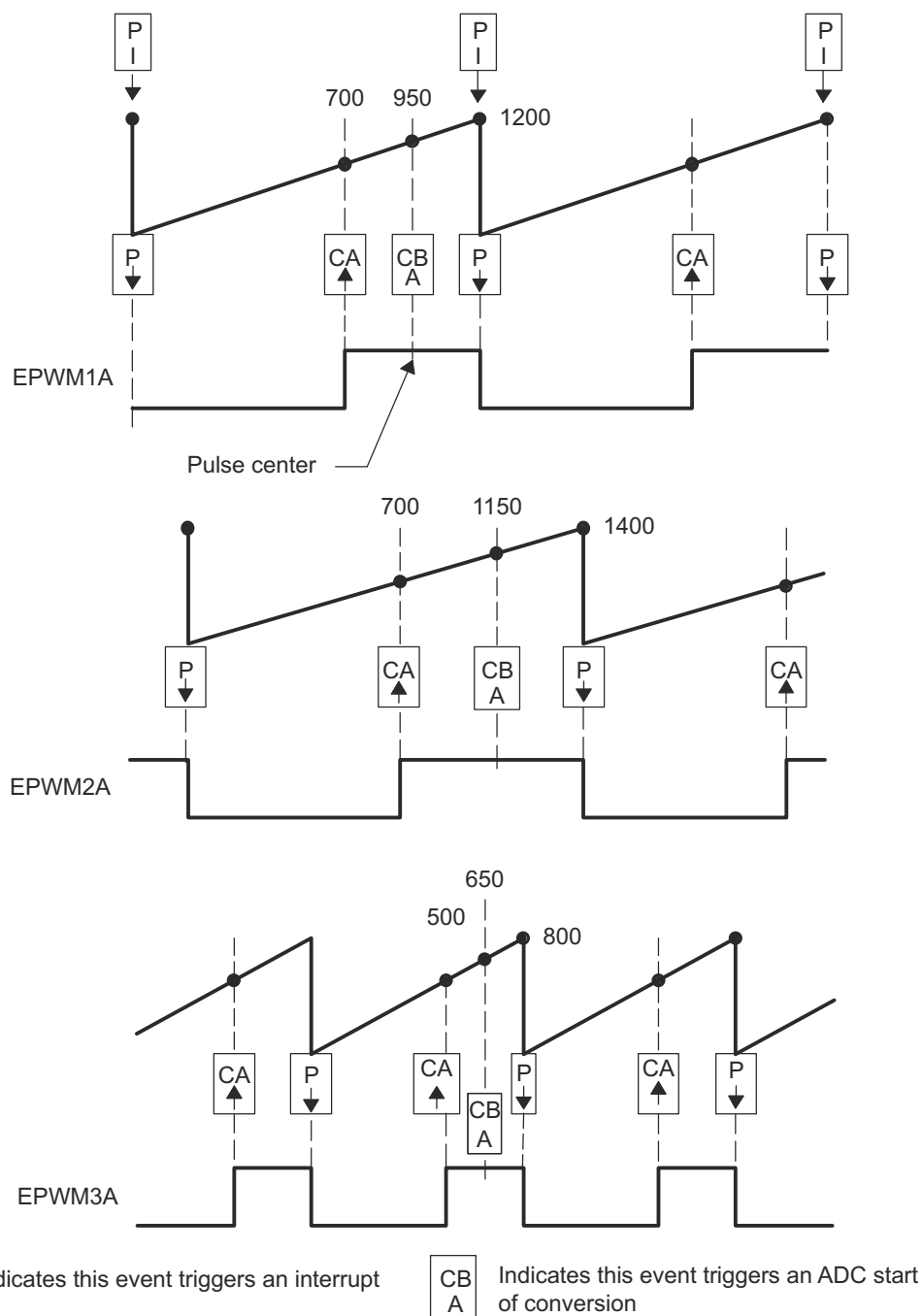
One of the simplest power converter topologies is the buck. A single ePWM module configured as a master can control two buck stages with the same PWM frequency. If independent frequency control is required for each buck converter, then one ePWM module must be allocated for each converter stage. Figure 26-60 shows four buck stages, each running at independent frequencies. In this case, all four ePWM modules are configured as Masters and no synchronization is used. Figure 26-61 shows the waveforms generated by the setup shown in Figure 26-60; note that only three waveforms are shown, although there are four stages.



A.  $\phi = X$  indicates value in phase register is a "don't care"

**Figure 26-60. Control of Four Buck Stages. Here  $F_{PWM1} \neq F_{PWM2} \neq F_{PWM3} \neq F_{PWM4}$**

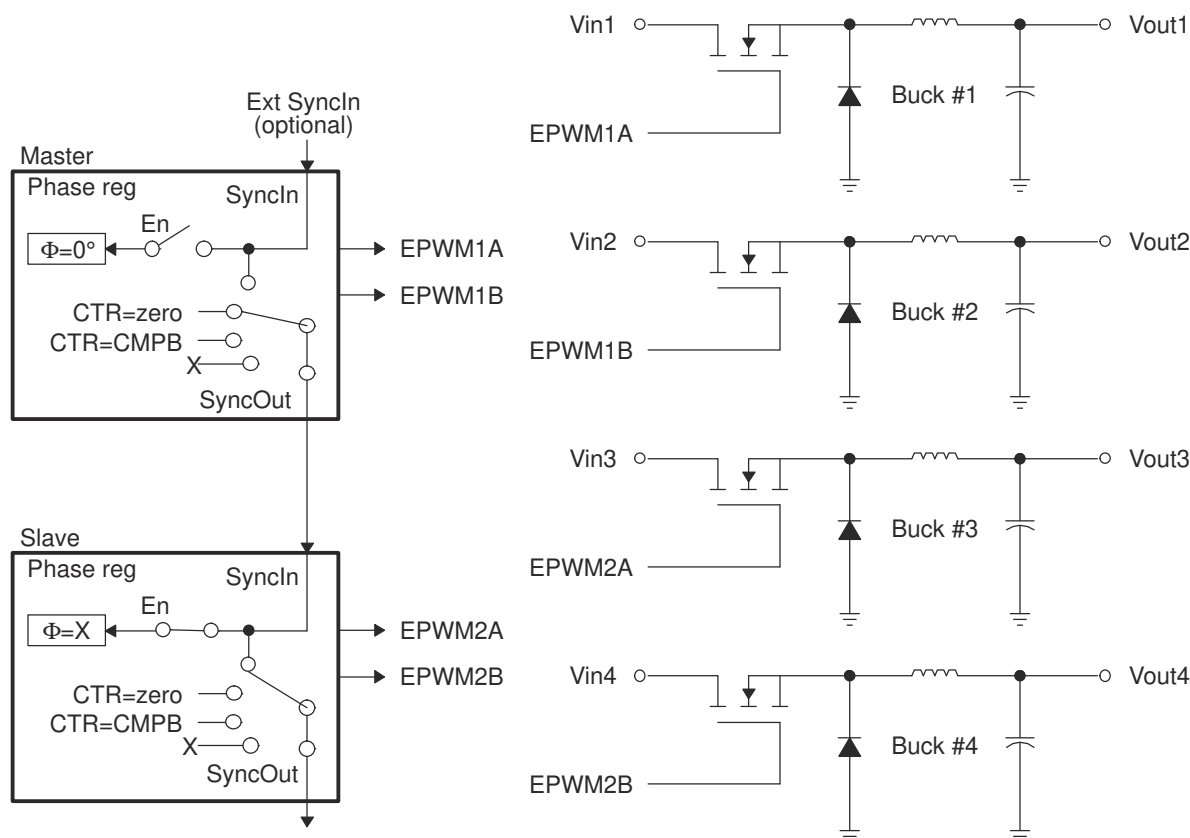




**Figure 26-61. Buck Waveforms for Control of Four Buck Stages (Note: Only three bucks shown here)**

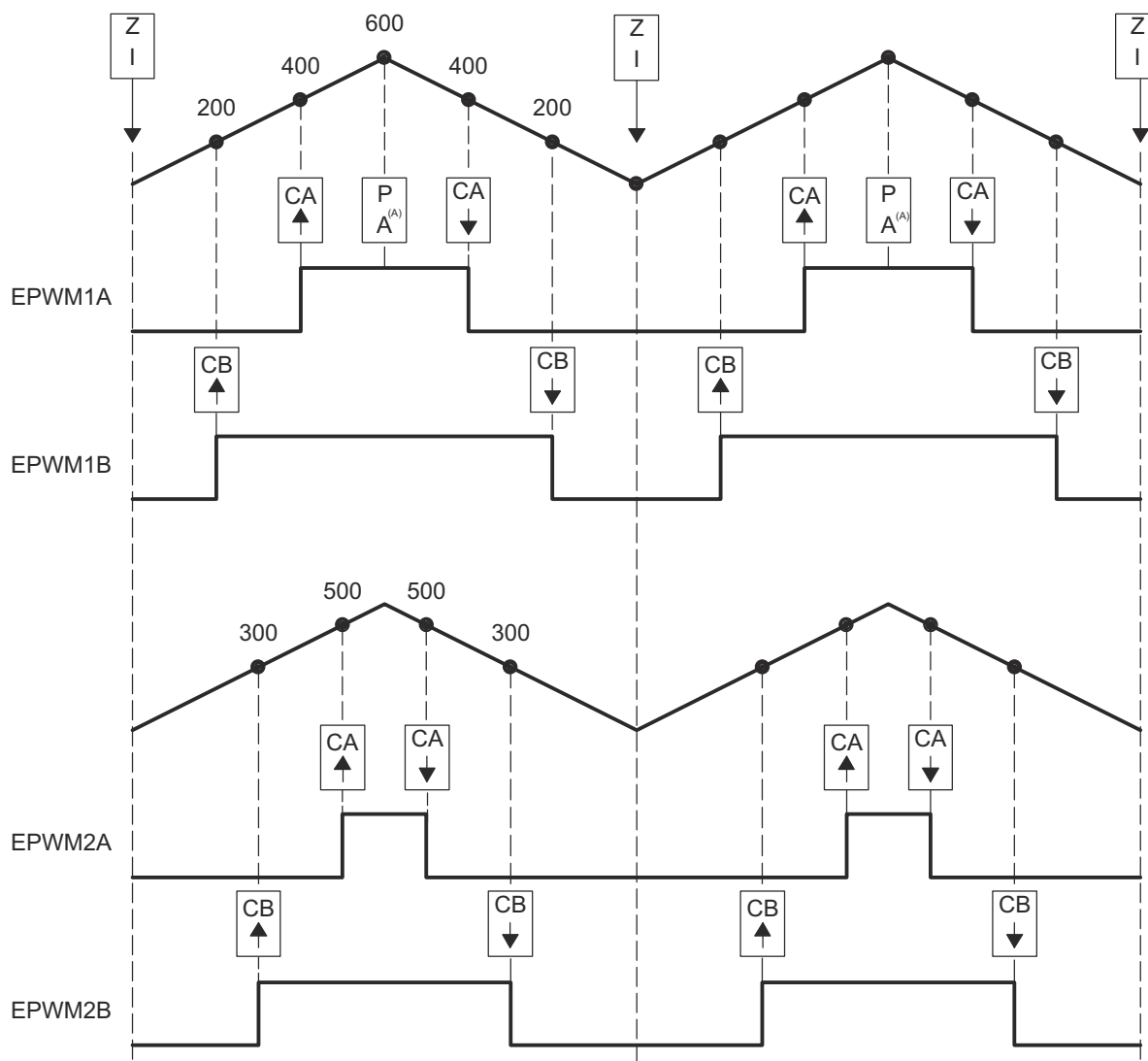
### 26.13.4 Controlling Multiple Buck Converters With Same Frequencies

If synchronization is a requirement, ePWM module 2 is configured as a slave and operates at integer multiple (N) frequencies of module 1. The sync signal from master to slave makes sure these modules remain locked. [Figure 26-62](#) shows such a configuration; [Figure 26-63](#) shows the waveforms generated by the configuration.



A.  $\phi = X$  indicates value in phase register is a "don't care"

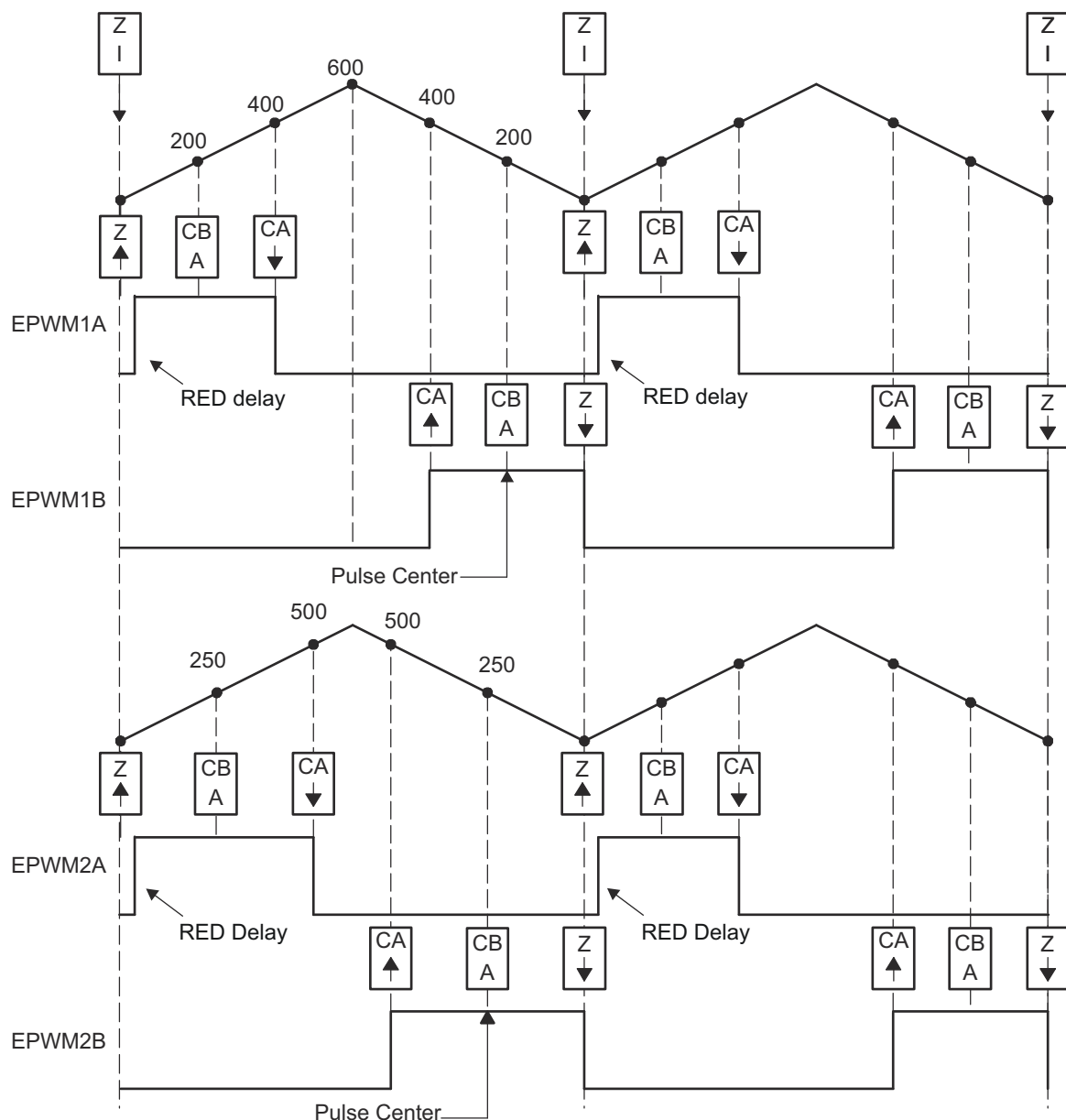
**Figure 26-62. Control of Four Buck Stages. (Note:  $F_{PWM2} = N \times F_{PWM1}$ )**



A. Starts ADC conversion.

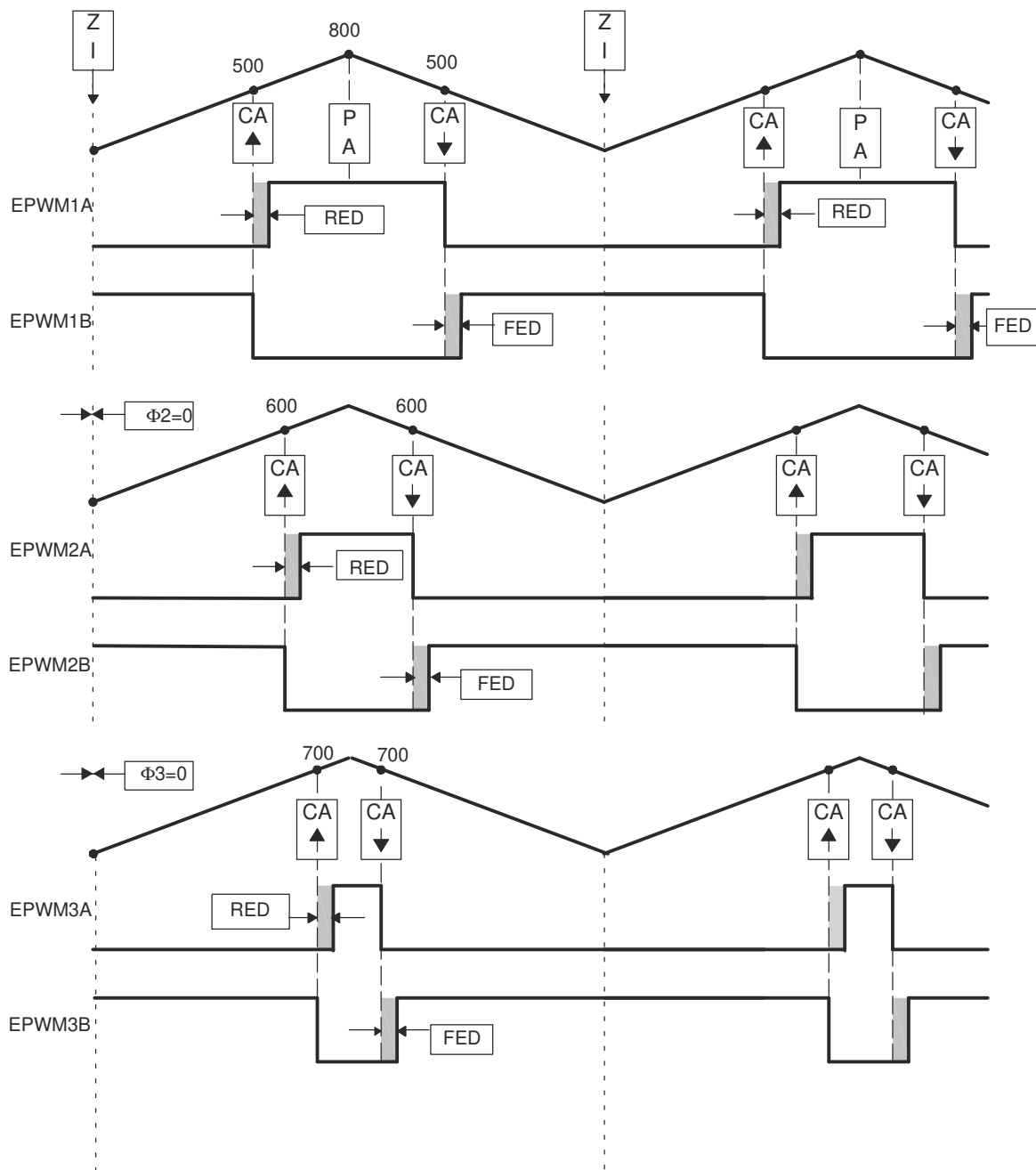
**Figure 26-63. Buck Waveforms for Control of Four Buck Stages (Note:  $F_{PWM2} = F_{PWM1}$ )**





**Figure 26-65. Half-H Bridge Waveforms for Control of Two Half-H Bridge Stages (Note: Here  $F_{PWM2} = F_{PWM1}$ )**

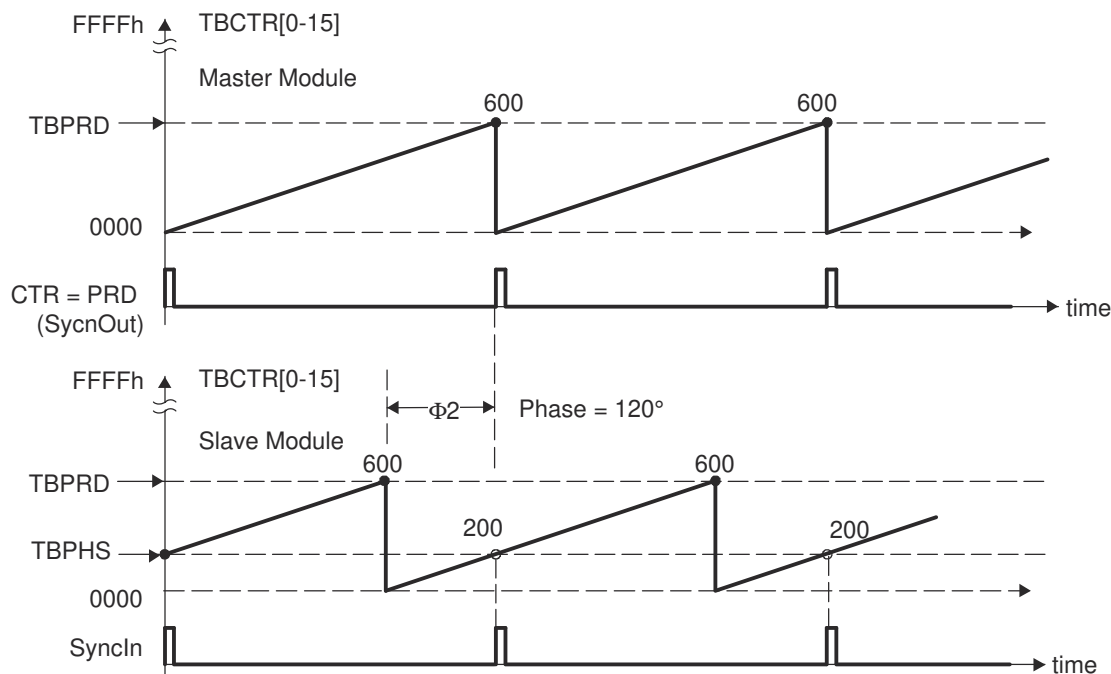




**Figure 26-67. 3-Phase Inverter Waveforms for Control of Dual 3-Phase Inverter Stages (Only One Inverter Shown)**







**Figure 26-69. Timing Waveforms Associated with Phase Control Between Two Modules**

### 26.13.8 Controlling a 3-Phase Interleaved DC/DC Converter

A popular power topology that makes use of phase-offset between modules is shown in [Figure 26-70](#). This system uses three PWM modules, with module 1 configured as the master. To work, the phase relationship between adjacent modules must be  $F = 120^\circ$ . This is achieved by setting the slave TBPHS registers 2 and 3 with values of 1/3 and 2/3 of the period value, respectively. For example, if the period register is loaded with a value of 600 counts, then TBPHS (slave 2) = 200 and TBPHS (slave 3) = 400. Both slave modules are synchronized to the master module 1.

This concept can be extended to four or more phases, by setting the TBPHS values appropriately. The following formula gives the TBPHS values for N phases:

$$TBPHS(N,M) = (TBPRD/N) \times (M-1)$$

Where:

N = number of phases

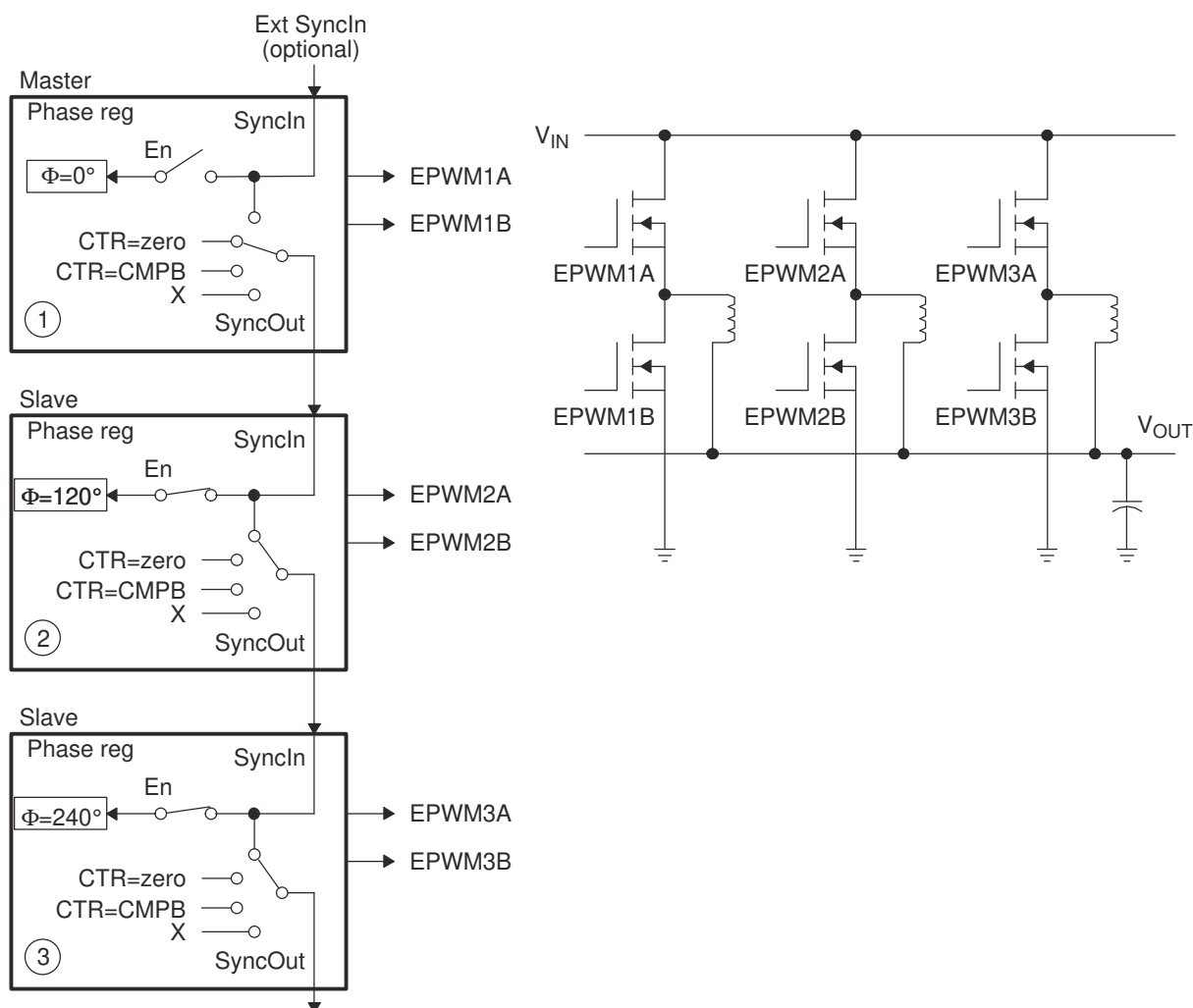
M = PWM module number

For example, for the 3-phase case (N=3), TBPRD = 600,

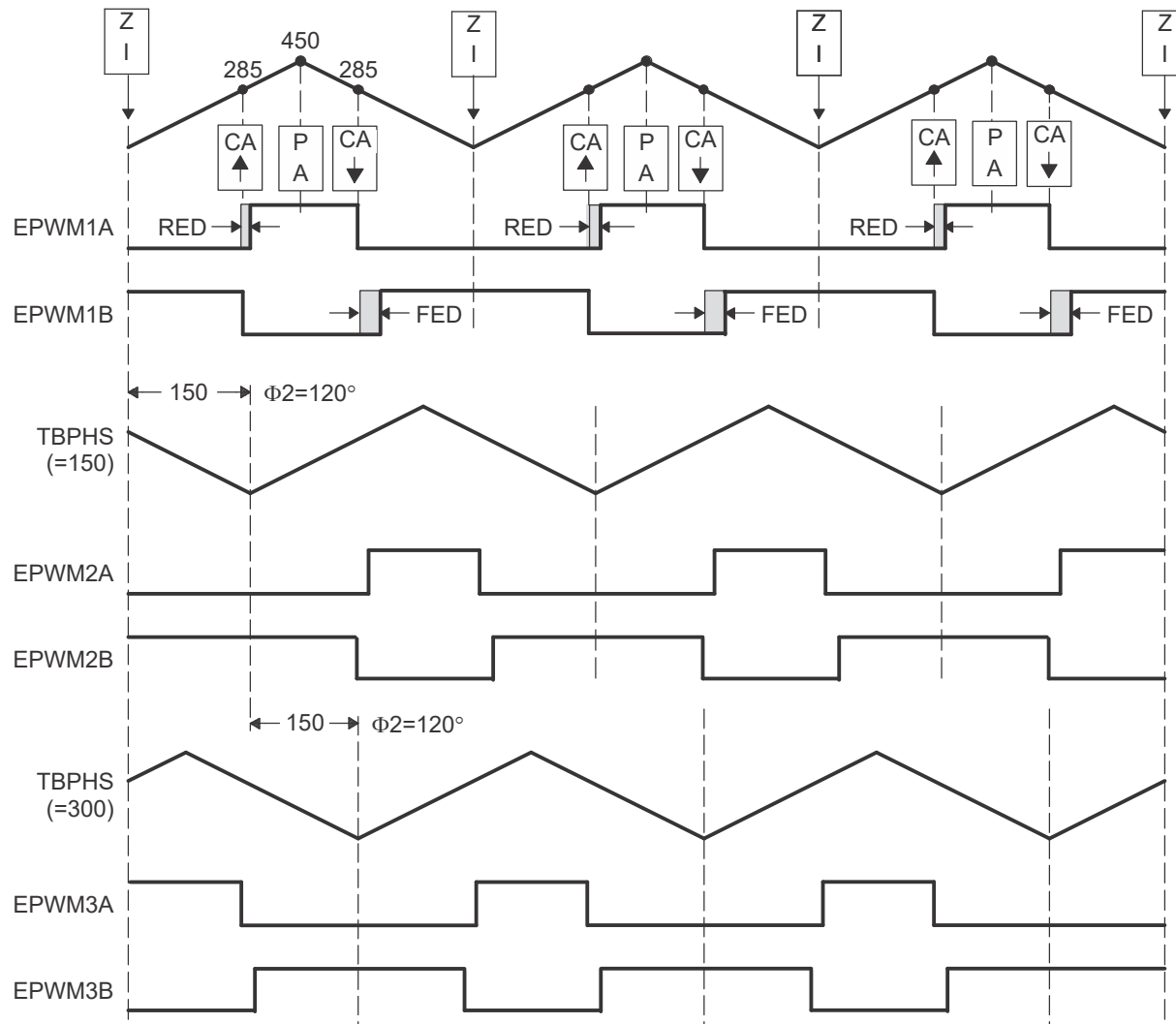
TBPHS(3,2) = (600/3) × (2-1) = 200 (that is, Phase value for Slave module 2)

TBPHS(3,3) = 400 (that is, Phase value for Slave module 3)

[Figure 26-71](#) shows the waveforms for the configuration in [Figure 26-70](#).



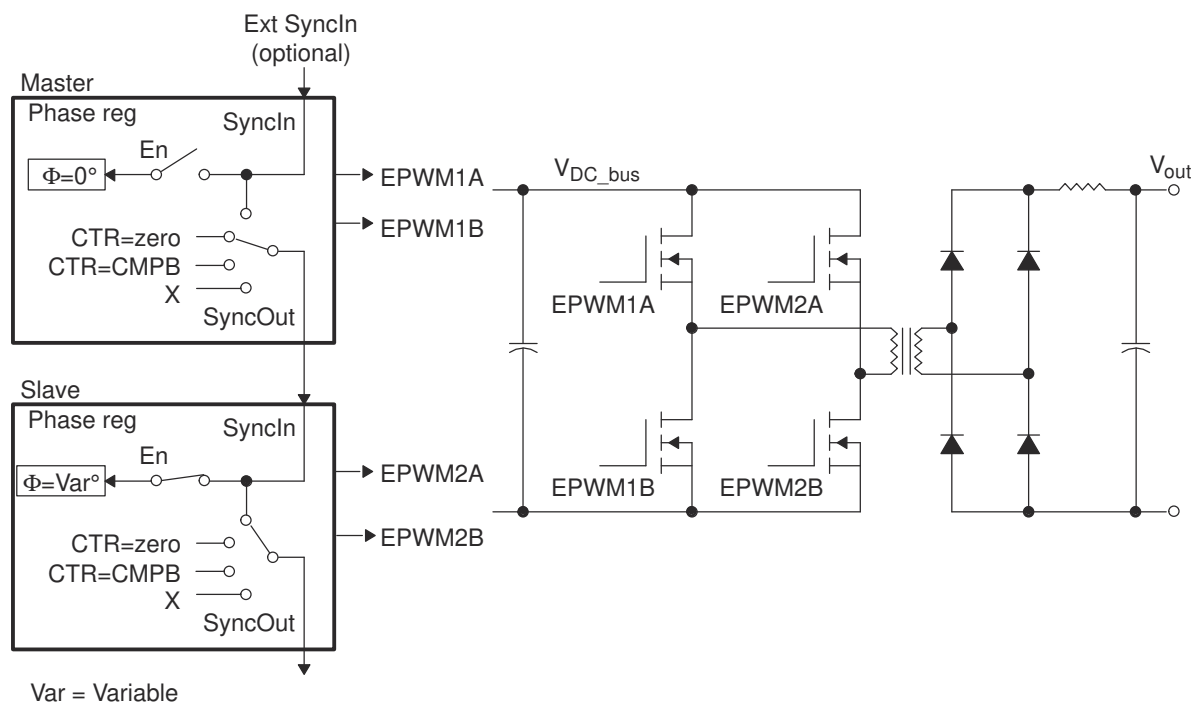
**Figure 26-70. Control of 3-Phase Interleaved DC/DC Converter**



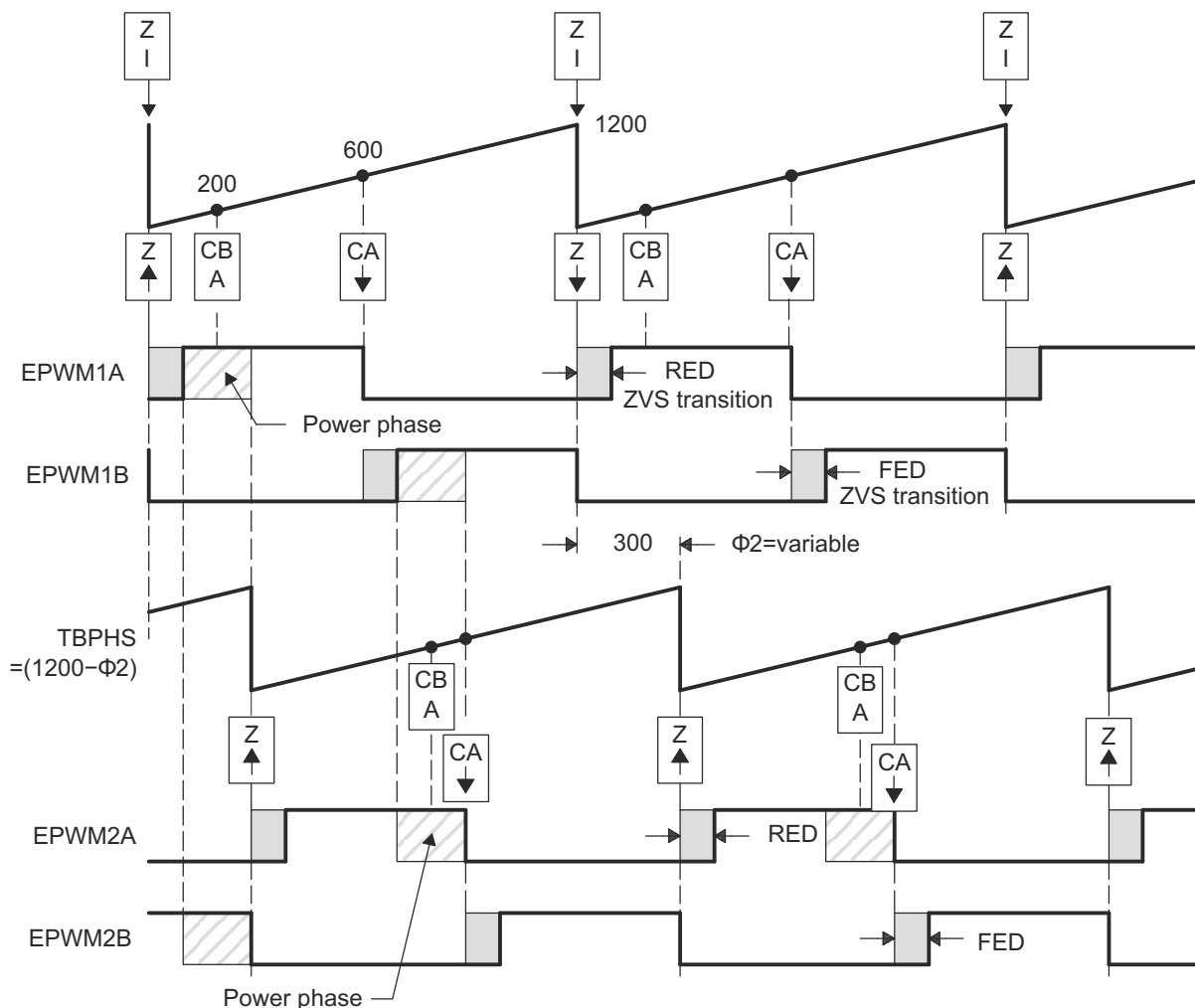
**Figure 26-71. 3-Phase Interleaved DC/DC Converter Waveforms for Control of 3-Phase Interleaved DC/DC Converter**

### 26.13.9 Controlling Zero Voltage Switched Full Bridge (ZVSFB) Converter

The example given in [Figure 26-72](#) assumes a static or constant phase relationship between legs (modules). In such a case, control is achieved by modulating the duty cycle. It is also possible to dynamically change the phase value on a cycle-by-cycle basis. This feature lends to controlling a class of power topologies known as *phase-shifted full bridge*, or *zero voltage switched full bridge*. Here the controlled parameter is not duty cycle (this is kept constant at approximately 50 percent); instead it is the phase relationship between legs. Such a system can be implemented by allocating the resources of two PWM modules to control a single power stage, which in turn requires control of four switching elements. [Figure 26-73](#) shows a master and slave module combination synchronized together to control a full H-bridge. In this case, both master and slave modules are required to switch at the same PWM frequency. The phase is controlled by using the slave phase register (TBPHS). The master phase register is not used and therefore can be initialized to zero.



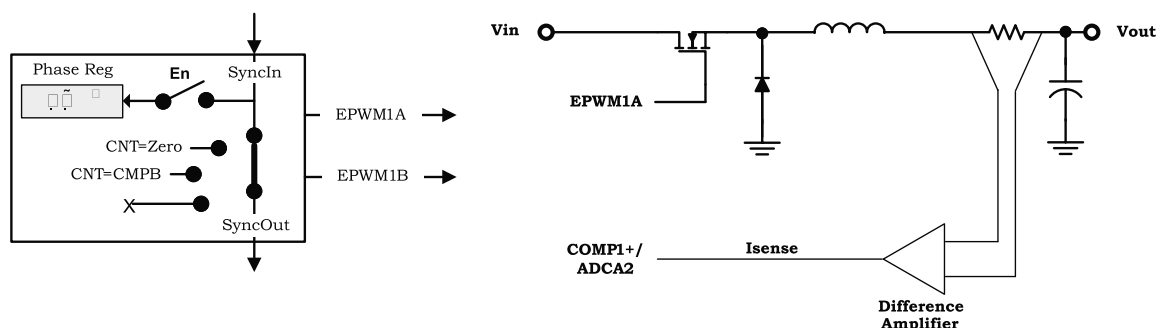
**Figure 26-72. Control of Full-H Bridge Stage ( $F_{PWM2} = F_{PWM1}$ )**



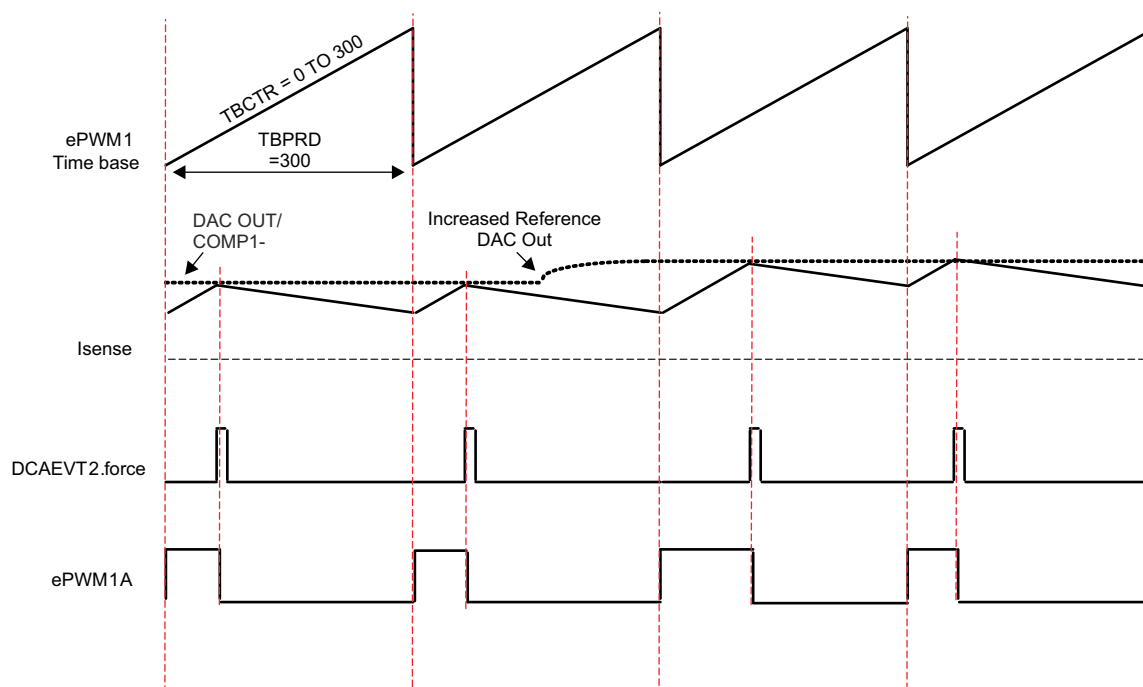
**Figure 26-73. ZVS Full-H Bridge Waveforms**

### 26.13.10 Controlling a Peak Current Mode Controlled Buck Module

Peak current control techniques offer a number of benefits like automatic over current limiting, fast correction for input voltage variations and reducing magnetic saturation. Figure 26-74 shows the use of ePWM1A along with the on-chip analog comparator for buck converter topology. The output current is sensed through a current sense resistor and fed to the positive terminal of the on-chip comparator. The internal programmable 12-bit DAC can be used to provide a reference peak current at the negative terminal of the comparator. Alternatively, an external reference can be connected at this input. The comparator output is an input to the Digital compare sub-module. The ePWM module is configured in such a way so as to trip the ePWM1A output as soon as the sensed current reaches the peak reference value. A cycle-by-cycle trip mechanism is used. Figure 26-75 shows the waveforms generated by the configuration.



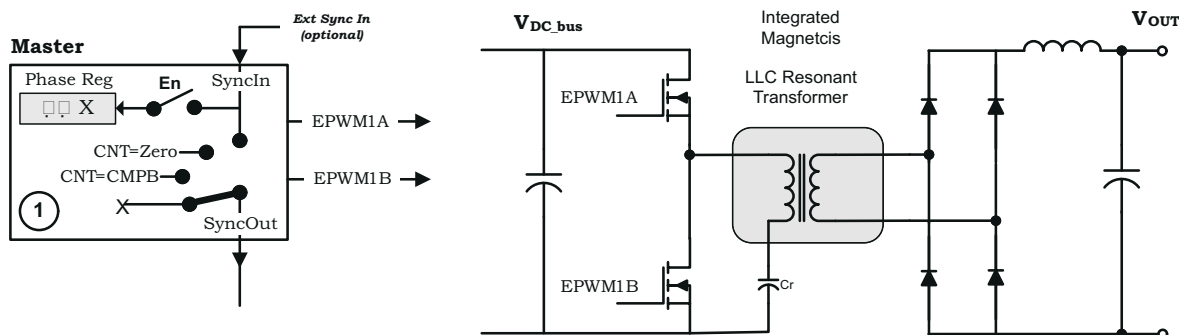
**Figure 26-74. Peak Current Mode Control of Buck Converter**



**Figure 26-75. Peak Current Mode Control Waveforms for Control of Buck Converter**

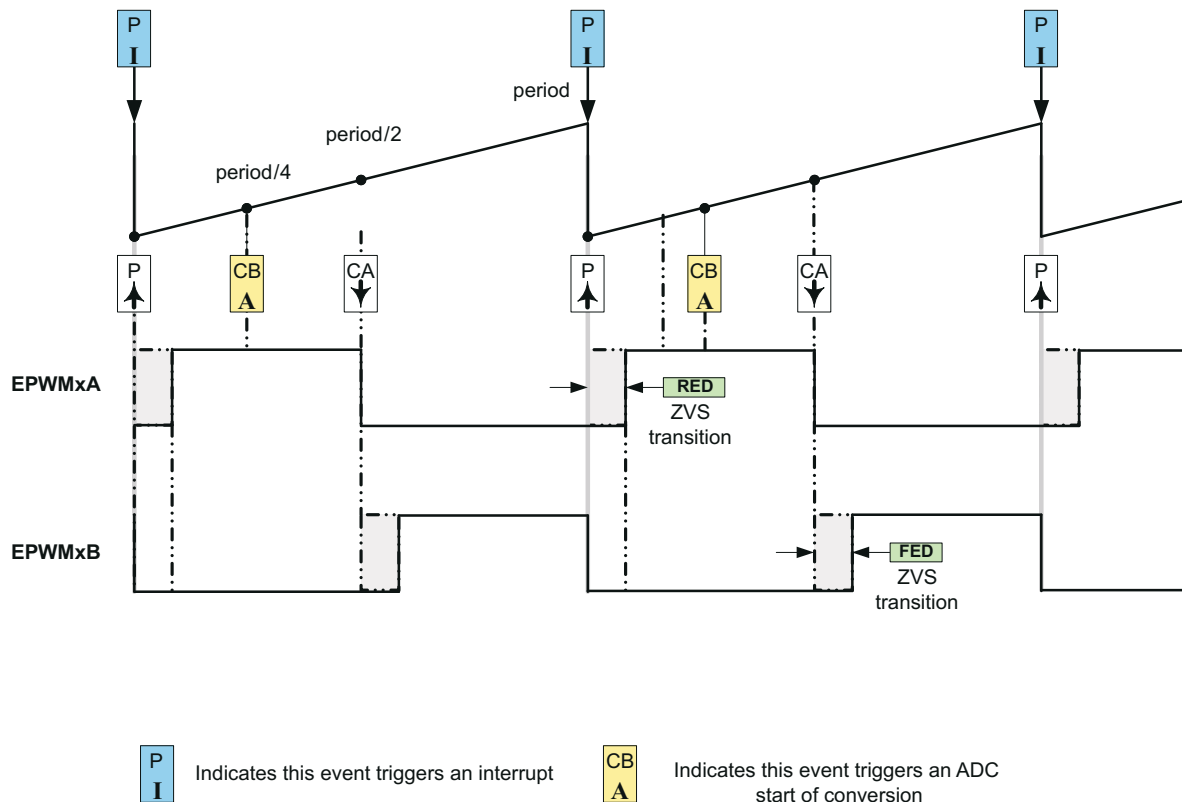
### 26.13.11 Controlling H-Bridge LLC Resonant Converter

Various topologies of resonant converters are well-known in the field of power electronics for many years. In addition to these, H-bridge LLC resonant converter topology has recently gained popularity in many consumer electronics applications where high efficiency and power density are required. In this example, single channel configuration of ePWM1 is detailed, yet the configuration can easily be extended to multichannel. Here the controlled parameter is not duty cycle (this is kept constant at approximately 50 percent); instead the parameter is frequency. Although the deadband is not controlled and kept constant as 300ns (that is, 30 at 100MHz TBCLK), the user can update the deadband in real time to enhance the efficiency by adjusting enough time delay for soft switching.



NOTE:  $\Theta = X$  indicates value in phase register is 'don't care'

**Figure 26-76. Control of Two Resonant Converter Stages**



**Figure 26-77. H-Bridge LLC Resonant Converter PWM Waveforms**

## 26.14 Register Lock Protection

The register lock protection mechanism is added to protect the critical ePWM registers from being corrupted by accidental writes in case of runaway code. The register EPWMLOCK contains the definition of Lock bits ([Table 26-14](#) shows the lock bits and the corresponding registers). This register also has a KEY field; writes to this register succeed only if the KEY field is written with a value of 0xa5a5. Refer to the register descriptions for more details.

**Table 26-14. Lock Bits and Corresponding Registers**

| Bit Field | Definition                        | Registers Locked                                                                                |
|-----------|-----------------------------------|-------------------------------------------------------------------------------------------------|
| HRLOCK    | HRPWM Register Set Lock           | HRCNFG, HRPWR, HRMSTEP, HRPCTL                                                                  |
| GLLOCK    | Global Load Register Set Lock     | GLDCTL, GLDCFG                                                                                  |
| TZCFGLOCK | TripZone Register Set Lock        | TZSEL, TZDCSEL, TZCTL, TZCTL2, TZCTLDCA, TZCTLDCB, TZEINT                                       |
| TZCLRLOCK | TripZone Clear Register Set Lock  | TZCLR, TZCBCCLR, TZOSTCLR, TZFRC                                                                |
| DCLOCK    | Digital Compare Register Set Lock | DCTRIPSEL, DCACTL, DCBCTL, DCFCTL, DCCAPCTL, DCAHTRIPSEL, DCALTRIPSEL, DCBHTRIPSEL, DCBLTRIPSEL |

### Note

Due to the presence of the KEY field in the same register, only 32-bit writes succeed if the KEY matches. The 16-bit writes to the upper or lower half of this register are ignored.



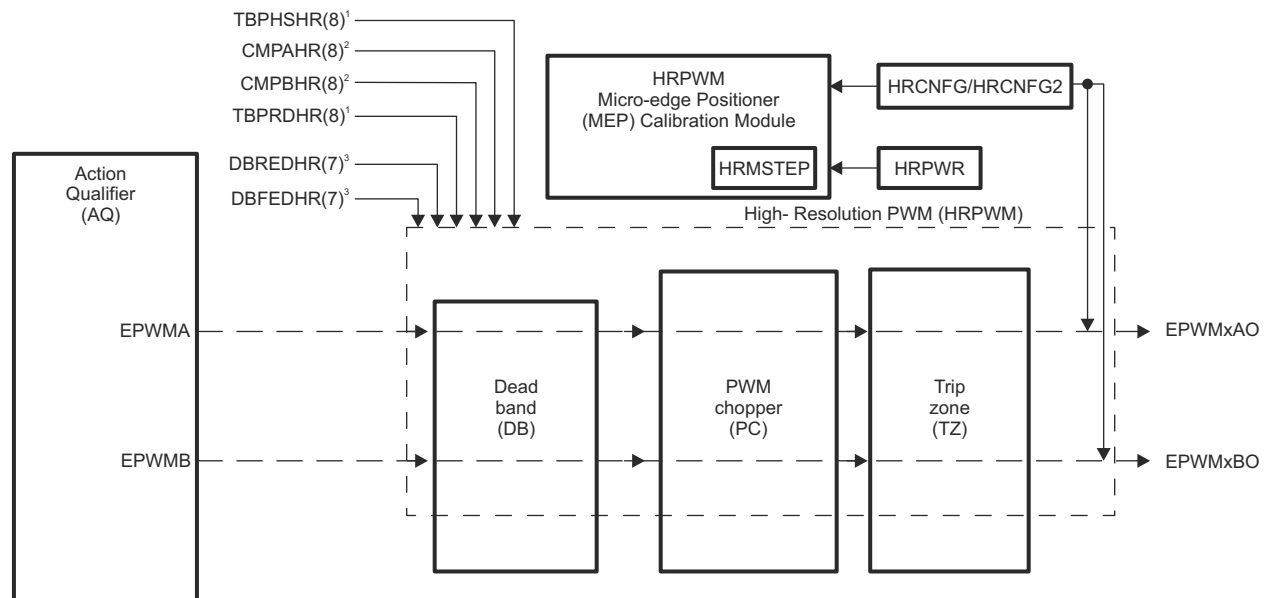
## 26.15 High-Resolution Pulse Width Modulator (HRPWM)

Figure 26-78 shows a block diagram of the HRPWM. This module extends the time resolution capabilities of the conventionally derived digital pulse width modulator (PWM). HRPWM is typically used when PWM resolution falls below approximately 9-10 bits. The key features of HRPWM are:

- Extended time resolution capability
- Used in both duty cycle and phase-shift control methods
- Finer time granularity control or edge positioning using extensions to the Compare A, Compare B and Phase registers
- Implemented using the A and B signal path of PWM, that is, on the EPWMxA and EPWMxB output
- Dead band high-resolution control for falling and rising edge delay in half cycle clocking operation
- Self-check diagnostics software mode to check if the micro edge positioner (MEP) logic is running how designed
- Enables high-resolution output swapping on the EPWMxA and EPWMxB output
- Enables high-resolution output on EPWMxB signal output using inversion of EPWMxA signal output
- Enables high-resolution period, duty and phase control on the EPWMxA and EPWMxB output on devices with an ePWM module

### Note

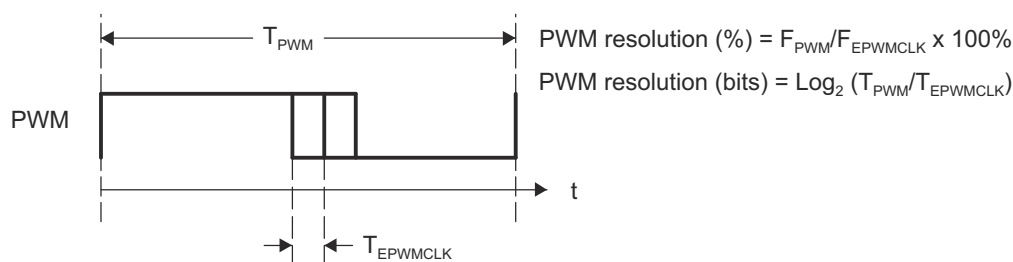
See the device data sheet to determine if your device has an ePWM module with high-resolution period support.



- A. From ePWM Time-base (TB) submodule  
 B. From ePWM counter-compare (CC) submodule  
 C. From ePWM Deadband (DB) submodule

**Figure 26-78. HRPWM Block Diagram**

The ePWM peripheral is used to perform a function mathematically equivalent to a digital-to-analog converter (DAC). As shown in Figure 26-79, the effective resolution for conventionally generated PWM is a function of PWM frequency (or period) and system clock frequency.



**Figure 26-79. Resolution Calculations for Conventionally Generated PWM**

If the required PWM operating frequency does not offer sufficient resolution in PWM mode, consider using HRPWM. As an example of improved performance offered by HRPWM, Table 26-15 shows resolution in bits for various PWM frequencies. These values assume a MEP step size of 180ps. See the device data sheet for typical and maximum performance specifications for the MEP.

**Table 26-15. Resolution for PWM and HRPWM**

| PWM Frequency<br>(kHz) | Regular Resolution (PWM)<br>100MHz EPWMCLK |      | High Resolution (HRPWM) |       |
|------------------------|--------------------------------------------|------|-------------------------|-------|
|                        | Bits                                       | %    | Bits                    | %     |
| 20                     | 12.3                                       | 0.02 | 18.1                    | 0.000 |
| 50                     | 11                                         | 0.05 | 16.8                    | 0.001 |
| 100                    | 10                                         | 0.1  | 15.8                    | 0.002 |
| 150                    | 9.4                                        | 0.15 | 15.2                    | 0.003 |
| 200                    | 9                                          | 0.2  | 14.8                    | 0.004 |
| 250                    | 8.6                                        | 0.25 | 14.4                    | 0.005 |
| 500                    | 7.6                                        | 0.5  | 13.4                    | 0.009 |
| 1000                   | 6.6                                        | 1    | 12.4                    | 0.018 |
| 1500                   | 6.1                                        | 1.5  | 11.9                    | 0.027 |
| 2000                   | 5.6                                        | 2    | 11.4                    | 0.036 |

Although each application can differ, typical low-frequency PWM operation (below 250kHz) does not require HRPWM. HRPWM capability is most useful for high-frequency PWM requirements of power conversion topologies such as:

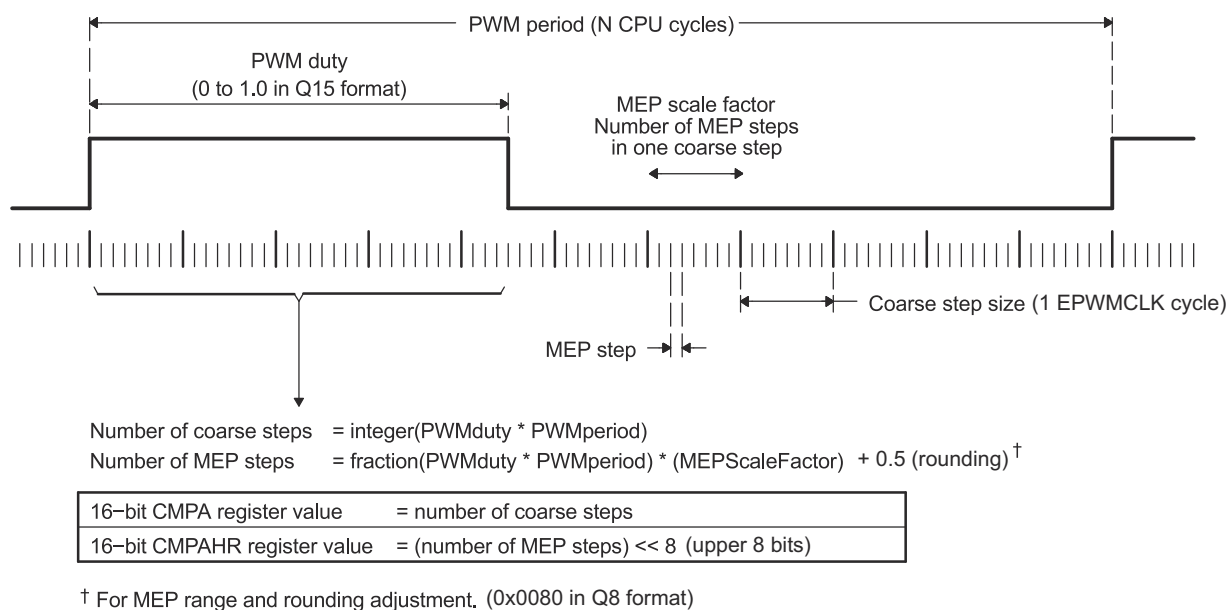
- Single-phase buck, boost, and flyback
- Multiphase buck, boost, and flyback
- Phase-shifted full bridge
- Direct modulation of D-Class power amplifiers

### 26.15.1 Operational Description of HRPWM

The HRPWM is based on micro-edge positioner (MEP) technology. MEP logic is capable of positioning an edge very finely by sub-dividing one coarse system clock of a conventional PWM generator. The time step accuracy is on the order of 150ps. See the device data sheet for the typical MEP step size on a particular device. The HRPWM also has a self-check software diagnostics mode to check if the MEP logic is running as designed under all operating conditions. Details on software diagnostics and functions are in [Section 26.15.1.7](#).

[Figure 26-80](#) shows the relationship between one coarse system clock and edge position in terms of MEP steps, which are controlled using an 8-bit field in the Compare A extension register (CMPAHR). The same operating logic applies to CMPBHR as well.

To generate an HRPWM waveform, configure the ePWM registers to generate a conventional PWM of a given frequency and polarity. The HRPWM works together with the ePWM registers to extend edge resolution. Although many programming combinations are possible, only a few are needed and practical. These methods are described in [Section 26.15.1.8](#).

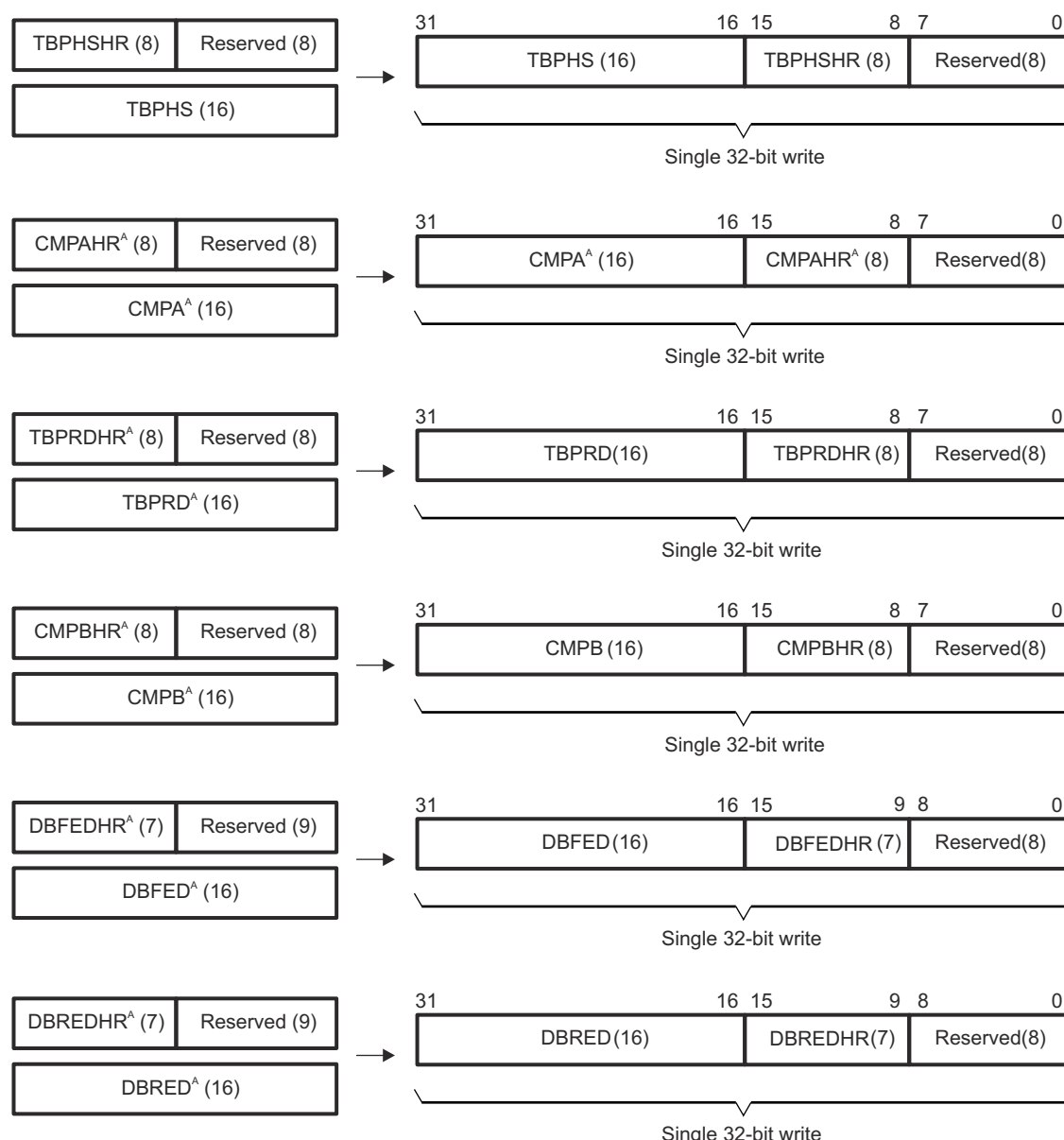


**Figure 26-80. Operating Logic Using MEP**

#### 26.15.1.1 Controlling the HRPWM Capabilities

The MEP of the HRPWM is controlled by six extension registers. These HRPWM registers are concatenated with the 16-bit TBPHS, TBPRD, CMPA, CMPBM, DBREDM, and DBFEDM registers used to control PWM operation.

- TBPHSHR - Time Base Phase High Resolution Register
- CMPAHR - Counter Compare A High Resolution Register; CMPAHR is for use with the AQ output of Channel A, and is not related to CMPA
- TBPRDHR - Time Base Period High Resolution Register. (available on some devices)
- CMPBHR - Counter Compare B High Resolution Register; CMPBHR is for use with the AQ output of Channel B, and is not related to CMPB
- DBREDHR - Dead-band Generator Rising Edge Delay High Resolution Register
- DBFEDHR - Dead-band Generator Falling Edge Delay High Resolution Register



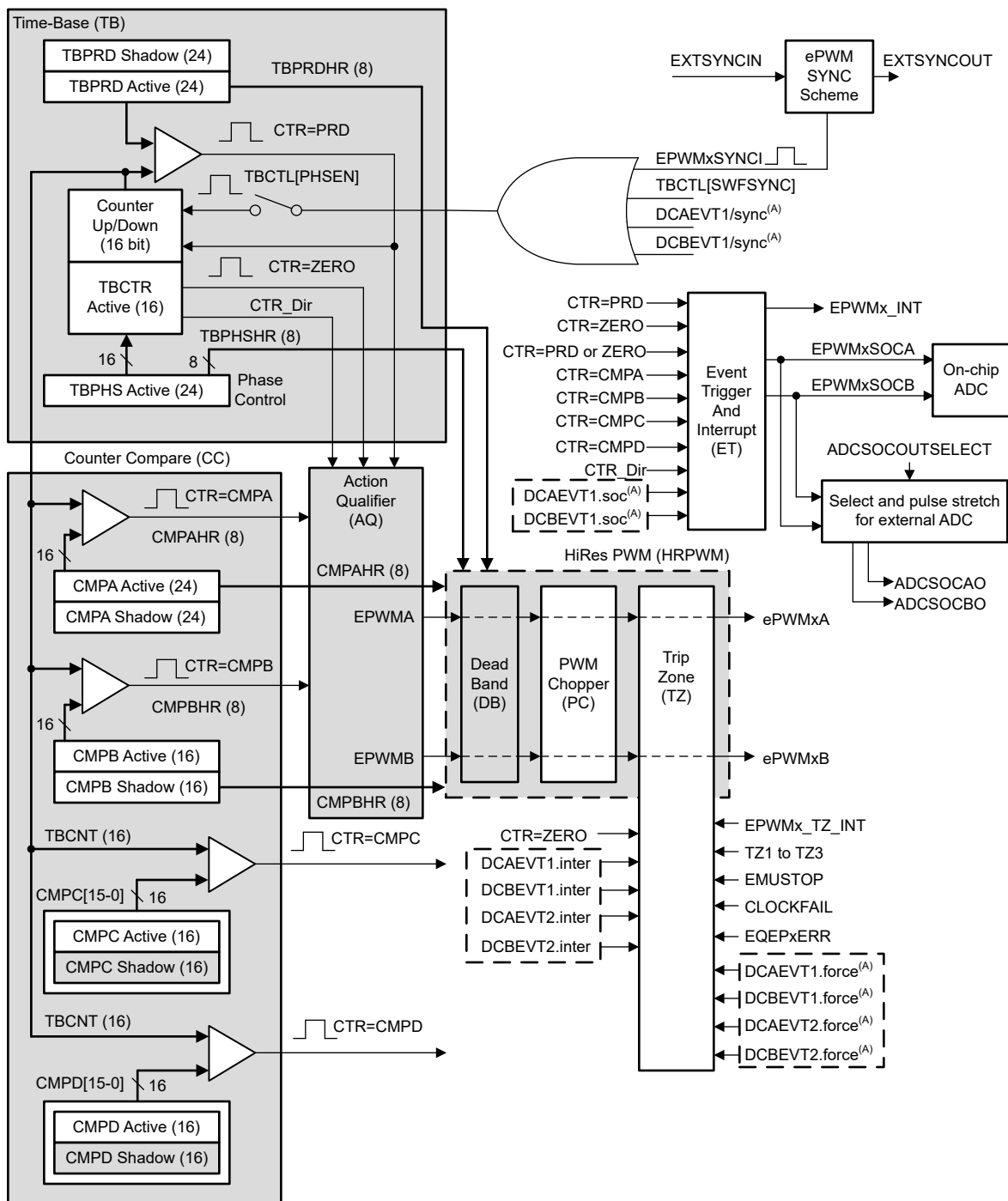
A. Dependent upon your device, these registers can be mirrored and can be written to at two different memory locations.

**Figure 26-81. HRPWM Extension Registers and Memory Configuration**

### Note

HRPWM capabilities on Deadband Rising Edge Delay and Falling Edge Delay is applicable only during dead band half cycle clocking Operation. The number of MEP steps is half in size [bits 15:9] than duty and phase high-resolution registers for the same reason.

HRPWM capabilities are controlled using the Channel A and B PWM signal path. HRPWM support on the Dead band signal path is available by properly configuring the HRCNFG2 register. [Figure 26-82](#) shows how the HRPWM interfaces with the 8-bit extension registers.

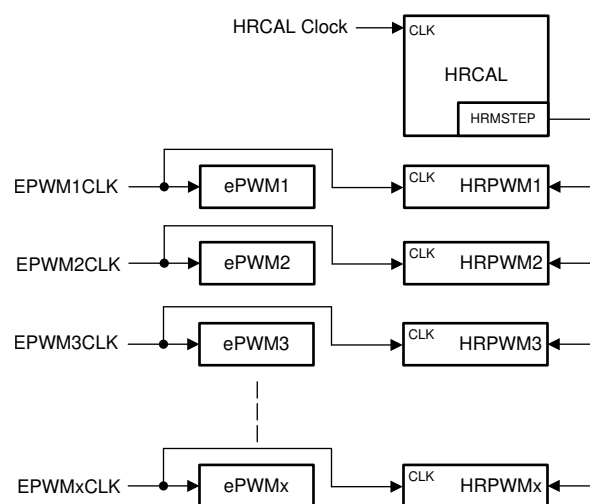


A. These events are generated by the ePWM Digital Compare (DC) submodule based on the levels of the TRIPIN inputs.

**Figure 26-82. HRPWM System Interface**

### 26.15.1.2 HRPWM Source Clock

Each HRPWM module is clocked from the respective EPWMxCLK. HRCAL has a separate clock. For example, HRPWM1 is sourced from EPWM1CLK while HRPWM2 is clocked from the EPWM2CLK. Figure 26-83 shows the HRCAL and HRPWM modules are sourced from their respective ePWM clock source.



**Figure 26-83. HRPWM and HRCAL Source Clock**

#### Note

HRCAL registers can not be configured using CPU2 through the registers available on ePWM2 to ePWM8 modules.

### 26.15.1.3 Configuring the HRPWM

Once the ePWM has been configured to provide conventional PWM of a given frequency and polarity, the HRPWM is configured by programming the HRCNFG register in that particular ePWM module's register space. This register provides the following configuration options:

- Edge Mode** The MEP can be programmed to provide precise position control on the rising edge (RE), falling edge (FE) or both edges (BE) at the same time. FE and RE are used for power topologies requiring duty cycle control (CMPA or CMPB high-resolution control), while BE is used for topologies requiring phase shifting, for example, phase shifted full bridge (TBPHS or TBPRD high-resolution control).
- Control Mode** The MEP is programmed to be controlled either from the CMPAHR/CMPBHR register in case of duty cycle control or the TBPHSHR register (phase control). RE or FE control mode can be used with the CMPAHR or CMPBHR register. BE control mode can be used with the TBPHSHR register. When the MEP is controlled from the TBPRDHR register (period control), the duty cycle and phase can also be controlled using their respective high-resolution registers.
- Shadow Mode** This mode provides the same shadowing (double buffering) option as in regular PWM mode. This option is valid only when operating from the CMPAHR, CMPBHR, and TBPRDHR registers and can be chosen to be the same as the regular load option for the CMPA/CMPB register. If TBPHSHR is used, then this option has no effect.
- High-Resolution B Signal Control** The B signal path of an ePWM channel can generate a high-resolution output by outputting an inverted version of the high-resolution ePWMxA signal on the ePWMxB pin. A Type 2 or Type 4 HRPWM module can also enable high-resolution features on the B signal path independently of the A signal path as well.

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Swap ePWMxA and ePWMxB Outputs</b> | This mode enables the swapping of the high-resolution A and B outputs. The mode selection allows either A and B Outputs Unchanged or A Output Comes Out On B and B Output Comes Out On A.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Auto-conversion Mode</b>           | This mode is used in conjunction with the scale factor optimization (SFO) software only. For a type 4 HRPWM module, below is a description of the Auto-conversion Mode taking CMPAHR as an example. If auto-conversion is enabled, $\text{CMPAHR} = \text{fraction}(\text{PWMduty} * \text{PWMperiod}) < 8$ . The scale factor optimization software calculates the MEP scale factor in the background code and automatically updates the HRMSTEP register with the calculated number of MEP steps per coarse step. The MEP Calibration Module then uses the values in the HRMSTEP and CMPAHR registers to automatically calculate the appropriate number of MEP steps represented by the fractional duty cycle and moves the high-resolution ePWM signal edge accordingly. If auto-conversion is disabled, the CMPAHR register behaves like a type 0 HRPWM module and $\text{CMPAHR} = (\text{fraction}(\text{PWMduty} * \text{PWMperiod}) * \text{MEP Scale Factor} + 0.5) < 8$ . All calculations need to be performed by your code in this mode, and the HRMSTEP register is ignored. Auto-conversion for high-resolution period has the same behavior as auto-conversion for high-resolution duty cycle. Auto-conversion must always be enabled for high-resolution period mode. |

### Note

If the HRPWM module is configured in UP-DOWN counter mode, the shadow mode for the HRPWM registers must be set to load on both ZERO AND PERIOD. New values from the user are loaded to the shadow registers only at CTR=ZERO, but the shadow mode of for the registers must be set to both ZERO AND PERIOD. The CTR=PRD event is used for specific internal logic inside the HRPWM module.

Auto-conversion Mode performs the calculation for CMPBHR, DBREDHR, and DBFEDHR. The scale factor optimization software calculates the MEP scale factor in the background code and automatically updates the HRMSTEP register with the calculated number of MEP steps per coarse step. The MEP Calibration Module then uses the values in the HRMSTEP and CMPBHR or DBREDHR/DBFEDHR register to automatically calculate the appropriate number of MEP steps represented by the fractional components and moves the high-resolution ePWM signal edge accordingly. If auto-conversion is disabled, CMPBHR behaves the same as CMPAHR.  $\text{CMPBHR} = (\text{fraction}(\text{PWMduty} * \text{PWMperiod}) * \text{MEP Scale Factor} + 0.5) < 8$ .

You are expected to disable protection for both ePWM1 and HRPWM when the application requires access to either of these modules.

### 26.15.1.4 Configuring High-Resolution in Deadband Rising-Edge and Falling-Edge Delay

Once the ePWM has been configured to provide conventional PWM of a given frequency, polarity, and dead band enabled in half-cycle clocking mode, the high-resolution operation on dead band RED and FED lines are enabled by programming the HRCNFG2 register in that particular ePWM module register space. This register provides the following configuration options:

- Edge Mode** The MEP can be programmed to provide precise position control on the dead band rising edge (RED), dead band falling edge (FED), or both edges (rising edge of DBRED signal and falling edge of DBFED signal) at the same time.
- Control Mode** Selects the time event that loads the shadow value in the active register for DBRED and DBFED in high-resolution mode. Select the pulse to match the selection in the ePWM DBCTL[LOADREDMODE] and DBCTL[LOADFEDMODE] bits.

### 26.15.1.5 Principle of Operation

The MEP logic is capable of placing an edge in one of 255 (8 bits) discrete time steps (see the device data sheet for typical MEP step size). The MEP works with the TBM and CCM registers to be certain that time steps are applied and that edge placement accuracy is maintained over a wide range of PWM frequencies, system clock frequencies, and other operating conditions. Table 26-16 shows the typical range of operating frequencies supported by the HRPWM.

**Table 26-16. Relationship Between MEP Steps, PWM Frequency, and Resolution**

| System (MHz) | MEP Steps Per EPWMCLK <sup>(1) (2) (3)</sup> | PWM Minimum (Hz) <sup>(4)</sup> | PWM Maximum (MHz) | Resolution at Maximum (Bits) <sup>(5)</sup> |
|--------------|----------------------------------------------|---------------------------------|-------------------|---------------------------------------------|
| 60.0         | 93                                           | 916                             | 3.00              | 10.9                                        |
| 70.0         | 79                                           | 1068                            | 3.50              | 10.6                                        |
| 80.0         | 69                                           | 1221                            | 4.00              | 10.4                                        |
| 90.0         | 62                                           | 1373                            | 4.50              | 10.3                                        |
| 100.0        | 56                                           | 1526                            | 5.00              | 10.1                                        |

(1) TBCLK = EPWMCLK.

(2) Table data based on a MEP time resolution of 180ps (this is an example value. See the device data sheet for MEP limits)

(3) MEP steps applied =  $T_{EPWMCLK}/180ps$  in this example.

(4) PWM minimum frequency is based on a maximum period value, (TBPRD = 65535). PWM mode is asymmetrical up-count.

(5) Resolution in bits is given for the maximum PWM frequency stated.



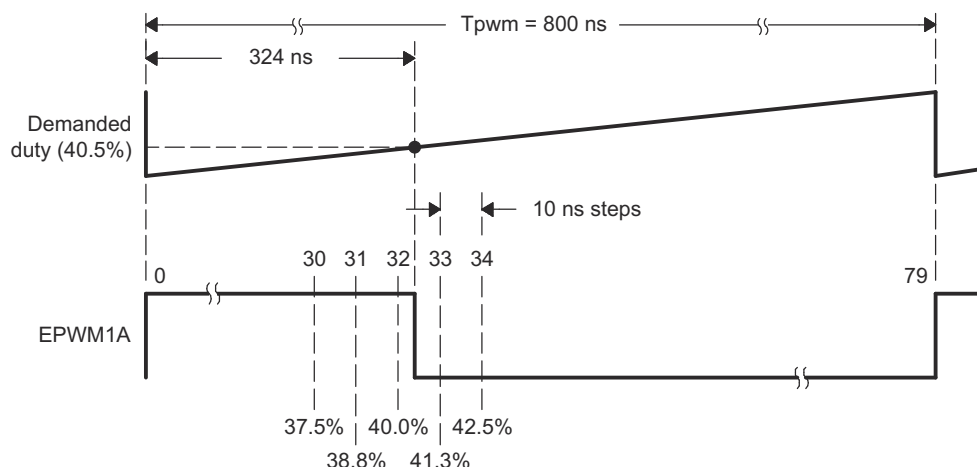
### 26.15.1.5.1 Edge Positioning

#### Note

The following example is presented using the [CMPA:CMPAHR] register combination. The theory of operation and equations are the same, if intending to use the [CMPBM:CMPBHRM] for duty cycle control.

In a typical power control loop, a digital controller issues a duty command, usually expressed in a per unit or percentage terms. Assume that for a particular operating point, the demanded duty cycle is 0.405 or 40.5% on time and the required converter PWM frequency is 1.25MHz. In conventional PWM generation with a system clock of 100MHz, the duty cycle choices are in the vicinity of 40.5%. As shown in Figure 26-84, a compare value of 32 counts (duty = 40%) is the closest to 40.5% that can be attained. This is equivalent to an edge position of 320ns instead of the desired 324ns. This data is shown in Table 26-17.

By utilizing the MEP, an edge position much closer to the desired point of 324ns can be achieved. Table 26-17 shows that in addition to the CMPA value, 22 steps of the MEP (CMPAHR register) positions the edge at 323.96ns, resulting in almost zero error. In this example, assume that the MEP has a step resolution of 180ps.



**Figure 26-84. Required PWM Waveform for a Requested Duty = 40.5%**

**Table 26-17. CMPA versus Duty (left), and [CMPA:CMPAHR] versus Duty (right)**

| CMPA<br>(count) <sup>(1) (2) (3)</sup> | Duty<br>(%) | High Time<br>(ns) | CMPA<br>(count) | CMPAHR<br>(count) | Duty<br>(%) | High Time<br>(ns) |
|----------------------------------------|-------------|-------------------|-----------------|-------------------|-------------|-------------------|
| 28                                     | 35.0        | 280               | 32              | 18                | 40.405      | 323.24            |
| 29                                     | 36.3        | 290               | 32              | 19                | 40.428      | 323.42            |
| 30                                     | 37.5        | 300               | 32              | 20                | 40.450      | 323.60            |
| 31                                     | 38.8        | 310               | 32              | 21                | 40.473      | 323.78            |
| 32                                     | 40.0        | 320               | 32              | 22                | 40.495      | 323.96            |
| 33                                     | 41.3        | 330               | 32              | 23                | 40.518      | 324.14            |
| 34                                     | 42.5        | 340               | 32              | 24                | 40.540      | 324.32            |
|                                        |             |                   | 32              | 25                | 40.563      | 324.50            |
| Required                               |             |                   | 32              | 26                | 40.585      | 324.68            |
| 32.40                                  | 40.5        | 324               | 32              | 27                | 40.608      | 324.86            |

(1) Assumed MEP step size for the above example = 180ps. See the device-specific data sheet for typical and maximum MEP values.

(2) TBCLK = 100MHz, 10ns

(3) For a PWM Period register value of 80 counts, PWM Period = 80 × 10ns = 800ns, PWM frequency = 1/800ns = 1.25MHz

### 26.15.1.5.2 Scaling Considerations

The mechanics of how to position an edge precisely in time has been demonstrated using the resources of the standard CMPA and MEP (CMPAHR) registers. In a practical application, however, it is necessary to seamlessly provide the CPU a mapping function from a per-unit (fractional) duty cycle to a final integer (non-fractional) representation that is written to the [CMPA:CMPAHR] register combination.

To do this, first examine the scaling or mapping steps involved. It is common in control software to express duty cycle in a per-unit or percentage basis. This has the advantage of performing all needed math calculations without concern for the final absolute duty cycle, expressed in clock counts or high time in nanoseconds (ns). Furthermore, it makes the code more transportable across multiple converter types running different PWM frequencies.

To implement the mapping scheme, a two-step scaling procedure is required.

#### Assumptions for this example:

|                                                                                                 |                            |
|-------------------------------------------------------------------------------------------------|----------------------------|
| TBCLK                                                                                           | = 10ns (100MHz)            |
| PWM frequency                                                                                   | = 1.25MHz (1/800ns)        |
| Required PWM duty cycle, <b>PWMDuty</b>                                                         | = 0.405 (40.5%)            |
| PWM period in terms of coarse steps, <b>PWMPeriod</b> (800ns/10ns)                              | = 80                       |
| Number of MEP steps per coarse step at 180ps (10ns/180ps), <b>MEP_ScaleFactor</b>               | = 55                       |
| Value to keep CMPAHR within the range of 1-255 and fractional rounding constant (default value) | = 0.5 (0080h in Q8 format) |

#### Step 1: Percentage Integer Duty value conversion for CMPA register

|                     |                                                                            |
|---------------------|----------------------------------------------------------------------------|
| CMPA register value | = $\text{int}(\text{PWMDuty} * \text{PWMPeriod})$ ; int means integer part |
|                     | = $\text{int}(0.405 * 80)$                                                 |
|                     | = $\text{int}(32.4)$                                                       |
| CMPA register value | = 32 (20h)                                                                 |

#### Step 2: Fractional value conversion for CMPAHR register

|        |                                                                                                                        |
|--------|------------------------------------------------------------------------------------------------------------------------|
| CMPAHR | = $(\text{frac}(\text{PWMDuty} * \text{PWMPeriod}) * \text{MEP\_ScaleFactor} + 0.5) << 8$ ; frac means fractional part |
|        | = $(\text{frac}(32.4) * 55 + 0.5) << 8$ ; Shifting is to move the value to the high byte of CMPAHR.                    |
|        | = $(0.4 * 55 + 0.5) << 8$                                                                                              |
|        | = $(22 + 0.5) << 8$                                                                                                    |
|        | = $22.5 * 256$ ; Shifting left by 8 is the same as multiplying by 256.                                                 |
|        | = 5760 (1680h)                                                                                                         |
| CMPAHR | = 1680h CMPAHR value = 1600h (lower 8 bits are ignored by hardware).                                                   |

### Note

If the AUTOCONV bit (HRCNFG.6) is set and the MEP\_ScaleFactor is in the HRMSTEP register, then CMPAHR / CMPBHR register value =  $\text{frac}(\text{PWMDuty} \times \text{PWMperiod} < 8)$ . The rest of the conversion calculations are performed automatically in hardware, and the correct MEP-scaled signal edge appears on the ePWM channel output. If AUTOCONV is not set, the above calculations must be performed by software.

The MEP scale factor (MEP\_ScaleFactor) varies with the system clock and DSP operating conditions. TI provides an MEP scale factor optimizing (SFO) software C function, which uses the built in diagnostics in each HRPWM and returns the best scale factor for a given operating point.

The scale factor varies slowly over a limited range so the optimizing C function can be run very slowly in a background loop.

The CMPA, CMPB, CMPAHR and CMPBHR registers are configured in memory so that the 32-bit data capability of the CPU can write this as a single concatenated value, that is, [CMPA:CMPAHR], [CMPB:CMPBHR], and so on.

The mapping scheme has been implemented in both C and assembly, as shown in [Section 26.15.1.8](#). The actual implementation takes advantage of the 32-bit CPU architecture and is somewhat different from the steps shown in [Section 26.15.1.5.2](#).

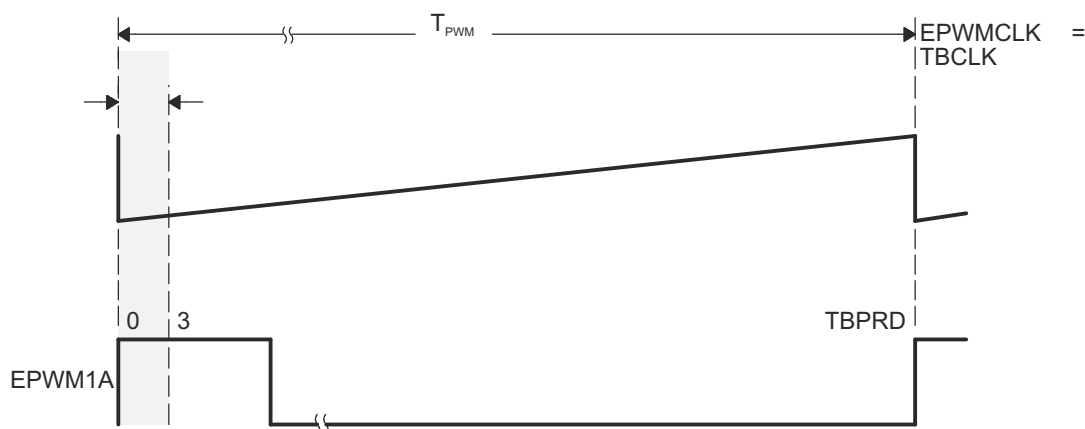
For time-critical control loops where every cycle counts, the assembly version is recommended. This is a cycle optimized function (11 EPWMCLK cycles) that takes a Q15 duty value as input and writes a single [CMPA:CMPAHR] value.

#### 26.15.1.5.3 Duty Cycle Range Limitation

In high-resolution mode, the MEP is not active for 100% of the PWM period and becomes operational:

- Three EPWMCLK cycles after the period starts when high-resolution period (TBPRDHR) control is not enabled.
- When high-resolution period (TBPRDHR) control is enabled using the HRPCTL register:
  - In up-count mode: three EPWMCLK cycles after the period starts until three EPWMCLK cycles before the period ends.
  - In up-down count mode: when counting up, three cycles after CTR = 0 until three cycles before CTR = PRD, and when counting down, three cycles after CTR = PRD until three cycles before CTR = 0.
- When using DBREDHR or DBFEDHR, DBRED or DBFED (the register corresponding to the edge with high-resolution displacement) must be greater than or equal to 7.

Duty cycle range limitations are illustrated in [Figure 26-85](#) to [Figure 26-88](#). This limitation imposes a duty cycle limit on the MEP. For example, precision edge control is not available all the way down to 0% duty cycle. When high-resolution period control is disabled, regular PWM duty control is fully operational down to 0% duty cycle despite the unavailability of HRPWM features in the first three cycles. In most applications, this cannot be an issue as the controller regulation point is usually not designed to be close to 0% duty cycle. To better understand the useable duty cycle range, see [Table 26-18](#). When high-resolution period control is enabled (HRPCTL[HRPE] = 1), the duty cycle must not fall within the restricted range; otherwise, there can be undefined behavior on the ePWMxA output.



**Figure 26-85. Low % Duty Cycle Range Limitation Example (HRPCTL[HRPE] = 0)**

**Table 26-18. Duty Cycle Range Limitation for Three EPWMCLK/TBCLK Cycles**

| PWM Frequency <sup>(1)</sup><br>(kHz) | 3 Cycles<br>Minimum Duty | 3 Cycles<br>Maximum Duty <sup>(2)</sup> |
|---------------------------------------|--------------------------|-----------------------------------------|
| 200                                   | 0.6%                     | 99.4%                                   |
| 400                                   | 1.2%                     | 98.8%                                   |
| 600                                   | 1.8%                     | 98.2%                                   |
| 800                                   | 2.4%                     | 97.6%                                   |
| 1000                                  | 3%                       | 97%                                     |
| 1200                                  | 3.6%                     | 96.4%                                   |
| 1400                                  | 4.2%                     | 95.8%                                   |
| 1600                                  | 4.8%                     | 95.2%                                   |
| 1800                                  | 5.4%                     | 94.6%                                   |
| 2000                                  | 6%                       | 94%                                     |

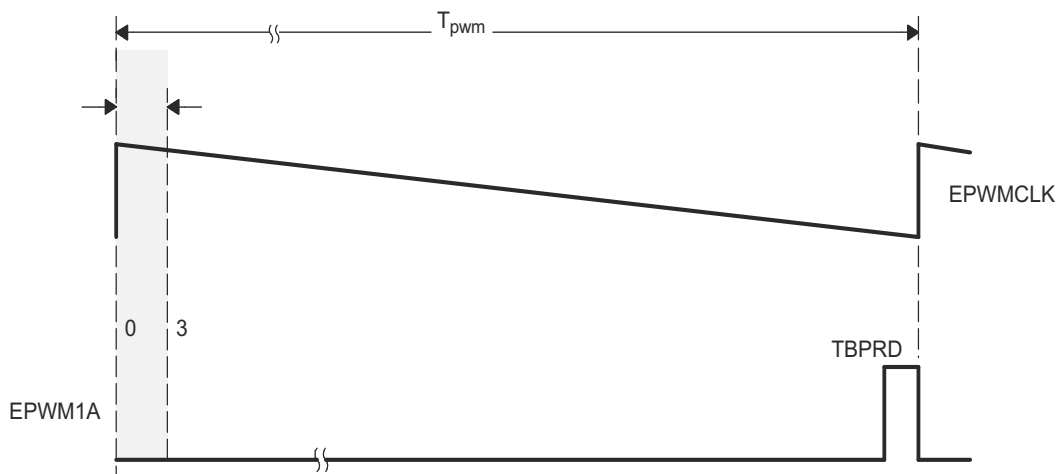
(1) EPWMCLK = TBCLK = 100MHz

(2) This limitation applies only if high-resolution period (TBPRDHR) control is enabled.

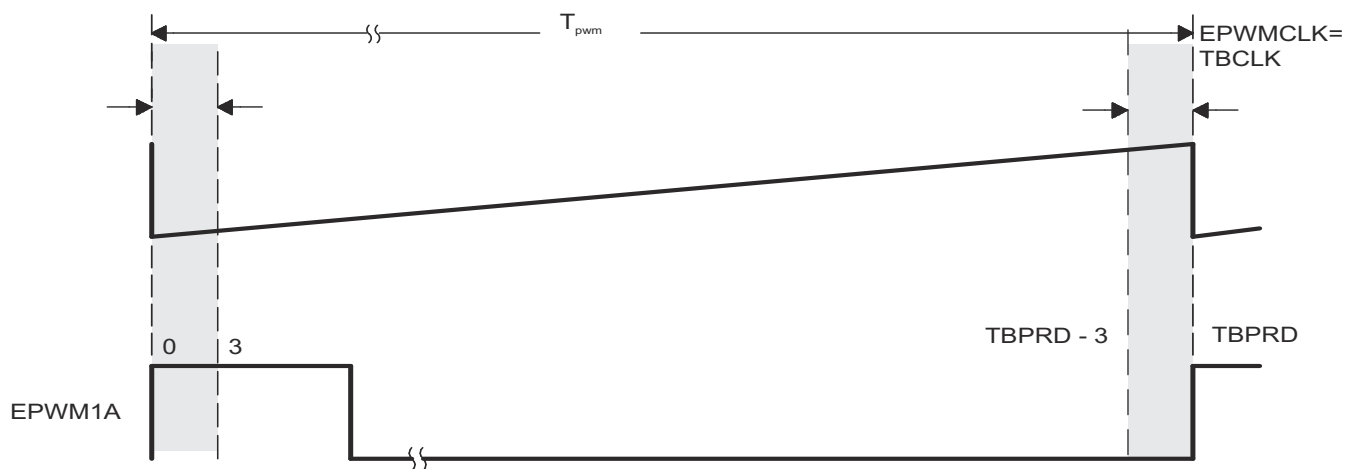
If the application demands HRPWM operation below the minimum duty cycle limitation, then the HRPWM can be configured to operate in count-down mode with the rising edge position (REP) controlled by the MEP when high-resolution period is disabled (HRPCTL[HRPE] = 0). This is illustrated in [Figure 26-86](#). In this configuration, the minimum duty cycle limitation is no longer an issue. However, there is a maximum duty limitation with same percent numbers as given in [Table 26-18](#).

### CAUTION

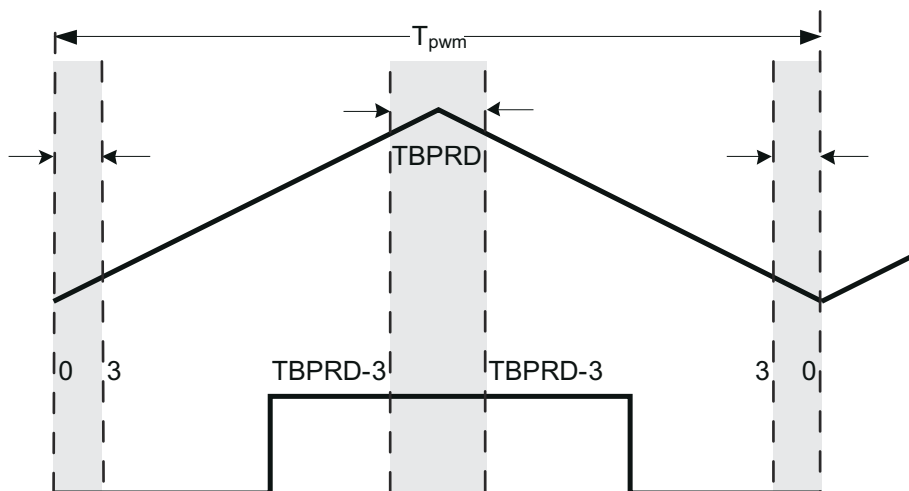
If the application has enabled high-resolution period control (HRPCTL[HRPE] = 1), the duty cycle must not fall within the restricted range; otherwise, there can be undefined behavior on the ePWM output.



**Figure 26-86. High % Duty Cycle Range Limitation Example (HRPCTL[HRPE] = 0)**



**Figure 26-87. Up-Count Duty Cycle Range Limitation Example (HRPCTL[HRPE] = 1)**



**Figure 26-88. Up-Down Count Duty Cycle Range Limitation Example (HRPCTL[HRPE] = 1)**

### 26.15.1.5.4 High-Resolution Period

High-resolution period control using the MEP logic is supported on devices with a Type 1 ePWM module or greater.

#### Note

When high-resolution period control is enabled, on ePWMxA only, and not ePWMxB output and conversely, the non high-resolution output has  $\pm 1$  TBCLK cycle jitter in up-count mode and  $\pm 2$  TBCLK cycle jitter in up-down count mode.

The scaling procedure described for duty cycle in [Section 26.15.1.5.2](#) applies for high-resolution period as well:

#### Assumptions for this example:

|                                                                                              |                              |
|----------------------------------------------------------------------------------------------|------------------------------|
| TBCLK                                                                                        | = 10ns (100MHz)              |
| Required PWM frequency                                                                       | = 175kHz (period of 571.428) |
| Number of MEP steps per coarse step at 180ps (10ns/180ps), (MEP_ScaleFactor)                 | = 55                         |
| Value to keep TBPRDHR within range of 1-255 and fractional rounding constant (default value) | = 0.5 (0080h in Q8 format)   |

#### Problem:

In up-count mode:

- If TBPRD = 571, then PWM frequency = 174.82kHz (period =  $(571+1) \cdot T_{TBCLK}$ ).
- If TBPRD = 570, then PWM frequency = 175.13kHz (period =  $(570+1) \cdot T_{TBCLK}$ ).

In up-down count mode:

- If TBPRD = 286, then PWM frequency = 174.82kHz (period =  $(286*2) \cdot T_{TBCLK}$ ).
- If TBPRD = 285, then PWM frequency = 175.44kHz (period =  $(285*2) \cdot T_{TBCLK}$ ).

#### Solution:

With 55 MEP steps per coarse step at 180ps each:

#### Step 1: Percentage Integer Period value conversion for TBPRD register

|                        |                                                                                                                        |
|------------------------|------------------------------------------------------------------------------------------------------------------------|
| Integer period value   | $= 571 \cdot T_{TBCLK}$<br>$= \text{int}(571.428) \cdot T_{TBCLK}$<br>$= \text{int}(\text{PWMperiod}) \cdot T_{TBCLK}$ |
| In up-count mode:      |                                                                                                                        |
| TBPRD                  | $= 570$ (TBPRD = period value - 1)<br>$= 023Ah$                                                                        |
| In up-down count mode: |                                                                                                                        |
| TBPRD                  | $= 285$ (TBPRD = period value/2)<br>$= 011Dh$                                                                          |

## Step 2: Fractional value conversion for TBPRDHR register

In up-count mode:

$$\text{TBPRDHR register value} = (\text{frac}(\text{PWMperiod}) * \text{MEP\_ScaleFactor} + 0.5)$$

If auto-conversion enabled and HRMSTEP =

$$\text{MEP\_ScaleFactor value (55):} = \text{frac}(\text{PWMperiod}) \ll 8 \text{ (Shifting is to move the value to the high byte of TBPRDHR)}$$

$$\begin{aligned} \text{TBPRDHR register value} &= \text{frac}(571.428) \ll 8 \\ &= 0.428 \times 256 \\ &= 6D00h \end{aligned}$$

The auto-conversion then automatically performs the calculation, such that TBPRDHR MEP delay is scaled by hardware to:

$$\begin{aligned} &= ((\text{TBPRDHR}(15:0) \gg 8) \times \text{HRMSTEP} + 80h) \ll 8 \\ &= (006Dh \times 55 + 80h) \gg 8 \\ &= (17EBh) \gg 8 \end{aligned}$$

$$\text{Period MEP delay} = 0017h \text{ MEP Steps}$$

In up-down count mode:

$$\text{TBPRDHR register value} = (\text{frac}(\text{PWMperiod}) * \text{MEP\_ScaleFactor} + 0.5)$$

If auto-conversion enabled and HRMSTEP =

$$\text{MEP\_ScaleFactor value (55):} = \text{frac}(\text{PWMperiod} / 2) \ll 8 \text{ (Shifting is to move the value to the high byte of TBPRDHR)}$$

$$\begin{aligned} \text{TBPRDHR register value} &= \text{frac}(285.714) \ll 8 \\ &= 0.714 \times 256 \\ &= B600h \end{aligned}$$

The auto-conversion then automatically performs the calculation, such that TBPRDHR MEP delay is scaled by hardware to:

$$\begin{aligned} &= (00B6h \times 55 + 80h) \gg 8 \\ &= (279Ah) \gg 8 \end{aligned}$$

$$\text{Period MEP delay} = 0027h \text{ MEP Steps}$$

#### 26.15.1.5.4.1 High-Resolution Period Configuration

To use high-resolution period, the ePWMx module must be initialized in the exact order presented.

The following steps use CMPA with shadow registers and the corresponding HRCNFG bits for high-resolution operation on EPWMxA. For high-resolution operation on EPWMxB, make the appropriate substitutions with the B channel fields.

1. Enable ePWMx clock
2. Enable HRPWM clock
3. Disable TBCLKSYNC
4. Configure ePWMx registers - AQ, TBPRD, CC, and so on.
  - ePWMx can only be configured for up-count or up-down count modes. High-resolution period is not compatible with down-count mode.
  - TBPRD and CC registers must be configured for shadow loads.
  - CMPCTL[LOADAMODE]
    - In up-count mode: CMPCTL[LOADAMODE] = 1 (load on CTR = PRD)
    - In up-down count mode: CMPCTL[LOADAMODE] = 2 (load on CTR=0 or CTR=PRD)
5. Configure the HRCNFG register such that:
  - HRCNFG[HRLOAD] = 2 (load on either CTR = 0 or CTR = PRD)
  - HRCNFG[AUTOCONV] = 1 (Enable auto-conversion)
  - HRCNFG[EDGMODE] = 3 (MEP control on both edges)
6. For TBPHS:TBPHSHR synchronization with high-resolution period, set both HRPCTL[TBPSHRLOADE] = 1 and TBCTL[PHSEN] = 1. In up-down count mode these bits must be set to 1 regardless of the contents of TBPHSHR.
7. Enable high-resolution period control (HRPCTL[HRPE] = 1)
8. Enable TBCLKSYNC
9. TBCTL[SWFSYNC] = 1
10. HRMSTEP must contain an accurate MEP scale factor (# of MEP steps per EPWMCLK coarse step) because auto-conversion is enabled. The MEP scale factor can be acquired using the SFO() function described in [Section 26.15.2](#).
11. To control high-resolution period, write to the TBPRDHR(M) registers.

#### Note

When high-resolution period mode is enabled, an EPWMxSYNC pulse introduces  $\pm 1$ -2 cycle jitter to the PWM ( $\pm 1$  cycle in up-count mode and  $\pm 2$  cycle in up-down count mode). For this reason, EPWMxSYNCO source cannot be set to CTR = 0 or CTR = CMPB. Otherwise, the jitter occurs on every PWM cycle with the synchronization pulse.

When EPWMxSYNCl is EPWMxSYNCO source, a software synchronization pulse can be issued only once during high-resolution period initialization. If a software sync pulse is applied while the PWM is running, the jitter appears on the PWM output at the time of the sync pulse.



### 26.15.1.6 Deadband High-Resolution Operation

---

**Note**

In up-count mode, the dead-band module is not available when any high-resolution mode is enabled.

---

**Assumptions for this example:**

|                                                      |                            |
|------------------------------------------------------|----------------------------|
| System clock                                         | = 10ns (100MHz)            |
| Deadband enabled in half-cycle mode, TBCLK = EPWMCLK |                            |
| Required PWM frequency                               | 1.33MHz (1/750ns)          |
| Required PWM duty cycle                              | 0.5 (50%)                  |
| Required Deadband Rising-Edge Delay                  | 5% over duty               |
| Required Deadband Rising-Edge Delay in ns            | $(0.05 * 375ns) = 18.75ns$ |

---

**Note**

Similar to the duty cycle restrictions when using HRPWM, the DBRED and DBFED values must be greater than 3 to use high-resolution deadband.

---

**Deadband delay values as a function of DBFED and DBRED:**

When half-cycle clocking is enabled, the formula to calculate the falling-edge delay (FED) and rising-edge delay (RED) becomes:

$$FED = DBFED * TBCLK / 2$$

$$RED = DBRED * TBCLK / 2$$

**DBRED and DBFED calculated values:**

Required Deadband Rising-Edge Delay in ns = 18.75ns

$$DBRED = RED / (TBCLK / 2)$$

$$DBRED = 18.75ns/5ns$$

$$DBRED \text{ Required} = 3.75ns$$

With 55 MEP steps per coarse step at 180ps each:

### Step 1: Integer Deadband value conversion for DBREDM register

|                     |                           |
|---------------------|---------------------------|
| Integer DBRED value | = int (RED / (TBCLK / 2)) |
|                     | = int (3.75)              |
| DBRED               | = 3                       |

### Step 2: Fractional value conversion for Deadband high-resolution register DBREDHR

|                        |                                                                                                                   |
|------------------------|-------------------------------------------------------------------------------------------------------------------|
| DBREDHR register value | = (frac(DBRED Required) * MEP_ScaleFactor + 0.5) << 8 (Shifting is to move the value to the high byte of DBREDHR) |
|                        | = (frac (3.75) * 55 + 0.5) << 8                                                                                   |
|                        | = (0.75 * 55 + 0.5) << 8                                                                                          |
|                        | = (41.75) * 256 Shifting left by 8 is the same as multiplying by 256.                                             |
| DBREDHR value          | = 29C0h MEP Steps                                                                                                 |
|                        | Hardware ignores lower 9 bits in the above calculated DBREDHR value                                               |

#### Note

If the AUTOCONV bit (HRCNFG.6) is set and the MEP\_ScaleFactor is in the HRMSTEP register, then DBREDHR:DBRED = frac((required DB value) < <8). The rest of the conversion calculations are performed automatically in hardware, and the correct MEP-scaled signal edge appears on the ePWM channel output. If AUTOCONV is not set, the above calculations must be performed by software.

#### 26.15.1.7 Scale Factor Optimizing Software (SFO)

The micro edge positioner (MEP) logic is capable of placing an edge in one of 255 discrete time steps. As previously mentioned, the size of these steps is on the order of 150ps (see the device data sheet for typical MEP step size on your device). The MEP step size varies based on worst-case process parameters, operating temperature, and voltage. MEP step size increases with decreasing voltage and increasing temperature and decreases with increasing voltage and decreasing temperature. Applications that use the HRPWM feature can use the TI-supplied MEP scale factor optimization (SFO) software function. The SFO function helps to dynamically determine the number of MEP steps per EPWMCLK period while the HRPWM is in operation.

To utilize the MEP capabilities effectively, the correct value for the MEP scaling factor needs to be known by the software. To accomplish this, the HRPWM module has built in self-check and diagnostic capabilities that can be used to determine the optimum MEP scale factor value for any operating condition. TI provides a C-callable library containing one SFO function that utilizes this hardware and determines the optimum MEP scale factor. As such, MEP control and diagnostics registers are reserved for TI use.

A detailed description of the SFO library - SFO\_TI\_Build\_V8.lib software can be found in [SFO Library Software - SFO TI\\_Build\\_V8.lib](#).

### 26.15.1.8 HRPWM Examples Using Optimized Assembly Code

The best way to understand how to use the HRPWM capabilities is through two real examples:

1. Simple buck converter using asymmetrical PWM (count-up) with active high polarity.
2. DAC function using simple R+C reconstruction filter.

The following examples all have initialization and configuration code written in C. To make these easier to understand, the #defines shown below are used.

[Example 26-2](#) assumes MEP step size of 150ps and does not use the SFO library.

#### **Example 26-2. #Defines for HRPWM Header Files**

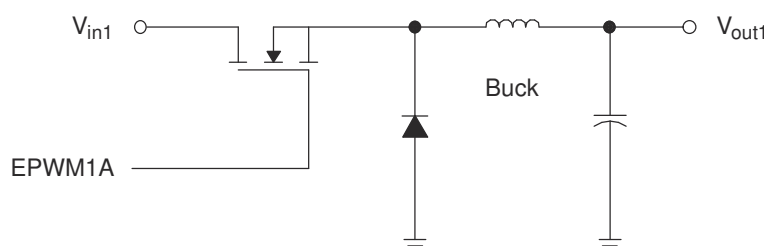
```
// HRPWM (High Resolution PWM) //
=====
// HRCNFG
#define HR_Disable 0x0
#define HR_REP 0x1           // Rising Edge position
#define HR_FEP 0x2           // Falling Edge position
#define HR_BEP 0x3           // Both Edge position #define HR_CMP 0x0 // CMPAHR controlled
#define HR_PHS 0x1           // TBPHSHR controlled #define HR_CTR_ZERO 0x0 // CTR = Zero event
#define HR_CTR_PRD 0x1        // CTR = Period event
#define HR_CTR_ZERO_PRD 0x2   // CTR = ZERO or Period event
#define HR_NORM_B 0x0         // Normal ePWMxB output
#define HR_INVERT_B 0x1       // ePWMxB is inverted ePWMxA output
```

### 26.15.1.8.1 Implementing a Simple Buck Converter

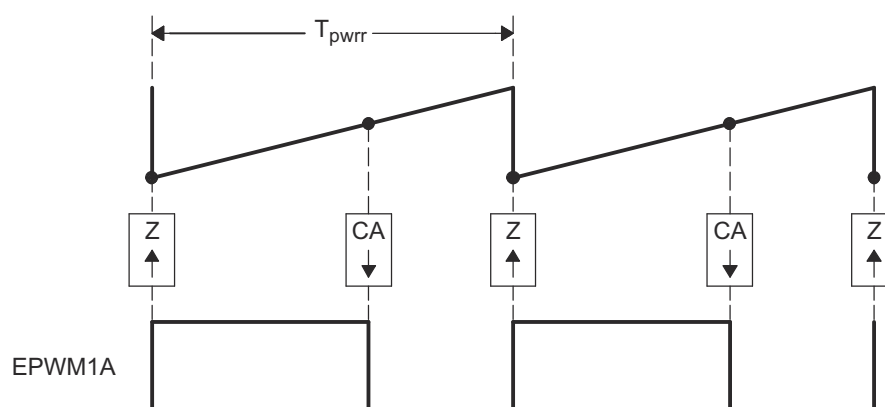
In this example, the PWM requirements are:

- PWM frequency = 1MHz (that is, TBPRD = 100)
- PWM mode = asymmetrical, up-count
- Resolution = 12.7 bits (with a MEP step size of 150ps)

Figure 26-89 and Figure 26-90 show the required PWM waveform. As explained previously, configuration for the ePWM1 module is almost identical to the normal case except that the appropriate MEP options need to be enabled/selected.



**Figure 26-89. Simple Buck Controlled Converter Using a Single PWM**



**Figure 26-90. PWM Waveform Generated for Simple Buck Controlled Converter**

The example code shown consists of two main parts:

- Initialization code (executed once)
- Run time code (typically executed within an ISR)

Example 26-3 shows the Initialization code. The first part is configured for conventional PWM. The second part sets up the HRPWM resources.

This example assumes MEP step size of 150ps and does not use the SFO library.

Example 26-4 shows an assembly example of run-time code for the HRPWM buck converter.

### Example 26-3. HRPWM Buck Converter Initialization Code

```
void HrBuckDrvCnf(void)
{
// Config for conventional PWM first
EPwm1Regs.TBCTL.bit.PRDL = TB_IMMEDIATE;           // set Immediate load
EPwm1Regs.TBPRD = 100;                             // Period set for 1000kHz PWM
hrbuck_period = 200;                                // Used for Q15 to Q0 scaling
EPwm1Regs.TBCTL.bit.CTRMODE = TB_COUNT_UP;
EPwm1Regs.TBCTL.bit.PHSEN = TB_DISABLE;             // EPWM1 is the Master

EPwm1Regs.EPWSYNCOUTEN.all = TB_SYNC_DISABLE;
EPwm1Regs.TBCTL.bit.HSPCLKDIV = TB_DIV1;
EPwm1Regs.TBCTL.bit.CLKDIV = TB_DIV1;
// Note: ChB is initialized here only for comparison purposes, it is not required

EPwm1Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO;
EPwm1Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.LOADBMODE = CC_CTR_ZERO;        // optional
EPwm1Regs.CMPCTL.bit.SHDWBMODE = CC_SHADOW;          // optional
EPwm1Regs.AQCTLA.bit.ZRO = AQ_SET;
EPwm1Regs.AQCTLA.bit.CAU = AQ_CLEAR;
EPwm1Regs.AQCTLB.bit.ZRO = AQ_SET;                  // optional
EPwm1Regs.AQCTLB.bit.CBU = AQ_CLEAR;                // optional
// Now configure the HRPWM resources
EALLOW;                                              // Note these registers are protected
                                              // and act only on ChA
EPwm1Regs.HRCNFG.all = 0x0;                         // clear all bits first
EPwm1Regs.HRCNFG.bit.EDGMODE = HR_FEP;              // Control Falling Edge Position
EPwm1Regs.HRCNFG.bit.CTLMODE = HR_CMP;              // CMPAHR controls the MEP
EPwm1Regs.HRCNFG.bit.HRLOAD = HR_CTR_ZERO;          // Shadow load on CTR=Zero
EDIS;
MEP_ScaleFactor = 66*256;                           // Start with typical Scale Factor
                                              // value for 100MHz
                                              // Note: Use SFO functions to update
                                              // MEP_ScaleFactor dynamically
}
```

### Example 26-4. HRPWM Buck Converter Run-Time Code

```
EPWM1_BASE .set 0x6800
CMPAHR1 .set EPWM1_BASE+0x8
;=====
HRBUCK_DRV; (can execute within an ISR or loop)
;=====
    MOVW DP, #_HRBUCK_In
    MOVL XAR2,@_HRBUCK_In      ; Pointer to Input Q15 Duty (XAR2)
    MOVL XAR3,#CMPAHR1         ; Pointer to HRPWM CMPA reg (XAR3)

; Output for EPWM1A (HRPWM)
    MOV T,*XAR2 ; T <= Duty
    MPYU ACC,T,@_hrbuck_period ; Q15 to Q0 scaling based on Period
    MOV T,@_MEP_ScaleFactor    ; MEP scale factor (from optimizer s/w)
    MPYU P,T,@AL               ; P <= T * AL, Optimizer scaling
    MOVH @AL,P                 ; AL <= P, move result back to ACC
    ADD ACC, #0x080            ; MEP range and rounding adjustment
    MOVL *XAR3,ACC             ; CMPA: CMPAHR(31:8) <= ACC

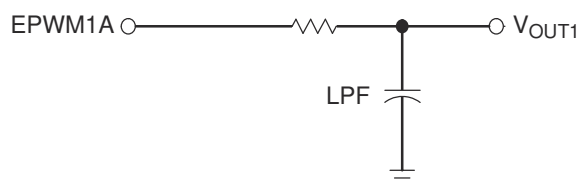
; Output for EPWM1B (Regular Res) Optional - for comparison purpose only
    MOV *+XAR3[2],AH           ; Store ACCH to regular CMPB
```

### 26.15.1.8.2 Implementing a DAC Function Using an R+C Reconstruction Filter

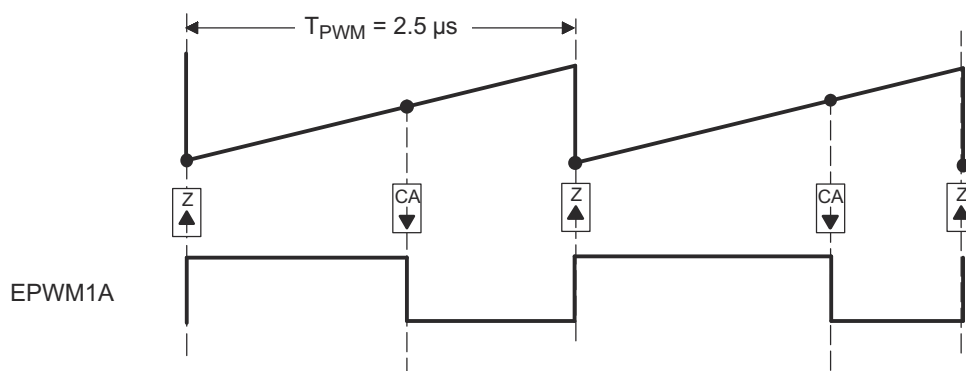
In this example, the PWM requirements are:

- PWM frequency = 400kHz (that is, TBPRD = 250)
- PWM mode = Asymmetrical, Up-count
- Resolution = 14 bits (MEP step size = 150ps)

Figure 26-91 and Figure 26-92 show the DAC function and the required PWM waveform. As explained previously, configuration for the ePWM1 module is almost identical to the normal case except that the appropriate MEP options need to be enabled/selected.



**Figure 26-91. Simple Reconstruction Filter for a PWM-based DAC**



**Figure 26-92. PWM Waveform Generated for the PWM DAC Function**

The example code shown consists of two main parts:

- Initialization code (executed once)
- Run time code (typically executed within an ISR)

This example assumes a typical MEP\_SP and does not use the SFO library.

Example 26-5 shows the Initialization code. The first part is configured for conventional PWM. The second part sets up the HRPWM resources.

Example 26-6 shows an assembly example of run-time code that can execute in a high-speed ISR loop.

### Example 26-5. PWM DAC Function Initialization Code

```
void HrPwmDacDrvCnf(void)
{
// Config for conventional PWM first
EPwm1Regs.TBCTL.bit.PRDL = TB_IMMEDIATE;           // Set Immediate load
EPwm1Regs.TBPRD = 250;                             // Period set for 400kHz PWM
HrDAC_period = 250;                                 // Used for Q15 to Q0 scaling
EPwm1Regs.TBCTL.bit.CTRMODE = TB_COUNT_UP;
EPwm1Regs.TBCTL.bit.PHSEN = TB_DISABLE;             // EPWM1 is the Master

EPwm1Regs.EPWSYNCOUTEN.all = TB_SYNC_DISABLE;
EPwm1Regs.TBCTL.bit.HSPCLKDIV = TB_DIV1;
EPwm1Regs.TBCTL.bit.CLKDIV = TB_DIV1;
// Note: ChB is initialized here only for comparison purposes, it is not required

EPwm1Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO;
EPwm1Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
EPwm1Regs.CMPCTL.bit.LOADBMODE = CC_CTR_ZERO;      // optional
EPwm1Regs.CMPCTL.bit.SHDWBMODE = CC_SHADOW;        // optional

EPwm1Regs.AQCTLA.bit.ZRO = AQ_SET;
EPwm1Regs.AQCTLA.bit.CAU = AQ_CLEAR;
EPwm1Regs.AQCTLB.bit.ZRO = AQ_SET;                 // optional
EPwm1Regs.AQCTLB.bit.CBU = AQ_CLEAR;               // optional
// Now configure the HRPWM resources
EALLOW;                                             // Note these registers are protected
                                                    // and act only on ChA.
EPwm1Regs.HRCNFG.all = 0x0;                        // Clear all bits first
EPwm1Regs.HRCNFG.bit.EDGMODE = HR_FEP;             // Control falling edge position
EPwm1Regs.HRCNFG.bit.CTLMODE = HR_CMP;             // CMPAHR controls the MEP.
EPwm1Regs.HRCNFG.bit.HRLOAD = HR_CTR_ZERO;         // Shadow load on CTR=Zero.
EDIS;
MEP_ScaleFactor = 66*256;                          // Start with typical scale factor
                                                    // value for 100MHz.
                                                    // Use SFO functions to update MEP_ScaleFactor
                                                    // dynamically.
}
```

### Example 26-6. PWM DAC Function Run-Time Code

```
EPWM1_BASE .set 0x6800
CMPAHR1 .set EPWM1_BASE+0x8
;=====
HRPWM_DAC_DRV; (can execute within an ISR or loop)
;=====
    MOVW DP, #_HRDAC_In
    MOVL XAR2,@_HRDAC_In          ; Pointer to input Q15 duty (XAR2)
    MOVL XAR3,#CMPAHR1            ; Pointer to HRPWM CMPA reg (XAR3)

; Output for EPWM1A (HRPWM
    MOV T,*XAR2                   ; T <= duty
    MPY ACC,T,@_HrDAC_period      ; Q15 to Q0 scaling based on period
    ADD ACC,@_HrDAC_period<<15   ; Offset for bipolar operation
    MOV T,@_MEP_ScaleFactor       ; MEP scale factor (from optimizer s/w)
    MPYU P,T,@AL                  ; P <= T * AL, optimizer scaling
    MOVH @AL,P                    ; AL <= P, move result back to ACC
    ADD ACC,#0x080                ; MEP range and rounding adjustment
    MOVL *XAR3,ACC                ; CMPA: CMPAHR(31:8) <= ACC

; Output for EPWM1B (Regular Res) Optional - for comparison purpose only
    MOV *+XAR3[2],AH              ; Store ACCH to regular CMPB
```

## 26.15.2 SFO Library Software - SFO\_TI\_Build\_V8.lib

Table 26-19 lists several features of the SFO\_TI\_Build\_V8.lib library.

**Table 26-19. SFO Library Features**

|                                                                                                       | SFO_TI_Build_V8.lib | Unit                  |
|-------------------------------------------------------------------------------------------------------|---------------------|-----------------------|
| Completion-checking?                                                                                  | Yes                 | Function return value |
| Typical cycles required for SFO() to update MEP_ScaleFactor if called repetitively without interrupts | 130,000             | EPWMCLK cycles        |

### 26.15.2.1 Scale Factor Optimizer Function - int SFO()

This routine drives the micro-edge positioner (MEP) calibration module to run SFO diagnostics and determine the appropriate MEP scale factor (number of MEP steps per coarse EPWMCLK step) for a device at any given time.

If EPWMCLK = TBCLK = 100MHz and assuming the MEP step size is 150ps, the typical scale factor value at 100MHz = 66 MEP steps per TBCLK unit (10ns)

The function returns a MEP scale factor value:

MEP\_ScaleFactor = Number of MEP steps per EPWMCLK

#### Constraints when using this function:

- SFO() can be used with a minimum EPWMCLK = TBCLK = 50MHz. MEP diagnostics logic uses EPWMCLK and not TBCLK, so the EPWMCLK restriction is an important constraint. Below 50MHz with device process variation, the MEP step size can decrease under cold temperature and high core voltage conditions to such a point that 255 MEP steps do not span an entire EPWMCLK cycle.
- At any time, SFO() can be called to run SFO diagnostics on the MEP calibration module.

#### Usage:

- SFO() can be called at any time in the background while the ePWM channels are running in HRPWM mode. The scale factor result obtained can be applied to all ePWM channels running in HRPWM mode because the function makes use of the diagnostics logic in the MEP calibration module (which runs independently of ePWM channels).
- This routine returns a 1 when calibration is finished and a new scale factor has been calculated or returns a 0 if calibration is still running. The routine returns a 2 if there is an error, and the MEP\_ScaleFactor is greater than the maximum 255 fine steps per coarse EPWMCLK cycle. In this case, the HRMSTEP register maintains the last MEP scale factor value less than 256 for auto conversion.
- All ePWM modules operating in HRPWM incur only a 3 EPWMCLK cycle minimum duty cycle limitation when high-resolution period control is not used. If high-resolution period control is enabled, there is an additional duty cycle limitation 3-EPWMCLK cycles before the end of the PWM period (see [Section 26.15.1.5.3](#)).
- The SFO() function also updates the HRMSTEP register with the scale factor result. If the HRCNFG[AUTOCONV] bit is set, the application software is responsible only for setting CMPAHR = fraction(PWMduty\*PWMperiod) << 8 or CMPBHR = fraction(PWMduty\*PWMperiod) << 8 or TBPRDHR = fraction(PWMperiod) while running SFO() in the background. The MEP Calibration Module then uses the values in the HRMSTEP and CMPAHR/CMPBHR/TBPRDHR register to automatically calculate the appropriate number of MEP steps represented by the fractional duty cycle or period and move the high-resolution ePWM signal edge accordingly.
- If the HRCNFG[AUTOCONV] bit is clear, the HRMSTEP register is ignored. The application software needs to perform the necessary calculations manually so that:
  - CMPAHR = (fraction(PWMduty \* PWMperiod) \* MEP Scale Factor) << 8 + 0x080.
  - Similar behavior applies for TBPHSHR, CMPBHR, DBREDHR, and DBFEDHR. Auto-conversion must be enabled when using TBPRDHR.



The following code snippet shows how to use the HRPWM DUTY using driverlib functions.

```
float32_t dutyFine = 85.62;
float32_t count = (dutyFine * (float32_t)(EPWM_TIMER_TBPRD << 8))/100;
uint32_t compCount = (count);
HRPWM_setCounterCompareValue(EPWM1_BASE, HRPWM_COUNTER_COMPARE_A, compCount);
HRPWM_setCounterCompareValue(EPWM1_BASE, HRPWM_COUNTER_COMPARE_B, compCount);
```

The routine can be run as a background task in a slow loop requiring negligible CPU cycles. The repetition rate at which an SFO function needs to be executed depends on the application's operating environment. As with all digital CMOS devices, temperature and supply voltage variations have an effect on MEP operation. However, in most applications these parameters vary slowly and therefore is often sufficient to execute the SFO function once every 5 to 10 seconds. If more rapid variations are expected, then execution can be performed more frequently to match the application. Note there is no high limit restriction on the SFO function repetition rate; hence, the SFO function can execute as quickly as the background loop is capable.

While using the HRPWM feature, HRPWM logic is not active for the first 3 EPWMCLK cycles of the PWM period (and the last 3 EPWMCLK cycles of the PWM period if TBPRDHR is used). While running the application in this configuration, if high-resolution period control is disabled (HRPCTL[HRPE=0]) and the CMPA/CMPB register value is less than three cycles, then the CMPAHR/CMPBHR register must be cleared to zero. If high-resolution period control is enabled (HRPCTL[HRPE=1]), the CMPA register value must not fall below 3 or above TBPRD-3. This can avoid any unexpected transitions on the PWM signal.

### 26.15.2.2 Software Usage

The software library function SFO(), calculates the MEP scale factor for the HRPWM-supported ePWM modules. The scale factor is an integer value in the range 1-255, and represents the number of micro step edge positions available for a system clock period. The scale factor value is returned in an integer variable called MEP\_ScaleFactor. For example, see [Table 26-20](#).

**Table 26-20. Factor Values**

| Software Function call | Functional Description                           | Updated Variables                     |
|------------------------|--------------------------------------------------|---------------------------------------|
| SFO()                  | Returns MEP scale factor in the HRMSTEP register | MEP_ScaleFactor and HRMSTEP register. |

To use the HRPWM feature of the ePWMs, it is recommended that the SFO function be used as described here.

#### Step 1. Add "Include" Files

The SFO\_V8.h file needs to be included as follows. This include file is mandatory while using the SFO library function. For the SFO() to operate, the appropriate (Device)\_Device.h and (Device)\_Epwm\_defines.h must be included in the project. These include files are optional if customized header files are used in the end applications.

#### Example 26-7. A Sample of How to Add "Include" Files

```
#include "F28x7x_Device.h"           // F28x7x Headerfile
#include "F28x7x_EPwm_defines.h"     // init defines
#include "SFO_V8.h"                  // SFO lib functions (needed for HRPWM)
```

## Step 2. Element Declaration

Declare an integer variable for the scale factor value as shown below.

### Example 26-8. Declaring an Element

```
int MEP_ScaleFactor = 0;    //scale factor value
volatile struct EPWM_REGS *ePWM[] = {0, &EPwm1Regs, &EPwm2Regs, &EPwm3Regs,
&EPwm4Regs};
```

## Step 3. MEP\_ScaleFactor Initialization

The SFO() function does not require a starting scale factor value in MEP\_ScaleFactor. Prior to using the MEP\_ScaleFactor variable in application code, SFO() can be called to drive the MEP calibration module to calculate an MEP\_ScaleFactor value.

As part of the one-time initialization code prior to using MEP\_ScaleFactor, include the following:

### Example 26-9. Initializing With a Scale Factor Value

```
MEP_ScaleFactor initialized using function SFO ()
while (SFO() == 0) {} // MEP_ScaleFactor calculated by MEP Cal Module
```

## Step 4. Application Code

While the application is running, fluctuations in both device temperature and supply voltage can be expected. To be sure that good Scale Factors are used for each ePWM module, the SFO function can be re-run periodically as part of a slower back-ground loop. Some examples of this are shown here.

### Note

See the HRPWM\_SFO example in the device-specific C/C++ header files and peripheral examples available from the TI website.

### Example 26-10. SFO Function Calls

```
main ()
{
    int status;
    // User code
    // ePWM1, 2, 3, 4 are running in HRPWM mode
    // The status variable returns 1 once a new MEP_ScaleFactor has been
    // calculated by the MEP Calibration Module running SFO
    // diagnostics.
    status = SFO();
    if(status==2) {ESTOP0;}    // The function returns a 2 if MEP_ScaleFactor is greater
                             // than the maximum 255 allowed (error condition)
}
```

## 26.16 Software

### 26.16.1 EPWM Examples

NOTE: These examples are located in the [C2000Ware](#) installation at the following location:  
C2000Ware\_VERSION#/driverlib/DEVICE\_GPN/examples/CORE\_IF\_MULTICORE/epwm

Cloud access to these examples is available at the following link: [dev.ti.com](http://dev.ti.com) [C2000Ware Examples](#).

#### 26.16.1.1 ePWM Trip Zone

FILE: epwm\_ex1\_trip\_zone.c

This example configures ePWM1 and ePWM2 as follows

- ePWM1 has TZ1 as one shot trip source
- ePWM2 has TZ1 as cycle by cycle trip source

Initially tie TZ1 high. During the test, monitor ePWM1 or ePWM2 outputs on a scope. Pull TZ1 low to see the effect.

##### *External Connections*

- ePWM1A is on GPIO0
- ePWM2A is on GPIO2
- TZ1 is on GPIO12

This example also makes use of the Input X-BAR. GPIO12 (the external trigger) is routed to the input X-BAR, from which it is routed to TZ1.

The TZ-Event is defined such that ePWM1A will undergo a One-Shot Trip and ePWM2A will undergo a Cycle-By-Cycle Trip.

#### 26.16.1.2 ePWM Up Down Count Action Qualifier

FILE: epwm\_ex2\_updown\_aq.c

This example configures ePWM1, ePWM2, ePWM3 to produce a waveform with independent modulation on ePWMxA and ePWMxB.

The compare values CMPA and CMPB are modified within the ePWM's ISR.

The TB counter is in up/down count mode for this example.

View the ePWM1A/B(GPIO0 & GPIO1), ePWM2A/B(GPIO2 & GPIO3) and ePWM3A/B(GPIO4 & GPIO5) waveforms on oscilloscope.

#### 26.16.1.3 ePWM Synchronization

FILE: epwm\_ex3\_synchronization.c

This example configures ePWM1, ePWM2, ePWM3 and ePWM4 as follows

- ePWM1 without phase shift as sync source
- ePWM2 with phase shift of 300 TBCLKs
- ePWM3 with phase shift of 600 TBCLKs
- ePWM4 with phase shift of 900 TBCLKs

##### *External Connections*

- GPIO0 EPWM1A
- GPIO1 EPWM1B
- GPIO2 EPWM2A
- GPIO3 EPWM2B
- GPIO4 EPWM3A
- GPIO5 EPWM3B
- GPIO6 EPWM4A

- GPIO7 EPWM4B

*Watch Variables*

- None.

**26.16.1.4 ePWM Digital Compare**

FILE: epwm\_ex4\_digital\_compare.c

This example configures ePWM1 as follows

- ePWM1 with DCAEVT1 forcing the ePWM output LOW
- GPIO25 is used as the input to the INPUT XBAR INPUT1
- INPUT1 (from INPUT XBAR) is used as the source for DCAEVT1
- GPIO25's PULL-UP resistor is enabled, in order to test the trip, PULL this pin to GND

*External Connections*

- GPIO0 EPWM1A
- GPIO1 EPWM1B
- GPIO25 TZ1, pull this pin low to trip the ePWM

*Watch Variables*

- None.

**26.16.1.5 ePWM Digital Compare Event Filter Blanking Window**

FILE: epwm\_ex5\_digital\_compare\_event\_filter.c

This example configures ePWM1 as follows

- ePWM1 with DCAEVT1 forcing the ePWM output LOW
- GPIO25 is used as the input to the INPUT XBAR INPUT1
- INPUT1 (from INPUT XBAR) is used as the source for DCAEVT1
- GPIO25's PULL-UP resistor is enabled, in order to test the trip, PULL this pin to GND
- ePWM1 with DCBEVT1 forcing the ePWM output LOW
- GPIO25 is used as the input to the INPUT XBAR INPUT1
- INPUT1 (from INPUT XBAR) is used as the source for DCAEVT1
- GPIO25's PULL-UP resistor is enabled, in order to test the trip, PULL this pin to GND
- DCBEVT1 uses the filtered version of DCBEVT1
- The DCFILT signal uses the blanking window to ignore the DCBEVT1 for the duration of DC Blanking window

*External Connections*

- GPIO0 EPWM1A
- GPIO1 EPWM1B
- GPIO25 TRIPIN1, pull this pin low to trip the ePWM

*Watch Variables*

- None.

**26.16.1.6 ePWM Valley Switching**

FILE: epwm\_ex6\_valley\_switching.c

This example configures ePWM1 as follows

- ePWM1 with DCAEVT1 forcing the ePWM output LOW
- GPIO25 is used as the input to the INPUT XBAR INPUT1
- INPUT1 (from INPUT XBAR) is used as the source for DCAEVT1
- GPIO25 is set to output and toggled in the main loop to trip the PWM
- ePWM1 with DCBEVT1 forcing the ePWM output LOW
- GPIO25 is used as the input to the INPUT XBAR INPUT1
- INPUT1 (from INPUT XBAR) is used as the source for DCAEVT1

- GPIO25 is set to output and toggled in the main loop to trip the PWM
- DCBEVT1 uses the filtered version of DCBEVT1
- The DCFILT signal uses the valley switching module to delay the
- DCFILT signal by a software defined DELAY value.

#### *External Connections*

- GPIO0 EPWM1A
- GPIO1 EPWM1B
- GPIO25 TRIPIN1 (Output Pin, toggled through software)

#### *Watch Variables*

- None.

### **26.16.1.7 ePWM Digital Compare Edge Filter**

FILE: epwm\_ex7\_edge\_filter.c

This example configures ePWM1 as follows

- ePWM1 with DCBEVT2 forcing the ePWM output LOW as a CBC source
- GPIO25 is used as the input to the INPUT XBAR INPUT1
- INPUT1 (from INPUT XBAR) is used as the source for DCBEVT2
- GPIO25 is set to output and toggled in the main loop to trip the PWM
- The DCBEVT2 is the source for DCFILT
- The DCFILT will count edges of the DCBEVT2 and generate a signal to trip the ePWM on the 4th edge of DCBEVT2

#### *External Connections*

- GPIO0 EPWM1A
- GPIO1 EPWM1B
- GPIO25 TRIPIN1 (Output Pin, toggled through software)

#### *Watch Variables*

- None.

### **26.16.1.8 ePWM Deadband**

FILE: epwm\_ex8\_deadband.c

This example configures ePWM1 through ePWM6 as follows

- ePWM1 with Deadband disabled (Reference)
- ePWM2 with Deadband Active High
- ePWM3 with Deadband Active Low
- ePWM4 with Deadband Active High Complimentary
- ePWM5 with Deadband Active Low Complimentary
- ePWM6 with Deadband Output Swap (switch A and B outputs)

#### *External Connections*

- GPIO0 EPWM1A
- GPIO1 EPWM1B
- GPIO2 EPWM2A
- GPIO3 EPWM2B
- GPIO4 EPWM3A
- GPIO5 EPWM3B
- GPIO6 EPWM4A
- GPIO7 EPWM4B
- GPIO8 EPWM5A
- GPIO9 EPWM5B
- GPIO10 EPWM6A

- GPIO11 EPWM6B

*Watch Variables*

- None.

**26.16.1.9 ePWM DMA**

FILE: epwm\_ex9\_dma.c

This example configures ePWM1 and DMA as follows:

- ePWM1 is set up to generate PWM waveforms
- DMA5 is set up to update the CMPAHR, CMPA, CMPBHR and CMPB every period with the next value in the configuration array. This allows the user to create a DMA enabled fifo for all the CMPx and CMPxHR registers to generate unconventional PWM waveforms.
- DMA6 is set up to update the TBPHSHR, TBPHS, TBPRDHR and TBPRD every period with the next value in the configuration array.
- Other registers such as AQCTL can be controlled through the DMA as well by following the same procedure. (Not used in this example)

*External Connections*

- GPIO0 EPWM1A
- GPIO1 EPWM1B

*Watch Variables*

- None.

**26.16.1.10 ePWM Chopper**

FILE: epwm\_ex10\_chopper.c

This example configures ePWM1, ePWM2, ePWM3 and ePWM4 as follows

- ePWM1 with Chopper disabled (Reference)
- ePWM2 with chopper enabled at 1/8 duty cycle
- ePWM3 with chopper enabled at 6/8 duty cycle
- ePWM4 with chopper enabled at 1/2 duty cycle with One-Shot Pulse enabled

*External Connections*

- GPIO0 EPWM1A
- GPIO1 EPWM1B
- GPIO2 EPWM2A
- GPIO3 EPWM2B
- GPIO4 EPWM3A
- GPIO5 EPWM3B
- GPIO6 EPWM4A
- GPIO7 EPWM4B

*Watch Variables*

- None.

**26.16.1.11 EPWM Configure Signal**

FILE: epwm\_ex11\_configure\_signal.c

This example configures ePWM1, ePWM2, ePWM3 to produce signal of desired frequency and duty. It also configures phase between the configured modules.

Signal of 10kHz with duty of 0.5 is configured on ePWMxA & ePWMxB with ePWMxB inverted. Also, phase of 120 degree is configured between ePWM1 to ePWM3 signals.

During the test, monitor ePWM1, ePWM2, and/or ePWM3 outputs on an oscilloscope.

- ePWM1A is on GPIO0
- ePWM1B is on GPIO1
- ePWM2A is on GPIO2
- ePWM2B is on GPIO3
- ePWM3A is on GPIO4
- ePWM3B is on GPIO5

#### 26.16.1.12 Realization of Monoshot mode

FILE: epwm\_ex12\_monoshot\_mode.c

This example showcases how to generate monoshot PWM output based on external trigger, that is, generating just a single pulse output on receipt of an external trigger. And the next pulse is generated only when the next trigger comes. The example utilizes external synchronization and T1 action qualifier event features to achieve the desired output.

ePWM1 is used to generate the monoshot output and ePWM2 is used as an external trigger for that. No external connections are required as ePWM2A is fed as the trigger using Input X-BAR automatically.

ePWM1 is configured to generate a single pulse of 0.5  $\mu$ s when received an external trigger. This is achieved by enabling the phase synchronization feature and configuring EPWMxSYNCl as EXTSYNCIN1. And this EPWMxSYNCl is also configured as T1 event of action qualifier to set output HIGH while CTR = PRD action is used to set output LOW.

ePWM2 is configured to generate a 100 KHz signal with a duty of 1% (to simulate a rising edge trigger) which is routed to EXTSYNCIN1 using Input XBAR.

Observe GPIO0 (EPWM1A : Monoshot Output) and GPIO2 (EPWM2 : External Trigger) on oscilloscope.

**NOTE :** In the following example, the ePWM timer is still running in a continuous mode rather than a one-shot mode thus for more reliable implementation, refer to CLB based one shot PWM implementation demonstrated in "clb\_ex17\_one\_shot\_pwm" example

#### 26.16.1.13 EPWM Action Qualifier (epwm\_up\_aq)

FILE: epwm\_ex13\_up\_aq.c

This example configures ePWM1, ePWM2, ePWM3 to produce an waveform with independent modulation on EPWMxA and EPWMxB.

The compare values CMPA and CMPB are modified within the ePWM's ISR.

The TB counter is in up count mode for this example.

View the EPWM1A/B(GPIO0 & GPIO1), EPWM2A/B(GPIO2 & GPIO3) and EPWM3A/B(GPIO4 & GPIO5) waveforms via an oscilloscope.

### 26.16.2 HRPWM Examples

**NOTE:** These examples are located in the [C2000Ware](#) installation at the following location: C2000Ware\_VERSION#/driverlib/DEVICE\_GPN/examples/CORE\_IF\_MULTICORE/hrpwm

Cloud access to these examples is available at the following link: [dev.ti.com C2000Ware Examples](https://dev.ti.com/C2000Ware/Examples).

#### 26.16.2.1 HRPWM Duty Control with SFO

FILE: hrpwm\_ex1\_duty\_sfo.c

This example modifies the MEP control registers to show edge displacement for high-resolution period with ePWM in Up count mode due to the HRPWM control extension of the respective ePWM module.

This example calls the following TI's MEP Scale Factor Optimizer (SFO) software library V8 functions:

```
int SFO();
```



- updates MEP\_ScaleFactor dynamically when HRPWM is in use
- updates HRMSTEP register (exists only in EPwm1Regs register space) with MEP\_ScaleFactor value
- returns 2 if error: MEP\_ScaleFactor is greater than maximum value of 255 (Auto-conversion may not function properly under this condition)
- returns 1 when complete for the specified channel
- returns 0 if not complete for the specified channel

This example is intended to explain the HRPWM capabilities. The code can be optimized for code efficiency. Refer to TI's Digital power application examples and TI Digital Power Supply software libraries for details.

#### External Connections

- Monitor ePWM1/2/3/4 A/B pins on an oscilloscope.

#### 26.16.2.2 HRPWM Slider

FILE: hrpwm\_ex2\_slider.c

This example modifies the MEP control registers to show edge displacement due to HRPWM. Control blocks of the respective ePWM module channel A and B have fine edge movement due to HRPWM logic.

Monitor ePWM1 A/B pins on an oscilloscope.

#### 26.16.2.3 HRPWM Period Control

FILE: hrpwm\_ex3\_prd\_updown\_sfo.c

This example modifies the MEP control registers to show edge displacement for high-resolution period with ePWM in Up-Down count mode due to the HRPWM control extension of the respective ePWM module.

This example calls the following TI's MEP Scale Factor Optimizer (SFO) software library V8 functions:

*int SFO();*

- updates MEP\_ScaleFactor dynamically when HRPWM is in use
- updates HRMSTEP register (exists only in EPwm1Regs register space) with MEP\_ScaleFactor value
- returns 2 if error: MEP\_ScaleFactor is greater than maximum value of 255 (Auto-conversion may not function properly under this condition)
- returns 1 when complete for the specified channel
- returns 0 if not complete for the specified channel

This example is intended to explain the HRPWM capabilities. The code can be optimized for code efficiency. Refer to TI's Digital power application examples and TI Digital Power Supply software libraries for details.

#### External Connections

- Monitor ePWM1/2/3/4 A/B pins on an oscilloscope.

#### 26.16.2.4 HRPWM Duty Control with UPDOWN Mode

FILE: hrpwm\_ex4\_duty\_updown\_sfo.c

This example calls the following TI's MEP Scale Factor Optimizer (SFO) software library V8 functions:

*int SFO();*

- updates MEP\_ScaleFactor dynamically when HRPWM is in use
- updates HRMSTEP register (exists only in EPwm1Regs register space) with MEP\_ScaleFactor value
- returns 2 if error: MEP\_ScaleFactor is greater than maximum value of 255 (Auto-conversion may not function properly under this condition)
- returns 1 when complete for the specified channel
- returns 0 if not complete for the specified channel

This example is intended to explain the HRPWM capabilities. The code can be optimized for code efficiency. Refer to TI's Digital power application examples and TI Digital Power Supply software libraries for details.

#### External Connections



- Monitor ePWM1/2/3/4 A/B pins on an oscilloscope.

#### 26.16.2.5 HRPWM Slider Test

FILE: hrpwm\_ex5\_slider\_qformat.c

This example modifies the MEP control registers to show edge displacement due to HRPWM. Control blocks of the respective ePWM module channel A and B will have fine edge movement due to HRPWM logic. Load the hrpwm\_slider.gel file. Select the HRPWM\_eval from the GEL menu. A FineDuty slider graphics will show up in CCS. Load the program and run. Use the Slider to and observe the EPWM edge displacement for each slider step change. This explains the MEP control on the EPwmxA channels.

Monitor ePWM1 & ePWM2 A/B pins on an oscilloscope.

#### 26.16.2.6 HRPWM Duty Up Count

FILE: hrpwm\_ex6\_duty\_sfo\_qformat.c

This example modifies the MEP control registers to show edge displacement for high-resolution period with ePWM in Up count mode due to the HRPWM control extension of the respective ePWM module.

This example calls the following TI's MEP Scale Factor Optimizer (SFO) software library V8 functions:

*int SFO();*

- updates MEP\_ScaleFactor dynamically when HRPWM is in use
- updates HRMSTEP register (exists only in EPwm1Regs register space) with MEP\_ScaleFactor value
- returns 2 if error: MEP\_ScaleFactor is greater than maximum value of 255 (Auto-conversion may not function properly under this condition)
- returns 1 when complete for the specified channel
- returns 0 if not complete for the specified channel

This example is intended to explain the HRPWM capabilities. The code can be optimized for code efficiency. Refer to TI's Digital power application examples and TI Digital Power Supply software libraries for details.

To run this example:

1. Run this example at maximum SYSCLKOUT
2. Activate Real time mode
3. Run the code

#### External Connections

- Monitor ePWM1/2 A/B pins on an oscilloscope.

#### Watch Variables

- status - Example run status
- updateFine - Set to 1 use HRPWM capabilities and observe in fine MEP steps(default) Set to 0 to disable HRPWM capabilities and observe in coarse SYSCLKOUT cycle steps

#### 26.16.2.7 HRPWM Period Up-Down Count

FILE: hrpwm\_ex7\_prd\_updown\_sfo\_qformat.c

This example modifies the MEP control registers to show edge displacement for high-resolution period with ePWM in Up-Down count mode due to the HRPWM control extension of the respective ePWM module.

This example calls the following TI's MEP Scale Factor Optimizer (SFO) software library V8 functions:

*int SFO();*

- updates MEP\_ScaleFactor dynamically when HRPWM is in use
- updates HRMSTEP register (exists only in EPwm1Regs register space) with MEP\_ScaleFactor value
- returns 2 if error: MEP\_ScaleFactor is greater than maximum value of 255 (Auto-conversion may not function properly under this condition)
- returns 1 when complete for the specified channel

- returns 0 if not complete for the specified channel

This example is intended to explain the HRPWM capabilities. The code can be optimized for code efficiency. Refer to TI's Digital power application examples and TI Digital Power Supply software libraries for details.

To run this example:

1. Run this example at maximum SYSCLKOUT
2. Activate Real time mode
3. Run the code

#### External Connections

- Monitor ePWM1/2 A/B pins on an oscilloscope.

#### Watch Variables

- updateFine - Set to 1 use HRPWM capabilities and observe in fine MEP steps(default) Set to 0 to disable HRPWM capabilities and observe in coarse SYSCLKOUT cycle steps

## 26.17 ePWM Registers

This section describes the Enhanced Pulse Width Modulator registers.

### 26.17.1 EPWM Base Address Table (C28)

**Table 26-21. EPWM Base Address Table (C28)**

| Bit Field Name |           | DriverLib Name | Base Address | CPU1 | CPU2 | DMA | CLA | Pipeline Protected |
|----------------|-----------|----------------|--------------|------|------|-----|-----|--------------------|
| Instance       | Structure |                |              |      |      |     |     |                    |
| EPwm1Regs      | EPWM_REGS | EPWM1_BASE     | 0x0000_4000  | YES  | YES  | YES | YES | YES                |
| EPwm2Regs      | EPWM_REGS | EPWM2_BASE     | 0x0000_4100  | YES  | YES  | YES | YES | YES                |
| EPwm3Regs      | EPWM_REGS | EPWM3_BASE     | 0x0000_4200  | YES  | YES  | YES | YES | YES                |
| EPwm4Regs      | EPWM_REGS | EPWM4_BASE     | 0x0000_4300  | YES  | YES  | YES | YES | YES                |
| EPwm5Regs      | EPWM_REGS | EPWM5_BASE     | 0x0000_4400  | YES  | YES  | YES | YES | YES                |
| EPwm6Regs      | EPWM_REGS | EPWM6_BASE     | 0x0000_4500  | YES  | YES  | YES | YES | YES                |
| EPwm7Regs      | EPWM_REGS | EPWM7_BASE     | 0x0000_4600  | YES  | YES  | YES | YES | YES                |
| EPwm8Regs      | EPWM_REGS | EPWM8_BASE     | 0x0000_4700  | YES  | YES  | YES | YES | YES                |
| EPwm9Regs      | EPWM_REGS | EPWM9_BASE     | 0x0000_4800  | YES  | YES  | YES | YES | YES                |
| EPwm10Regs     | EPWM_REGS | EPWM10_BASE    | 0x0000_4900  | YES  | YES  | YES | YES | YES                |
| EPwm11Regs     | EPWM_REGS | EPWM11_BASE    | 0x0000_4A00  | YES  | YES  | YES | YES | YES                |
| EPwm12Regs     | EPWM_REGS | EPWM12_BASE    | 0x0000_4B00  | YES  | YES  | YES | YES | YES                |
| EPwm13Regs     | EPWM_REGS | EPWM13_BASE    | 0x0000_4C00  | YES  | YES  | YES | YES | YES                |
| EPwm14Regs     | EPWM_REGS | EPWM14_BASE    | 0x0000_4D00  | YES  | YES  | YES | YES | YES                |
| EPwm15Regs     | EPWM_REGS | EPWM15_BASE    | 0x0000_4E00  | YES  | YES  | YES | YES | YES                |
| EPwm16Regs     | EPWM_REGS | EPWM16_BASE    | 0x0000_4F00  | YES  | YES  | YES | YES | YES                |

### 26.17.2 EPWM\_REGS Registers

Table 26-22 lists the memory-mapped registers for the EPWM\_REGS registers. All register offset addresses not listed in Table 26-22 should be considered as reserved locations and the register contents should not be modified.

**Table 26-22. EPWM\_REGS Registers**

| Offset | Acronym       | Register Name                                                         | Write Protection | Section            |
|--------|---------------|-----------------------------------------------------------------------|------------------|--------------------|
| 0h     | TBCTL         | Time Base Control Register                                            |                  | <a href="#">Go</a> |
| 1h     | TBCTL2        | Time Base Control Register 2                                          |                  | <a href="#">Go</a> |
| 3h     | EPWMSYNCINSEL | EPWMxSYNCIN Source Select Register                                    |                  | <a href="#">Go</a> |
| 4h     | TBCTR         | Time Base Counter Register                                            |                  | <a href="#">Go</a> |
| 5h     | TBSTS         | Time Base Status Register                                             |                  | <a href="#">Go</a> |
| 6h     | EPWMSYNCOUTEN | EPWMxSYNCOUT Source Enable Register                                   |                  | <a href="#">Go</a> |
| 7h     | TBCTL3        | Time Base Control Register 3                                          |                  | <a href="#">Go</a> |
| 8h     | CMPCTL        | Counter Compare Control Register                                      |                  | <a href="#">Go</a> |
| 9h     | CMPCTL2       | Counter Compare Control Register 2                                    |                  | <a href="#">Go</a> |
| Ch     | DBCTL         | Dead-Band Generator Control Register                                  |                  | <a href="#">Go</a> |
| Dh     | DBCTL2        | Dead-Band Generator Control Register 2                                |                  | <a href="#">Go</a> |
| 10h    | AQCTL         | Action Qualifier Control Register                                     |                  | <a href="#">Go</a> |
| 11h    | AQTSRCSEL     | Action Qualifier Trigger Event Source Select Register                 |                  | <a href="#">Go</a> |
| 14h    | PCCTL         | PWM Chopper Control Register                                          |                  | <a href="#">Go</a> |
| 18h    | VCAPCTL       | Valley Capture Control Register                                       |                  | <a href="#">Go</a> |
| 19h    | VCNTCFG       | Valley Counter Config Register                                        |                  | <a href="#">Go</a> |
| 20h    | HRCNFG        | HRPWM Configuration Register                                          | EALLOW           | <a href="#">Go</a> |
| 21h    | HRPWR         | HRPWM Power Register                                                  | EALLOW           | <a href="#">Go</a> |
| 26h    | HRMSTEP       | HRPWM MEP Step Register                                               | EALLOW           | <a href="#">Go</a> |
| 27h    | HRCNFG2       | HRPWM Configuration 2 Register                                        | EALLOW           | <a href="#">Go</a> |
| 2Dh    | HRPCTL        | High Resolution Period Control Register                               | EALLOW           | <a href="#">Go</a> |
| 2Eh    | TRREM         | HRPWM High Resolution Remainder Register                              | EALLOW           | <a href="#">Go</a> |
| 34h    | GLDCTL        | Global PWM Load Control Register                                      | EALLOW           | <a href="#">Go</a> |
| 35h    | GLDCFG        | Global PWM Load Config Register                                       | EALLOW           | <a href="#">Go</a> |
| 38h    | EPWMXLINK     | EPWMx Link Register                                                   |                  | <a href="#">Go</a> |
| 40h    | AQCTLA        | Action Qualifier Control Register For Output A                        |                  | <a href="#">Go</a> |
| 41h    | AQCTLA2       | Additional Action Qualifier Control Register For Output A             |                  | <a href="#">Go</a> |
| 42h    | AQCTLB        | Action Qualifier Control Register For Output B                        |                  | <a href="#">Go</a> |
| 43h    | AQCTLB2       | Additional Action Qualifier Control Register For Output B             |                  | <a href="#">Go</a> |
| 47h    | AQSFR         | Action Qualifier Software Force Register                              |                  | <a href="#">Go</a> |
| 49h    | AQCSFR        | Action Qualifier Continuous S/W Force Register                        |                  | <a href="#">Go</a> |
| 50h    | DBREDHR       | Dead-Band Generator Rising Edge Delay High Resolution Mirror Register |                  | <a href="#">Go</a> |
| 51h    | DBRED         | Dead-Band Generator Rising Edge Delay High Resolution Mirror Register |                  | <a href="#">Go</a> |
| 52h    | DBFEDHR       | Dead-Band Generator Falling Edge Delay High Resolution Register       |                  | <a href="#">Go</a> |
| 53h    | DBFED         | Dead-Band Generator Falling Edge Delay Count Register                 |                  | <a href="#">Go</a> |
| 60h    | TBPHS         | Time Base Phase High                                                  |                  | <a href="#">Go</a> |

**Table 26-22. EPWM\_REGS Registers (continued)**

| Offset | Acronym      | Register Name                                         | Write Protection | Section            |
|--------|--------------|-------------------------------------------------------|------------------|--------------------|
| 62h    | TBPRDHR      | Time Base Period High Resolution Register             |                  | <a href="#">Go</a> |
| 63h    | TBPRD        | Time Base Period Register                             |                  | <a href="#">Go</a> |
| 6Ah    | CMPA         | Counter Compare A Register                            |                  | <a href="#">Go</a> |
| 6Ch    | CMPB         | Compare B Register                                    |                  | <a href="#">Go</a> |
| 6Fh    | CMPC         | Counter Compare C Register                            |                  | <a href="#">Go</a> |
| 71h    | CMPD         | Counter Compare D Register                            |                  | <a href="#">Go</a> |
| 74h    | GLDCTL2      | Global PWM Load Control Register 2                    |                  | <a href="#">Go</a> |
| 77h    | SWVDELVAL    | Software Valley Mode Delay Register                   |                  | <a href="#">Go</a> |
| 80h    | TZSEL        | Trip Zone Select Register                             | EALLOW           | <a href="#">Go</a> |
| 82h    | TZDCSEL      | Trip Zone Digital Comparator Select Register          | EALLOW           | <a href="#">Go</a> |
| 84h    | TZCTL        | Trip Zone Control Register                            | EALLOW           | <a href="#">Go</a> |
| 85h    | TZCTL2       | Additional Trip Zone Control Register                 | EALLOW           | <a href="#">Go</a> |
| 86h    | TZCTLDCA     | Trip Zone Control Register Digital Compare A          | EALLOW           | <a href="#">Go</a> |
| 87h    | TZCTLDCB     | Trip Zone Control Register Digital Compare B          | EALLOW           | <a href="#">Go</a> |
| 8Dh    | TZEINT       | Trip Zone Enable Interrupt Register                   | EALLOW           | <a href="#">Go</a> |
| 93h    | TZFLG        | Trip Zone Flag Register                               |                  | <a href="#">Go</a> |
| 94h    | TZCBCFLG     | Trip Zone CBC Flag Register                           |                  | <a href="#">Go</a> |
| 95h    | TZOSTFLG     | Trip Zone OST Flag Register                           |                  | <a href="#">Go</a> |
| 97h    | TZCLR        | Trip Zone Clear Register                              | EALLOW           | <a href="#">Go</a> |
| 98h    | TZCBCCLR     | Trip Zone CBC Clear Register                          | EALLOW           | <a href="#">Go</a> |
| 99h    | TZOSTCLR     | Trip Zone OST Clear Register                          | EALLOW           | <a href="#">Go</a> |
| 9Bh    | TZFRC        | Trip Zone Force Register                              | EALLOW           | <a href="#">Go</a> |
| A4h    | ETSEL        | Event Trigger Selection Register                      |                  | <a href="#">Go</a> |
| A6h    | ETPS         | Event Trigger Pre-Scale Register                      |                  | <a href="#">Go</a> |
| A8h    | ETFLG        | Event Trigger Flag Register                           |                  | <a href="#">Go</a> |
| AAh    | ETCLR        | Event Trigger Clear Register                          |                  | <a href="#">Go</a> |
| ACh    | ETFRC        | Event Trigger Force Register                          |                  | <a href="#">Go</a> |
| AEh    | ETINTPS      | Event-Trigger Interrupt Pre-Scale Register            |                  | <a href="#">Go</a> |
| B0h    | ETSOCPS      | Event-Trigger SOC Pre-Scale Register                  |                  | <a href="#">Go</a> |
| B2h    | ETCNTINITCTL | Event-Trigger Counter Initialization Control Register |                  | <a href="#">Go</a> |
| B4h    | ETCNTINIT    | Event-Trigger Counter Initialization Register         |                  | <a href="#">Go</a> |
| C0h    | DCTRISEL     | Digital Compare Trip Select Register                  | EALLOW           | <a href="#">Go</a> |
| C3h    | DCACTL       | Digital Compare A Control Register                    | EALLOW           | <a href="#">Go</a> |
| C4h    | DCBCTL       | Digital Compare B Control Register                    | EALLOW           | <a href="#">Go</a> |
| C7h    | DCFCTL       | Digital Compare Filter Control Register               | EALLOW           | <a href="#">Go</a> |
| C8h    | DCCAPCTL     | Digital Compare Capture Control Register              | EALLOW           | <a href="#">Go</a> |
| C9h    | DCFOFFSET    | Digital Compare Filter Offset Register                |                  | <a href="#">Go</a> |
| CAh    | DCFOFFSETCNT | Digital Compare Filter Offset Counter Register        |                  | <a href="#">Go</a> |
| CBh    | DCFWINDOW    | Digital Compare Filter Window Register                |                  | <a href="#">Go</a> |
| CCh    | DCFWINDOWCNT | Digital Compare Filter Window Counter Register        |                  | <a href="#">Go</a> |
| CFh    | DCCAP        | Digital Compare Counter Capture Register              |                  | <a href="#">Go</a> |
| D2h    | DCAHTRIPSEL  | Digital Compare AH Trip Select                        | EALLOW           | <a href="#">Go</a> |
| D3h    | DCALTRIPSEL  | Digital Compare AL Trip Select                        | EALLOW           | <a href="#">Go</a> |
| D4h    | DCBHTRIPSEL  | Digital Compare BH Trip Select                        | EALLOW           | <a href="#">Go</a> |

**Table 26-22. EPWM\_REGS Registers (continued)**

| Offset | Acronym     | Register Name                       | Write Protection | Section            |
|--------|-------------|-------------------------------------|------------------|--------------------|
| D5h    | DCBLTRIPSEL | Digital Compare BL Trip Select      | EALLOW           | <a href="#">Go</a> |
| FAh    | EPWMLOCK    | EPWM Lock Register                  |                  | <a href="#">Go</a> |
| FDh    | HWVDELVAL   | Hardware Valley Mode Delay Register |                  | <a href="#">Go</a> |
| FEh    | VCNTVAL     | Hardware Valley Counter Register    |                  | <a href="#">Go</a> |

Complex bit access types are encoded to fit into small table cells. [Table 26-23](#) shows the codes that are used for access types in this section.

**Table 26-23. EPWM\_REGS Access Type Codes**

| Access Type              | Code      | Description                                                                                                                                                                                                                                                                          |
|--------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Read Type                |           |                                                                                                                                                                                                                                                                                      |
| R                        | R         | Read                                                                                                                                                                                                                                                                                 |
| R-0                      | R<br>-0   | Read<br>Returns 0s                                                                                                                                                                                                                                                                   |
| Write Type               |           |                                                                                                                                                                                                                                                                                      |
| W                        | W         | Write                                                                                                                                                                                                                                                                                |
| W1C                      | W<br>1C   | Write<br>1 to clear                                                                                                                                                                                                                                                                  |
| W1S                      | W<br>1S   | Write<br>1 to set                                                                                                                                                                                                                                                                    |
| WOnce                    | W<br>Once | Write<br>Write once                                                                                                                                                                                                                                                                  |
| Reset or Default Value   |           |                                                                                                                                                                                                                                                                                      |
| -n                       |           | Value after reset or the default value                                                                                                                                                                                                                                               |
| Register Array Variables |           |                                                                                                                                                                                                                                                                                      |
| i,j,k,l,m,n              |           | When these variables are used in a register name, an offset, or an address, they refer to the value of a register array where the register is part of a group of repeating registers. The register groups form a hierarchical structure and the array is represented with a formula. |
| y                        |           | When this variable is used in a register name, an offset, or an address it refers to the value of a register array.                                                                                                                                                                  |

### 26.17.2.1 TBCTL Register (Offset = 0h) [Reset = 0083h]

TBCTL is shown in [Figure 26-93](#) and described in [Table 26-24](#).

Return to the [Summary Table](#).

Time Base Control Register

**Figure 26-93. TBCTL Register**

| 15        | 14         | 13       | 12     | 11     | 10        | 9       | 8 |
|-----------|------------|----------|--------|--------|-----------|---------|---|
| FREE_SOFT |            | PHSDIR   | CLKDIV |        | HSPCLKDIV |         |   |
| R/W-0h    |            | R/W-0h   | R/W-0h |        | R/W-1h    |         |   |
| 7         | 6          | 5        | 4      | 3      | 2         | 1       | 0 |
| HSPCLKDIV | SWFSYNC    | RESERVED |        | PRDLD  | PHSEN     | CTRMODE |   |
| R/W-1h    | R-0/W1S-0h | R-0h     |        | R/W-0h | R/W-0h    | R/W-3h  |   |

**Table 26-24. TBCTL Register Field Descriptions**

| Bit   | Field     | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------|-----------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-14 | FREE_SOFT | R/W  | 0h    | Emulation Mode Bits. These bits select the behavior of the ePWM time-base counter during emulation events<br>00: Stop after the next time-base counter increment or decrement<br>01: Stop when counter completes a whole cycle:<br>- Up-count mode: stop when the time-base counter = period (TBCTR = TBPRD)<br>- Down-count mode: stop when the time-base counter = 0x00 (TBCTR = 0x00)<br>- Up-down-count mode: stop when the time-base counter = 0x00 (TBCTR = 0x00)<br>1x: Free run<br>Reset type: SYSRSn                                                                       |
| 13    | PHSDIR    | R/W  | 0h    | Phase Direction Bit<br>This bit is only used when the time-base counter is configured in the up-down-count mode. The PHSDIR bit indicates the direction the time-base counter (TBCTR) will count after a synchronization event occurs and a new phase value is loaded from the phase (TBPHS) register. This is irrespective of the direction of the counter before the synchronization event..<br>In the up-count and down-count modes this bit is ignored.<br>0: Count down after the synchronization event.<br>1: Count up after the synchronization event.<br>Reset type: SYSRSn |
| 12-10 | CLKDIV    | R/W  | 0h    | Time Base Clock Pre-Scale Bits<br>These bits select the time base clock pre-scale value (TBCLK = EPWMCLK/(HSPCLKDIV * CLKDIV):<br>000: /1 (default on reset)<br>001: /2<br>010: /4<br>011: /8<br>100: /16<br>101: /32<br>110: /64<br>111: /128<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                |

**Table 26-24. TBCTL Register Field Descriptions (continued)**

| Bit | Field     | Type    | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----|-----------|---------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9-7 | HSPCLKDIV | R/W     | 1h    | High Speed Time Base Clock Pre-Scale Bits<br>These bits determine part of the time-base clock prescale value.<br>$TBCLK = EPWMCLK / (HSPCLKDIV \times CLKDIV)$ . This divisor emulates the HSPCLK in the TMS320x281x system as used on the Event Manager (EV) peripheral.<br>000: /1<br>001: /2 (default on reset)<br>010: /4<br>011: /6<br>100: /8<br>101: /10<br>110: /12<br>111: /14<br>Reset type: SYSRSn                                                                                                                 |
| 6   | SWFSYNC   | R-0/W1S | 0h    | Software Forced Sync Pulse<br>0: Writing a 0 has no effect and reads always return a 0.<br>1: Writing a 1 forces a one-time synchronization pulse to be generated.<br>SWFSYNC can be enabled to affect EPWMxSYNCO by setting the EPWMSYNCOOUTEN.SWEN bit.<br>Reset type: SYSRSn                                                                                                                                                                                                                                               |
| 5-4 | RESERVED  | R       | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3   | PRDLD     | R/W     | 0h    | Active Period Reg Load from Shadow Select<br>0: The period register (TBPRD) is loaded from its shadow register when the time-base counter, TBCTR, is equal to zero and/or a sync event as determined by the TBCTL2[PRDLDSYNC] bit.<br>A write/read to the TBPRD register accesses the shadow register.<br>1: Immediate Mode (Shadow register bypassed): A write or read to the TBPRD register accesses the active register.<br>Reset type: SYSRSn                                                                             |
| 2   | PHSEN     | R/W     | 0h    | Counter Reg Load from Phase Reg Enable<br>0: Do not load the time-base counter (TBCTR) from the time-base phase register (TBPHS).<br>1: Allow Counter to be loaded from the Phase register (TBPHS) and shadow to active load events when an EPWMxSYNCl input signal occurs or a software-forced sync signal, see bit 6.<br>Reset type: SYSRSn                                                                                                                                                                                 |
| 1-0 | CTRMODE   | R/W     | 3h    | Counter Mode<br>The time-base counter mode is normally configured once and not changed during normal operation. If you change the mode of the counter, the change will take effect at the next TBCLK edge and the current counter value shall increment or decrement from the value before the mode change. These bits set the time-base counter mode of operation as follows:<br>00: Up-count mode<br>01: Down-count mode<br>10: Up-down count mode<br>11: Freeze counter operation (default on reset)<br>Reset type: SYSRSn |

### 26.17.2.2 TBCTL2 Register (Offset = 1h) [Reset = 0000h]

TBCTL2 is shown in [Figure 26-94](#) and described in [Table 26-25](#).

Return to the [Summary Table](#).

Time Base Control Register 2

**Figure 26-94. TBCTL2 Register**

|            |              |          |          |    |          |   |   |
|------------|--------------|----------|----------|----|----------|---|---|
| 15         | 14           | 13       | 12       | 11 | 10       | 9 | 8 |
| PRDLDSYNC  |              | RESERVED |          |    | RESERVED |   |   |
| R/W-0h     |              | R-0h     |          |    | R-0-0h   |   |   |
| 7          | 6            | 5        | 4        | 3  | 2        | 1 | 0 |
| OSHTSYNC   | OSHTSYNCMODE | RESERVED | RESERVED |    |          |   |   |
| R-0/W1S-0h | R/W-0h       | R/W-0h   | R-0-0h   |    |          |   |   |

**Table 26-25. TBCTL2 Register Field Descriptions**

| Bit   | Field        | Type    | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|--------------|---------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-14 | PRDLDSYNC    | R/W     | 0h    | Shadow to Active Period Register Load on SYNC event<br>00: Shadow to Active Load of TBPRD occurs only when TBCTR = 0 (same as legacy).<br>01: Shadow to Active Load of TBPRD occurs both when TBCTR = 0 and when SYNC occurs.<br>10: Shadow to Active Load of TBPRD occurs only when a SYNC is received.<br>11: Reserved<br>Note: This bit selection is valid only if TBCTL[PRDLD]=0.<br>Reset type: SYSRSn |
| 13-12 | RESERVED     | R       | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                    |
| 11-8  | RESERVED     | R-0     | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                    |
| 7     | OSHTSYNC     | R-0/W1S | 0h    | Oneshot sync bit<br>0: Writing a '0' has no effect.<br>1: Allow one sync pulse to propagate.<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                          |
| 6     | OSHTSYNCMODE | R/W     | 0h    | Oneshot sync enable bit<br>0: Oneshot sync mode disabled<br>1: Oneshot sync mode enabled<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                              |
| 5     | RESERVED     | R/W     | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                    |
| 4-0   | RESERVED     | R-0     | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                    |



### 26.17.2.3 EPWMSYNCINSEL Register (Offset = 3h) [Reset = 0001h]

EPWMSYNCINSEL is shown in [Figure 26-95](#) and described in [Table 26-26](#).

Return to the [Summary Table](#).

EPWMxSYNCIN Source Select Register

**Figure 26-95. EPWMSYNCINSEL Register**

|          |    |    |    |        |    |   |   |
|----------|----|----|----|--------|----|---|---|
| 15       | 14 | 13 | 12 | 11     | 10 | 9 | 8 |
| RESERVED |    |    |    |        |    |   |   |
| R-0h     |    |    |    |        |    |   |   |
| 7        | 6  | 5  | 4  | 3      | 2  | 1 | 0 |
| RESERVED |    |    |    | SEL    |    |   |   |
| R-0h     |    |    |    | R/W-1h |    |   |   |

**Table 26-26. EPWMSYNCINSEL Register Field Descriptions**

| Bit  | Field    | Type | Reset | Description                                                                                                                                                 |
|------|----------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-5 | RESERVED | R    | 0h    | Reserved                                                                                                                                                    |
| 4-0  | SEL      | R/W  | 1h    | These bits determine the source of the EPWMxSYNCl signal.<br>0x00 Disabled<br>Other Values defined in the 'ePWM SYNC Selection' table<br>Reset type: SYSRSn |

#### 26.17.2.4 TBCTR Register (Offset = 4h) [Reset = 0000h]

TBCTR is shown in [Figure 26-96](#) and described in [Table 26-27](#).

Return to the [Summary Table](#).

Time Base Counter Register

**Figure 26-96. TBCTR Register**

|        |    |    |    |    |    |   |   |
|--------|----|----|----|----|----|---|---|
| 15     | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| TBCTR  |    |    |    |    |    |   |   |
| R/W-0h |    |    |    |    |    |   |   |
| 7      | 6  | 5  | 4  | 3  | 2  | 1 | 0 |
| TBCTR  |    |    |    |    |    |   |   |
| R/W-0h |    |    |    |    |    |   |   |

**Table 26-27. TBCTR Register Field Descriptions**

| Bit  | Field | Type | Reset | Description                                      |
|------|-------|------|-------|--------------------------------------------------|
| 15-0 | TBCTR | R/W  | 0h    | Time Base Counter Register<br>Reset type: SYSRSn |

### 26.17.2.5 TBSTS Register (Offset = 5h) [Reset = 0001h]

TBSTS is shown in [Figure 26-97](#) and described in [Table 26-28](#).

Return to the [Summary Table](#).

Time Base Status Register

**Figure 26-97. TBSTS Register**

|          |    |    |    |    |          |          |       |
|----------|----|----|----|----|----------|----------|-------|
| 15       | 14 | 13 | 12 | 11 | 10       | 9        | 8     |
| RESERVED |    |    |    |    |          |          |       |
| R-0-0h   |    |    |    |    |          |          |       |
| 7        | 6  | 5  | 4  | 3  | 2        | 1        | 0     |
| RESERVED |    |    |    |    | CTRMAX   | SYNCI    | CTDIR |
| R-0-0h   |    |    |    |    | R/W1C-0h | R/W1C-0h | R-1h  |

**Table 26-28. TBSTS Register Field Descriptions**

| Bit  | Field    | Type  | Reset | Description                                                                                                                                                                                                                                                                                                                                 |
|------|----------|-------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-3 | RESERVED | R-0   | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                    |
| 2    | CTRMAX   | R/W1C | 0h    | Time-Base Counter Max Latched Status Bit<br>0: Reading a 0 indicates the time-base counter never reached its maximum value. Writing a 0 will have no effect.<br>1: Reading a 1 on this bit indicates that the time-base counter reached the max value 0xFFFF. Writing a 1 to this bit will clear the latched event.<br>Reset type: SYSRSn   |
| 1    | SYNCI    | R/W1C | 0h    | Input Synchronization Latched Status Bit<br>0: Writing a 0 will have no effect. Reading a 0 indicates no external synchronization event has occurred.<br>1: Reading a 1 on this bit indicates that an external synchronization event has occurred (EPWMxSYNCI). Writing a 1 to this bit will clear the latched event.<br>Reset type: SYSRSn |
| 0    | CTDIR    | R     | 1h    | Time Base Counter Direction Status Bit<br>0: Time-Base Counter is currently counting down.<br>1: Time-Base Counter is currently counting up.<br>Note: This bit is only valid when the counter is not frozen.<br>Reset type: SYSRSn                                                                                                          |

### 26.17.2.6 EPWMSYNCOUEN Register (Offset = 6h) [Reset = 0001h]

EPWMSYNCOUEN is shown in [Figure 26-98](#) and described in [Table 26-29](#).

Return to the [Summary Table](#).

EPWMxSYNCOU Source Enable Register

**Figure 26-98. EPWMSYNCOUEN Register**

|          |           |           |        |        |        |        |        |
|----------|-----------|-----------|--------|--------|--------|--------|--------|
| 15       | 14        | 13        | 12     | 11     | 10     | 9      | 8      |
| RESERVED |           |           |        |        |        |        |        |
| R-0h     |           |           |        |        |        |        |        |
| 7        | 6         | 5         | 4      | 3      | 2      | 1      | 0      |
| RESERVED | DCBEVT1EN | DCAEVT1EN | CMPDEN | CMPCEN | CMPBEN | ZEROEN | SWEN   |
| R-0h     | R/W-0h    | R/W-0h    | R/W-0h | R/W-0h | R/W-0h | R/W-0h | R/W-1h |

**Table 26-29. EPWMSYNCOUEN Register Field Descriptions**

| Bit  | Field     | Type | Reset | Description                                                                                                                                                                                                                                      |
|------|-----------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-8 | RESERVED  | R    | 0h    | Reserved                                                                                                                                                                                                                                         |
| 7    | RESERVED  | R    | 0h    | Reserved                                                                                                                                                                                                                                         |
| 6    | DCBEVT1EN | R/W  | 0h    | This bit enables the DCBEVT1.sync event to set the EPWMxSYNCO signal.<br>0 Disabled<br>1 The EPWMxSYNCO signal is pulsed for one PWM clock period upon a DCBEVT1.sync event<br>Reset type: SYSRSn                                                |
| 5    | DCAEVT1EN | R/W  | 0h    | This bit enables the DCAEVT1.sync event to set the EPWMxSYNCOU signal.<br>0 Disabled<br>1 The EPWMxSYNCOU signal is pulsed for one PWM clock period upon a DCAEVT1.sync event<br>Reset type: SYSRSn                                              |
| 4    | CMPDEN    | R/W  | 0h    | This bit enables the TBCTR = CMPD event to set the EPWMxSYNCO signal.<br>0 Disabled<br>1 The EPWMxSYNCO signal is pulsed for one PWM clock period upon a time-base counter equal to counter compare D event (TBCTR = CMPD)<br>Reset type: SYSRSn |
| 3    | CMPCEN    | R/W  | 0h    | This bit enables the TBCTR = CMPC event to set the EPWMxSYNCO signal.<br>0 Disabled<br>1 The EPWMxSYNCO signal is pulsed for one PWM clock period upon a time-base counter equal to counter compare C event (TBCTR = CMPC)<br>Reset type: SYSRSn |
| 2    | CMPBEN    | R/W  | 0h    | This bit enables the TBCTR = CMPB event to set the EPWMxSYNCO signal.<br>0 Disabled<br>1 The EPWMxSYNCO signal is pulsed for one PWM clock period upon a time-base counter equal to counter compare B event (TBCTR = CMPB)<br>Reset type: SYSRSn |
| 1    | ZEROEN    | R/W  | 0h    | This bit enables the TBCTR = 0x0000 event to set the EPWMxSYNCOU signal.<br>0 Disabled<br>1 The EPWMxSYNCOU signal is pulsed for one PWM clock period upon the value of TBCTR changing to 0x0000<br>Reset type: SYSRSn                           |

**Table 26-29. EPWMSYNCOUEN Register Field Descriptions (continued)**

| Bit | Field | Type | Reset | Description                                                                                                                                                                                                                 |
|-----|-------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0   | SWEN  | R/W  | 1h    | <p>This bit enables the TBCTL.SWFSYNC bit to set the EPWMxSYNCO signal.</p> <p>0 Disabled</p> <p>1 The EPWMxSYNCO signal is pulsed for one PWM clock period when the TBCTL.SWFSYNC bit is set</p> <p>Reset type: SYSRSn</p> |

### 26.17.2.7 TBCTL3 Register (Offset = 7h) [Reset = 0000h]

TBCTL3 is shown in [Figure 26-99](#) and described in [Table 26-30](#).

Return to the [Summary Table](#).

Time Base Control Register 3

**Figure 26-99. TBCTL3 Register**

|          |    |    |    |    |    |   |          |
|----------|----|----|----|----|----|---|----------|
| 15       | 14 | 13 | 12 | 11 | 10 | 9 | 8        |
| RESERVED |    |    |    |    |    |   |          |
| R-0h     |    |    |    |    |    |   |          |
| 7        | 6  | 5  | 4  | 3  | 2  | 1 | 0        |
| RESERVED |    |    |    |    |    |   | OSSFRGEN |
| R-0h     |    |    |    |    |    |   | R/W-0h   |

**Table 26-30. TBCTL3 Register Field Descriptions**

| Bit  | Field    | Type | Reset | Description                                                                                                                                                                         |
|------|----------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-1 | RESERVED | R    | 0h    | Reserved                                                                                                                                                                            |
| 0    | OSSFRGEN | R/W  | 0h    | This bit determines which bit sets the EPWMxSYNCOU One Shot Latch.<br>0 TBCTL2[OSHTSYNC] sets the One Shot Latch<br>1 GLDCTL2[OSHTLD] sets the One Shot Latch<br>Reset type: SYSRSn |

### 26.17.2.8 CMPCTL Register (Offset = 8h) [Reset = 0000h]

CMPCTL is shown in [Figure 26-100](#) and described in [Table 26-31](#).

Return to the [Summary Table](#).

Counter Compare Control Register

**Figure 26-100. CMPCTL Register**

| 15       | 14        | 13        | 12        | 11        | 10 | 9         | 8         |
|----------|-----------|-----------|-----------|-----------|----|-----------|-----------|
| RESERVED |           | LOADBSYNC |           | LOADASYNC |    | SHDWBFULL | SHDWAFULL |
| R-0-0h   |           | R/W-0h    |           | R/W-0h    |    | R-0h      | R-0h      |
| 7        | 6         | 5         | 4         | 3         | 2  | 1         | 0         |
| RESERVED | SHDWBMODE | RESERVED  | SHDWAMODE | LOADBMODE |    | LOADAMODE |           |
| R-0-0h   | R/W-0h    | R-0-0h    | R/W-0h    | R/W-0h    |    | R/W-0h    |           |

**Table 26-31. CMPCTL Register Field Descriptions**

| Bit   | Field     | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|-----------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-14 | RESERVED  | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 13-12 | LOADBSYNC | R/W  | 0h    | Shadow to Active CMPB Register Load on SYNC event<br>00: Shadow to Active Load of CMPB:CMPBHR occurs according to LOADBMODE (bits 1,0) (same as legacy)<br>01: Shadow to Active Load of CMPB:CMPBHR occurs both according to LOADBMODE bits and when SYNC occurs<br>10: Shadow to Active Load of CMPB:CMPBHR occurs only when a SYNC is received<br>11: Reserved<br>Note: This bit is valid only if CMPCTL[SHDWBMODE] = 0.<br>Reset type: SYSRSn         |
| 11-10 | LOADASYNC | R/W  | 0h    | Shadow to Active CMPA Register Load on SYNC event<br>00: Shadow to Active Load of CMPA:CMPAHR occurs according to LOADAMODE (bits 1,0) (same as legacy)<br>01: Shadow to Active Load of CMPA:CMPAHR occurs both according to LOADAMODE bits and when SYNC occurs<br>10: Shadow to Active Load of CMPA:CMPAHR occurs only when a SYNC is received<br>11: Reserved<br>Note: This bit is valid only if CMPCTL[SHDWAMODE] = 0.<br>Reset type: SYSRSn         |
| 9     | SHDWBFULL | R    | 0h    | Counter-compare B (CMPB) Shadow Register Full Status Flag<br>This bit self clears once a loadstrobe occurs.<br>0: CMPB shadow register not full yet<br>1: Indicates the CMPB shadow register is full<br>a CPU write will overwrite current shadow value<br>Reset type: SYSRSn                                                                                                                                                                            |
| 8     | SHDWAFULL | R    | 0h    | Counter-compare A (CMPA) Shadow Register Full Status Flag<br>The flag bit is set when a 32-bit write to CMPA:CMPAHR register or a 16-bit write to CMPA register is made. A 16-bit write to CMPAHR register will not affect the flag. This bit self clears once a load-strobe occurs.<br>0: CMPA shadow register not full yet<br>1: Indicates the CMPA shadow register is full, a CPU write will overwrite the current shadow value<br>Reset type: SYSRSn |
| 7     | RESERVED  | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

**Table 26-31. CMPCTL Register Field Descriptions (continued)**

| Bit | Field     | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|-----------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6   | SHDWBMODE | R/W  | 0h    | Counter-compare B (CMPB) Register Operating Mode<br>0: Shadow mode. Operates as a double buffer. All writes via the CPU access the shadow register<br>1: Immediate mode. Only the active compare B register is used. All writes and reads directly access the active register for immediate compare action<br>Reset type: SYSRSn                                                                |
| 5   | RESERVED  | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                        |
| 4   | SHDWAMODE | R/W  | 0h    | Counter-compare A (CMPA) Register Operating Mode<br>0: Shadow mode. Operates as a double buffer. All writes via the CPU access the shadow register<br>1: Immediate mode. Only the active compare register is used. All writes and reads directly access the active register for immediate compare action<br>Reset type: SYSRSn                                                                  |
| 3-2 | LOADBMODE | R/W  | 0h    | Active Counter-Compare B (CMPB) Load From Shadow Select Mode<br>This bit has no effect in immediate mode (CMPCTL[SHDWBMODE] = 1).<br>00: Load on CTR = Zero: Time-base counter equal to zero (TBCTR = 0x0000)<br>01: Load on CTR = PRD: Time-base counter equal to period (TBCTR = TBPRD)<br>10: Load on either CTR = Zero or CTR = PRD<br>11: Freeze (no loads possible)<br>Reset type: SYSRSn |
| 1-0 | LOADAMODE | R/W  | 0h    | Active Counter-Compare A (CMPA) Load From Shadow Select Mode<br>This bit has no effect in immediate mode (CMPCTL[SHDWAMODE] = 1).<br>00: Load on CTR = Zero: Time-base counter equal to zero (TBCTR = 0x0000)<br>01: Load on CTR = PRD: Time-base counter equal to period (TBCTR = TBPRD)<br>10: Load on either CTR = Zero or CTR = PRD<br>11: Freeze (no loads possible)<br>Reset type: SYSRSn |



### 26.17.2.9 CMPCTL2 Register (Offset = 9h) [Reset = 0000h]

CMPCTL2 is shown in [Figure 26-101](#) and described in [Table 26-32](#).

Return to the [Summary Table](#).

Counter Compare Control Register 2

**Figure 26-101. CMPCTL2 Register**

|          |           |           |           |           |    |           |   |
|----------|-----------|-----------|-----------|-----------|----|-----------|---|
| 15       | 14        | 13        | 12        | 11        | 10 | 9         | 8 |
| RESERVED |           | LOADDSYNC |           | LOADCSYNC |    | RESERVED  |   |
| R-0-0h   |           | R/W-0h    |           | R/W-0h    |    | R-0-0h    |   |
| 7        | 6         | 5         | 4         | 3         | 2  | 1         | 0 |
| RESERVED | SHDWDMODE | RESERVED  | SHDWCMODE | LOADDMODE |    | LOADCMODE |   |
| R-0-0h   | R/W-0h    | R-0-0h    | R/W-0h    | R/W-0h    |    | R/W-0h    |   |

**Table 26-32. CMPCTL2 Register Field Descriptions**

| Bit   | Field     | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|-----------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-14 | RESERVED  | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                         |
| 13-12 | LOADDSYNC | R/W  | 0h    | Shadow to Active CMPD Register Load on SYNC event<br>00: Shadow to Active Load of CMPD occurs according to LOADDMODE<br>01: Shadow to Active Load of CMPD occurs both according to LOADDMODE bits and when SYNC occurs<br>10: Shadow to Active Load of CMPD occurs only when a SYNC is received<br>11: Reserved<br>Note: This bit is valid only if CMPCTL2[SHDWDMODE] = 0.<br>Reset type: SYSRSn |
| 11-10 | LOADCSYNC | R/W  | 0h    | Shadow to Active CMPC Register Load on SYNC event<br>00: Shadow to Active Load of CMPC occurs according to LOADCMODE<br>01: Shadow to Active Load of CMPC occurs both according to LOADCMODE bits and when SYNC occurs<br>10: Shadow to Active Load of CMPC occurs only when a SYNC is received<br>11: Reserved<br>Note: This bit is valid only if CMPCTL2[SHDWCMODE] = 0.<br>Reset type: SYSRSn |
| 9-7   | RESERVED  | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                         |
| 6     | SHDWDMODE | R/W  | 0h    | Counter-Compare D Register Operating Mode<br>0: Shadow mode - operates as a double buffer. All writes via the CPU access Shadow register.<br>1: Immediate mode - only the Active compare register is used. All writes/reads via the CPU directly access the Active register for immediate Compare action.<br>Reset type: SYSRSn                                                                  |
| 5     | RESERVED  | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                         |
| 4     | SHDWCMODE | R/W  | 0h    | Counter-Compare C Register Operating Mode<br>0: Shadow mode - operates as a double buffer. All writes via the CPU access Shadow register.<br>1: Immediate mode - only the Active compare register is used. All writes/reads via the CPU directly access the Active register for immediate Compare action.<br>Reset type: SYSRSn                                                                  |

**Table 26-32. CMPCTL2 Register Field Descriptions (continued)**

| Bit | Field     | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                          |
|-----|-----------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3-2 | LOADDMODE | R/W  | 0h    | Active Counter-Compare D (CMPD) Load from Shadow Select Mode<br>00: Load on CTR = Zero: Time-base counter equal to zero (TBCTR = 0x0000)<br>01: Load on CTR = PRD: Time-base counter equal to period (TBCTR = TBPRD)<br>10: Load on either CTR = Zero or CTR = PRD<br>11: Freeze (no loads possible)<br>Note: Has no effect in Immediate mode.<br>Reset type: SYSRSn |
| 1-0 | LOADCMODE | R/W  | 0h    | Active Counter-Compare C (CMPC) Load from Shadow Select Mode<br>00: Load on CTR = Zero: Time-base counter equal to zero (TBCTR = 0x0000)<br>01: Load on CTR = PRD: Time-base counter equal to period (TBCTR = TBPRD)<br>10: Load on either CTR = Zero or CTR = PRD<br>11: Freeze (no loads possible)<br>Note: Has no effect in Immediate mode.<br>Reset type: SYSRSn |

### 26.17.2.10 DBCTL Register (Offset = Ch) [Reset = 0000h]

DBCTL is shown in [Figure 26-102](#) and described in [Table 26-33](#).

Return to the [Summary Table](#).

Dead-Band Generator Control Register

**Figure 26-102. DBCTL Register**

| 15          | 14        | 13      | 12                | 11                | 10          | 9      | 8 |
|-------------|-----------|---------|-------------------|-------------------|-------------|--------|---|
| HALFCYCLE   | DEDB_MODE | OUTSWAP | SHDWDBFED<br>MODE | SHDWDBRED<br>MODE | LOADFEDMODE |        |   |
| R/W-0h      | R/W-0h    | R/W-0h  | R/W-0h            | R/W-0h            | R/W-0h      | R/W-0h |   |
| 7           | 6         | 5       | 4                 | 3                 | 2           | 1      | 0 |
| LOADREDMODE | IN_MODE   |         | POLSEL            |                   | OUT_MODE    |        |   |
| R/W-0h      | R/W-0h    |         | R/W-0h            |                   | R/W-0h      |        |   |

**Table 26-33. DBCTL Register Field Descriptions**

| Bit   | Field         | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|---------------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15    | HALFCYCLE     | R/W  | 0h    | Half Cycle Clocking Enable Bit<br>0: Full cycle clocking enabled. The dead-band counters are clocked at the TBCLK rate.<br>1: Half cycle clocking enabled. The dead-band counters are clocked at TBCLK*2.<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 14    | DEDB_MODE     | R/W  | 0h    | Dead Band Dual-Edge B Mode Control (S8 switch)<br>0: Rising edge delay applied to InA/InB as selected by S4 switch (IN-MODE bits) on A signal path only. Falling edge delay applied to InA/InB as selected by S5 switch (INMODE bits) on B signal path only.<br>1: Rising edge delay and falling edge delay applied to source selected by S4 switch (INMODE bits) and output to B signal path only. Note: When this bit is set to 1, user should always either set OUT_MODE bits such that Apath = InA OR OUTSWAP bits such that OutA=Bpath otherwise, OutA will be invalid.<br>Reset type: SYSRSn                                                                                                      |
| 13-12 | OUTSWAP       | R/W  | 0h    | Dead Band Output Swap Control<br>Bit 13 controls the S6 switch and bit 12 controls the S7 switch.<br>00: OutA and OutB signals are as defined by OUT-MODE bits.<br>01: OutA = A-path as defined by OUT-MODE bits.<br>OutB = A-path as defined by OUT-MODE bits (rising edge delay or delay-bypassed A signal path).<br>10: OutA = B-path as defined by OUT-MODE bits (falling edge delay or delay-bypassed B signal path).<br>OutB = B-path as defined by OUT-MODE bits.<br>11: OutA = B-path as defined by OUT-MODE bits (falling edge delay or delay-bypassed B signal path).<br>OutB = A-path as defined by OUT-MODE bits (rising edge delay or delay-bypassed A signal path).<br>Reset type: SYSRSn |
| 11    | SHDWDBFEDMODE | R/W  | 0h    | FED Dead-Band Load Mode<br>0: Immediate mode. Only the active DBFED register is used. All writes/reads via the CPU directly access the active register for immediate 'FED dead-band action.'<br>1: Shadow mode. Operates as a double buffer. All writes via the CPU access Shadow register. Default at Reset is Immediate mode (for compatibility with legacy).<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                   |

**Table 26-33. DBCTL Register Field Descriptions (continued)**

| Bit | Field         | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|---------------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10  | SHDWDBREDMODE | R/W  | 0h    | RED Dead-Band Load Mode<br>0: Immediate mode. Only the active DBRED register is used. All writes/reads via the CPU directly access the active register for immediate 'RED dead-band action.'<br>1: Shadow mode. Operates as a double buffer. All writes via the CPU access Shadow register. Default at Reset is Immediate mode (for compatibility with legacy).<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 9-8 | LOADFEDMODE   | R/W  | 0h    | Active DBFED Load from Shadow Select Mode<br>00: Load on Counter = 0 (CNT_eq)<br>01: Load on Counter = Period (PRD_eq)<br>10: Load on either Counter = 0, or Counter = Period<br>11: Freeze (no loads possible)<br>Note: has no effect in Immediate mode.<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 7-6 | LOADREDMODE   | R/W  | 0h    | Active DBRED Load from Shadow Select Mode<br>00: Load on Counter = 0 (CNT_eq)<br>01: Load on Counter = Period (PRD_eq)<br>10: Load on either Counter = 0, or Counter = Period<br>11: Freeze (no loads possible)<br>Note: has no effect in Immediate mode.<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5-4 | IN_MODE       | R/W  | 0h    | Dead-Band Input Mode Control<br>Bit 5 controls the S5 switch and bit 4 controls the S4 switch shown. This allows you to select the input source to the falling-edge and rising-edge delay. To produce classical dead-band waveforms the default is EPWMxA In is the source for both falling and rising-edge delays.<br>00: EPWMxA In (from the action-qualifier) is the source for both falling-edge and rising-edge delay.<br>01: EPWMxB In (from the action-qualifier) is the source for rising-edge delayed signal.<br>EPWMxA In (from the action-qualifier) is the source for falling-edge delayed signal.<br>10: EPWMxA In (from the action-qualifier) is the source for rising-edge delayed signal.<br>EPWMxB In (from the action-qualifier) is the source for falling-edge delayed signal.<br>11: EPWMxB In (from the action-qualifier) is the source for both rising-edge delay and falling-edge delayed signal.<br>Reset type: SYSRSn |
| 3-2 | POLSEL        | R/W  | 0h    | Polarity Select Control<br>Bit 3 controls the S3 switch and bit 2 controls the S2 switch. This allows you to selectively invert one of the delayed signals before it is sent out of the dead-band submodule. The following descriptions correspond to classical upper/lower switch control as found in one leg of a digital motor control inverter. These assume that DBCTL[OUT_MODE] = 1,1 and DBCTL[IN_MODE] = 0x0. Other enhanced modes are also possible, but not regarded as typical usage modes.<br>00: Active high (AH) mode. Neither EPWMxA nor EPWMxB is inverted (default).<br>01: Active low complementary (ALC) mode. EPWMxA is inverted.<br>10: Active high complementary (AHC). EPWMxB is inverted.<br>11: Active low (AL) mode. Both EPWMxA and EPWMxB are inverted.<br>Reset type: SYSRSn                                                                                                                                      |

**Table 26-33. DBCTL Register Field Descriptions (continued)**

| Bit | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----|----------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-0 | OUT_MODE | R/W  | 0h    | <p>Dead-Band Output Mode Control</p> <p>Bit 1 controls the S1 switch and bit 0 controls the S0 switch.</p> <p>00: DBM is fully disabled or by-passed. In this mode the POLSEL and IN-MODE bits have no effect.</p> <p>01: Apath = InA (delay is by-passed for A signal path)</p> <p>Bpath = FED (Falling Edge Delay in B signal path)</p> <p>10: Apath = RED (Rising Edge Delay in A signal path)</p> <p>Bpath = InB (delay is by-passed for B signal path)</p> <p>11: DBM is fully enabled (i.e. both RED and FED active)</p> <p>Reset type: SYSRSn</p> |

### 26.17.2.11 DBCTL2 Register (Offset = Dh) [Reset = 0000h]

DBCTL2 is shown in [Figure 26-103](#) and described in [Table 26-34](#).

Return to the [Summary Table](#).

Dead-Band Generator Control Register 2

**Figure 26-103. DBCTL2 Register**

|          |    |    |    |    |               |               |   |
|----------|----|----|----|----|---------------|---------------|---|
| 15       | 14 | 13 | 12 | 11 | 10            | 9             | 8 |
| RESERVED |    |    |    |    |               |               |   |
| R-0-0h   |    |    |    |    |               |               |   |
| 7        | 6  | 5  | 4  | 3  | 2             | 1             | 0 |
| RESERVED |    |    |    |    | SHDWDBCTLMODE | LOADDBCTLMODE |   |
| R-0-0h   |    |    |    |    | R/W-0h        | R/W-0h        |   |

**Table 26-34. DBCTL2 Register Field Descriptions**

| Bit  | Field         | Type | Reset | Description                                                                                                                                                                                                                                                                                                        |
|------|---------------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-3 | RESERVED      | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                           |
| 2    | SHDWDBCTLMODE | R/W  | 0h    | DBCTL Load Mode<br>0: Immediate mode - only the Active DBCTL register is used. All writes/reads via the CPU directly access the Active register.<br>1: Shadow mode - All writes and reads to bits [5:0] of the DBCTL register are shadowed. All other bits still access the active register.<br>Reset type: SYSRSn |
| 1-0  | LOADDBCTLMODE | R/W  | 0h    | Active DBCTL Load from Shadow Select Mode<br>00: Load on Counter = 0 (CNT_eq)<br>01: Load on Counter = Period (PRD_eq)<br>10: Load on either Counter = 0, or Counter = Period<br>11: Freeze (no loads possible)<br>Note: has no effect in Immediate mode<br>Reset type: SYSRSn                                     |

### 26.17.2.12 AQCTL Register (Offset = 10h) [Reset = 0000h]

AQCTL is shown in [Figure 26-104](#) and described in [Table 26-35](#).

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Action Qualifier Control Register

**Figure 26-104. AQCTL Register**

|          |             |          |             |           |    |           |   |
|----------|-------------|----------|-------------|-----------|----|-----------|---|
| 15       | 14          | 13       | 12          | 11        | 10 | 9         | 8 |
| RESERVED |             |          |             | LDAQBSYNC |    | LDAQASYNC |   |
| R-0-0h   |             |          |             | R/W-0h    |    | R/W-0h    |   |
| 7        | 6           | 5        | 4           | 3         | 2  | 1         | 0 |
| RESERVED | SHDWAQBMODE | RESERVED | SHDWAQAMODE | LDAQBMODE |    | LDAQAMODE |   |
| R-0-0h   | R/W-0h      | R-0-0h   | R/W-0h      | R/W-0h    |    | R/W-0h    |   |

**Table 26-35. AQCTL Register Field Descriptions**

| Bit   | Field       | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|-------------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-12 | RESERVED    | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                   |
| 11-10 | LDAQBSYNC   | R/W  | 0h    | Shadow to Active AQCTLB Register Load on SYNC event<br>00: Shadow to Active Load of AQCTLB occurs according to LDAQBMODE<br>01: Shadow to Active Load of AQCTLB occurs both according to LDAQBMODE bits and when SYNC occurs.<br>10: Shadow to Active Load of AQCTLB occurs only when a SYNC is received.<br>11: Reserved<br>Note: This bit is valid only if AQCTL[SHDWAQBMODE] = 1.<br>Reset type: SYSRSn |
| 9-8   | LDAQASYNC   | R/W  | 0h    | Shadow to Active AQCTLA Register Load on SYNC event<br>00: Shadow to Active Load of AQCTLA occurs according to LDAQAMODE<br>01: Shadow to Active Load of AQCTLA occurs both according to LDAQAMODE bits and when SYNC occurs.<br>10: Shadow to Active Load of AQCTLA occurs only when a SYNC is received.<br>11: Reserved<br>Note: This bit is valid only if AQCTL[SHDWAQAMODE] = 1.<br>Reset type: SYSRSn |
| 7     | RESERVED    | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                   |
| 6     | SHDWAQBMODE | R/W  | 0h    | Action Qualifier B Register operating mode<br>1: Shadow mode - operates as a double buffer. All writes via the CPU access Shadow register.<br>0: Immediate mode - only the Active action qualifier register is used. All writes/reads via the CPU directly access the Active register.<br>Reset type: SYSRSn                                                                                               |
| 5     | RESERVED    | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4     | SHDWAQAMODE | R/W  | 0h    | Action Qualifier A Register operating mode<br>1: Shadow mode - operates as a double buffer. All writes via the CPU access Shadow register.<br>0: Immediate mode - only the Active action qualifier register is used. All writes/reads via the CPU directly access the Active register.<br>Reset type: SYSRSn                                                                                               |

**Table 26-35. AQCTL Register Field Descriptions (continued)**

| Bit | Field     | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                    |
|-----|-----------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3-2 | LDAQBMODE | R/W  | 0h    | Active Action Qualifier B Load from Shadow Select Mode<br>00: Load on CTR = Zero: Time-base counter equal to zero (TBCTR = 0x0000)<br>01: Load on CTR = PRD: Time-base counter equal to period (TBCTR = TBPRD)<br>10: Load on either CTR = Zero or CTR = PRD<br>11: Freeze (no loads possible)<br>Note: has no effect in Immediate mode.<br>Reset type: SYSRSn |
| 1-0 | LDAQAMODE | R/W  | 0h    | Active Action Qualifier A Load from Shadow Select Mode<br>00: Load on CTR = Zero: Time-base counter equal to zero (TBCTR = 0x0000)<br>01: Load on CTR = PRD: Time-base counter equal to period (TBCTR = TBPRD)<br>10: Load on either CTR = Zero or CTR = PRD<br>11: Freeze (no loads possible)<br>Note: has no effect in Immediate mode.<br>Reset type: SYSRSn |



### 26.17.2.13 AQTSRCSEL Register (Offset = 11h) [Reset = 0000h]

AQTSRCSEL is shown in [Figure 26-105](#) and described in [Table 26-36](#).

Return to the [Summary Table](#).

Action Qualifier Trigger Event Source Select Register

**Figure 26-105. AQTSRCSEL Register**

|          |    |    |    |        |    |   |   |
|----------|----|----|----|--------|----|---|---|
| 15       | 14 | 13 | 12 | 11     | 10 | 9 | 8 |
| RESERVED |    |    |    |        |    |   |   |
| R-0-0h   |    |    |    |        |    |   |   |
| 7        | 6  | 5  | 4  | 3      | 2  | 1 | 0 |
| T2SEL    |    |    |    | T1SEL  |    |   |   |
| R/W-0h   |    |    |    | R/W-0h |    |   |   |

**Table 26-36. AQTSRCSEL Register Field Descriptions**

| Bit  | Field    | Type | Reset | Description                                                                                                                                                                                                             |
|------|----------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-8 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                |
| 7-4  | T2SEL    | R/W  | 0h    | T2 Event Source Select Bits<br>0000: DCAEVT1<br>0001: DCAEVT2<br>0010: DCBEVT1<br>0011: DCBEVT2<br>0100: TZ1<br>0101: TZ2<br>0110: TZ3<br>0111: EPWMxSYNCl<br>1000: DCEVTFILT<br>Others: Reserved<br>Reset type: SYSRSn |
| 3-0  | T1SEL    | R/W  | 0h    | T1 Event Source Select Bits<br>0000: DCAEVT1<br>0001: DCAEVT2<br>0010: DCBEVT1<br>0011: DCBEVT2<br>0100: TZ1<br>0101: TZ2<br>0110: TZ3<br>0111: EPWMxSYNCl<br>1000: DCEVTFILT<br>Others: Reserved<br>Reset type: SYSRSn |

### 26.17.2.14 PCCTL Register (Offset = 14h) [Reset = 0000h]

PCCTL is shown in [Figure 26-106](#) and described in [Table 26-37](#).

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PWM Chopper Control Register

**Figure 26-106. PCCTL Register**

|          |    |    |         |    |    |         |   |
|----------|----|----|---------|----|----|---------|---|
| 15       | 14 | 13 | 12      | 11 | 10 | 9       | 8 |
| RESERVED |    |    |         |    |    | CHPDUTY |   |
| R-0-0h   |    |    |         |    |    | R/W-0h  |   |
| 7        | 6  | 5  | 4       | 3  | 2  | 1       | 0 |
| CHPFREQ  |    |    | OSHTWTH |    |    | CHPEN   |   |
| R/W-0h   |    |    | R/W-0h  |    |    | R/W-0h  |   |

**Table 26-37. PCCTL Register Field Descriptions**

| Bit   | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|----------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-11 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 10-8  | CHPDUTY  | R/W  | 0h    | Chopping Clock Duty Cycle<br>000: Duty = 1/8 (12.5%)<br>001: Duty = 2/8 (25.0%)<br>010: Duty = 3/8 (37.5%)<br>011: Duty = 4/8 (50.0%)<br>100: Duty = 5/8 (62.5%)<br>101: Duty = 6/8 (75.0%)<br>110: Duty = 7/8 (87.5%)<br>111: Reserved<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 7-5   | CHPFREQ  | R/W  | 0h    | Chopping Clock Frequency<br>000: Divide by 1 (no prescale, = 12.5 MHz at 100 MHz TBCLK)<br>001: Divide by 2 (6.25 MHz at 100 MHz TBCLK)<br>010: Divide by 3 (4.16 MHz at 100 MHz TBCLK)<br>011: Divide by 4 (3.12 MHz at 100 MHz TBCLK)<br>100: Divide by 5 (2.50 MHz at 100 MHz TBCLK)<br>101: Divide by 6 (2.08 MHz at 100 MHz TBCLK)<br>110: Divide by 7 (1.78 MHz at 100 MHz TBCLK)<br>111: Divide by 8 (1.56 MHz at 100 MHz TBCLK)<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 4-1   | OSHTWTH  | R/W  | 0h    | One-Shot Pulse Width<br>0000: 1 x EPWMCLK / 8 wide ( = 80 ns at 100 MHz EPWMCLK)<br>0001: 2 x EPWMCLK / 8 wide ( = 160 ns at 100 MHz EPWMCLK)<br>0010: 3 x EPWMCLK / 8 wide ( = 240 ns at 100 MHz EPWMCLK)<br>0011: 4 x EPWMCLK / 8 wide ( = 320 ns at 100 MHz EPWMCLK)<br>0100: 5 x EPWMCLK / 8 wide ( = 400 ns at 100 MHz EPWMCLK)<br>0101: 6 x EPWMCLK / 8 wide ( = 480 ns at 100 MHz EPWMCLK)<br>0110: 7 x EPWMCLK / 8 wide ( = 560 ns at 100 MHz EPWMCLK)<br>0111: 8 x EPWMCLK / 8 wide ( = 640 ns at 100 MHz EPWMCLK)<br>1000: 9 x EPWMCLK / 8 wide ( = 720 ns at 100 MHz EPWMCLK)<br>1001: 10 x EPWMCLK / 8 wide ( = 800 ns at 100 MHz EPWMCLK)<br>1010: 11 x EPWMCLK / 8 wide ( = 880 ns at 100 MHz EPWMCLK)<br>1011: 12 x EPWMCLK / 8 wide ( = 960 ns at 100 MHz EPWMCLK)<br>1100: 13 x EPWMCLK / 8 wide ( = 1040 ns at 100 MHz EPWMCLK)<br>1101: 14 x EPWMCLK / 8 wide ( = 1120 ns at 100 MHz EPWMCLK)<br>1110: 15 x EPWMCLK / 8 wide ( = 1200 ns at 100 MHz EPWMCLK)<br>1111: 16 x EPWMCLK / 8 wide ( = 1280 ns at 100 MHz EPWMCLK)<br>Reset type: SYSRSn |

**Table 26-37. PCCTL Register Field Descriptions (continued)**

| Bit | Field | Type | Reset | Description                                                                                                           |
|-----|-------|------|-------|-----------------------------------------------------------------------------------------------------------------------|
| 0   | CHPEN | R/W  | 0h    | PWM-Chopping Enable<br>0: Disable (bypass) PWM chopping function<br>1: Enable chopping function<br>Reset type: SYSRSn |

### 26.17.2.15 VCAPCTL Register (Offset = 18h) [Reset = 0000h]

VCAPCTL is shown in [Figure 26-107](#) and described in [Table 26-38](#).

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Valley Capture Control Register

**Figure 26-107. VCAPCTL Register**

|           |          |    |         |    |                    |            |        |
|-----------|----------|----|---------|----|--------------------|------------|--------|
| 15        | 14       | 13 | 12      | 11 | 10                 | 9          | 8      |
| RESERVED  |          |    |         |    | EDGEFILTDLY<br>SEL | VDELAYDIV  |        |
| R-0-0h    |          |    |         |    | R/W-0h             | R/W-0h     |        |
| 7         | 6        | 5  | 4       | 3  | 2                  | 1          | 0      |
| VDELAYDIV | RESERVED |    | TRIGSEL |    |                    | VCAPSTART  | VCAPE  |
| R/W-0h    | R-0-0h   |    | R/W-0h  |    |                    | R-0/W1S-0h | R/W-0h |

**Table 26-38. VCAPCTL Register Field Descriptions**

| Bit   | Field          | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|----------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-11 | RESERVED       | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 10    | EDGEFILTDLYSEL | R/W  | 0h    | Valley Switching Mode Delay Selection<br>0: No delay applied to the edge filter output<br>1: HWDELAYVAL delay applied to the edge filter output<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 9-7   | VDELAYDIV      | R/W  | 0h    | Valley Delay Mode Divide Enable<br>000: HWVDELVAL = SWVDELVAL<br>001: HWVDELVAL = VCNTVAL+SWVDELVAL<br>010: HWVDELVAL = VCNTVAL>>1+SWVDELVAL<br>011: HWVDELVAL = VCNTVAL>>2+SWVDELVAL<br>100: HWVDELVAL = VCNTVAL>>4+SWVDELVAL<br>Note: Delay value between the consecutive edge captures can optionally be divided by using these bits.<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 6-5   | RESERVED       | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4-2   | TRIGSEL        | R/W  | 0h    | Status of Numbered of Captured Events<br>000: Capture sequence is triggered by software via writes to VCAPCTL[VCAPSTART].<br>001: Capture sequence is triggered by CNT_zero event.<br>010: Capture sequence is triggered by PRD_eq event.<br>011: Capture sequence is triggered by CNT_zero or PRD_eq event.<br>100: Capture sequence is triggered by DCAEVT1 event.<br>101: Capture sequence is triggered by DCAEVT2 event.<br>110: Capture sequence is triggered by DCBEVT1 event.<br>111: Capture sequence is triggered by DCBEVT2 event.<br>Note: Valley capture sequence triggered by the selected event in this register field. Once the chosen event occurs the capture sequence is armed. Event captures occur based of the event chosen in DCFCTL[SRCSEL] register.<br>Note: Same event may not be chosen in both DCFCTL[SRCSEL] and VCAPCTL[TRIGSEL] registers.<br>Note: Once the chosen event in VCAPCTL[TRIGSEL] occurs, irrespective of the current capture status, capture sequence is retriggered.<br>Reset type: SYSRSn |

**Table 26-38. VCAPCTL Register Field Descriptions (continued)**

| Bit | Field     | Type    | Reset | Description                                                                                                                                                                                                                                                                                                                                                               |
|-----|-----------|---------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | VCAPSTART | R-0/W1S | 0h    | Valley Capture Start<br>0: Writing a 0 has no effect<br>1: Trigger the capture sequence once if VCAPCTL[TRIGSEL]=0x0<br>Note: This bit is used to start valley capture sequence through software. VCAPCTL[TRIGSEL] has to be chosen for software trigger for this bit to have any effect. Writing of 1 will result in one capture sequence trigger.<br>Reset type: SYSRSn |
| 0   | VCAPE     | R/W     | 0h    | Valley Capture Enable/Disable<br>0: Disabled<br>1: Enabled<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                          |

### 26.17.2.16 VCNTCFG Register (Offset = 19h) [Reset = 0000h]

VCNTCFG is shown in [Figure 26-108](#) and described in [Table 26-39](#).

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Valley Counter Config Register

**Figure 26-108. VCNTCFG Register**

|              |          |    |    |    |           |   |   |
|--------------|----------|----|----|----|-----------|---|---|
| 15           | 14       | 13 | 12 | 11 | 10        | 9 | 8 |
| STOPEDGESTS  | RESERVED |    |    |    | STOPEDGE  |   |   |
| R-0h         | R-0-0h   |    |    |    | R/W-0h    |   |   |
| 7            | 6        | 5  | 4  | 3  | 2         | 1 | 0 |
| STARTEDGESTS | RESERVED |    |    |    | STARTEDGE |   |   |
| R-0h         | R-0-0h   |    |    |    | R/W-0h    |   |   |

**Table 26-39. VCNTCFG Register Field Descriptions**

| Bit   | Field        | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------|--------------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15    | STOPEDGESTS  | R    | 0h    | Stop Edge Status Bit<br>0: Stop edge has not occurred<br>1: Stop edge occurred<br>Note: This bit is set only after the trigger sequence is armed (upon occurrence of trigger pulse selected through VCAPCTL[TRIGSEL]) and STOPEDGE occurs.<br>Note: This bit is reset by the occurrence of the trigger pulse selected through VCAPCTL[TRIGSEL]<br>Reset type: SYSRSn                                                      |
| 14-12 | RESERVED     | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 11-8  | STOPEDGE     | R/W  | 0h    | Counter Stop Edge Selection<br>Once the counter operation is armed, upon occurrence of trigger pulse selected through VCAPCTL[TRIGSEL] pulse - valley counter would stop counting upon the occurrence of chosen number of events thorough this bit field. Stop counting on occurrence of:<br>0000: Do not stop<br>0001<br>1st edge<br>0010: 2nd edge<br>0011: 3rd edge<br>...<br>1,1,1,1: 15th edge<br>Reset type: SYSRSn |
| 7     | STARTEDGESTS | R    | 0h    | Start Edge Status Bit<br>0: Start edge has not occurred<br>1: Start edge occurred<br>Note: This bit is set only after the trigger sequence is armed (upon occurrence of trigger pulse selected through VCAPCTL[TRIGSEL]) and STARTEDGE occurs.<br>Note: This bit is reset by the occurrence of the trigger pulse selected through VCAPCTL[TRIGSEL]<br>Reset type: SYSRSn                                                  |
| 6-4   | RESERVED     | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                  |

**Table 26-39. VCNTCFG Register Field Descriptions (continued)**

| Bit | Field     | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|-----------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3-0 | STARTEDGE | R/W  | 0h    | <p>Counter Start Edge Selection</p> <p>Once the counter operation is armed, upon occurrence of trigger pulse selected through VCAPCTL[TRIGSEL] pulse - valley counter would start counting upon the occurrence of chosen number of events thorough this bit field. Start counting on occurrence of</p> <p>0000: Do not start</p> <p>0001: 1st edge</p> <p>0010: 2nd edge</p> <p>0011: 3rd edge</p> <p>...</p> <p>1111: 15th edge</p> <p>Reset type: SYSRSn</p> |

### 26.17.2.17 HRCNFG Register (Offset = 20h) [Reset = 0000h]

HRCNFG is shown in [Figure 26-109](#) and described in [Table 26-40](#).

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HRPWM Configuration Register

This register is only accessible on EPWM modules with HRPWM capabilities.

**Figure 26-109. HRCNFG Register**

| 15       | 14       | 13       | 12      | 11 | 10       | 9        | 8 |
|----------|----------|----------|---------|----|----------|----------|---|
| RESERVED |          | RESERVED | HRLOADB |    | CTLMODEB | EDGMODEB |   |
| R/W-0h   |          | R-0-0h   | R/W-0h  |    | R/W-0h   | R/W-0h   |   |
| 7        | 6        | 5        | 4       | 3  | 2        | 1        | 0 |
| SWAPAB   | AUTOCONV | SELOUTB  | HRLOAD  |    | CTLMODE  | EDGMODE  |   |
| R/W-0h   | R/W-0h   | R/W-0h   | R/W-0h  |    | R/W-0h   | R/W-0h   |   |

**Table 26-40. HRCNFG Register Field Descriptions**

| Bit   | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                        |
|-------|----------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-14 | RESERVED | R/W  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                           |
| 13    | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                           |
| 12-11 | HRLOADB  | R/W  | 0h    | Shadow Mode Bit<br>Selects the time event that loads the CMPBHR shadow value into the active register.<br>00: Load on CTR = Zero: Time-base counter equal to zero (TBCTR = 0x0000)<br>01: Load on CTR = PRD: Time-base counter equal to period (TBCTR = TBPRD)<br>10: Load on either CTR = Zero or CTR = PRD<br>11: Reserved<br>Reset type: SYSRSn |
| 10    | CTLMODEB | R/W  | 0h    | Control Mode Bits<br>Selects the register (CMP/TBPRD or TBPHS) that controls the MEP:<br>0: CMPBHR(8) or TBPRDHR(8) Register controls the edge position (i.e., this is duty or period control mode). (Default on Reset)<br>1: TBPHSHR(8) Register controls the edge position (i.e., this is phase control mode).<br>Reset type: SYSRSn             |
| 9-8   | EDGMODEB | R/W  | 0h    | Edge Mode Bits<br>Selects the edge of the PWM that is controlled by the micro-edge position (MEP) logic:<br>00: HRPWM capability is disabled (default on reset)<br>01: MEP control of rising edge (CMPBHR)<br>10: MEP control of falling edge (CMPBHR)<br>11: MEP control of both edges (TBPHSHR or TBPRDHR)<br>Reset type: SYSRSn                 |
| 7     | SWAPAB   | R/W  | 0h    | Swap ePWM A & B Output Signals<br>This bit enables the swapping of the A & B signal outputs. The selection is as follows:<br>0: ePWMxA and ePWMxB outputs are unchanged.<br>1: ePWMxA signal appears on ePWMxB output and ePWMxB signal appears on ePWMxA output.<br>Reset type: SYSRSn                                                            |



**Table 26-40. HRCNFG Register Field Descriptions (continued)**

| Bit | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|----------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6   | AUTOCONV | R/W  | 0h    | <p>Auto Convert Delay Line Value</p> <p>Selects whether the fractional duty cycle/period/phase in the CMPAHR/TBPRDHR/TBPHSHR register is automatically scaled by the MEP scale factor in the HRMSTEP register or manually scaled by calculations in application software. The SFO library function automatically updates the HRMSTEP register with the appropriate MEP scale factor.</p> <p>0: Automatic HRMSTEP scaling is disabled.</p> <p>1: Automatic HRMSTEP scaling is enabled.</p> <p>If application software is manually scaling the fractional duty cycle, or phase (i.e. software sets <math>CMPAHR = (\text{fraction}(PWMduty * PWMperiod) * MEP \text{ Scale Factor}) \ll 8 + 0x080</math> for duty cycle), then this mode must be disabled.</p> <p>Reset type: SYSRSn</p> |
| 5   | SELOUTB  | R/W  | 0h    | <p>EPWMxB Output Select Bit</p> <p>This bit selects which signal is output on the ePWMxB channel output.</p> <p>The inversion will take the high resolution mode into account and the inverted signal will contain any high resolution modification. The inversion takes place as the last step in modifying the ePWMxB signal.</p> <p>0: ePWMxB output is normal.</p> <p>1: ePWMxB output is inverted version of ePWMxA signal.</p> <p>Reset type: SYSRSn</p>                                                                                                                                                                                                                                                                                                                         |
| 4-3 | HRLOAD   | R/W  | 0h    | <p>Shadow Mode Bit</p> <p>Selects the time event that loads the CMPAHR shadow value into the active register.</p> <p>00: Load on CTR = Zero: Time-base counter equal to zero (TBCTR = 0x0000)</p> <p>01: Load on CTR = PRD: Time-base counter equal to period (TBCTR = TBPRD)</p> <p>10: Load on either CTR = Zero or CTR = PRD</p> <p>11: Reserved</p> <p>Reset type: SYSRSn</p>                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2   | CTLMODE  | R/W  | 0h    | <p>Control Mode Bits</p> <p>Selects the register (CMP/TBPRD or TBPHS) that controls the MEP:</p> <p>0: CMPAHR(8) or TBPRDHR(8) Register controls the edge position (i.e., this is duty or period control mode). (Default on Reset)</p> <p>1: TBPHSHR(8) Register controls the edge position (i.e., this is phase control mode).</p> <p>Reset type: SYSRSn</p>                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 1-0 | EDGMODE  | R/W  | 0h    | <p>Edge Mode Bits</p> <p>Selects the edge of the PWM that is controlled by the micro-edge position (MEP) logic:</p> <p>00: HRPWM capability is disabled (default on reset)</p> <p>01: MEP control of rising edge (CMPAHR)</p> <p>10: MEP control of falling edge (CMPAHR)</p> <p>11: MEP control of both edges (TBPHSHR or TBPRDHR)</p> <p>Reset type: SYSRSn</p>                                                                                                                                                                                                                                                                                                                                                                                                                      |

### 26.17.2.18 HRPWR Register (Offset = 21h) [Reset = 0000h]

HRPWR is shown in [Figure 26-110](#) and described in [Table 26-41](#).

Return to the [Summary Table](#).

HRPWM Power Register

This register is only accessible on EPWM modules with HRPWM capabilities.

**Figure 26-110. HRPWR Register**

|          |          |          |          |          |          |          |   |
|----------|----------|----------|----------|----------|----------|----------|---|
| 15       | 14       | 13       | 12       | 11       | 10       | 9        | 8 |
| CALPWRON | RESERVED |          |          |          |          | RESERVED |   |
| R/W-0h   | R-0-0h   |          |          |          |          | R/W-0h   |   |
| 7        | 6        | 5        | 4        | 3        | 2        | 1        | 0 |
| RESERVED |          | RESERVED | RESERVED | RESERVED | RESERVED | RESERVED |   |
| R/W-0h   |          | R/W-0h   | R-0h     | R/W-0h   | R/W-0h   | R/W-0h   |   |

**Table 26-41. HRPWR Register Field Descriptions**

| Bit   | Field    | Type | Reset | Description                                                                                                                                                                                     |
|-------|----------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15    | CALPWRON | R/W  | 0h    | MEP Calibration Power Bits (only available on ePWM1)<br>0: Disables MEP calibration logic in the HRPWM and reduces power consumption.<br>1: Enables MEP calibration logic<br>Reset type: SYSRSn |
| 14-10 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                        |
| 9-6   | RESERVED | R/W  | 0h    | Reserved                                                                                                                                                                                        |
| 5     | RESERVED | R/W  | 0h    | Reserved                                                                                                                                                                                        |
| 4     | RESERVED | R    | 0h    | Reserved                                                                                                                                                                                        |
| 3     | RESERVED | R/W  | 0h    | Reserved                                                                                                                                                                                        |
| 2     | RESERVED | R/W  | 0h    | Reserved                                                                                                                                                                                        |
| 1-0   | RESERVED | R/W  | 0h    | Reserved                                                                                                                                                                                        |

### 26.17.2.19 HRMSTEP Register (Offset = 26h) [Reset = 0000h]

HRMSTEP is shown in [Figure 26-111](#) and described in [Table 26-42](#).

Return to the [Summary Table](#).

#### HRPWM MEP Step Register

This register is only accessible on EPWM modules with HRPWM capabilities. Only 16 bit accesses are allowed for this register. Debugger access in 32 bit mode may display incorrect values.

**Figure 26-111. HRMSTEP Register**

|          |    |    |    |    |    |   |   |
|----------|----|----|----|----|----|---|---|
| 15       | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| RESERVED |    |    |    |    |    |   |   |
| R-0-0h   |    |    |    |    |    |   |   |
| 7        | 6  | 5  | 4  | 3  | 2  | 1 | 0 |
| HRMSTEP  |    |    |    |    |    |   |   |
| R/W-0h   |    |    |    |    |    |   |   |

**Table 26-42. HRMSTEP Register Field Descriptions**

| Bit  | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------|----------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-8 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 7-0  | HRMSTEP  | R/W  | 0h    | High Resolution MEP Step<br>When auto-conversion is enabled (HRCNFG[AUTOCONV] = 1), This 8-bit field contains the MEP_ScaleFactor (number of MEP steps per coarse steps) used by the hardware to automatically convert the value in the CMPAHR, CMPBHR, DBFEDHR, DBREDHR, TBPHSHR, or TBPRDHR register to a scaled micro-edge delay on the high-resolution ePWM output. The value in this register is written by the SFO calibration software at the end of each calibration run.<br>Reset type: SYSRSn |

### 26.17.2.20 HRCNFG2 Register (Offset = 27h) [Reset = 0000h]

HRCNFG2 is shown in [Figure 26-112](#) and described in [Table 26-43](#).

Return to the [Summary Table](#).

#### HRPWM Configuration 2 Register

This register is only accessible on EPWM modules with HRPWM capabilities. Only 16 bit accesses are allowed for this register. Debugger access in 32 bit mode may display incorrect values.

**Figure 26-112. HRCNFG2 Register**

|          |            |              |    |              |    |           |   |
|----------|------------|--------------|----|--------------|----|-----------|---|
| 15       | 14         | 13           | 12 | 11           | 10 | 9         | 8 |
| RESERVED | RESERVED   | RESERVED     |    |              |    |           |   |
| R/W-0h   | R-0/W1S-0h | R-0-0h       |    |              |    |           |   |
| 7        | 6          | 5            | 4  | 3            | 2  | 1         | 0 |
| RESERVED |            | CTLMODEDBFED |    | CTLMODEDBRED |    | EDGMODEDB |   |
| R-0-0h   |            | R/W-0h       |    | R/W-0h       |    | R/W-0h    |   |

**Table 26-43. HRCNFG2 Register Field Descriptions**

| Bit  | Field        | Type    | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                 |
|------|--------------|---------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | RESERVED     | R/W     | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                    |
| 14   | RESERVED     | R-0/W1S | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                    |
| 13-6 | RESERVED     | R-0     | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                    |
| 5-4  | CTLMODEDBFED | R/W     | 0h    | Shadow Mode Bit - selection should match DBCTL[LOADFEDMODE]<br>Selects the time event that loads the DBFEDHR shadow value into the active register.<br>00 Load on CTR = Zero: Time-base counter equal to zero (TBCTR = 0x0000)<br>01 Load on CTR = PRD: Time-base counter equal to period (TBCTR = TBPRD)<br>10 Load on either CTR = Zero or CTR = PRD<br>11 Reserved<br>Reset type: SYSRSn |
| 3-2  | CTLMODEDBRED | R/W     | 0h    | Shadow Mode Bit - selection should match DBCTL[LOADREDMODE]<br>Selects the time event that loads the DBREDHR shadow value into the active register.<br>00 Load on CTR = Zero: Time-base counter equal to zero (TBCTR = 0x0000)<br>01 Load on CTR = PRD: Time-base counter equal to period (TBCTR = TBPRD)<br>10 Load on either CTR = Zero or CTR = PRD<br>11 Reserved<br>Reset type: SYSRSn |
| 1-0  | EDGMODEDB    | R/W     | 0h    | Edge Mode Bits<br>Selects the edge of the PWM that is controlled by the micro-edge position (MEP) logic:<br>00 HRPWM capability is disabled (default on reset)<br>01 MEP control of rising edge (DBREDHR)<br>10 MEP control of falling edge (DBFEDHR)<br>11 MEP control of both edges (rising edge of DBREDHR or falling edge of DBFEDHR)<br>Reset type: SYSRSn                             |

### 26.17.2.21 HRPCTL Register (Offset = 2Dh) [Reset = 0000h]

HRPCTL is shown in [Figure 26-113](#) and described in [Table 26-44](#).

Return to the [Summary Table](#).

High Resolution Period Control Register

This register is only accessible on EPWM modules with HRPWM capabilities.

**Figure 26-113. HRPCTL Register**

|          |              |    |    |          |              |             |        |
|----------|--------------|----|----|----------|--------------|-------------|--------|
| 15       | 14           | 13 | 12 | 11       | 10           | 9           | 8      |
| RESERVED |              |    |    |          |              |             |        |
| R-0-0h   |              |    |    |          |              |             |        |
| 7        | 6            | 5  | 4  | 3        | 2            | 1           | 0      |
| RESERVED | PWMSYNCSSELX |    |    | RESERVED | TBPHSHRLOADE | PWMSYNCSSEL | HRPE   |
| R-0-0h   | R/W-0h       |    |    | R/W-0h   | R/W-0h       | R/W-0h      | R/W-0h |

**Table 26-44. HRPCTL Register Field Descriptions**

| Bit  | Field        | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|--------------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-7 | RESERVED     | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 6-4  | PWMSYNCSSELX | R/W  | 0h    | Extended selection bits for EPWMSYNCSSEL<br>000: EPWMSYNCSSEL is defined by PWMSYNCSSEL - > default condition (compatible with previous EPWM versions)<br>001: Reserved<br>010: Reserved<br>011: Reserved<br>100: CTR = CMPC, Count direction Up<br>101: CTR = CMPC, Count direction Down<br>110: CTR = CMPD, Count direction Up<br>111: CTR = CMPD, Count direction Down<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 3    | RESERVED     | R/W  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2    | TBPHSHRLOADE | R/W  | 0h    | TBPHSHR Load Enable<br>This bit allows you to synchronize ePWM modules with a high-resolution phase on a SYNCIN, TBCTL[SWFSYNC] or digital compare event. This allows for multiple ePWM modules operating at the same frequency to be phase aligned with high-resolution.<br>0: Disables synchronization of high-resolution phase on a SYNCIN, TBCTL[SWFSYNC] or digital compare event:<br>1: Synchronize the high-resolution phase on a SYNCIN, TBCTL[SWFSYNC] or digital comparator synchronization event. The phase is synchronized using the contents of the high-resolution phase TBPHSHR register. The TBCTL[PHSEN] bit which enables the loading of the TBCTR register with TBPHS register value on a SYNCIN or TBCTL[SWFSYNC] event works independently. However, users need to enable this bit also if they want to control phase in conjunction with the high-resolution period feature. This bit and the TBCTL[PHSEN] bit must be set to 1 when high-resolution period is enabled for up-down count mode even if TBPHSHR = 0x0000. This bit does not need to be set when only high-resolution duty is enabled.<br>Reset type: SYSRSn |
| 1    | PWMSYNCSSEL  | R/W  | 0h    | PWMSYNCS Source Select Bit: This bit selects the source for the EPWMSYNCSSEL signal that goes to the CMPSS and GPDAC:<br>0 CTR = PRD: Time-base counter equal to period (TBCTR = TBPRD)<br>1 CTR = zero: Time-base counter equal to zero (TBCTR = 0x00)<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

**Table 26-44. HRPCTL Register Field Descriptions (continued)**

| Bit | Field | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                |
|-----|-------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0   | HRPE  | R/W  | 0h    | <p>High Resolution Period Enable Bit</p> <p>0: High resolution period feature disabled. In this mode the ePWM behaves as a Type 4 ePWM.</p> <p>1: High resolution period enabled. In this mode the HRPWM module can control high-resolution of both the duty and frequency. When high-resolution period is enabled, TBCTL[CTRMODE] = 0,1 (down-count mode) is not supported.</p> <p>Reset type: SYSRSn</p> |

### 26.17.2.22 TRREM Register (Offset = 2Eh) [Reset = 0000h]

TRREM is shown in [Figure 26-114](#) and described in [Table 26-45](#).

Return to the [Summary Table](#).

HRPWM High Resolution Remainder Register

This register is only accessible on EPWM modules with HRPWM capabilities.

**Figure 26-114. TRREM Register**

|          |    |    |    |    |    |        |   |
|----------|----|----|----|----|----|--------|---|
| 15       | 14 | 13 | 12 | 11 | 10 | 9      | 8 |
| RESERVED |    |    |    |    |    | TRREM  |   |
| R-0-0h   |    |    |    |    |    | R/W-0h |   |
| 7        | 6  | 5  | 4  | 3  | 2  | 1      | 0 |
| TRREM    |    |    |    |    |    |        |   |
| R/W-0h   |    |    |    |    |    |        |   |

**Table 26-45. TRREM Register Field Descriptions**

| Bit   | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------|----------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-11 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 10-0  | TRREM    | R/W  | 0h    | <p>HRPWM Remainder Bits: This 11-bit value keeps track of the remainder portion of the HRPWM algorithm calculations. This value keeps track of the remainder portion of the HRPWM hardware calculations.</p> <p>Notes:</p> <ol style="list-style-type: none"> <li>The lower 8-bits of the TRREM register can be automatically initialized with the TBPHSHR value on a SYNCIN or TBCTL[SWFSYNC] event or DC event (if enabled). The user can also write a value with the CPU.</li> <li>Priority of TRREM register updates:<br/>Sync (software or hardware) TBPHSHR copied to TRREM : Highest Priority<br/>HRPWM Hardware (updates TRREM register): Next priority<br/>CPU Write To TRREM Register: Lowest Priority</li> <li>Bit 10 of TRREM register is not used in asymmetrical mode. This bit can be forced to zero.<br/>TRREM will be initialized to 0x0 and 0x100 in Up and Up-down modes respectively.</li> </ol> <p>Asymmetrical Mode:<br/>TRREM[7:0] = TBPHSHR[15:8]<br/>TRREM[10,9,8] = 0,0,0</p> <p>Symmetrical Mode:<br/>TRREM[7:0] = TBPHSHR[15:8]<br/>TRREM[10,9,8] = 0,0,1</p> <p>Reset type: SYSRSn</p> |

### 26.17.2.23 GLDCTL Register (Offset = 34h) [Reset = 0000h]

GLDCTL is shown in [Figure 26-115](#) and described in [Table 26-46](#).

Return to the [Summary Table](#).

Global PWM Load Control Register

**Figure 26-115. GLDCTL Register**

|          |          |          |         |    |    |        |   |
|----------|----------|----------|---------|----|----|--------|---|
| 15       | 14       | 13       | 12      | 11 | 10 | 9      | 8 |
| RESERVED |          |          | GLDCNT  |    |    | GLDPRD |   |
| R-0-0h   |          |          | R-0h    |    |    | R/W-0h |   |
| 7        | 6        | 5        | 4       | 3  | 2  | 1      | 0 |
| GLDPRD   | RESERVED | OSHTMODE | GLDMODE |    |    | GLD    |   |
| R/W-0h   | R-0-0h   | R/W-0h   | R/W-0h  |    |    | R/W-0h |   |

**Table 26-46. GLDCTL Register Field Descriptions**

| Bit   | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------|----------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-13 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 12-10 | GLDCNT   | R    | 0h    | Global Load Strobe Counter Register<br>These bits indicate how many selected events have occurred:<br>000: No events<br>001: 1 event<br>010: 2 events<br>011: 3 events<br>100: 4 events<br>101: 5 events<br>110: 6 events<br>111: 7 events<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                  |
| 9-7   | GLDPRD   | R/W  | 0h    | Global Load Strobe Period Select Register<br>These bits select how many selected events need to occur before a load strobe is generated<br>000: Disable counter<br>001: Generate strobe on GLDCNT = 001 (1st event)<br>010: Generate strobe on GLDCNT = 010 (2nd event)<br>011: Generate strobe on GLDCNT = 011 (3rd event)<br>100: Generate strobe on GLDCNT = 011 (4th event)<br>101: Generate strobe on GLDCNT = 001 (5th event)<br>110: Generate strobe on GLDCNT = 010 (6th event)<br>111: Generate strobe on GLDCNT = 011 (7th event)<br>Reset type: SYSRSn |
| 6     | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5     | OSHTMODE | R/W  | 0h    | One Shot Load Mode Control Bit<br>0: One shot load mode is disabled and shadow to active loading happens continuously on all the chosen load strobes.<br>1: One shot mode is active. All load strobes are blocked until GLDCTL2[OSHTLD] is written with 1.<br>Note: One Shot mode can only be used with global shadow to active load mode enabled (GLDCTL[GLD]=1)<br>Reset type: SYSRSn                                                                                                                                                                           |



**Table 26-46. GLDCTL Register Field Descriptions (continued)**

| Bit | Field   | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----|---------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4-1 | GLDMODE | R/W  | 0h    | Global Load Pulse selection for Shadow to Active Mode Reloads<br>0000: Load on Counter = 0 (CNT_ZRO)<br>0001: Load on Counter = Period (PRD_EQ)<br>0010: Load on either Counter = 0, or Counter = Period<br>0011: Load on SYNCEVT - this is logical OR of DCAEVT1.sync, DCBEVT1.sync, EPWMxSYNCl and TBCTL[SWFSYNC]<br>0100: Load on SYNCEVT or CNT_ZRO<br>0101: Load on SYNCEVT or PRD_EQ<br>0110: Load on SYNCEVT or CNT_ZRO or PRD_EQ<br>1000: Reserved<br>...<br>1110: Reserved<br>1111: Load on GLDCTL2[GFRCLD] write<br>Reset type: SYSRSn |
| 0   | GLD     | R/W  | 0h    | Global Shadow to Active Load Event Control<br>0: Shadow to active reload for all shadowed registers happens as per the individual reload control bits specified (Compatible with previous EPWM versions).<br>1: When set, all the shadow to active reload events are defined by GLDMODE bits in GLDCTL register. All the shadow registers use same reload pulse from shadow to active reloading. Individual LOADMODE bits are ignored.<br>Reset type: SYSRSn                                                                                     |

### 26.17.2.24 GLDCFG Register (Offset = 35h) [Reset = 0000h]

GLDCFG is shown in [Figure 26-116](#) and described in [Table 26-47](#).

Return to the [Summary Table](#).

Global PWM Load Config Register

**Figure 26-116. GLDCFG Register**

| 15       | 14                | 13                | 12     | 11     | 10              | 9                  | 8                  |
|----------|-------------------|-------------------|--------|--------|-----------------|--------------------|--------------------|
| RESERVED |                   |                   |        |        | AQCSFRC         | AQCTLB_AQC<br>TLB2 | AQCTLA_AQC<br>TLA2 |
| R-0-0h   |                   |                   |        |        | R/W-0h          | R/W-0h             | R/W-0h             |
| 7        | 6                 | 5                 | 4      | 3      | 2               | 1                  | 0                  |
| DBCTL    | DBFED_DBFE<br>DHR | DBRED_DBRE<br>DHR | CMPD   | CMPC   | CMPB_CMPBH<br>R | CMPA_CMPAH<br>R    | TBPRD_TBPR<br>DHR  |
| R/W-0h   | R/W-0h            | R/W-0h            | R/W-0h | R/W-0h | R/W-0h          | R/W-0h             | R/W-0h             |

**Table 26-47. GLDCFG Register Field Descriptions**

| Bit   | Field          | Type | Reset | Description                                                                                                                                                                                                                                                                 |
|-------|----------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-11 | RESERVED       | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                    |
| 10    | AQCSFRC        | R/W  | 0h    | Global load event configuration for AQCSFRC<br>0: Registers use local reload configuration even if GLDCTL(GLD)=1 (reload is compatible with previous EPWMs)<br>1: Registers use global load configuration if this bit is set and GLDCTL(GLD)=1<br>Reset type: SYSRSn        |
| 9     | AQCTLB_AQCTLB2 | R/W  | 0h    | Global load event configuration for AQCTLB_AQCTLB2<br>0: Registers use local reload configuration even if GLDCTL(GLD)=1 (reload is compatible with previous EPWMs)<br>1: Registers use global load configuration if this bit is set and GLDCTL(GLD)=1<br>Reset type: SYSRSn |
| 8     | AQCTLA_AQCTLA2 | R/W  | 0h    | Global load event configuration for AQCTLA_AQCTLA2<br>0: Registers use local reload configuration even if GLDCTL(GLD)=1 (reload is compatible with previous EPWMs)<br>1: Registers use global load configuration if this bit is set and GLDCTL(GLD)=1<br>Reset type: SYSRSn |
| 7     | DBCTL          | R/W  | 0h    | Global load event configuration for DBCTL<br>0: Registers use local reload configuration even if GLDCTL(GLD)=1 (reload is compatible with previous EPWMs)<br>1: Registers use global load configuration if this bit is set and GLDCTL(GLD)=1<br>Reset type: SYSRSn          |
| 6     | DBFED_DBFEDHR  | R/W  | 0h    | Global load event configuration for DBFED_DBFEDHR<br>0: Registers use local reload configuration even if GLDCTL(GLD)=1 (reload is compatible with previous EPWMs)<br>1: Registers use global load configuration if this bit is set and GLDCTL(GLD)=1<br>Reset type: SYSRSn  |
| 5     | DBRED_DBREDHR  | R/W  | 0h    | Global load event configuration for DBRED_DBREDHR<br>0: Registers use local reload configuration even if GLDCTL(GLD)=1 (reload is compatible with previous EPWMs)<br>1: Registers use global load configuration if this bit is set and GLDCTL(GLD)=1<br>Reset type: SYSRSn  |

**Table 26-47. GLDCFG Register Field Descriptions (continued)**

| Bit | Field         | Type | Reset | Description                                                                                                                                                                                                                                                                |
|-----|---------------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4   | CMPD          | R/W  | 0h    | Global load event configuration for CMPD<br>0: Registers use local reload configuration even if GLDCTL(GLD)=1 (reload is compatible with previous EPWMs)<br>1: Registers use global load configuration if this bit is set and GLDCTL(GLD)=1<br>Reset type: SYSRSn          |
| 3   | CMPC          | R/W  | 0h    | Global load event configuration for CMPC<br>0: Registers use local reload configuration even if GLDCTL(GLD)=1 (reload is compatible with previous EPWMs)<br>1: Registers use global load configuration if this bit is set and GLDCTL(GLD)=1<br>Reset type: SYSRSn          |
| 2   | CMPB_CMPBHR   | R/W  | 0h    | Global load event configuration for CMPB_CMPBHR<br>0: Registers use local reload configuration even if GLDCTL(GLD)=1 (reload is compatible with previous EPWMs)<br>1: Registers use global load configuration if this bit is set and GLDCTL(GLD)=1<br>Reset type: SYSRSn   |
| 1   | CMPA_CMPAHR   | R/W  | 0h    | Global load event configuration for CMPA_CMPAHR<br>0: Registers use local reload configuration even if GLDCTL(GLD)=1 (reload is compatible with previous EPWMs)<br>1: Registers use global load configuration if this bit is set and GLDCTL(GLD)=1<br>Reset type: SYSRSn   |
| 0   | TBPRD_TBPRDHR | R/W  | 0h    | Global load event configuration for TBPRD_TBPRDHR<br>0: Registers use local reload configuration even if GLDCTL(GLD)=1 (reload is compatible with previous EPWMs)<br>1: Registers use global load configuration if this bit is set and GLDCTL(GLD)=1<br>Reset type: SYSRSn |

### 26.17.2.25 EPWMXLINK Register (Offset = 38h) [Reset = 000XXXXXh]

EPWMXLINK is shown in [Figure 26-117](#) and described in [Table 26-48](#).

Return to the [Summary Table](#).

#### EPWMx Link Register

This register controls which EPWMs are linked to other EPWM modules. The default reset value will vary for each module. The reset value will link each EPWM module to itself to prevent unintentional linking of modules.

**Figure 26-117. EPWMXLINK Register**

|             |    |    |    |          |    |    |    |          |    |    |    |           |    |    |    |
|-------------|----|----|----|----------|----|----|----|----------|----|----|----|-----------|----|----|----|
| 31          | 30 | 29 | 28 | 27       | 26 | 25 | 24 | 23       | 22 | 21 | 20 | 19        | 18 | 17 | 16 |
| GLDCTL2LINK |    |    |    | RESERVED |    |    |    |          |    |    |    | CMPDLINK  |    |    |    |
| R/W-0h      |    |    |    | R-0-0h   |    |    |    |          |    |    |    | R/W-X     |    |    |    |
| 15          | 14 | 13 | 12 | 11       | 10 | 9  | 8  | 7        | 6  | 5  | 4  | 3         | 2  | 1  | 0  |
| CMPCLINK    |    |    |    | CMPBLINK |    |    |    | CMPALINK |    |    |    | TBPRDLINK |    |    |    |
| R/W-X       |    |    |    | R/W-X    |    |    |    | R/W-X    |    |    |    | R/W-X     |    |    |    |

**Table 26-48. EPWMXLINK Register Field Descriptions**

| Bit   | Field       | Type | Reset | Description                                                                                                                                                                                                                                                                                                                          |
|-------|-------------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-28 | GLDCTL2LINK | R/W  | 0h    | GLDCTL2 Link Bits<br>Writes to the GLDCTL2 registers in the ePWM module selected by the following bit selections results in a simultaneous write to the current ePWM module's GLDCTL2 registers.<br>0000: ePWM1<br>0001: ePWM2<br>...<br>Up to the last instance of ePWM. All others are reserved.<br>Reset type: SYSRSn             |
| 27-20 | RESERVED    | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                             |
| 19-16 | CMPDLINK    | R/W  | X     | CMPD Link Bits<br>Writes to the CMPD registers in the ePWM module selected by the following bit selections results in a simultaneous write to the current ePWM module's CMPD registers.<br>0000: ePWM1<br>0001: ePWM2<br>...<br>Up to the last instance of ePWM. All others are reserved.<br>Reset type: SYSRSn                      |
| 15-12 | CMPCLINK    | R/W  | X     | CMPC Link Bits<br>Writes to the CMPC registers in the ePWM module selected by the following bit selections results in a simultaneous write to the current ePWM module's CMPC registers.<br>0000: ePWM1<br>0001: ePWM2<br>...<br>Up to the last instance of ePWM. All others are reserved.<br>Reset type: SYSRSn                      |
| 11-8  | CMPBLINK    | R/W  | X     | CMPB_CMPBHR Link Bits<br>Writes to the CMPB_CMPBHR registers in the ePWM module selected by the following bit selections results in a simultaneous write to the current ePWM module's CMPB_CMPBHR registers.<br>0000: ePWM1<br>0001: ePWM2<br>...<br>Up to the last instance of ePWM. All others are reserved.<br>Reset type: SYSRSn |

**Table 26-48. EPWMXLINK Register Field Descriptions (continued)**

| Bit | Field     | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                |
|-----|-----------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7-4 | CMPALINK  | R/W  | X     | CMPA_CMPAHR Link Bits<br>Writes to the CMPA_CMPAHR registers in the ePWM module selected by the following bit selections results in a simultaneous write to the current ePWM module's CMPA_CMPAHR registers.<br>0000: ePWM1<br>0001: ePWM2<br>...<br>Up to the last instance of ePWM. All others are reserved.<br>Reset type: SYSRSn       |
| 3-0 | TBPRDLINK | R/W  | X     | TBPRD_TBPRDHR Link Bits<br>Writes to the TBPRD:TBPDRHR registers in the ePWM module selected by the following bit selections results in a simultaneous write to the current ePWM module's TBPRD_TBPRDHR registers.<br>0000: ePWM1<br>0001: ePWM2<br>...<br>Up to the last instance of ePWM. All others are reserved.<br>Reset type: SYSRSn |

### 26.17.2.26 AQCTLA Register (Offset = 40h) [Reset = 0000h]

AQCTLA is shown in [Figure 26-118](#) and described in [Table 26-49](#).

Return to the [Summary Table](#).

Action Qualifier Control Register For Output A

**Figure 26-118. AQCTLA Register**

|          |    |        |    |        |    |        |   |
|----------|----|--------|----|--------|----|--------|---|
| 15       | 14 | 13     | 12 | 11     | 10 | 9      | 8 |
| RESERVED |    |        |    | CBD    |    | CBU    |   |
| R-0-0h   |    |        |    | R/W-0h |    | R/W-0h |   |
| 7        | 6  | 5      | 4  | 3      | 2  | 1      | 0 |
| CAD      |    | CAU    |    | PRD    |    | ZRO    |   |
| R/W-0h   |    | R/W-0h |    | R/W-0h |    | R/W-0h |   |

**Table 26-49. AQCTLA Register Field Descriptions**

| Bit   | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|----------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-12 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                      |
| 11-10 | CBD      | R/W  | 0h    | Action When TBCTR = CMPB on Down Count<br>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.<br>00: Do nothing (action disabled)<br>01: Clear: force EPWMxA output low.<br>10: Set: force EPWMxA output high.<br>11: Toggle EPWMxA output: low output signal will be forced high, and a high signal will be forced low.<br>Reset type: SYSRSn |
| 9-8   | CBU      | R/W  | 0h    | Action When TBCTR = CMPB on Up Count<br>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.<br>00: Do nothing (action disabled)<br>01: Clear: force EPWMxA output low.<br>10: Set: force EPWMxA output high.<br>11: Toggle EPWMxA output: low output signal will be forced high, and a high signal will be forced low.<br>Reset type: SYSRSn   |
| 7-6   | CAD      | R/W  | 0h    | Action When TBCTR = CMPA on Down Count<br>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.<br>00: Do nothing (action disabled)<br>01: Clear: force EPWMxA output low.<br>10: Set: force EPWMxA output high.<br>11: Toggle EPWMxA output: low output signal will be forced high, and a high signal will be forced low.<br>Reset type: SYSRSn |
| 5-4   | CAU      | R/W  | 0h    | Action When TBCTR = CMPA on Up Count<br>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.<br>00: Do nothing (action disabled)<br>01: Clear: force EPWMxA output low.<br>10: Set: force EPWMxA output high.<br>11: Toggle EPWMxA output: low output signal will be forced high, and a high signal will be forced low.<br>Reset type: SYSRSn   |

**Table 26-49. AQCTLA Register Field Descriptions (continued)**

| Bit | Field | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|-------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3-2 | PRD   | R/W  | 0h    | <p>Action When TBCTR = TBPRD</p> <p>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.</p> <p>00: Do nothing (action disabled)</p> <p>01: Clear: force EPWMxA output low.</p> <p>10: Set: force EPWMxA output high.</p> <p>11: Toggle EPWMxA output: low output signal will be forced high, and a high signal will be forced low.</p> <p>Reset type: SYSRSn</p> |
| 1-0 | ZRO   | R/W  | 0h    | <p>Action When TBCTR = 0</p> <p>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.</p> <p>00: Do nothing (action disabled)</p> <p>01: Clear: force EPWMxA output low.</p> <p>10: Set: force EPWMxA output high.</p> <p>11: Toggle EPWMxA output: low output signal will be forced high, and a high signal will be forced low.</p> <p>Reset type: SYSRSn</p>     |

### 26.17.2.27 AQCTLA2 Register (Offset = 41h) [Reset = 0000h]

AQCTLA2 is shown in [Figure 26-119](#) and described in [Table 26-50](#).

Return to the [Summary Table](#).

Additional Action Qualifier Control Register For Output A

**Figure 26-119. AQCTLA2 Register**

|          |    |        |    |        |    |        |   |
|----------|----|--------|----|--------|----|--------|---|
| 15       | 14 | 13     | 12 | 11     | 10 | 9      | 8 |
| RESERVED |    |        |    |        |    |        |   |
| R-0-0h   |    |        |    |        |    |        |   |
| 7        | 6  | 5      | 4  | 3      | 2  | 1      | 0 |
| T2D      |    | T2U    |    | T1D    |    | T1U    |   |
| R/W-0h   |    | R/W-0h |    | R/W-0h |    | R/W-0h |   |

**Table 26-50. AQCTLA2 Register Field Descriptions**

| Bit  | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|----------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-8 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7-6  | T2D      | R/W  | 0h    | Action when event occurs on T2 in DOWN-Count<br>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.<br>00: Do nothing (action disabled)<br>01: Clear: force EPWMxA output low.<br>10: Set: force EPWMxA output high.<br>11: Toggle EPWMxA output: low output signal will be forced high, and a high signal will be forced low.<br>Reset type: SYSRSn |
| 5-4  | T2U      | R/W  | 0h    | Action when event occurs on T2 in UP-Count<br>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.<br>00: Do nothing (action disabled)<br>01: Clear: force EPWMxA output low.<br>10: Set: force EPWMxA output high.<br>11: Toggle EPWMxA output: low output signal will be forced high, and a high signal will be forced low.<br>Reset type: SYSRSn   |
| 3-2  | T1D      | R/W  | 0h    | Action when event occurs on T1 in DOWN-Count<br>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.<br>00: Do nothing (action disabled)<br>01: Clear: force EPWMxA output low.<br>10: Set: force EPWMxA output high.<br>11: Toggle EPWMxA output: low output signal will be forced high, and a high signal will be forced low.<br>Reset type: SYSRSn |
| 1-0  | T1U      | R/W  | 0h    | Action when event occurs on T1 in UP-Count<br>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.<br>00: Do nothing (action disabled)<br>01: Clear: force EPWMxA output low.<br>10: Set: force EPWMxA output high.<br>11: Toggle EPWMxA output: low output signal will be forced high, and a high signal will be forced low.<br>Reset type: SYSRSn   |



### 26.17.2.28 AQCTLB Register (Offset = 42h) [Reset = 0000h]

AQCTLB is shown in [Figure 26-120](#) and described in [Table 26-51](#).

Return to the [Summary Table](#).

Action Qualifier Control Register For Output B

**Figure 26-120. AQCTLB Register**

|          |    |        |    |        |    |        |   |
|----------|----|--------|----|--------|----|--------|---|
| 15       | 14 | 13     | 12 | 11     | 10 | 9      | 8 |
| RESERVED |    |        |    | CBD    |    | CBU    |   |
| R-0-0h   |    |        |    | R/W-0h |    | R/W-0h |   |
| 7        | 6  | 5      | 4  | 3      | 2  | 1      | 0 |
| CAD      |    | CAU    |    | PRD    |    | ZRO    |   |
| R/W-0h   |    | R/W-0h |    | R/W-0h |    | R/W-0h |   |

**Table 26-51. AQCTLB Register Field Descriptions**

| Bit   | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|----------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-12 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                      |
| 11-10 | CBD      | R/W  | 0h    | Action When TBCTR = CMPB on Down Count<br>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.<br>00: Do nothing (action disabled)<br>01: Clear: force EPWMxB output low.<br>10: Set: force EPWMxB output high.<br>11: Toggle EPWMxB output: low output signal will be forced high, and a high signal will be forced low.<br>Reset type: SYSRSn |
| 9-8   | CBU      | R/W  | 0h    | Action When TBCTR = CMPB on Up Count<br>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.<br>00: Do nothing (action disabled)<br>01: Clear: force EPWMxB output low.<br>10: Set: force EPWMxB output high.<br>11: Toggle EPWMxB output: low output signal will be forced high, and a high signal will be forced low.<br>Reset type: SYSRSn   |
| 7-6   | CAD      | R/W  | 0h    | Action When TBCTR = CMPA on Down Count<br>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.<br>00: Do nothing (action disabled)<br>01: Clear: force EPWMxB output low.<br>10: Set: force EPWMxB output high.<br>11: Toggle EPWMxB output: low output signal will be forced high, and a high signal will be forced low.<br>Reset type: SYSRSn |
| 5-4   | CAU      | R/W  | 0h    | Action When TBCTR = CMPA on Up Count<br>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.<br>00: Do nothing (action disabled)<br>01: Clear: force EPWMxB output low.<br>10: Set: force EPWMxB output high.<br>11: Toggle EPWMxB output: low output signal will be forced high, and a high signal will be forced low.<br>Reset type: SYSRSn   |

**Table 26-51. AQCTLB Register Field Descriptions (continued)**

| Bit | Field | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|-------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3-2 | PRD   | R/W  | 0h    | <p>Action When TBCTR = TBPRD</p> <p>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.</p> <p>00: Do nothing (action disabled)</p> <p>01: Clear: force EPWMxB output low.</p> <p>10: Set: force EPWMxB output high.</p> <p>11: Toggle EPWMxB output: low output signal will be forced high, and a high signal will be forced low.</p> <p>Reset type: SYSRSn</p> |
| 1-0 | ZRO   | R/W  | 0h    | <p>Action When TBCTR = 0</p> <p>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.</p> <p>00: Do nothing (action disabled)</p> <p>01: Clear: force EPWMxB output low.</p> <p>10: Set: force EPWMxB output high.</p> <p>11: Toggle EPWMxB output: low output signal will be forced high, and a high signal will be forced low.</p> <p>Reset type: SYSRSn</p>     |

### 26.17.2.29 AQCTLB2 Register (Offset = 43h) [Reset = 0000h]

AQCTLB2 is shown in [Figure 26-121](#) and described in [Table 26-52](#).

Return to the [Summary Table](#).

Additional Action Qualifier Control Register For Output B

**Figure 26-121. AQCTLB2 Register**

|          |    |        |    |        |    |        |   |
|----------|----|--------|----|--------|----|--------|---|
| 15       | 14 | 13     | 12 | 11     | 10 | 9      | 8 |
| RESERVED |    |        |    |        |    |        |   |
| R-0-0h   |    |        |    |        |    |        |   |
| 7        | 6  | 5      | 4  | 3      | 2  | 1      | 0 |
| T2D      |    | T2U    |    | T1D    |    | T1U    |   |
| R/W-0h   |    | R/W-0h |    | R/W-0h |    | R/W-0h |   |

**Table 26-52. AQCTLB2 Register Field Descriptions**

| Bit  | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|----------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-8 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7-6  | T2D      | R/W  | 0h    | Action when event occurs on T2 in DOWN-Count<br>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.<br>00: Do nothing (action disabled)<br>01: Clear: force EPWMxB output low.<br>10: Set: force EPWMxB output high.<br>11: Toggle EPWMxB output: low output signal will be forced high, and a high signal will be forced low.<br>Reset type: SYSRSn |
| 5-4  | T2U      | R/W  | 0h    | Action when event occurs on T2 in UP-Count<br>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.<br>00: Do nothing (action disabled)<br>01: Clear: force EPWMxB output low.<br>10: Set: force EPWMxB output high.<br>11: Toggle EPWMxB output: low output signal will be forced high, and a high signal will be forced low.<br>Reset type: SYSRSn   |
| 3-2  | T1D      | R/W  | 0h    | Action when event occurs on T1 in DOWN-Count<br>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.<br>00: Do nothing (action disabled)<br>01: Clear: force EPWMxB output low.<br>10: Set: force EPWMxB output high.<br>11: Toggle EPWMxB output: low output signal will be forced high, and a high signal will be forced low.<br>Reset type: SYSRSn |
| 1-0  | T1U      | R/W  | 0h    | Action when event occurs on T1 in UP-Count<br>Note: By definition, in count up-down mode when the counter equals 0 the direction is defined as 1 or counting up.<br>00: Do nothing (action disabled)<br>01: Clear: force EPWMxB output low.<br>10: Set: force EPWMxB output high.<br>11: Toggle EPWMxB output: low output signal will be forced high, and a high signal will be forced low.<br>Reset type: SYSRSn   |

### 26.17.2.30 AQSFR Register (Offset = 47h) [Reset = 0000h]

AQSFR is shown in [Figure 26-122](#) and described in [Table 26-53](#).

Return to the [Summary Table](#).

Action Qualifier Software Force Register

**Figure 26-122. AQSFR Register**

|          |    |            |        |        |       |            |   |
|----------|----|------------|--------|--------|-------|------------|---|
| 15       | 14 | 13         | 12     | 11     | 10    | 9          | 8 |
| RESERVED |    |            |        |        |       |            |   |
| R-0-0h   |    |            |        |        |       |            |   |
| 7        | 6  | 5          | 4      | 3      | 2     | 1          | 0 |
| RLDCSF   |    | OTSFB      | ACTSFB |        | OTSFA | ACTSFA     |   |
| R/W-0h   |    | R-0/W1S-0h |        | R/W-0h |       | R-0/W1S-0h |   |
|          |    |            |        |        |       |            |   |

**Table 26-53. AQSFR Register Field Descriptions**

| Bit  | Field    | Type    | Reset | Description                                                                                                                                                                                                                                                                                                                                                                            |
|------|----------|---------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-8 | RESERVED | R-0     | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                               |
| 7-6  | RLDCSF   | R/W     | 0h    | AQCSFR Active Register Reload From Shadow Options<br>00: Load on time-base counter equals zero<br>01: Load on time-base counter equals period<br>10: Load on time-base counter equals zero or counter equals period<br>11: Load immediately (the active register is directly accessed by the CPU and is not loaded from the shadow register).<br>Reset type: SYSRSn                    |
| 5    | OTSFB    | R-0/W1S | 0h    | One-Time Software Forced Event on Output B<br>0: Writing a 0 (zero) has no effect. Always reads back a 0. This bit is auto cleared once a write to this register is complete (i.e., a forced event is initiated.). This is a one-shot forced event. It can be overridden by another subsequent event on output B.<br>1: Initiates a single software forced event<br>Reset type: SYSRSn |
| 4-3  | ACTSFB   | R/W     | 0h    | Action When One-Time Software Force B is Invoked<br>00: Does nothing (action disabled)<br>01: Clear (low)<br>10: Set (high)<br>11: Toggle (Low -> High, High -> Low)<br>Note: This action is not qualified by counter direction (CNT_dir)<br>Reset type: SYSRSn                                                                                                                        |
| 2    | OTSFA    | R-0/W1S | 0h    | One-Time Software Forced Event on Output A<br>0: Writing a 0 (zero) has no effect. Always reads back a 0. This bit is auto cleared once a write to this register is complete ( i.e., a forced event is initiated). This is a one-shot forced event. It can be overridden by another subsequent event on output A.<br>1: Initiates a single software forced event<br>Reset type: SYSRSn |
| 1-0  | ACTSFA   | R/W     | 0h    | Action When One-Time Software Force A Is Invoked<br>00: Does nothing (action disabled)<br>01: Clear (low)<br>10: Set (high)<br>11: Toggle (Low -> High, High -> Low)<br>Note: This action is not qualified by counter direction (CNT_dir)<br>Reset type: SYSRSn                                                                                                                        |

### 26.17.2.31 AQCSFRC Register (Offset = 49h) [Reset = 0000h]

AQCSFRC is shown in [Figure 26-123](#) and described in [Table 26-54](#).

Return to the [Summary Table](#).

Action Qualifier Continuous S/W Force Register

**Figure 26-123. AQCSFRC Register**

|          |    |    |    |        |    |        |   |
|----------|----|----|----|--------|----|--------|---|
| 15       | 14 | 13 | 12 | 11     | 10 | 9      | 8 |
| RESERVED |    |    |    |        |    |        |   |
| R-0-0h   |    |    |    |        |    |        |   |
| 7        | 6  | 5  | 4  | 3      | 2  | 1      | 0 |
| RESERVED |    |    |    | CSFB   |    | CSFA   |   |
| R-0-0h   |    |    |    | R/W-0h |    | R/W-0h |   |

**Table 26-54. AQCSFRC Register Field Descriptions**

| Bit  | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|----------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-4 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 3-2  | CSFB     | R/W  | 0h    | Continuous Software Force on Output B<br>In immediate mode, a continuous force takes effect on the next TBCLK edge. In shadow mode, a continuous force takes effect on the next TBCLK edge after a shadow load into the active register. To configure shadow mode, use AQSFRC[RLDCSF].<br>00: Software forcing is disabled and has no effect<br>01: Forces a continuous low on output B<br>10: Forces a continuous high on output B<br>11: Software forcing is disabled and has no effect<br>Reset type: SYSRSn |
| 1-0  | CSFA     | R/W  | 0h    | Continuous Software Force on Output A<br>In immediate mode, a continuous force takes effect on the next TBCLK edge. In shadow mode, a continuous force takes effect on the next TBCLK edge after a shadow load into the active register.<br>00: Software forcing is disabled and has no effect<br>01: Forces a continuous low on output A<br>10: Forces a continuous high on output A<br>11: Software forcing is disabled and has no effect<br>Reset type: SYSRSn                                               |

### 26.17.2.32 DBREDHR Register (Offset = 50h) [Reset = 0000h]

DBREDHR is shown in [Figure 26-124](#) and described in [Table 26-55](#).

Return to the [Summary Table](#).

Dead-Band Generator Rising Edge Delay High Resolution Mirror Register

**Figure 26-124. DBREDHR Register**

|          |    |    |    |    |    |   |          |
|----------|----|----|----|----|----|---|----------|
| 15       | 14 | 13 | 12 | 11 | 10 | 9 | 8        |
| DBREDHR  |    |    |    |    |    |   | RESERVED |
| R/W-0h   |    |    |    |    |    |   | R-0h     |
| 7        | 6  | 5  | 4  | 3  | 2  | 1 | 0        |
| RESERVED |    |    |    |    |    |   | RESERVED |
| R-0h     |    |    |    |    |    |   | R-0h     |

**Table 26-55. DBREDHR Register Field Descriptions**

| Bit  | Field    | Type | Reset | Description                                                            |
|------|----------|------|-------|------------------------------------------------------------------------|
| 15-9 | DBREDHR  | R/W  | 0h    | Dead Band Rising Edge Delay High Resolution Bits<br>Reset type: SYSRSn |
| 8    | RESERVED | R    | 0h    | Reserved                                                               |
| 7-1  | RESERVED | R    | 0h    | Reserved                                                               |
| 0    | RESERVED | R    | 0h    | Reserved                                                               |

### 26.17.2.33 DBRED Register (Offset = 51h) [Reset = 0000h]

DBRED is shown in [Figure 26-125](#) and described in [Table 26-56](#).

Return to the [Summary Table](#).

Dead-Band Generator Rising Edge Delay High Resolution Mirror Register

**Figure 26-125. DBRED Register**

|          |    |    |    |        |    |   |   |
|----------|----|----|----|--------|----|---|---|
| 15       | 14 | 13 | 12 | 11     | 10 | 9 | 8 |
| RESERVED |    |    |    | DBRED  |    |   |   |
| R-0h     |    |    |    | R/W-0h |    |   |   |
| 7        | 6  | 5  | 4  | 3      | 2  | 1 | 0 |
| DBRED    |    |    |    |        |    |   |   |
| R/W-0h   |    |    |    |        |    |   |   |

**Table 26-56. DBRED Register Field Descriptions**

| Bit   | Field    | Type | Reset | Description                                   |
|-------|----------|------|-------|-----------------------------------------------|
| 15-14 | RESERVED | R    | 0h    | Reserved                                      |
| 13-0  | DBRED    | R/W  | 0h    | Rising edge delay value<br>Reset type: SYSRSn |

### 26.17.2.34 DBFEDHR Register (Offset = 52h) [Reset = 0000h]

DBFEDHR is shown in [Figure 26-126](#) and described in [Table 26-57](#).

Return to the [Summary Table](#).

Dead-Band Generator Falling Edge Delay High Resolution Register

**Figure 26-126. DBFEDHR Register**

|          |    |    |    |    |    |   |          |
|----------|----|----|----|----|----|---|----------|
| 15       | 14 | 13 | 12 | 11 | 10 | 9 | 8        |
| DBFEDHR  |    |    |    |    |    |   | RESERVED |
| R/W-0h   |    |    |    |    |    |   | R-0h     |
| 7        | 6  | 5  | 4  | 3  | 2  | 1 | 0        |
| RESERVED |    |    |    |    |    |   | RESERVED |
| R-0h     |    |    |    |    |    |   | R-0h     |

**Table 26-57. DBFEDHR Register Field Descriptions**

| Bit  | Field    | Type | Reset | Description                                                             |
|------|----------|------|-------|-------------------------------------------------------------------------|
| 15-9 | DBFEDHR  | R/W  | 0h    | Dead Band Falling Edge Delay High Resolution Bits<br>Reset type: SYSRSn |
| 8    | RESERVED | R    | 0h    | Reserved                                                                |
| 7-1  | RESERVED | R    | 0h    | Reserved                                                                |
| 0    | RESERVED | R    | 0h    | Reserved                                                                |



### 26.17.2.35 DBFED Register (Offset = 53h) [Reset = 0000h]

DBFED is shown in [Figure 26-127](#) and described in [Table 26-58](#).

Return to the [Summary Table](#).

Dead-Band Generator Falling Edge Delay Count Register

**Figure 26-127. DBFED Register**

|          |    |    |    |        |    |   |   |
|----------|----|----|----|--------|----|---|---|
| 15       | 14 | 13 | 12 | 11     | 10 | 9 | 8 |
| RESERVED |    |    |    | DBFED  |    |   |   |
| R-0h     |    |    |    | R/W-0h |    |   |   |
| 7        | 6  | 5  | 4  | 3      | 2  | 1 | 0 |
| DBFED    |    |    |    |        |    |   |   |
| R/W-0h   |    |    |    |        |    |   |   |

**Table 26-58. DBFED Register Field Descriptions**

| Bit   | Field    | Type | Reset | Description                                                      |
|-------|----------|------|-------|------------------------------------------------------------------|
| 15-14 | RESERVED | R    | 0h    | Reserved                                                         |
| 13-0  | DBFED    | R/W  | 0h    | Falling Edge Delay Count<br>14-bit counter<br>Reset type: SYSRSn |

### 26.17.2.36 TBPHS Register (Offset = 60h) [Reset = 00000000h]

TBPHS is shown in [Figure 26-128](#) and described in [Table 26-59](#).

Return to the [Summary Table](#).

Time Base Phase High

**Figure 26-128. TBPHS Register**

|        |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |         |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31     | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15      | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| TBPHS  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | TBPHSHR |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| R/W-0h |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | R/W-0h  |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |

**Table 26-59. TBPHS Register Field Descriptions**

| Bit   | Field   | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------|---------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | TBPHS   | R/W  | 0h    | <p>Phase Offset Register</p> <p>These bits set time-base counter phase of the selected ePWM relative to the time-base that is supplying the synchronization input signal.</p> <ul style="list-style-type: none"> <li>- If TBCTL[PHSEN] = 0, then the synchronization event is ignored and the time-base counter is not loaded with the phase.</li> <li>- If TBCTL[PHSEN] = 1, then the time-base counter (TBCTR) will be loaded with the phase (TBPHS) when a synchronization event occurs. The synchronization event can be initiated by the input synchronization signal (EPWMxSYNCI) or by a software forced synchronization.</li> </ul> <p>Reset type: SYSRSn</p> |
| 15-0  | TBPHSHR | R/W  | 0h    | <p>Phase Offset (High Resolution) Register.</p> <p>TBPHSHR must not be used. Instead TRREM (HRPWM remainder register) must be used to mimic the functionality of TBPHSHR. The lower 8 bits in this register are ignored - writes are ignored and reads return zero</p> <p>Reset type: SYSRSn</p>                                                                                                                                                                                                                                                                                                                                                                      |

### 26.17.2.37 TBPRDHR Register (Offset = 62h) [Reset = 0000h]

TBPRDHR is shown in [Figure 26-129](#) and described in [Table 26-60](#).

Return to the [Summary Table](#).

Time Base Period High Resolution Register

**Figure 26-129. TBPRDHR Register**

|         |    |    |    |    |    |   |   |
|---------|----|----|----|----|----|---|---|
| 15      | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| TBPRDHR |    |    |    |    |    |   |   |
| R/W-0h  |    |    |    |    |    |   |   |
| 7       | 6  | 5  | 4  | 3  | 2  | 1 | 0 |
| TBPRDHR |    |    |    |    |    |   |   |
| R/W-0h  |    |    |    |    |    |   |   |

**Table 26-60. TBPRDHR Register Field Descriptions**

| Bit  | Field   | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------|---------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-0 | TBPRDHR | R/W  | 0h    | <p>Period High Resolution Bits</p> <p>The upper 8-bits contain the high-resolution portion of the period value. The TBPRDHR register is not affected by the TBCTL[PRDL] bit. Reads from this register always reflect the shadow register. Likewise writes are also to the shadow register. The TBPRDHR register is only used when the high resolution period feature is enabled. This register is only available with ePWM modules which support high-resolution period control.</p> <p>The lower 8 bits in this register are ignored - writes are ignored and reads return zero</p> <p>Reset type: SYSRSn</p> |

### 26.17.2.38 TBPRD Register (Offset = 63h) [Reset = 0000h]

TBPRD is shown in [Figure 26-130](#) and described in [Table 26-61](#).

Return to the [Summary Table](#).

Time Base Period Register

**Figure 26-130. TBPRD Register**

|        |    |    |    |    |    |   |   |
|--------|----|----|----|----|----|---|---|
| 15     | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| TBPRD  |    |    |    |    |    |   |   |
| R/W-0h |    |    |    |    |    |   |   |
| 7      | 6  | 5  | 4  | 3  | 2  | 1 | 0 |
| TBPRD  |    |    |    |    |    |   |   |
| R/W-0h |    |    |    |    |    |   |   |

**Table 26-61. TBPRD Register Field Descriptions**

| Bit  | Field | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------|-------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-0 | TBPRD | R/W  | 0h    | <p>Time Base Period Register</p> <p>These bits determine the period of the time-base counter. This sets the PWM frequency. Shadowing of this register is enabled and disabled by the TBCTL[PRDLD] bit. By default this register is shadowed.</p> <ul style="list-style-type: none"> <li>- If TBCTL[PRDLD] = 0, then the shadow is enabled and any write or read will automatically go to the shadow register. In this case, the active register will be loaded from the shadow register when the time-base counter equals zero.</li> <li>- If TBCTL[PRDLD] = 1, then the shadow is disabled and any write or read will go directly to the active register, that is the register actively controlling the hardware.</li> <li>- The active and shadow registers share the same memory map address.</li> </ul> <p>Reset type: SYSRSn</p> |

### 26.17.2.39 CMPA Register (Offset = 6Ah) [Reset = 00000000h]

CMPA is shown in [Figure 26-131](#) and described in [Table 26-62](#).

Return to the [Summary Table](#).

Counter Compare A Register

**Figure 26-131. CMPA Register**

|        |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31     | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15     | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| CMPA   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | CMPAHR |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| R/W-0h |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | R/W-0h |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |

**Table 26-62. CMPA Register Field Descriptions**

| Bit   | Field  | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|--------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | CMPA   | R/W  | 0h    | <p>Compare A Register</p> <p>The value in the active CMPA register is continuously compared to the time-base counter (TBCTR). When the values are equal, the counter-compare module generates a 'time-base counter equal to counter compare A' event. This event is sent to the action-qualifier where it is qualified and converted it into one or more actions. These actions can be applied to either the EPWMxA or the EPWMxB output depending on the configuration of the AQCTLA and AQCTLB registers. The actions that can be defined in the AQCTLA and AQCTLB registers include:</p> <ul style="list-style-type: none"> <li>- Do nothing</li> <li>- Clear: Pull the EPWMxA and/or EPWMxB signal low</li> <li>- Set: Pull the EPWMxA and/or EPWMxB signal high</li> <li>- Toggle the EPWMxA and/or EPWMxB signal</li> </ul> <p>Shadowing of this register is enabled and disabled by the CMPCTL[SHDWAMODE] bit. By default this register is shadowed.</p> <ul style="list-style-type: none"> <li>- If CMPCTL[SHDWAMODE] = 0, then the shadow is enabled and any write or read will automatically go to the shadow register. In this case, the CMPCTL[LOADAMODE] bit field determines which event will load the active register from the shadow register.</li> <li>- Before a write, the CMPCTL[SHDWAFULL] bit can be read to determine if the shadow register is currently full.</li> <li>- If CMPCTL[SHDWAMODE] = 1, then the shadow register is disabled and any write or read will go directly to the active register, that is the register actively controlling the hardware.</li> <li>- In either mode, the active and shadow registers share the same memory map address.</li> </ul> <p>Reset type: SYSRSn</p> |
| 15-0  | CMPAHR | R/W  | 0h    | <p>Compare A HRPWM Extension Register</p> <p>The UPPER 8-bits contain the high-resolution portion (most significant 8-bits) of the counter-compare A value. CMPA:CMPAHR can be accessed in a single 32-bit read/write. Shadowing is enabled and disabled by the CMPCTL[SHDWAMODE] bit as described for the CMPA register.</p> <p>The lower 8 bits in this register are ignored</p> <p>Reset type: SYSRSn</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

### 26.17.2.40 CMPB Register (Offset = 6Ch) [Reset = 00000000h]

CMPB is shown in [Figure 26-132](#) and described in [Table 26-63](#).

Return to the [Summary Table](#).

Compare B Register

**Figure 26-132. CMPB Register**

|        |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31     | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15     | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| CMPB   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | CMPBHR |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| R/W-0h |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | R/W-0h |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |

**Table 26-63. CMPB Register Field Descriptions**

| Bit   | Field  | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|--------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | CMPB   | R/W  | 0h    | <p>Compare B Register</p> <p>The value in the active CMPB register is continuously compared to the time-base counter (TBCTR). When the values are equal, the counter-compare module generates a 'time-base counter equal to counter compare B' event. This event is sent to the action-qualifier where it is qualified and converted it into one or more actions. These actions can be applied to either the EPWMxA or the EPWMxB output depending on the configuration of the AQCTLA and AQCTLB registers. The actions that can be defined in the AQCTLA and AQCTLB registers include:</p> <ul style="list-style-type: none"> <li>- Do nothing</li> <li>- Clear: Pull the EPWMxA and/or EPWMxB signal low</li> <li>- Set: Pull the EPWMxA and/or EPWMxB signal high</li> <li>- Toggle the EPWMxA and/or EPWMxB signal</li> </ul> <p>Shadowing of this register is enabled and disabled by the CMPCTL[SHDWBMODE] bit. By default this register is shadowed.</p> <ul style="list-style-type: none"> <li>- If CMPCTL[SHDWBMODE] = 0, then the shadow is enabled and any write or read will automatically go to the shadow register. In this case, the CMPCTL[LOADBMODE] bit field determines which event will load the active register from the shadow register.</li> <li>- Before a write, the CMPCTL[SHDWBFULL] bit can be read to determine if the shadow register is currently full.</li> <li>- If CMPCTL[SHDWBMODE] = 1, then the shadow register is disabled and any write or read will go directly to the active register, that is the register actively controlling the hardware.</li> <li>- In either mode, the active and shadow registers share the same memory map address.</li> </ul> <p>Reset type: SYSRSn</p> |
| 15-0  | CMPBHR | R/W  | 0h    | <p>Compare B High Resolution Bits</p> <p>The lower 8 bits in this register are ignored</p> <p>Reset type: SYSRSn</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

### 26.17.2.41 CMPC Register (Offset = 6Fh) [Reset = 0000h]

CMPC is shown in [Figure 26-133](#) and described in [Table 26-64](#).

Return to the [Summary Table](#).

Counter Compare C Register

LINK feature access should always be 16-bit

**Figure 26-133. CMPC Register**

|        |    |    |    |    |    |   |   |
|--------|----|----|----|----|----|---|---|
| 15     | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| CMPC   |    |    |    |    |    |   |   |
| R/W-0h |    |    |    |    |    |   |   |
| 7      | 6  | 5  | 4  | 3  | 2  | 1 | 0 |
| CMPC   |    |    |    |    |    |   |   |
| R/W-0h |    |    |    |    |    |   |   |

**Table 26-64. CMPC Register Field Descriptions**

| Bit  | Field | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------|-------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-0 | CMPC  | R/W  | 0h    | <p>Compare C Register</p> <p>The value in the active CMPC register is continuously compared to the time-base counter (TBCTR). When the values are equal, the counter-compare module generates a 'time-base counter equal to counter compare C' event.</p> <p>Shadowing of this register is enabled and disabled by the CMPCTL2[SHDWCMODE] bit. By default this register is shadowed.</p> <ul style="list-style-type: none"> <li>- If CMPCTL2[SHDWCMODE] = 0, then the shadow is enabled and any write or read will automatically go to the shadow register. In this case, the CMPCTL2[LOADCMODE] bit field determines which event will load the active register from the shadow register:</li> <li>- If CMPCTL2[SHDWCMODE] = 1, then the shadow register is disabled and any write or read will go directly to the active register that is, the register actively controlling the hardware.</li> <li>- In either mode, the active and shadow registers share the same memory map address.</li> </ul> <p>Reset type: SYSRSn</p> |

### 26.17.2.42 CMPD Register (Offset = 71h) [Reset = 0000h]

CMPD is shown in [Figure 26-134](#) and described in [Table 26-65](#).

Return to the [Summary Table](#).

Counter Compare D Register

LINK feature access should always be 16-bit

**Figure 26-134. CMPD Register**

|        |    |    |    |    |    |   |   |
|--------|----|----|----|----|----|---|---|
| 15     | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| CMPD   |    |    |    |    |    |   |   |
| R/W-0h |    |    |    |    |    |   |   |
| 7      | 6  | 5  | 4  | 3  | 2  | 1 | 0 |
| CMPD   |    |    |    |    |    |   |   |
| R/W-0h |    |    |    |    |    |   |   |

**Table 26-65. CMPD Register Field Descriptions**

| Bit  | Field | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|-------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-0 | CMPD  | R/W  | 0h    | <p>Compare D Register</p> <p>The value in the active CMPD register is continuously compared to the time-base counter (TBCTR). When the values are equal, the counter-compare module generates a 'time-base counter equal to counter compare D' event.</p> <p>Shadowing of this register is enabled and disabled by the CMPCTL2[SHDWDMODE] bit. By default this register is shadowed.</p> <ul style="list-style-type: none"> <li>- If CMPCTL2[SHDWDMODE] = 0, then the shadow is enabled and any write or read will automatically go to the shadow register. In this case, the CMPCTL2[LOADDMODE] bit field determines which event will load the active register from the shadow register:</li> <li>- If CMPCTL2[SHDWDMODE] = 1, then the shadow register is disabled and any write or read will go directly to the active register that is, the register actively controlling the hardware.</li> <li>- In either mode, the active and shadow registers share the same memory map address.</li> </ul> <p>Reset type: SYSRStn</p> |



### 26.17.2.43 GLDCTL2 Register (Offset = 74h) [Reset = 0000h]

GLDCTL2 is shown in [Figure 26-135](#) and described in [Table 26-66](#).

Return to the [Summary Table](#).

Global PWM Load Control Register 2

**Figure 26-135. GLDCTL2 Register**

|          |    |    |    |    |    |            |            |
|----------|----|----|----|----|----|------------|------------|
| 15       | 14 | 13 | 12 | 11 | 10 | 9          | 8          |
| RESERVED |    |    |    |    |    |            |            |
| R-0-0h   |    |    |    |    |    |            |            |
| 7        | 6  | 5  | 4  | 3  | 2  | 1          | 0          |
| RESERVED |    |    |    |    |    | GFRCLD     | OSHTLD     |
| R-0-0h   |    |    |    |    |    | R-0/W1S-0h | R-0/W1S-0h |

**Table 26-66. GLDCTL2 Register Field Descriptions**

| Bit  | Field    | Type    | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                  |
|------|----------|---------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-2 | RESERVED | R-0     | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                     |
| 1    | GFRCLD   | R-0/W1S | 0h    | Force Load Event in One Shot Mode<br>0: Writing of 0 will be ignored. Always reads back a 0.<br>1: Force one load event at the input of the event pre-scale counter. This bit is intended to be used for testing and/or software force loading of the events in global load mode.<br>Reset type: SYSRSn                                                                                      |
| 0    | OSHTLD   | R-0/W1S | 0h    | Enable Reload Event in One Shot Mode<br>0: Writing of 0 will be ignored. Always reads back a 0.<br>1: Turns the one shot latch condition ON. Upon occurrence of a chosen load strobe, one shadow to active reload occurs and the latch will be cleared. Hence writing 1 to this bit would allow one load strobe event to pass through and block further strobe events.<br>Reset type: SYSRSn |

### 26.17.2.44 SWVDELVAL Register (Offset = 77h) [Reset = 0000h]

SWVDELVAL is shown in [Figure 26-136](#) and described in [Table 26-67](#).

Return to the [Summary Table](#).

Software Valley Mode Delay Register

**Figure 26-136. SWVDELVAL Register**

|           |    |    |    |    |    |   |   |
|-----------|----|----|----|----|----|---|---|
| 15        | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| SWVDELVAL |    |    |    |    |    |   |   |
| R/W-0h    |    |    |    |    |    |   |   |
| 7         | 6  | 5  | 4  | 3  | 2  | 1 | 0 |
| SWVDELVAL |    |    |    |    |    |   |   |
| R/W-0h    |    |    |    |    |    |   |   |

**Table 26-67. SWVDELVAL Register Field Descriptions**

| Bit  | Field     | Type | Reset | Description                                                                                                                                                                                                |
|------|-----------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-0 | SWVDELVAL | R/W  | 0h    | Software Valley Delay Value Register<br>This register can be optionally used define offset value for the hardware calculated delay HWDELAYVAL as defined in VCAPCTL[VDELAYDIV] bits.<br>Reset type: SYSRSn |

### 26.17.2.45 TZSEL Register (Offset = 80h) [Reset = 0000h]

TZSEL is shown in [Figure 26-137](#) and described in [Table 26-68](#).

Return to the [Summary Table](#).

Trip Zone Select Register

**Figure 26-137. TZSEL Register**

| 15      | 14      | 13     | 12     | 11     | 10     | 9      | 8      |
|---------|---------|--------|--------|--------|--------|--------|--------|
| DCBEVT1 | DCAEVT1 | OSHT6  | OSHT5  | OSHT4  | OSHT3  | OSHT2  | OSHT1  |
| R/W-0h  | R/W-0h  | R/W-0h | R/W-0h | R/W-0h | R/W-0h | R/W-0h | R/W-0h |
| 7       | 6       | 5      | 4      | 3      | 2      | 1      | 0      |
| DCBEVT2 | DCAEVT2 | CBC6   | CBC5   | CBC4   | CBC3   | CBC2   | CBC1   |
| R/W-0h  | R/W-0h  | R/W-0h | R/W-0h | R/W-0h | R/W-0h | R/W-0h | R/W-0h |

**Table 26-68. TZSEL Register Field Descriptions**

| Bit | Field   | Type | Reset | Description                                                                                                                                                                                          |
|-----|---------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15  | DCBEVT1 | R/W  | 0h    | Digital Compare Output B Event 1 Select<br>0: Disable DCBEVT1 as one-shot-trip source for this ePWM module.<br>1: Enable DCBEVT1 as one-shot-trip source for this ePWM module.<br>Reset type: SYSRSn |
| 14  | DCAEVT1 | R/W  | 0h    | Digital Compare Output A Event 1 Select<br>0: Disable DCAEVT1 as one-shot-trip source for this ePWM module.<br>1: Enable DCAEVT1 as one-shot-trip source for this ePWM module.<br>Reset type: SYSRSn |
| 13  | OSHT6   | R/W  | 0h    | Trip-zone 6 (TZ6) Select<br>0: Disable TZ6 as a one-shot trip source for this ePWM module<br>1: Enable TZ6 as a one-shot trip source for this ePWM module<br>Reset type: SYSRSn                      |
| 12  | OSHT5   | R/W  | 0h    | Trip-zone 5 (TZ5) Select<br>0: Disable TZ5 as a one-shot trip source for this ePWM module<br>1: Enable TZ5 as a one-shot trip source for this ePWM module<br>Reset type: SYSRSn                      |
| 11  | OSHT4   | R/W  | 0h    | Trip-zone 4 (TZ4) Select<br>0: Disable TZ4 as a one-shot trip source for this ePWM module<br>1: Enable TZ4 as a one-shot trip source for this ePWM module<br>Reset type: SYSRSn                      |
| 10  | OSHT3   | R/W  | 0h    | Trip-zone 3 (TZ3) Select<br>0: Disable TZ3 as a one-shot trip source for this ePWM module<br>1: Enable TZ3 as a one-shot trip source for this ePWM module<br>Reset type: SYSRSn                      |
| 9   | OSHT2   | R/W  | 0h    | Trip-zone 2 (TZ2) Select<br>0: Disable TZ2 as a one-shot trip source for this ePWM module<br>1: Enable TZ2 as a one-shot trip source for this ePWM module<br>Reset type: SYSRSn                      |
| 8   | OSHT1   | R/W  | 0h    | Trip-zone 1 (TZ1) Select<br>0: Disable TZ1 as a one-shot trip source for this ePWM module<br>1: Enable TZ1 as a one-shot trip source for this ePWM module<br>Reset type: SYSRSn                      |
| 7   | DCBEVT2 | R/W  | 0h    | Digital Compare Output B Event 2 Select<br>0: Disable DCBEVT2 as a CBC trip source for this ePWM module<br>1: Enable DCBEVT2 as a CBC trip source for this ePWM module<br>Reset type: SYSRSn         |

**Table 26-68. TZSEL Register Field Descriptions (continued)**

| Bit | Field   | Type | Reset | Description                                                                                                                                                                                  |
|-----|---------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6   | DCAEVT2 | R/W  | 0h    | Digital Compare Output A Event 2 Select<br>0: Disable DCAEVT2 as a CBC trip source for this ePWM module<br>1: Enable DCAEVT2 as a CBC trip source for this ePWM module<br>Reset type: SYSRSn |
| 5   | CBC6    | R/W  | 0h    | Trip-zone 6 (TZ6) Select<br>0: Disable TZ6 as a CBC trip source for this ePWM module<br>1: Enable TZ6 as a CBC trip source for this ePWM module<br>Reset type: SYSRSn                        |
| 4   | CBC5    | R/W  | 0h    | Trip-zone 5 (TZ5) Select<br>0: Disable TZ5 as a CBC trip source for this ePWM module<br>1: Enable TZ5 as a CBC trip source for this ePWM module<br>Reset type: SYSRSn                        |
| 3   | CBC4    | R/W  | 0h    | Trip-zone 4 (TZ4) Select<br>0: Disable TZ4 as a CBC trip source for this ePWM module<br>1: Enable TZ4 as a CBC trip source for this ePWM module<br>Reset type: SYSRSn                        |
| 2   | CBC3    | R/W  | 0h    | Trip-zone 3 (TZ3) Select<br>0: Disable TZ3 as a CBC trip source for this ePWM module<br>1: Enable TZ3 as a CBC trip source for this ePWM module<br>Reset type: SYSRSn                        |
| 1   | CBC2    | R/W  | 0h    | Trip-zone 2 (TZ2) Select<br>0: Disable TZ2 as a CBC trip source for this ePWM module<br>1: Enable TZ2 as a CBC trip source for this ePWM module<br>Reset type: SYSRSn                        |
| 0   | CBC1    | R/W  | 0h    | Trip-zone 1 (TZ1) Select<br>0: Disable TZ1 as a CBC trip source for this ePWM module<br>1: Enable TZ1 as a CBC trip source for this ePWM module<br>Reset type: SYSRSn                        |

### 26.17.2.46 TZDCSEL Register (Offset = 82h) [Reset = 0000h]

TZDCSEL is shown in [Figure 26-138](#) and described in [Table 26-69](#).

Return to the [Summary Table](#).

Trip Zone Digital Comparator Select Register

**Figure 26-138. TZDCSEL Register**

|          |    |         |    |         |         |   |         |
|----------|----|---------|----|---------|---------|---|---------|
| 15       | 14 | 13      | 12 | 11      | 10      | 9 | 8       |
| RESERVED |    |         |    | DCBEVT2 |         |   | DCBEVT1 |
| R-0-0h   |    |         |    | R/W-0h  |         |   | R/W-0h  |
| 7        | 6  | 5       | 4  | 3       | 2       | 1 | 0       |
| DCBEVT1  |    | DCAEVT2 |    |         | DCAEVT1 |   |         |
| R/W-0h   |    | R/W-0h  |    |         | R/W-0h  |   |         |

**Table 26-69. TZDCSEL Register Field Descriptions**

| Bit   | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                         |
|-------|----------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-12 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                            |
| 11-9  | DCBEVT2  | R/W  | 0h    | Digital Compare Output B Event 2 Selection<br>000: Event disabled<br>001: DCBH = low, DCBL = don't care<br>010: DCBH = high, DCBL = don't care<br>011: DCBL = low, DCBH = don't care<br>100: DCBL = high, DCBH = don't care<br>101: DCBL = high, DCBH = low<br>110: Reserved<br>111: Reserved<br>Reset type: SYSRSn |
| 8-6   | DCBEVT1  | R/W  | 0h    | Digital Compare Output B Event 1 Selection<br>000: Event disabled<br>001: DCBH = low, DCBL = don't care<br>010: DCBH = high, DCBL = don't care<br>011: DCBL = low, DCBH = don't care<br>100: DCBL = high, DCBH = don't care<br>101: DCBL = high, DCBH = low<br>110: Reserved<br>111: Reserved<br>Reset type: SYSRSn |
| 5-3   | DCAEVT2  | R/W  | 0h    | Digital Compare Output A Event 2 Selection<br>000: Event disabled<br>001: DCAH = low, DCAL = don't care<br>010: DCAH = high, DCAL = don't care<br>011: DCAL = low, DCAH = don't care<br>100: DCAL = high, DCAH = don't care<br>101: DCAL = high, DCAH = low<br>110: Reserved<br>111: Reserved<br>Reset type: SYSRSn |
| 2-0   | DCAEVT1  | R/W  | 0h    | Digital Compare Output A Event 1 Selection<br>000: Event disabled<br>001: DCAH = low, DCAL = don't care<br>010: DCAH = high, DCAL = don't care<br>011: DCAL = low, DCAH = don't care<br>100: DCAL = high, DCAH = don't care<br>101: DCAL = high, DCAH = low<br>110: Reserved<br>111: Reserved<br>Reset type: SYSRSn |

### 26.17.2.47 TZCTL Register (Offset = 84h) [Reset = 0000h]

TZCTL is shown in [Figure 26-139](#) and described in [Table 26-70](#).

Return to the [Summary Table](#).

Trip Zone Control Register

**Figure 26-139. TZCTL Register**

|          |    |         |    |         |    |         |   |
|----------|----|---------|----|---------|----|---------|---|
| 15       | 14 | 13      | 12 | 11      | 10 | 9       | 8 |
| RESERVED |    |         |    | DCBEVT2 |    | DCBEVT1 |   |
| R-0-0h   |    |         |    | R/W-0h  |    | R/W-0h  |   |
| 7        | 6  | 5       | 4  | 3       | 2  | 1       | 0 |
| DCAEVT2  |    | DCAEVT1 |    | TZB     |    | TZA     |   |
| R/W-0h   |    | R/W-0h  |    | R/W-0h  |    | R/W-0h  |   |

**Table 26-70. TZCTL Register Field Descriptions**

| Bit   | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|----------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-12 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                         |
| 11-10 | DCBEVT2  | R/W  | 0h    | Digital Compare Output B Event 2 Action On EPWMxB<br>00: High-impedance (EPWMxB = High-impedance state)<br>01: Force EPWMxB to a high state.<br>10: Force EPWMxB to a low state.<br>11: Do Nothing, trip action is disabled<br>Reset type: SYSRSn                                                                                                                                                                |
| 9-8   | DCBEVT1  | R/W  | 0h    | Digital Compare Output B Event 1 Action On EPWMxB<br>00: High-impedance (EPWMxB = High-impedance state)<br>01: Force EPWMxB to a high state.<br>10: Force EPWMxB to a low state.<br>11: Do Nothing, trip action is disabled<br>Reset type: SYSRSn                                                                                                                                                                |
| 7-6   | DCAEVT2  | R/W  | 0h    | Digital Compare Output A Event 2 Action On EPWMxA<br>00: High-impedance (EPWMxA = High-impedance state)<br>01: Force EPWMxA to a high state.<br>10: Force EPWMxA to a low state.<br>11: Do Nothing, trip action is disabled<br>Reset type: SYSRSn                                                                                                                                                                |
| 5-4   | DCAEVT1  | R/W  | 0h    | Digital Compare Output A Event 1 Action On EPWMxA<br>00: High-impedance (EPWMxA = High-impedance state)<br>01: Force EPWMxA to a high state.<br>10: Force EPWMxA to a low state.<br>11: Do Nothing, trip action is disabled<br>Reset type: SYSRSn                                                                                                                                                                |
| 3-2   | TZB      | R/W  | 0h    | TZ1 to TZ6, DCAEVT1/2, DCBEVT1/2 Trip Action On EPWMxB<br>When a trip event occurs the following action is taken on output EPWMxB. Which trip-zone pins can cause an event is defined in the TZSEL register.<br>00: High-impedance (EPWMxB = High-impedance state)<br>01: Force EPWMxB to a high state<br>10: Force EPWMxB to a low state<br>11: Do nothing, no action is taken on EPWMxB.<br>Reset type: SYSRSn |

**Table 26-70. TZCTL Register Field Descriptions (continued)**

| Bit | Field | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|-------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-0 | TZA   | R/W  | 0h    | <p>TZ1 to TZ6, DCAEVT1/2, DCBEVT1/2 Trip Action On EPWMxA</p> <p>When a trip event occurs the following action is taken on output EPWMxA. Which trip-zone pins can cause an event is defined in the TZSEL register.</p> <p>00: High-impedance (EPWMxA = High-impedance state)</p> <p>01: Force EPWMxA to a high state</p> <p>10: Force EPWMxA to a low state</p> <p>11: Do nothing, no action is taken on EPWMxA.</p> <p>Reset type: SYSRSn</p> |

### 26.17.2.48 TZCTL2 Register (Offset = 85h) [Reset = 0000h]

TZCTL2 is shown in [Figure 26-140](#) and described in [Table 26-71](#).

Return to the [Summary Table](#).

Additional Trip Zone Control Register

**Figure 26-140. TZCTL2 Register**

|        |          |        |    |    |        |   |        |
|--------|----------|--------|----|----|--------|---|--------|
| 15     | 14       | 13     | 12 | 11 | 10     | 9 | 8      |
| ETZE   | RESERVED |        |    |    | TZBD   |   | TZBU   |
| R/W-0h | R-0-0h   |        |    |    | R/W-0h |   | R/W-0h |
| 7      | 6        | 5      | 4  | 3  | 2      | 1 | 0      |
| TZBU   |          | TZAD   |    |    | TZAU   |   |        |
| R/W-0h |          | R/W-0h |    |    | R/W-0h |   |        |

**Table 26-71. TZCTL2 Register Field Descriptions**

| Bit   | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                        |
|-------|----------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15    | ETZE     | R/W  | 0h    | TZCTL2 Enable<br>0: Use trip action from TZCTL (legacy EPWM compatibility)<br>1: Use trip action defined in TZCTL2, TZCTLDCA and TZCTLDCB.<br>Settings in TZCTL are ignored<br>Reset type: SYSRSn                                                                                                                                                                  |
| 14-12 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                           |
| 11-9  | TZBD     | R/W  | 0h    | TZ1 to TZ6 Trip Action On EPWMxB while Count direction is DOWN<br>000: HiZ (EPWMxB = HiZ state)<br>001: Forced Hi (EPWMxB = High state)<br>010: Forced Lo (EPWMxB = Lo state)<br>011: Toggle (Low -> High, High -> Low)<br>100: Reserved<br>101: Reserved<br>110: Reserved<br>111: Do Nothing, trip action is disabled<br>Reset type: SYSRSn                       |
| 8-6   | TZBU     | R/W  | 0h    | TZ1 to TZ6, DCAEVT1/2, DCBEVT1/2 Trip Action On EPWMxB while Count direction is UP<br>000: HiZ (EPWMxB = HiZ state)<br>001: Forced Hi (EPWMxB = High state)<br>010: Forced Lo (EPWMxB = Lo state)<br>011: Toggle (Low -> High, High -> Low)<br>100: Reserved<br>101: Reserved<br>110: Reserved<br>111: Do Nothing, trip action is disabled<br>Reset type: SYSRSn   |
| 5-3   | TZAD     | R/W  | 0h    | TZ1 to TZ6, DCAEVT1/2, DCBEVT1/2 Trip Action On EPWMxA while Count direction is DOWN<br>000: HiZ (EPWMxA = HiZ state)<br>001: Forced Hi (EPWMxA = High state)<br>010: Forced Lo (EPWMxA = Lo state)<br>011: Toggle (Low -> High, High -> Low)<br>100: Reserved<br>101: Reserved<br>110: Reserved<br>111: Do Nothing, trip action is disabled<br>Reset type: SYSRSn |



**Table 26-71. TZCTL2 Register Field Descriptions (continued)**

| Bit | Field | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                      |
|-----|-------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2-0 | TZAU  | R/W  | 0h    | TZ1 to TZ6, DCAEVT1/2, DCBEVT1/2 Trip Action On EPWMxA while Count direction is UP<br>000: HiZ (EPWMxA = HiZ state)<br>001: Forced Hi (EPWMxA = High state)<br>010: Forced Lo (EPWMxA = Lo state)<br>011: Toggle (Low -> High, High -> Low)<br>100: Reserved<br>101: Reserved<br>110: Reserved<br>111: Do Nothing, trip action is disabled<br>Reset type: SYSRSn |

### 26.17.2.49 TZCTLDCA Register (Offset = 86h) [Reset = 0000h]

TZCTLDCA is shown in [Figure 26-141](#) and described in [Table 26-72](#).

Return to the [Summary Table](#).

Trip Zone Control Register Digital Compare A

**Figure 26-141. TZCTLDCA Register**

|          |    |          |    |          |          |          |   |
|----------|----|----------|----|----------|----------|----------|---|
| 15       | 14 | 13       | 12 | 11       | 10       | 9        | 8 |
| RESERVED |    |          |    | DCAEVT2D |          | DCAEVT2U |   |
| R-0-0h   |    |          |    | R/W-0h   |          | R/W-0h   |   |
| 7        | 6  | 5        | 4  | 3        | 2        | 1        | 0 |
| DCAEVT2U |    | DCAEVT1D |    |          | DCAEVT1U |          |   |
| R/W-0h   |    | R/W-0h   |    |          | R/W-0h   |          |   |

**Table 26-72. TZCTLDCA Register Field Descriptions**

| Bit   | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                   |
|-------|----------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-12 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                      |
| 11-9  | DCAEVT2D | R/W  | 0h    | Digital Compare Output A Event 2 Action On EPWMxA while Count direction is DOWN<br>000: HiZ (EPWMxA = HiZ state)<br>001: Forced Hi (EPWMxA = High state)<br>010: Forced Lo (EPWMxA = Lo state)<br>011: Toggle (Low -> High, High -> Low)<br>100: Reserved<br>101: Reserved<br>110: Reserved<br>111: Do Nothing, trip action is disabled<br>Reset type: SYSRSn |
| 8-6   | DCAEVT2U | R/W  | 0h    | Digital Compare Output A Event 2 Action On EPWMxA while Count direction is UP<br>000: HiZ (EPWMxA = HiZ state)<br>001: Forced Hi (EPWMxA = High state)<br>010: Forced Lo (EPWMxA = Lo state)<br>011: Toggle (Low -> High, High -> Low)<br>100: Reserved<br>101: Reserved<br>110: Reserved<br>111: Do Nothing, trip action is disabled<br>Reset type: SYSRSn   |
| 5-3   | DCAEVT1D | R/W  | 0h    | Digital Compare Output A Event 1 Action On EPWMxA while Count direction is DOWN<br>000: HiZ (EPWMxA = HiZ state)<br>001: Forced Hi (EPWMxA = High state)<br>010: Forced Lo (EPWMxA = Lo state)<br>011: Toggle (Low -> High, High -> Low)<br>100: Reserved<br>101: Reserved<br>110: Reserved<br>111: Do Nothing, trip action is disabled<br>Reset type: SYSRSn |

**Table 26-72. TZCTLDCA Register Field Descriptions (continued)**

| Bit | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                 |
|-----|----------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2-0 | DCAEVT1U | R/W  | 0h    | Digital Compare Output A Event 1 Action On EPWMxA while Count direction is UP<br>000: HiZ (EPWMxA = HiZ state)<br>001: Forced Hi (EPWMxA = High state)<br>010: Forced Lo (EPWMxA = Lo state)<br>011: Toggle (Low -> High, High -> Low)<br>100: Reserved<br>101: Reserved<br>110: Reserved<br>111: Do Nothing, trip action is disabled<br>Reset type: SYSRSn |

### 26.17.2.50 TZCTLDCB Register (Offset = 87h) [Reset = 0000h]

TZCTLDCB is shown in [Figure 26-142](#) and described in [Table 26-73](#).

Return to the [Summary Table](#).

Trip Zone Control Register Digital Compare B

**Figure 26-142. TZCTLDCB Register**

|          |    |          |    |          |          |          |   |
|----------|----|----------|----|----------|----------|----------|---|
| 15       | 14 | 13       | 12 | 11       | 10       | 9        | 8 |
| RESERVED |    |          |    | DCBEVT2D |          | DCBEVT2U |   |
| R-0-0h   |    |          |    | R/W-0h   |          | R/W-0h   |   |
| 7        | 6  | 5        | 4  | 3        | 2        | 1        | 0 |
| DCBEVT2U |    | DCBEVT1D |    |          | DCBEVT1U |          |   |
| R/W-0h   |    | R/W-0h   |    |          | R/W-0h   |          |   |

**Table 26-73. TZCTLDCB Register Field Descriptions**

| Bit   | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                   |
|-------|----------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-12 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                      |
| 11-9  | DCBEVT2D | R/W  | 0h    | Digital Compare Output B Event 2 Action On EPWMxB while Count direction is DOWN<br>000: HiZ (EPWMxB = HiZ state)<br>001: Forced Hi (EPWMxB = High state)<br>010: Forced Lo (EPWMxB = Lo state)<br>011: Toggle (Low -> High, High -> Low)<br>100: Reserved<br>101: Reserved<br>110: Reserved<br>111: Do Nothing, trip action is disabled<br>Reset type: SYSRSn |
| 8-6   | DCBEVT2U | R/W  | 0h    | Digital Compare Output B Event 2 Action On EPWMxB while Count direction is UP<br>000: HiZ (EPWMxB = HiZ state)<br>001: Forced Hi (EPWMxB = High state)<br>010: Forced Lo (EPWMxB = Lo state)<br>011: Toggle (Low -> High, High -> Low)<br>100: Reserved<br>101: Reserved<br>110: Reserved<br>111: Do Nothing, trip action is disabled<br>Reset type: SYSRSn   |
| 5-3   | DCBEVT1D | R/W  | 0h    | Digital Compare Output B Event 1 Action On EPWMxB while Count direction is DOWN<br>000: HiZ (EPWMxB = HiZ state)<br>001: Forced Hi (EPWMxB = High state)<br>010: Forced Lo (EPWMxB = Lo state)<br>011: Toggle (Low -> High, High -> Low)<br>100: Reserved<br>101: Reserved<br>110: Reserved<br>111: Do Nothing, trip action is disabled<br>Reset type: SYSRSn |

**Table 26-73. TZCTLDCB Register Field Descriptions (continued)**

| Bit | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                 |
|-----|----------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2-0 | DCBEVT1U | R/W  | 0h    | Digital Compare Output B Event 1 Action On EPWMxB while Count direction is UP<br>000: HiZ (EPWMxB = HiZ state)<br>001: Forced Hi (EPWMxB = High state)<br>010: Forced Lo (EPWMxB = Lo state)<br>011: Toggle (Low -> High, High -> Low)<br>100: Reserved<br>101: Reserved<br>110: Reserved<br>111: Do Nothing, trip action is disabled<br>Reset type: SYSRSn |

### 26.17.2.51 TZEINT Register (Offset = 8Dh) [Reset = 0000h]

TZEINT is shown in [Figure 26-143](#) and described in [Table 26-74](#).

Return to the [Summary Table](#).

Trip Zone Enable Interrupt Register

**Figure 26-143. TZEINT Register**

|          |         |         |         |         |        |        |          |
|----------|---------|---------|---------|---------|--------|--------|----------|
| 15       | 14      | 13      | 12      | 11      | 10     | 9      | 8        |
| RESERVED |         |         |         |         |        |        |          |
| R-0-0h   |         |         |         |         |        |        |          |
| 7        | 6       | 5       | 4       | 3       | 2      | 1      | 0        |
| RESERVED | DCBEVT2 | DCBEVT1 | DCAEVT2 | DCAEVT1 | OST    | CBC    | RESERVED |
| R-0-0h   | R/W-0h  | R/W-0h  | R/W-0h  | R/W-0h  | R/W-0h | R/W-0h | R-0-0h   |

**Table 26-74. TZEINT Register Field Descriptions**

| Bit  | Field    | Type | Reset | Description                                                                                                                                                                                                                  |
|------|----------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-7 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                     |
| 6    | DCBEVT2  | R/W  | 0h    | Digital Compare Output B Event 2 Interrupt Enable<br>0: Disabled<br>1: Enabled<br>Reset type: SYSRSn                                                                                                                         |
| 5    | DCBEVT1  | R/W  | 0h    | Digital Compare Output B Event 1 Interrupt Enable<br>0: Disabled<br>1: Enabled<br>Reset type: SYSRSn                                                                                                                         |
| 4    | DCAEVT2  | R/W  | 0h    | Digital Compare Output A Event 2 Interrupt Enable<br>0: Disabled<br>1: Enabled<br>Reset type: SYSRSn                                                                                                                         |
| 3    | DCAEVT1  | R/W  | 0h    | Digital Compare Output A Event 1 Interrupt Enable<br>0: Disabled<br>1: Enabled<br>Reset type: SYSRSn                                                                                                                         |
| 2    | OST      | R/W  | 0h    | Trip-zone One-Shot Interrupt Enable<br>0: Disable one-shot interrupt generation<br>1: Enable Interrupt generation<br>a one-shot trip event will cause a EPWMx_TZINT PIE interrupt.<br>Reset type: SYSRSn                     |
| 1    | CBC      | R/W  | 0h    | Trip-zone Cycle-by-Cycle Interrupt Enable<br>0: Disable cycle-by-cycle interrupt generation.<br>1: Enable interrupt generation<br>a cycle-by-cycle trip event will cause an EPWMx_TZINT PIE interrupt.<br>Reset type: SYSRSn |
| 0    | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                     |

### 26.17.2.52 TZFLG Register (Offset = 93h) [Reset = 0000h]

TZFLG is shown in [Figure 26-144](#) and described in [Table 26-75](#).

Return to the [Summary Table](#).

Trip Zone Flag Register

**Figure 26-144. TZFLG Register**

|          |         |         |         |         |      |      |      |
|----------|---------|---------|---------|---------|------|------|------|
| 15       | 14      | 13      | 12      | 11      | 10   | 9    | 8    |
| RESERVED |         |         |         |         |      |      |      |
| R-0-0h   |         |         |         |         |      |      |      |
| 7        | 6       | 5       | 4       | 3       | 2    | 1    | 0    |
| RESERVED | DCBEVT2 | DCBEVT1 | DCAEVT2 | DCAEVT1 | OST  | CBC  | INT  |
| R-0-0h   | R-0h    | R-0h    | R-0h    | R-0h    | R-0h | R-0h | R-0h |

**Table 26-75. TZFLG Register Field Descriptions**

| Bit  | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                          |
|------|----------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-7 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                             |
| 6    | DCBEVT2  | R    | 0h    | Latched Status Flag for Digital Compare Output B Event 2<br>0: Indicates no trip event has occurred on DCBEVT2<br>1: Indicates a trip event has occurred for the event defined for DCBEVT2<br>Reset type: SYSRSn                                                                     |
| 5    | DCBEVT1  | R    | 0h    | Latched Status Flag for Digital Compare Output B Event 1<br>0: Indicates no trip event has occurred on DCBEVT1<br>1: Indicates a trip event has occurred for the event defined for DCBEVT1<br>Reset type: SYSRSn                                                                     |
| 4    | DCAEVT2  | R    | 0h    | Latched Status Flag for Digital Compare Output A Event 2<br>0: Indicates no trip event has occurred on DCAEVT2<br>1: Indicates a trip event has occurred for the event defined for DCAEVT2<br>Reset type: SYSRSn                                                                     |
| 3    | DCAEVT1  | R    | 0h    | Latched Status Flag for Digital Compare Output A Event 1<br>0: Indicates no trip event has occurred on DCAEVT1<br>1: Indicates a trip event has occurred for the event defined for DCAEVT1<br>Reset type: SYSRSn                                                                     |
| 2    | OST      | R    | 0h    | Latched Status Flag for A One-Shot Trip Event<br>0: No one-shot trip event has occurred.<br>1: Indicates a trip event has occurred on a pin selected as a one-shot trip source.<br>This bit is cleared by writing the appropriate value to the TZCLR register.<br>Reset type: SYSRSn |

**Table 26-75. TZFLG Register Field Descriptions (continued)**

| Bit | Field | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----|-------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | CBC   | R    | 0h    | <p>Latched Status Flag for Cycle-By-Cycle Trip Event</p> <p>0: No cycle-by-cycle trip event has occurred.</p> <p>1: Indicates a trip event has occurred on a signal selected as a cycle-by-cycle trip source. The TZFLG[CBC] bit will remain set until it is manually cleared by the user. If the cycle-by-cycle trip event is still present when the CBC bit is cleared, then CBC will be immediately set again. The specified condition on the signal is automatically cleared when the ePWM time-base counter reaches zero (TBCTR = 0x00) if the trip condition is no longer present. The condition on the signal is only cleared when the TBCTR = 0x00 no matter where in the cycle the CBC flag is cleared.</p> <p>This bit is cleared by writing the appropriate value to the TZCLR register.</p> <p>Reset type: SYSRSn</p> |
| 0   | INT   | R    | 0h    | <p>Latched Trip Interrupt Status Flag</p> <p>0: Indicates no interrupt has been generated.</p> <p>1: Indicates an EPWMx_TZINT PIE interrupt was generated because of a trip condition.</p> <p>No further EPWMx_TZINT PIE interrupts will be generated until this flag is cleared. If the interrupt flag is cleared when either CBC or OST is set, then another interrupt pulse will be generated. Clearing all flag bits will prevent further interrupts. This bit is cleared by writing the appropriate value to the TZCLR register.</p> <p>Reset type: SYSRSn</p>                                                                                                                                                                                                                                                               |



### 26.17.2.53 TZCBCFLG Register (Offset = 94h) [Reset = 0000h]

TZCBCFLG is shown in [Figure 26-145](#) and described in [Table 26-76](#).

Return to the [Summary Table](#).

Trip Zone CBC Flag Register

**Figure 26-145. TZCBCFLG Register**

|          |         |      |      |      |      |      |      |
|----------|---------|------|------|------|------|------|------|
| 15       | 14      | 13   | 12   | 11   | 10   | 9    | 8    |
| RESERVED |         |      |      |      |      |      |      |
| R-0-0h   |         |      |      |      |      |      |      |
| 7        | 6       | 5    | 4    | 3    | 2    | 1    | 0    |
| DCBEVT2  | DCAEVT2 | CBC6 | CBC5 | CBC4 | CBC3 | CBC2 | CBC1 |
| R-0h     | R-0h    | R-0h | R-0h | R-0h | R-0h | R-0h | R-0h |

**Table 26-76. TZCBCFLG Register Field Descriptions**

| Bit  | Field    | Type | Reset | Description                                                                                                                                                                                                                                |
|------|----------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-8 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                   |
| 7    | DCBEVT2  | R    | 0h    | Latched Status Flag for Digital Compare B Output Event 2 Trip Latch<br>0: Reading a 0 indicates that no trip has occurred on DCBEVT2.<br>1: Reading a 1 indicates a trip has occurred on the DCBEVT2 selected event.<br>Reset type: SYSRSn |
| 6    | DCAEVT2  | R    | 0h    | Latched Status Flag for Digital Compare A Output Event 2 Trip Latch<br>0: Reading a 0 indicates that no trip has occurred on DCAEVT2.<br>1: Reading a 1 indicates a trip has occurred on the DCAEVT2 selected event.<br>Reset type: SYSRSn |
| 5    | CBC6     | R    | 0h    | Latched Status Flag for CBC6 Trip Latch<br>0: Reading a 0 indicates that no trip has occurred on CBC6.<br>1: Reading a 1 indicates a trip has occurred on the CBC6 selected event.<br>Reset type: SYSRSn                                   |
| 4    | CBC5     | R    | 0h    | Latched Status Flag for CBC5 Trip Latch<br>0: Reading a 0 indicates that no trip has occurred on CBC5.<br>1: Reading a 1 indicates a trip has occurred on the CBC5 selected event.<br>Reset type: SYSRSn                                   |
| 3    | CBC4     | R    | 0h    | Latched Status Flag for CBC4 Trip Latch<br>0: Reading a 0 indicates that no trip has occurred on CBC4.<br>1: Reading a 1 indicates a trip has occurred on the CBC4 selected event.<br>Reset type: SYSRSn                                   |
| 2    | CBC3     | R    | 0h    | Latched Status Flag for CBC3 Trip Latch<br>0: Reading a 0 indicates that no trip has occurred on CBC3.<br>1: Reading a 1 indicates a trip has occurred on the CBC3 selected event.<br>Reset type: SYSRSn                                   |
| 1    | CBC2     | R    | 0h    | Latched Status Flag for CBC2 Trip Latch<br>0: Reading a 0 indicates that no trip has occurred on CBC2.<br>1: Reading a 1 indicates a trip has occurred on the CBC2 selected event.<br>Reset type: SYSRSn                                   |

**Table 26-76. TZCBCFLG Register Field Descriptions (continued)**

| Bit | Field | Type | Reset | Description                                                                                                                                                                                              |
|-----|-------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0   | CBC1  | R    | 0h    | Latched Status Flag for CBC1 Trip Latch<br>0: Reading a 0 indicates that no trip has occurred on CBC1.<br>1: Reading a 1 indicates a trip has occurred on the CBC1 selected event.<br>Reset type: SYSRSn |

### 26.17.2.54 TZOSTFLG Register (Offset = 95h) [Reset = 0000h]

TZOSTFLG is shown in [Figure 26-146](#) and described in [Table 26-77](#).

Return to the [Summary Table](#).

Trip Zone OST Flag Register

**Figure 26-146. TZOSTFLG Register**

|          |         |      |      |      |      |      |      |
|----------|---------|------|------|------|------|------|------|
| 15       | 14      | 13   | 12   | 11   | 10   | 9    | 8    |
| RESERVED |         |      |      |      |      |      |      |
| R-0-0h   |         |      |      |      |      |      |      |
| 7        | 6       | 5    | 4    | 3    | 2    | 1    | 0    |
| DCBEVT1  | DCAEVT1 | OST6 | OST5 | OST4 | OST3 | OST2 | OST1 |
| R-0h     | R-0h    | R-0h | R-0h | R-0h | R-0h | R-0h | R-0h |

**Table 26-77. TZOSTFLG Register Field Descriptions**

| Bit  | Field    | Type | Reset | Description                                                                                                                                                                                                                                |
|------|----------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-8 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                   |
| 7    | DCBEVT1  | R    | 0h    | Latched Status Flag for Digital Compare B Output Event 1 Trip Latch<br>0: Reading a 0 indicates that no trip has occurred on DCBEVT1.<br>1: Reading a 1 indicates a trip has occurred on the DCBEVT1 selected event.<br>Reset type: SYSRSn |
| 6    | DCAEVT1  | R    | 0h    | Latched Status Flag for Digital Compare A Output Event 1 Trip Latch<br>0: Reading a 0 indicates that no trip has occurred on DCAEVT1.<br>1: Reading a 1 indicates a trip has occurred on the DCAEVT1 selected event.<br>Reset type: SYSRSn |
| 5    | OST6     | R    | 0h    | Latched Status Flag for OST6 Trip Latch<br>0: Reading a 0 indicates that no trip has occurred on OST6.<br>1: Reading a 1 indicates a trip has occurred on the OST6 selected event.<br>Reset type: SYSRSn                                   |
| 4    | OST5     | R    | 0h    | Latched Status Flag for OST5 Trip Latch<br>0: Reading a 0 indicates that no trip has occurred on OST5.<br>1: Reading a 1 indicates a trip has occurred on the OST5 selected event.<br>Reset type: SYSRSn                                   |
| 3    | OST4     | R    | 0h    | Latched Status Flag for OST4 Trip Latch<br>0: Reading a 0 indicates that no trip has occurred on OST4.<br>1: Reading a 1 indicates a trip has occurred on the OST4 selected event.<br>Reset type: SYSRSn                                   |
| 2    | OST3     | R    | 0h    | Latched Status Flag for OST3 Trip Latch<br>0: Reading a 0 indicates that no trip has occurred on OST3.<br>1: Reading a 1 indicates a trip has occurred on the OST3 selected event.<br>Reset type: SYSRSn                                   |
| 1    | OST2     | R    | 0h    | Latched Status Flag for OST2 Trip Latch<br>0: Reading a 0 indicates that no trip has occurred on OST2.<br>1: Reading a 1 indicates a trip has occurred on the OST2 selected event.<br>Reset type: SYSRSn                                   |

**Table 26-77. TZOSTFLG Register Field Descriptions (continued)**

| Bit | Field | Type | Reset | Description                                                                                                                                                                                              |
|-----|-------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0   | OST1  | R    | 0h    | Latched Status Flag for OST1 Trip Latch<br>0: Reading a 0 indicates that no trip has occurred on OST1.<br>1: Reading a 1 indicates a trip has occurred on the OST1 selected event.<br>Reset type: SYSRSn |

### 26.17.2.55 TZCLR Register (Offset = 97h) [Reset = 0000h]

TZCLR is shown in [Figure 26-147](#) and described in [Table 26-78](#).

Return to the [Summary Table](#).

Trip Zone Clear Register

**Figure 26-147. TZCLR Register**

|          |            |            |            |            |            |            |            |
|----------|------------|------------|------------|------------|------------|------------|------------|
| 15       | 14         | 13         | 12         | 11         | 10         | 9          | 8          |
| CBCPULSE |            |            | RESERVED   |            |            |            |            |
| R/W-0h   |            |            | R-0-0h     |            |            |            |            |
| 7        | 6          | 5          | 4          | 3          | 2          | 1          | 0          |
| RESERVED | DCBEVT2    | DCBEVT1    | DCAEVT2    | DCAEVT1    | OST        | CBC        | INT        |
| R-0-0h   | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h |

**Table 26-78. TZCLR Register Field Descriptions**

| Bit   | Field    | Type    | Reset | Description                                                                                                                                                                                                                                                                                                                                                       |
|-------|----------|---------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-14 | CBCPULSE | R/W     | 0h    | Clear Pulse for Cycle-By-Cycle (CBC) Trip Latch<br>This bit field determines which pulse clears the CBC trip latch.<br>00: CTR = zero pulse clears CBC trip latch. (Same as legacy designs.)<br>01: CTR = PRD pulse clears CBC trip latch.<br>10: CTR = zero or CTR = PRD pulse clears CBC trip latch.<br>11: CBC trip latch is not cleared<br>Reset type: SYSRSn |
| 13-7  | RESERVED | R-0     | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                          |
| 6     | DCBEVT2  | R-0/W1S | 0h    | Clear Flag for Digital Compare Output B Event 2<br>0: Writing 0 has no effect. This bit always reads back 0.<br>1: Writing 1 clears the DCBEVT2 event trip condition.<br>Reset type: SYSRSn                                                                                                                                                                       |
| 5     | DCBEVT1  | R-0/W1S | 0h    | Clear Flag for Digital Compare Output B Event 1<br>0: Writing 0 has no effect. This bit always reads back 0.<br>1: Writing 1 clears the DCBEVT1 event trip condition.<br>Reset type: SYSRSn                                                                                                                                                                       |
| 4     | DCAEVT2  | R-0/W1S | 0h    | Clear Flag for Digital Compare Output A Event 2<br>0: Writing 0 has no effect. This bit always reads back 0.<br>1: Writing 1 clears the DCAEVT2 event trip condition.<br>Reset type: SYSRSn                                                                                                                                                                       |
| 3     | DCAEVT1  | R-0/W1S | 0h    | Clear Flag for Digital Compare Output A Event 1<br>0: Writing 0 has no effect. This bit always reads back 0.<br>1: Writing 1 clears the DCAEVT1 event trip condition.<br>Reset type: SYSRSn                                                                                                                                                                       |
| 2     | OST      | R-0/W1S | 0h    | Clear Flag for One-Shot Trip (OST) Latch<br>0: Has no effect. Always reads back a 0.<br>1: Clears this Trip (set) condition.<br>Reset type: SYSRSn                                                                                                                                                                                                                |
| 1     | CBC      | R-0/W1S | 0h    | Clear Flag for Cycle-By-Cycle (CBC) Trip Latch<br>0: Has no effect. Always reads back a 0.<br>1: Clears this Trip (set) condition.<br>Reset type: SYSRSn                                                                                                                                                                                                          |

**Table 26-78. TZCLR Register Field Descriptions (continued)**

| Bit | Field | Type    | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----|-------|---------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0   | INT   | R-0/W1S | 0h    | <p>Global Interrupt Clear Flag</p> <p>0: Has no effect. Always reads back a 0.</p> <p>1: Clears the trip-interrupt flag for this ePWM module (TZFLG[INT]).</p> <p>NOTE: No further EPWMx_TZINT PIE interrupts will be generated until the flag is cleared. If the TZFLG[INT] bit is cleared and any of the other flag bits are set, then another interrupt pulse will be generated. Clearing all flag bits will prevent further interrupts.</p> <p>Reset type: SYSRSn</p> |

### 26.17.2.56 TZCBCCLR Register (Offset = 98h) [Reset = 0000h]

TZCBCCLR is shown in [Figure 26-148](#) and described in [Table 26-79](#).

Return to the [Summary Table](#).

Trip Zone CBC Clear Register

**Figure 26-148. TZCBCCLR Register**

|            |            |            |            |            |            |            |            |
|------------|------------|------------|------------|------------|------------|------------|------------|
| 15         | 14         | 13         | 12         | 11         | 10         | 9          | 8          |
| RESERVED   |            |            |            |            |            |            |            |
| R-0-0h     |            |            |            |            |            |            |            |
| 7          | 6          | 5          | 4          | 3          | 2          | 1          | 0          |
| DCBEVT2    | DCAEVT2    | CBC6       | CBC5       | CBC4       | CBC3       | CBC2       | CBC1       |
| R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h |

**Table 26-79. TZCBCCLR Register Field Descriptions**

| Bit  | Field    | Type    | Reset | Description                                                                                                                                                                     |
|------|----------|---------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-8 | RESERVED | R-0     | 0h    | Reserved                                                                                                                                                                        |
| 7    | DCBEVT2  | R-0/W1S | 0h    | Clear Flag for Digital Compare Output B Event 2 selected for CBC<br>0: Writing a 0 has no effect.<br>1: Writing a 1 will clear the TZCBCFLG[DCBEVT2] bit.<br>Reset type: SYSRSn |
| 6    | DCAEVT2  | R-0/W1S | 0h    | Clear Flag for Digital Compare Output A Event 2 selected for CBC<br>0: Writing a 0 has no effect.<br>1: Writing a 1 will clear the TZCBCFLG[DCAEVT2] bit.<br>Reset type: SYSRSn |
| 5    | CBC6     | R-0/W1S | 0h    | Clear Flag for Cycle-By-Cycle (CBC6) Trip Latch<br>0: Writing a 0 has no effect.<br>1: Writing a 1 will clear the TZCBCFLG[CBC6] bit.<br>Reset type: SYSRSn                     |
| 4    | CBC5     | R-0/W1S | 0h    | Clear Flag for Cycle-By-Cycle (CBC5) Trip Latch<br>0: Writing a 0 has no effect.<br>1: Writing a 1 will clear the TZCBCFLG[CBC5] bit.<br>Reset type: SYSRSn                     |
| 3    | CBC4     | R-0/W1S | 0h    | Clear Flag for Cycle-By-Cycle (CBC4) Trip Latch<br>0: Writing a 0 has no effect.<br>1: Writing a 1 will clear the TZCBCFLG[CBC4] bit.<br>Reset type: SYSRSn                     |
| 2    | CBC3     | R-0/W1S | 0h    | Clear Flag for Cycle-By-Cycle (CBC3) Trip Latch<br>0: Writing a 0 has no effect.<br>1: Writing a 1 will clear the TZCBCFLG[CBC3] bit.<br>Reset type: SYSRSn                     |
| 1    | CBC2     | R-0/W1S | 0h    | Clear Flag for Cycle-By-Cycle (CBC2) Trip Latch<br>0: Writing a 0 has no effect.<br>1: Writing a 1 will clear the TZCBCFLG[CBC2] bit.<br>Reset type: SYSRSn                     |
| 0    | CBC1     | R-0/W1S | 0h    | Clear Flag for Cycle-By-Cycle (CBC1) Trip Latch<br>0: Writing a 0 has no effect.<br>1: Writing a 1 will clear the TZCBCFLG[CBC1] bit.<br>Reset type: SYSRSn                     |

### 26.17.2.57 TZOSTCLR Register (Offset = 99h) [Reset = 0000h]

TZOSTCLR is shown in [Figure 26-149](#) and described in [Table 26-80](#).

Return to the [Summary Table](#).

Trip Zone OST Clear Register

**Figure 26-149. TZOSTCLR Register**

|            |            |            |            |            |            |            |            |
|------------|------------|------------|------------|------------|------------|------------|------------|
| 15         | 14         | 13         | 12         | 11         | 10         | 9          | 8          |
| RESERVED   |            |            |            |            |            |            |            |
| R-0-0h     |            |            |            |            |            |            |            |
| 7          | 6          | 5          | 4          | 3          | 2          | 1          | 0          |
| DCBEVT1    | DCAEVT1    | OST6       | OST5       | OST4       | OST3       | OST2       | OST1       |
| R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h |

**Table 26-80. TZOSTCLR Register Field Descriptions**

| Bit  | Field    | Type    | Reset | Description                                                                                                                                                                     |
|------|----------|---------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-8 | RESERVED | R-0     | 0h    | Reserved                                                                                                                                                                        |
| 7    | DCBEVT1  | R-0/W1S | 0h    | Clear Flag for Digital Compare Output B Event 1 selected for OST<br>0: Writing a 0 has no effect.<br>1: Writing a 1 will clear the TZOSTFLG[DCBEVT1] bit.<br>Reset type: SYSRSn |
| 6    | DCAEVT1  | R-0/W1S | 0h    | Clear Flag for Digital Compare Output A Event 1 selected for OST<br>0: Writing a 0 has no effect.<br>1: Writing a 1 will clear the TZOSTFLG[DCAEVT1] bit.<br>Reset type: SYSRSn |
| 5    | OST6     | R-0/W1S | 0h    | Clear Flag for Oneshot (OST6) Trip Latch<br>0: Writing a 0 has no effect.<br>1: Writing a 1 will clear the TZOSTFLG[OST6] bit.<br>Reset type: SYSRSn                            |
| 4    | OST5     | R-0/W1S | 0h    | Clear Flag for Oneshot (OST5) Trip Latch<br>0: Writing a 0 has no effect.<br>1: Writing a 1 will clear the TZOSTFLG[OST5] bit.<br>Reset type: SYSRSn                            |
| 3    | OST4     | R-0/W1S | 0h    | Clear Flag for Oneshot (OST4) Trip Latch<br>0: Writing a 0 has no effect.<br>1: Writing a 1 will clear the TZOSTFLG[OST4] bit.<br>Reset type: SYSRSn                            |
| 2    | OST3     | R-0/W1S | 0h    | Clear Flag for Oneshot (OST3) Trip Latch<br>0: Writing a 0 has no effect.<br>1: Writing a 1 will clear the TZOSTFLG[OST3] bit.<br>Reset type: SYSRSn                            |
| 1    | OST2     | R-0/W1S | 0h    | Clear Flag for Oneshot (OST2) Trip Latch<br>0: Writing a 0 has no effect.<br>1: Writing a 1 will clear the TZOSTFLG[OST2] bit.<br>Reset type: SYSRSn                            |
| 0    | OST1     | R-0/W1S | 0h    | Clear Flag for Oneshot (OST1) Trip Latch<br>0: Writing a 0 has no effect.<br>1: Writing a 1 will clear the TZOSTFLG[OST1] bit.<br>Reset type: SYSRSn                            |



### 26.17.2.58 TZFRC Register (Offset = 9Bh) [Reset = 0000h]

TZFRC is shown in [Figure 26-150](#) and described in [Table 26-81](#).

Return to the [Summary Table](#).

Trip Zone Force Register

**Figure 26-150. TZFRC Register**

|          |            |            |            |            |            |            |          |
|----------|------------|------------|------------|------------|------------|------------|----------|
| 15       | 14         | 13         | 12         | 11         | 10         | 9          | 8        |
| RESERVED |            |            |            |            |            |            |          |
| R-0-0h   |            |            |            |            |            |            |          |
| 7        | 6          | 5          | 4          | 3          | 2          | 1          | 0        |
| RESERVED | DCBEVT2    | DCBEVT1    | DCAEVT2    | DCAEVT1    | OST        | CBC        | RESERVED |
| R-0-0h   | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0/W1S-0h | R-0-0h   |

**Table 26-81. TZFRC Register Field Descriptions**

| Bit  | Field    | Type    | Reset | Description                                                                                                                                                                                                                 |
|------|----------|---------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-7 | RESERVED | R-0     | 0h    | Reserved                                                                                                                                                                                                                    |
| 6    | DCBEVT2  | R-0/W1S | 0h    | Force Flag for Digital Compare Output B Event 2<br>0: Writing 0 has no effect. This bit always reads back 0.<br>1: Writing 1 forces the DCBEVT2 event trip condition and sets the TZFLG[DCBEVT2] bit.<br>Reset type: SYSRSn |
| 5    | DCBEVT1  | R-0/W1S | 0h    | Force Flag for Digital Compare Output B Event 1<br>0: Writing 0 has no effect. This bit always reads back 0.<br>1: Writing 1 forces the DCBEVT1 event trip condition and sets the TZFLG[DCBEVT1] bit.<br>Reset type: SYSRSn |
| 4    | DCAEVT2  | R-0/W1S | 0h    | Force Flag for Digital Compare Output A Event 2<br>0: Writing 0 has no effect. This bit always reads back 0.<br>1: Writing 1 forces the DCAEVT2 event trip condition and sets the TZFLG[DCAEVT2] bit.<br>Reset type: SYSRSn |
| 3    | DCAEVT1  | R-0/W1S | 0h    | Force Flag for Digital Compare Output A Event 1<br>0: Writing 0 has no effect. This bit always reads back 0.<br>1: Writing 1 forces the DCAEVT1 event trip condition and sets the TZFLG[DCAEVT1] bit.<br>Reset type: SYSRSn |
| 2    | OST      | R-0/W1S | 0h    | Force a One-Shot Trip Event via Software<br>0: Writing of 0 is ignored. Always reads back a 0.<br>1: Forces a one-shot trip event and sets the TZFLG[OST] bit.<br>Reset type: SYSRSn                                        |
| 1    | CBC      | R-0/W1S | 0h    | Force a Cycle-by-Cycle Trip Event via Software<br>0: Writing of 0 is ignored. Always reads back a 0.<br>1: Forces a cycle-by-cycle trip event and sets the TZFLG[CBC] bit.<br>Reset type: SYSRSn                            |
| 0    | RESERVED | R-0     | 0h    | Reserved                                                                                                                                                                                                                    |

### 26.17.2.59 ETSEL Register (Offset = A4h) [Reset = 0000h]

ETSEL is shown in [Figure 26-151](#) and described in [Table 26-82](#).

Return to the [Summary Table](#).

Event Trigger Selection Register

**Figure 26-151. ETSEL Register**

| 15       | 14        | 13         | 12         | 11     | 10      | 9 | 8 |
|----------|-----------|------------|------------|--------|---------|---|---|
| SOCBEN   | SOCBSEL   |            |            | SOCAEN | SOCASEL |   |   |
| R/W-0h   | R/W-0h    |            |            | R/W-0h | R/W-0h  |   |   |
| 7        | 6         | 5          | 4          | 3      | 2       | 1 | 0 |
| RESERVED | INTSELCMP | SOCBSELCMP | SOCASELCMP | INTEN  | INTSEL  |   |   |
| R-0-0h   | R/W-0h    | R/W-0h     | R/W-0h     | R/W-0h | R/W-0h  |   |   |

**Table 26-82. ETSEL Register Field Descriptions**

| Bit   | Field   | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|---------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15    | SOCBEN  | R/W  | 0h    | Enable the ADC Start of Conversion B (EPWMxSOCB) Pulse<br>0: Disable EPWMxSOCB.<br>1: Enable EPWMxSOCB pulse.<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 14-12 | SOCBSEL | R/W  | 0h    | EPWMxSOCB Selection Options<br>These bits determine when a EPWMxSOCB pulse will be generated.<br>000: Enable DCBEVT1.soc event<br>001: Enable event time-base counter equal to zero. (TBCTR = 0x00)<br>010: Enable event time-base counter equal to period (TBCTR = TBPRD)<br>011: Enable event time-base counter equal to zero or period (TBCTR = 0x00 or TBCTR = TBPRD). This mode is useful in up-down count mode.<br>100: Enable event time-base counter equal to CMPA when the timer is incrementing or CMPC when the timer is incrementing<br>101: Enable event time-base counter equal to CMPA when the timer is decrementing or CMPC when the timer is decrementing<br>110: Enable event: time-base counter equal to CMPB when the timer is incrementing or CMPD when the timer is incrementing<br>111: Enable event: time-base counter equal to CMPB when the timer is decrementing or CMPD when the timer is decrementing (*) Event selected is determined by SOCBSELCMP bit.<br>Reset type: SYSRSn |
| 11    | SOCAEN  | R/W  | 0h    | Enable the ADC Start of Conversion A (EPWMxSOCA) Pulse<br>0: Disable EPWMxSOCA.<br>1: Enable EPWMxSOCA pulse.<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

**Table 26-82. ETSEL Register Field Descriptions (continued)**

| Bit  | Field      | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|------------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10-8 | SOCASEL    | R/W  | 0h    | <p>EPWMxSOCA Selection Options</p> <p>These bits determine when a EPWMxSOCA pulse will be generated.</p> <p>000: Enable DCAEVT1.soc event</p> <p>001: Enable event time-base counter equal to zero. (TBCTR = 0x00)</p> <p>010: Enable event time-base counter equal to period (TBCTR = TBPRD)</p> <p>011: Enable event time-base counter equal to zero or period (TBCTR = 0x00 or TBCTR = TBPRD). This mode is useful in up-down count mode.</p> <p>100: Enable event time-base counter equal to CMPA when the timer is incrementing or CMPC when the timer is incrementing</p> <p>101: Enable event time-base counter equal to CMPA when the timer is decrementing or CMPC when the timer is decrementing</p> <p>110: Enable event: time-base counter equal to CMPB when the timer is incrementing or CMPD when the timer is incrementing</p> <p>111: Enable event: time-base counter equal to CMPB when the timer is decrementing or CMPD when the timer is decrementing (*) Event selected is determined by SOCASELCMP bit.</p> <p>Reset type: SYSRSn</p> |
| 7    | RESERVED   | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 6    | INTSELCMP  | R/W  | 0h    | <p>EPWMxINT Compare Register Selection Options</p> <p>0: Enable event time-base counter equal to CMPA when the timer is incrementing / Enable event time-base counter equal to CMPA when the timer is decrementing / Enable event: time-base counter equal to CMPB when the timer is incrementing / Enable event: time-base counter equal to CMPB when the timer is decrementing to INTSEL selection mux.</p> <p>1: Enable event time-base counter equal to CMPC when the timer is incrementing / Enable event time-base counter equal to CMPC when the timer is decrementing / Enable event: time-base counter equal to CMPD when the timer is incrementing / Enable event: time-base counter equal to CMPD when the timer is decrementing to INTSEL selection mux.</p> <p>Reset type: SYSRSn</p>                                                                                                                                                                                                                                                           |
| 5    | SOCBSELCMP | R/W  | 0h    | <p>EPWMxSOCB Compare Register Selection Options</p> <p>0: Enable event time-base counter equal to CMPA when the timer is incrementing / Enable event time-base counter equal to CMPA when the timer is decrementing / Enable event: time-base counter equal to CMPB when the timer is incrementing / Enable event: time-base counter equal to CMPB when the timer is decrementing to SOCBSEL selection mux.</p> <p>1: Enable event time-base counter equal to CMPC when the timer is incrementing / Enable event time-base counter equal to CMPC when the timer is decrementing / Enable event: time-base counter equal to CMPD when the timer is incrementing / Enable event: time-base counter equal to CMPD when the timer is decrementing to SOCBSEL selection mux.</p> <p>Reset type: SYSRSn</p>                                                                                                                                                                                                                                                        |
| 4    | SOCASELCMP | R/W  | 0h    | <p>EPWMxSOCA Compare Register Selection Options</p> <p>0: Enable event time-base counter equal to CMPA when the timer is incrementing / Enable event time-base counter equal to CMPA when the timer is decrementing / Enable event: time-base counter equal to CMPB when the timer is incrementing / Enable event: time-base counter equal to CMPB when the timer is decrementing to SOCASEL selection mux.</p> <p>1: Enable event time-base counter equal to CMPC when the timer is incrementing / Enable event time-base counter equal to CMPC when the timer is decrementing / Enable event: time-base counter equal to CMPD when the timer is incrementing / Enable event: time-base counter equal to CMPD when the timer is decrementing to SOCASEL selection mux.</p> <p>Reset type: SYSRSn</p>                                                                                                                                                                                                                                                        |

**Table 26-82. ETSEL Register Field Descriptions (continued)**

| Bit | Field  | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----|--------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3   | INTEN  | R/W  | 0h    | Enable ePWM Interrupt (EPWMx_INT) Generation<br>0: Disable EPWMx_INT generation<br>1: Enable EPWMx_INT generation<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2-0 | INTSEL | R/W  | 0h    | ePWM Interrupt (EPWMx_INT) Selection Options<br>000: Reserved<br>001: Enable event time-base counter equal to zero. (TBCTR = 0x00)<br>010: Enable event time-base counter equal to period (TBCTR = TBPRD)<br>011: Enable event time-base counter equal to zero or period (TBCTR = 0x00 or TBCTR = TBPRD). This mode is useful in up-down count mode.<br>100: Enable event time-base counter equal to CMPA when the timer is incrementing or CMPC when the timer is incrementing<br>101: Enable event time-base counter equal to CMPA when the timer is decrementing or CMPC when the timer is decrementing<br>110: Enable event: time-base counter equal to CMPB when the timer is incrementing or CMPD when the timer is incrementing<br>111: Enable event: time-base counter equal to CMPB when the timer is decrementing or CMPD when the timer is decrementing (*) Event selected is determined by INTSELCMP bit.<br>Reset type: SYSRSn |

### 26.17.2.60 ETPS Register (Offset = A6h) [Reset = 0000h]

ETPS is shown in [Figure 26-152](#) and described in [Table 26-83](#).

Return to the [Summary Table](#).

Event Trigger Pre-Scale Register

**Figure 26-152. ETPS Register**

|          |    |          |         |         |    |         |   |
|----------|----|----------|---------|---------|----|---------|---|
| 15       | 14 | 13       | 12      | 11      | 10 | 9       | 8 |
| SOCBCNT  |    | SOCBPRD  |         | SOCACNT |    | SOCAPRD |   |
| R-0h     |    | R/W-0h   |         | R-0h    |    | R/W-0h  |   |
| 7        | 6  | 5        | 4       | 3       | 2  | 1       | 0 |
| RESERVED |    | SOCPSSEL | INTPSEL | INTCNT  |    | INTPRD  |   |
| R-0-0h   |    | R/W-0h   | R/W-0h  | R-0h    |    | R/W-0h  |   |

**Table 26-83. ETPS Register Field Descriptions**

| Bit   | Field   | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------|---------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-14 | SOCBCNT | R    | 0h    | ePWM ADC Start-of-Conversion B Event (EPWMxSOCB) Counter Register<br>These bits indicate how many selected ETSEL[SOCBSEL] events have occurred:<br>00: No events have occurred.<br>01: 1 event has occurred.<br>10: 2 events have occurred.<br>11: 3 events have occurred.<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 13-12 | SOCBPRD | R/W  | 0h    | ePWM ADC Start-of-Conversion B Event (EPWMxSOCB) Period Select<br>These bits determine how many selected ETSEL[SOCBSEL] events need to occur before an EPWMxSOCB pulse is generated. To be generated, the pulse must be enabled (ETSEL[SOCBEN] = 1). The SOCB pulse will be generated even if the status flag is set from a previous start of conversion (ETFLG[SOCB] = 1). Once the SOCB pulse is generated, the ETPS[SOCBCNT] bits will automatically be cleared.<br>00: Disable the SOCB event counter. No EPWMxSOCB pulse will be generated<br>01: Generate the EPWMxSOCB pulse on the first event:<br>ETPS[SOCBCNT] = 0,1<br>10: Generate the EPWMxSOCB pulse on the second event:<br>ETPS[SOCBCNT] = 1,0<br>11: Generate the EPWMxSOCB pulse on the third event:<br>ETPS[SOCBCNT] = 1,1<br>Reset type: SYSRSn |
| 11-10 | SOCACNT | R    | 0h    | ePWM ADC Start-of-Conversion A Event (EPWMxSOCA) Counter Register<br>These bits indicate how many selected ETSEL[SOCASEL] events have occurred:<br>00: No events have occurred.<br>01: 1 event has occurred.<br>10: 2 events have occurred.<br>11: 3 events have occurred.<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

**Table 26-83. ETPS Register Field Descriptions (continued)**

| Bit | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|----------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9-8 | SOCAPRD  | R/W  | 0h    | <p>ePWM ADC Start-of-Conversion A Event (EPWMxSOCA) Period Select</p> <p>These bits determine how many selected ETSEL[SOCASEL] events need to occur before an EPWMxSOCA pulse is generated. To be generated, the pulse must be enabled (ETSEL[SOCASEN] = 1). The SOCA pulse will be generated even if the status flag is set from a previous start of conversion (ETFLG[SOCA] = 1). Once the SOCA pulse is generated, the ETPS[SOCACNT] bits will automatically be cleared.</p> <p>00: Disable the SOCA event counter. No EPWMxSOCA pulse will be generated</p> <p>01: Generate the EPWMxSOCA pulse on the first event:<br/>ETPS[SOCACNT] = 0,1</p> <p>10: Generate the EPWMxSOCA pulse on the second event:<br/>ETPS[SOCACNT] = 1,0</p> <p>11: Generate the EPWMxSOCA pulse on the third event:<br/>ETPS[SOCACNT] = 1,1</p> <p>Reset type: SYSRSn</p> |
| 7-6 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5   | SOCPSSEL | R/W  | 0h    | <p>EPWMxSOC A/B Pre-Scale Selection Bits</p> <p>0: Selects ETPS [SOCACNT/SOCBCNT] and [SOCAPRD/SOCBPRD] registers to determine frequency of events (SOC pulse once every 0-3 events).</p> <p>1: Selects ETSOCPS [SOCACNT2/SOCBCNT2] and [SOCAPRD2/SOCBPRD2] registers to determine frequency of events (SOC pulse once every 0-15 events).</p> <p>Reset type: SYSRSn</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 4   | INTPSEL  | R/W  | 0h    | <p>EPWMxINTn Pre-Scale Selection Bits</p> <p>0: Selects ETPS [INTCNT, and INTPRD] registers to determine frequency of events (interrupt once every 0-3 events).</p> <p>1: Selects ETINTPS [ INTCNT2, and INTPRD2 ] registers to determine frequency of events (interrupt once every 0-15 events).</p> <p>Reset type: SYSRSn</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 3-2 | INTCNT   | R    | 0h    | <p>ePWM Interrupt Event (EPWMx_INT) Counter Register</p> <p>These bits indicate how many selected ETSEL[INTSEL] events have occurred. These bits are automatically cleared when an interrupt pulse is generated. If interrupts are disabled, ETSEL[INT] = 0 or the interrupt flag is set, ETFLG[INT] = 1, the counter will stop counting events when it reaches the period value ETPS[INTCNT] = ETPS[INTPRD].</p> <p>00: No events have occurred.</p> <p>01: 1 event has occurred.</p> <p>10: 2 events have occurred.</p> <p>11: 3 events have occurred.</p> <p>Reset type: SYSRSn</p>                                                                                                                                                                                                                                                                 |

**Table 26-83. ETPS Register Field Descriptions (continued)**

| Bit | Field  | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----|--------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-0 | INTPRD | R/W  | 0h    | <p>ePWM Interrupt (EPWMx_INT) Period Select</p> <p>These bits determine how many selected ETSEL[INTSEL] events need to occur before an interrupt is generated. To be generated, the interrupt must be enabled (ETSEL[INT] = 1). If the interrupt status flag is set from a previous interrupt (ETFLG[INT] = 1) then no interrupt will be generated until the flag is cleared via the ETCLR[INT] bit. This allows for one interrupt to be pending while another is still being serviced. Once the interrupt is generated, the ETPS[INTCNT] bits will automatically be cleared.</p> <p>Writing a INTPRD value that is the same as the current counter value will trigger an interrupt if it is enabled and the status flag is clear.</p> <p>Writing a INTPRD value that is less than the current counter value will result in an undefined state. If a counter event occurs at the same instant as a new zero or non-zero INTPRD value is written, the counter is incremented.</p> <p>00: Disable the interrupt event counter. No interrupt will be generated and ETFRC[INT] is ignored.</p> <p>01: Generate an interrupt on the first event INTCNT = 01 (first event)</p> <p>10: Generate interrupt on ETPS[INTCNT] = 1,0 (second event)</p> <p>11: Generate interrupt on ETPS[INTCNT] = 1,1 (third event)</p> <p>Reset type: SYSRSn</p> |

### 26.17.2.61 ETFLG Register (Offset = A8h) [Reset = 0000h]

ETFLG is shown in [Figure 26-153](#) and described in [Table 26-84](#).

Return to the [Summary Table](#).

Event Trigger Flag Register

**Figure 26-153. ETFLG Register**

|          |    |    |    |      |      |          |      |
|----------|----|----|----|------|------|----------|------|
| 15       | 14 | 13 | 12 | 11   | 10   | 9        | 8    |
| RESERVED |    |    |    |      |      |          |      |
| R-0-0h   |    |    |    |      |      |          |      |
| 7        | 6  | 5  | 4  | 3    | 2    | 1        | 0    |
| RESERVED |    |    |    | SOCB | SOCA | RESERVED | INT  |
| R-0-0h   |    |    |    | R-0h | R-0h | R-0-0h   | R-0h |

**Table 26-84. ETFLG Register Field Descriptions**

| Bit  | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|----------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-4 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 3    | SOCB     | R    | 0h    | Latched ePWM ADC Start-of-Conversion A (EPWMxSOCB) Status Flag<br>Unlike the ETFLG[INT] flag, the EPWMxSOCB output will continue to pulse even if the flag bit is set.<br>0: Indicates no event occurred<br>1: Indicates that a start of conversion pulse was generated on EPWMxSOCB. The EPWMxSOCB output will continue to be generated even if the flag bit is set.<br>Reset type: SYSRSn                                |
| 2    | SOCA     | R    | 0h    | Latched ePWM ADC Start-of-Conversion A (EPWMxSOCA) Status Flag<br>Unlike the ETFLG[INT] flag, the EPWMxSOCA output will continue to pulse even if the flag bit is set.<br>0: Indicates no event occurred<br>1: Indicates that a start of conversion pulse was generated on EPWMxSOCA. The EPWMxSOCA output will continue to be generated even if the flag bit is set.<br>Reset type: SYSRSn                                |
| 1    | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 0    | INT      | R    | 0h    | Latched ePWM Interrupt (EPWMx_INT) Status Flag<br>0: Indicates no event occurred<br>1: Indicates that an ePWMx interrupt (EPWMx_INT) was generated. No further interrupts will be generated until the flag bit is cleared. Up to one interrupt can be pending while the ETFLG[INT] bit is still set. If an interrupt is pending, it will not be generated until after the ETFLG[INT] bit is cleared.<br>Reset type: SYSRSn |



### 26.17.2.62 ETCLR Register (Offset = AAh) [Reset = 0000h]

ETCLR is shown in [Figure 26-154](#) and described in [Table 26-85](#).

Return to the [Summary Table](#).

Event Trigger Clear Register

**Figure 26-154. ETCLR Register**

|          |    |    |    |            |            |          |            |
|----------|----|----|----|------------|------------|----------|------------|
| 15       | 14 | 13 | 12 | 11         | 10         | 9        | 8          |
| RESERVED |    |    |    |            |            |          |            |
| R-0-0h   |    |    |    |            |            |          |            |
| 7        | 6  | 5  | 4  | 3          | 2          | 1        | 0          |
| RESERVED |    |    |    | SOCB       | SOCA       | RESERVED | INT        |
| R-0-0h   |    |    |    | R-0/W1S-0h | R-0/W1S-0h | R-0-0h   | R-0/W1S-0h |

**Table 26-85. ETCLR Register Field Descriptions**

| Bit  | Field    | Type    | Reset | Description                                                                                                                                                                                                      |
|------|----------|---------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-4 | RESERVED | R-0     | 0h    | Reserved                                                                                                                                                                                                         |
| 3    | SOCB     | R-0/W1S | 0h    | ePWM ADC Start-of-Conversion A (EPWMxSOCB) Flag Clear Bit<br>0: Writing a 0 has no effect. Always reads back a 0<br>1: Clears the ETFLG[SOCB] flag bit<br>Reset type: SYSRSn                                     |
| 2    | SOCA     | R-0/W1S | 0h    | ePWM ADC Start-of-Conversion A (EPWMxSOCA) Flag Clear Bit<br>0: Writing a 0 has no effect. Always reads back a 0<br>1: Clears the ETFLG[SOCA] flag bit<br>Reset type: SYSRSn                                     |
| 1    | RESERVED | R-0     | 0h    | Reserved                                                                                                                                                                                                         |
| 0    | INT      | R-0/W1S | 0h    | ePWM Interrupt (EPWMx_INT) Flag Clear Bit<br>0: Writing a 0 has no effect. Always reads back a 0<br>1: Clears the ETFLG[INT] flag bit and enable further interrupts pulses to be generated<br>Reset type: SYSRSn |

### 26.17.2.63 ETFRC Register (Offset = ACh) [Reset = 0000h]

ETFRC is shown in [Figure 26-155](#) and described in [Table 26-86](#).

Return to the [Summary Table](#).

Event Trigger Force Register

**Figure 26-155. ETFRC Register**

|          |    |    |    |            |            |          |            |
|----------|----|----|----|------------|------------|----------|------------|
| 15       | 14 | 13 | 12 | 11         | 10         | 9        | 8          |
| RESERVED |    |    |    |            |            |          |            |
| R-0-0h   |    |    |    |            |            |          |            |
| 7        | 6  | 5  | 4  | 3          | 2          | 1        | 0          |
| RESERVED |    |    |    | SOCB       | SOCA       | RESERVED | INT        |
| R-0-0h   |    |    |    | R-0/W1S-0h | R-0/W1S-0h | R-0-0h   | R-0/W1S-0h |

**Table 26-86. ETFRC Register Field Descriptions**

| Bit  | Field    | Type    | Reset | Description                                                                                                                                                                                                                                                                                                                                                                |
|------|----------|---------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-4 | RESERVED | R-0     | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                   |
| 3    | SOCB     | R-0/W1S | 0h    | <p>SOCB Force Bit</p> <p>The SOCB pulse will only be generated if the event is enabled in the ETSEL register. The ETFLG[SOCB] flag bit will be set regardless.</p> <p>0: Writing 0 to this bit will be ignored. Always reads back a 0.</p> <p>1: Generates a pulse on EPWMxSOCB and set the SOCBFLG bit. This bit is used for test purposes.</p> <p>Reset type: SYSRSn</p> |
| 2    | SOCA     | R-0/W1S | 0h    | <p>SOCA Force Bit</p> <p>The SOCA pulse will only be generated if the event is enabled in the ETSEL register. The ETFLG[SOCA] flag bit will be set regardless.</p> <p>0: Writing 0 to this bit will be ignored. Always reads back a 0.</p> <p>1: Generates a pulse on EPWMxSOCA and set the SOCAFLG bit. This bit is used for test purposes.</p> <p>Reset type: SYSRSn</p> |
| 1    | RESERVED | R-0     | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                   |
| 0    | INT      | R-0/W1S | 0h    | <p>INT Force Bit</p> <p>The interrupt will only be generated if the event is enabled in the ETSEL register. The INT flag bit will be set regardless.</p> <p>0: Writing 0 to this bit will be ignored. Always reads back a 0.</p> <p>1: Generates an interrupt on EPWMxINT and set the INT flag bit. This bit is used for test purposes.</p> <p>Reset type: SYSRSn</p>      |

### 26.17.2.64 ETINTPS Register (Offset = AEh) [Reset = 0000h]

ETINTPS is shown in [Figure 26-156](#) and described in [Table 26-87](#).

Return to the [Summary Table](#).

Event-Trigger Interrupt Pre-Scale Register

**Figure 26-156. ETINTPS Register**

|          |    |    |    |         |    |   |   |
|----------|----|----|----|---------|----|---|---|
| 15       | 14 | 13 | 12 | 11      | 10 | 9 | 8 |
| RESERVED |    |    |    |         |    |   |   |
| R-0-0h   |    |    |    |         |    |   |   |
| 7        | 6  | 5  | 4  | 3       | 2  | 1 | 0 |
| INTCNT2  |    |    |    | INTPRD2 |    |   |   |
| R-0h     |    |    |    | R/W-0h  |    |   |   |

**Table 26-87. ETINTPS Register Field Descriptions**

| Bit  | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|----------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-8 | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 7-4  | INTCNT2  | R    | 0h    | EPWMxINT Counter 2<br>When ETPS[INTPSSEL]=1, these bits indicate how many selected events have occurred:<br>0000: No events<br>0001: 1 event<br>0010: 2 events<br>0011: 3 events<br>0100: 4 events<br>...<br>1111: 15 events<br>Reset type: SYSRSn                                                                                                                                                                                                                                                   |
| 3-0  | INTPRD2  | R/W  | 0h    | EPWMxINT Period 2 Select<br>When ETPS[INTPSSEL] = 1, these bits select how many selected events need to occur before an interrupt is generated:<br>0000: Disable counter<br>0001: Generate interrupt on INTCNT = 1 (first event)<br>0010: Generate interrupt on INTCNT = 2 (second event)<br>0011: Generate interrupt on INTCNT = 3 (third event)<br>0100: Generate interrupt on INTCNT = 4 (fourth event)<br>...<br>1111: Generate interrupt on INTCNT = 15 (fifteenth event)<br>Reset type: SYSRSn |

### 26.17.2.65 ETSOCPS Register (Offset = B0h) [Reset = 0000h]

ETSOCPS is shown in [Figure 26-157](#) and described in [Table 26-88](#).

Return to the [Summary Table](#).

Event-Trigger SOC Pre-Scale Register

**Figure 26-157. ETSOCPS Register**

|          |    |    |    |          |    |   |   |
|----------|----|----|----|----------|----|---|---|
| 15       | 14 | 13 | 12 | 11       | 10 | 9 | 8 |
| SOCBCNT2 |    |    |    | SOCBPRD2 |    |   |   |
| R-0h     |    |    |    | R/W-0h   |    |   |   |
| 7        | 6  | 5  | 4  | 3        | 2  | 1 | 0 |
| SOCACNT2 |    |    |    | SOCAPRD2 |    |   |   |
| R-0h     |    |    |    | R/W-0h   |    |   |   |

**Table 26-88. ETSOCPS Register Field Descriptions**

| Bit   | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------|----------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-12 | SOCBCNT2 | R    | 0h    | EPWMxSOCB Counter 2<br>When ETPS[SOCPSSEL] = 1, these bits indicate how many selected events have occurred:<br>0000: No events<br>0001: 1 event<br>0010: 2 events<br>0011: 3 events<br>0100: 4 events<br>...<br>1111: 15 events<br>Reset type: SYSRSn                                                                                                                                                                                                                                                           |
| 11-8  | SOCBPRD2 | R/W  | 0h    | EPWMxSOCB Period 2 Select<br>When ETPS[SOCPSSEL] = 1, these bits select how many selected event need to occur before an SOCB pulse is generated:<br>0000: Disable counter<br>0001: Generate SOC pulse on SOCBCNT2 = 1 (first event)<br>0010: Generate SOC pulse on SOCBCNT2 = 2 (second event)<br>0011: Generate SOC pulse on SOCBCNT2 = 3 (third event)<br>0100: Generate SOC pulse on SOCBCNT2 = 4 (fourth event)<br>...<br>1111: Generate SOC pulse on SOCBCNT2 = 15 (fifteenth event)<br>Reset type: SYSRSn |
| 7-4   | SOCACNT2 | R    | 0h    | EPWMxSOCA Counter 2<br>When ETPS[SOCPSSEL] = 1, these bits indicate how many selected events have occurred:<br>0000: No events<br>0001: 1 event<br>0010: 2 events<br>0011: 3 events<br>0100: 4 events<br>...<br>1111: 15 events<br>Reset type: SYSRSn                                                                                                                                                                                                                                                           |

**Table 26-88. ETSOCPS Register Field Descriptions (continued)**

| Bit | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----|----------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3-0 | SOCAPRD2 | R/W  | 0h    | <p>EPWMxSOCA Period 2 Select</p> <p>When ETPS[SOCPSSEL] = 1, these bits select how many selected event need to occur before an SOCA pulse is generated:</p> <p>0000: Disable counter</p> <p>0001: Generate SOC pulse on SOCACNT2 = 1 (first event)</p> <p>0010: Generate SOC pulse on SOCACNT2 = 2 (second event)</p> <p>0011: Generate SOC pulse on SOCACNT2 = 3 (third event)</p> <p>0100: Generate SOC pulse on SOCACNT2 = 4 (fourth event)</p> <p>...</p> <p>1111: Generate SOC pulse on SOCACNT2 = 15 (fifteenth event)</p> <p>Reset type: SYSRSn</p> |

### 26.17.2.66 ETCNTINITCTL Register (Offset = B2h) [Reset = 0000h]

ETCNTINITCTL is shown in [Figure 26-158](#) and described in [Table 26-89](#).

Return to the [Summary Table](#).

Event-Trigger Counter Initialization Control Register

**Figure 26-158. ETCNTINITCTL Register**

|            |            |           |             |             |            |          |   |
|------------|------------|-----------|-------------|-------------|------------|----------|---|
| 15         | 14         | 13        | 12          | 11          | 10         | 9        | 8 |
| SOCBINITEN | SOCAINITEN | INTINITEN | SOCBINITFRC | SOCAINITFRC | INTINITFRC | RESERVED |   |
| R/W-0h     | R/W-0h     | R/W-0h    | R-0/W1S-0h  | R-0/W1S-0h  | R-0/W1S-0h | R-0-0h   |   |
| 7          | 6          | 5         | 4           | 3           | 2          | 1        | 0 |
| RESERVED   |            |           |             |             |            |          |   |
| R-0-0h     |            |           |             |             |            |          |   |

**Table 26-89. ETCNTINITCTL Register Field Descriptions**

| Bit | Field       | Type    | Reset | Description                                                                                                                                                                                                   |
|-----|-------------|---------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15  | SOCBINITEN  | R/W     | 0h    | EPWMxSOCB Counter 2 Initialization Enable<br>0: Has no effect.<br>1: Enable initialization of EPWMxSOCB counter with contents of ETCNTINIT[SOCBINIT] on a SYNC event or software force.<br>Reset type: SYSRSn |
| 14  | SOCAINITEN  | R/W     | 0h    | EPWMxSOCA Counter 2 Initialization Enable<br>0: Has no effect.<br>1: Enable initialization of EPWMxSOCA counter with contents of ETCNTINIT[SOCAINIT] on a SYNC event or software force.<br>Reset type: SYSRSn |
| 13  | INTINITEN   | R/W     | 0h    | EPWMxINT Counter 2 Initialization Enable<br>0: Has no effect.<br>1: Enable initialization of EPWMxINT counter 2 with contents of ETCNTINIT[INTINIT] on a SYNC event or software force.<br>Reset type: SYSRSn  |
| 12  | SOCBINITFRC | R-0/W1S | 0h    | EPWMxSOCB Counter 2 Initialization Force<br>0: Has no effect.<br>1: This bit forces the ET EPWMxSOCB counter to be initialized with the contents of ETCNTINIT[SOCBINIT].<br>Reset type: SYSRSn                |
| 11  | SOCAINITFRC | R-0/W1S | 0h    | EPWMxSOCA Counter 2 Initialization Force<br>0: Has no effect.<br>1: This bit forces the ET EPWMxSOCA counter to be initialized with the contents of ETCNTINIT[SOCAINIT].<br>Reset type: SYSRSn                |
| 10  | INTINITFRC  | R-0/W1S | 0h    | EPWMxINT Counter 2 Initialization Force<br>0: Has no effect.<br>1: This bit forces the ET EPWMxINT counter to be initialized with the contents of ETCNTINIT[INTINIT].<br>Reset type: SYSRSn                   |
| 9-0 | RESERVED    | R-0     | 0h    | Reserved                                                                                                                                                                                                      |

### 26.17.2.67 ETCNTINIT Register (Offset = B4h) [Reset = 0000h]

ETCNTINIT is shown in [Figure 26-159](#) and described in [Table 26-90](#).

Return to the [Summary Table](#).

Event-Trigger Counter Initialization Register

**Figure 26-159. ETCNTINIT Register**

|          |    |    |    |          |    |   |   |
|----------|----|----|----|----------|----|---|---|
| 15       | 14 | 13 | 12 | 11       | 10 | 9 | 8 |
| RESERVED |    |    |    | SOCBINIT |    |   |   |
| R-0h     |    |    |    | R/W-0h   |    |   |   |
| 7        | 6  | 5  | 4  | 3        | 2  | 1 | 0 |
| SOCAINIT |    |    |    | INTINIT  |    |   |   |
| R/W-0h   |    |    |    | R/W-0h   |    |   |   |

**Table 26-90. ETCNTINIT Register Field Descriptions**

| Bit   | Field    | Type | Reset | Description                                                                                                                                                                            |
|-------|----------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-12 | RESERVED | R    | 0h    | Reserved                                                                                                                                                                               |
| 11-8  | SOCBINIT | R/W  | 0h    | EPWMxSOCB Counter 2 Initialization Bits<br>The ET EPWMxSOCB counter is initialized with the contents of this register on an ePWM SYNC event or a software force.<br>Reset type: SYSRSn |
| 7-4   | SOCAINIT | R/W  | 0h    | EPWMxSOCA Counter 2 Initialization Bits<br>The ET EPWMxSOCA counter is initialized with the contents of this register on an ePWM SYNC event or a software force.<br>Reset type: SYSRSn |
| 3-0   | INTINIT  | R/W  | 0h    | EPWMxINT Counter 2 Initialization Bits<br>The ET EPWMxINT counter is initialized with the contents of this register on an ePWM SYNC event or a software force.<br>Reset type: SYSRSn   |

### 26.17.2.68 DCTRIPSEL Register (Offset = C0h) [Reset = 0000h]

DCTRIPSEL is shown in [Figure 26-160](#) and described in [Table 26-91](#).

Return to the [Summary Table](#).

Digital Compare Trip Select Register

**Figure 26-160. DCTRIPSEL Register**

|             |    |    |    |             |    |   |   |
|-------------|----|----|----|-------------|----|---|---|
| 15          | 14 | 13 | 12 | 11          | 10 | 9 | 8 |
| DCBLCOMPSEL |    |    |    | DCBHCOMPSEL |    |   |   |
| R/W-0h      |    |    |    | R/W-0h      |    |   |   |
| 7           | 6  | 5  | 4  | 3           | 2  | 1 | 0 |
| DCALCOMPSEL |    |    |    | DCAHCOMPSEL |    |   |   |
| R/W-0h      |    |    |    | R/W-0h      |    |   |   |

**Table 26-91. DCTRIPSEL Register Field Descriptions**

| Bit   | Field       | Type | Reset | Description                                                                                                                                                                                                                                                                                                        |
|-------|-------------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-12 | DCBLCOMPSEL | R/W  | 0h    | Digital Compare B Low Input Select Bits<br>0000: TRIPIN1<br>0001: TRIPIN2<br>0010: TRIPIN3<br>0011: TRIPIN4<br>...<br>1011: TRIPIN12<br>1100: Reserved<br>1101: TRIPIN14<br>1110: TRIPIN15<br>1111: Trip combination input (all trip inputs selected by DCBLTRIPSEL register ORed together)<br>Reset type: SYSRSn  |
| 11-8  | DCBHCOMPSEL | R/W  | 0h    | Digital Compare B High Input Select Bits<br>0000: TRIPIN1<br>0001: TRIPIN2<br>0010: TRIPIN3<br>0011: TRIPIN4<br>...<br>1011: TRIPIN12<br>1100: Reserved<br>1101: TRIPIN14<br>1110: TRIPIN15<br>1111: Trip combination input (all trip inputs selected by DCBHTRIPSEL register ORed together)<br>Reset type: SYSRSn |
| 7-4   | DCALCOMPSEL | R/W  | 0h    | Digital Compare A Low Input Select Bits<br>0000: TRIPIN1<br>0001: TRIPIN2<br>0010: TRIPIN3<br>0011: TRIPIN4<br>...<br>1011: TRIPIN12<br>1100: Reserved<br>1101: TRIPIN14<br>1110: TRIPIN15<br>1111: Trip combination input (all trip inputs selected by DCALTRIPSEL register ORed together)<br>Reset type: SYSRSn  |



**Table 26-91. DCTRIPSEL Register Field Descriptions (continued)**

| Bit | Field       | Type | Reset | Description                                                                                                                                                                                                                                                                                                           |
|-----|-------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3-0 | DCAHCOMPSEL | R/W  | 0h    | Digital Compare A High Input Select Bits<br>0000: TRIPIN1<br>0001: TRIPIN2<br>0010: TRIPIN3<br>0011: TRIPIN4<br>...<br>1011: TRIPIN12<br>1100: Reserved<br>1101: TRIPIN14<br>1110: TRIPIN15<br>1111: Trip combination input (all trip inputs selected by<br>DCAHTRIPSEL register ORed together)<br>Reset type: SYSRSn |

### 26.17.2.69 DCACTL Register (Offset = C3h) [Reset = 0000h]

DCACTL is shown in [Figure 26-161](#) and described in [Table 26-92](#).

Return to the [Summary Table](#).

Digital Compare A Control Register

**Figure 26-161. DCACTL Register**

| 15      | 14            | 13         | 12        | 11       | 10              | 9          | 8      |
|---------|---------------|------------|-----------|----------|-----------------|------------|--------|
| EVT2LAT | EVT2LATCLRSEL | EVT2LATSEL | RESERVED  |          | EVT2FRCSYN CSEL | EVT2SRCSEL |        |
| R-0h    | R/W-0h        | R/W-0h     | R-0-0h    |          | R/W-0h          | R/W-0h     |        |
| 7       | 6             | 5          | 4         | 3        | 2               | 1          | 0      |
| EVT1LAT | EVT1LATCLRSEL | EVT1LATSEL | EVT1SYNCE | EVT1SOCE | EVT1FRCSYN CSEL | EVT1SRCSEL |        |
| R-0h    | R/W-0h        | R/W-0h     | R/W-0h    | R/W-0h   | R/W-0h          | R/W-0h     | R/W-0h |

**Table 26-92. DCACTL Register Field Descriptions**

| Bit   | Field           | Type | Reset | Description                                                                                                                                                                                                                    |
|-------|-----------------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15    | EVT2LAT         | R    | 0h    | Indicates the status of DCAEVT2LAT signal.<br>0 : The DCAEVT2LAT latch is cleared.<br>1 : The DCAEVT2LAT latch is set.<br>Reset type: SYSRSn                                                                                   |
| 14-13 | EVT2LATCLRSEL   | R/W  | 0h    | DCAEVT2 Latched clear source select:<br>00: CNT_ZERO event clears DCAEVT2 latch.<br>01: PRD_EQ event clears DCAEVT2 latch.<br>10: CNT_ZERO event or PRD_EQ event clears DCAEVT2 latch.<br>11: Reserved.<br>Reset type: SYSRSn  |
| 12    | EVT2LATSEL      | R/W  | 0h    | DCAEVT2 Latched signal select:<br>0: Does not select the DCAEVT2 latched signal as source of DCAEVT2.force.<br>1: Selects the DCAEVT2 latched signal as source of DCAEVT2.force.<br>Reset type: SYSRSn                         |
| 11-10 | RESERVED        | R-0  | 0h    | Reserved                                                                                                                                                                                                                       |
| 9     | EVT2FRCSYN CSEL | R/W  | 0h    | DCAEVT2 Force Synchronization Signal Select<br>0: Source is synchronized with EPWMCLK<br>1: Source is passed through asynchronously<br>Reset type: SYSRSn                                                                      |
| 8     | EVT2SRCSEL      | R/W  | 0h    | DCAEVT2 Source Signal Select<br>0: Source Is DCAEVT2 Signal<br>1: Source Is DCEVTFILT Signal<br>Reset type: SYSRSn                                                                                                             |
| 7     | EVT1LAT         | R    | 0h    | Indicates the status of DCAEVT1LAT signal.<br>0 : The DCAEVT1LAT latch is cleared.<br>1 : The DCAEVT1LAT latch is set.<br>Reset type: SYSRSn                                                                                   |
| 6-5   | EVT1LATCLRSEL   | R/W  | 0h    | DCAEVT1 Latched clear source select:<br>00: CNT_ZERO event clears DCAEVT1 latch.<br>01: PRD_EQ event clears DCAEVT1 latch.<br>10: CNT_ZERO event or PRD_EQ event clears DCAEVT1 latch.<br>11 : Reserved.<br>Reset type: SYSRSn |

**Table 26-92. DCACTL Register Field Descriptions (continued)**

| Bit | Field           | Type | Reset | Description                                                                                                                                                                                            |
|-----|-----------------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4   | EVT1LATSEL      | R/W  | 0h    | DCAEVT1 Latched signal select:<br>0: Does not select the DCAEVT1 latched signal as source of DCAEVT1.force.<br>1: Selects the DCAEVT1 latched signal as source of DCAEVT1.force.<br>Reset type: SYSRSn |
| 3   | EVT1SYNCE       | R/W  | 0h    | DCAEVT1 SYNC, Enable/Disable<br>0: SYNC Generation Disabled<br>1: SYNC Generation Enabled<br>Reset type: SYSRSn                                                                                        |
| 2   | EVT1SOCE        | R/W  | 0h    | DCAEVT1 SOC, Enable/Disable<br>0: SOC Generation Disabled<br>1: SOC Generation Enabled<br>Reset type: SYSRSn                                                                                           |
| 1   | EVT1FRCSYNCSSEL | R/W  | 0h    | DCAEVT1 Force Synchronization Signal Select<br>0: Source is synchronized with EPWMCLK<br>1: Source is passed through asynchronously<br>Reset type: SYSRSn                                              |
| 0   | EVT1SRCSEL      | R/W  | 0h    | DCAEVT1 Source Signal Select<br>0: Source Is DCAEVT1 Signal<br>1: Source Is DCEVT1FILT Signal<br>Reset type: SYSRSn                                                                                    |

### 26.17.2.70 DCBCTL Register (Offset = C4h) [Reset = 0000h]

DCBCTL is shown in [Figure 26-162](#) and described in [Table 26-93](#).

Return to the [Summary Table](#).

Digital Compare B Control Register

**Figure 26-162. DCBCTL Register**

| 15      | 14            | 13         | 12        | 11       | 10              | 9          | 8      |
|---------|---------------|------------|-----------|----------|-----------------|------------|--------|
| EVT2LAT | EVT2LATCLRSEL | EVT2LATSEL | RESERVED  |          | EVT2FRCSYN CSEL | EVT2SRCSEL |        |
| R-0h    | R/W-0h        | R/W-0h     | R-0-0h    |          | R/W-0h          | R/W-0h     |        |
| 7       | 6             | 5          | 4         | 3        | 2               | 1          | 0      |
| EVT1LAT | EVT1LATCLRSEL | EVT1LATSEL | EVT1SYNCE | EVT1SOCE | EVT1FRCSYN CSEL | EVT1SRCSEL |        |
| R-0h    | R/W-0h        | R/W-0h     | R/W-0h    | R/W-0h   | R/W-0h          | R/W-0h     | R/W-0h |

**Table 26-93. DCBCTL Register Field Descriptions**

| Bit   | Field           | Type | Reset | Description                                                                                                                                                                                                                                                                    |
|-------|-----------------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15    | EVT2LAT         | R    | 0h    | Indicates the status of DCBEVT2LAT signal.<br>0 The DCBEVT2LAT latch is cleared.<br>1 The DCBEVT2LAT latch is set.<br>Reset type: SYSRSn                                                                                                                                       |
| 14-13 | EVT2LATCLRSEL   | R/W  | 0h    | DCBEVT2 Latched clear source select:<br>00 CNT_ZERO event clears DCBEVT2 latch.<br>01 PRD_EQ event clears DCBEVT2 latch.<br>10 CNT_ZERO event or PRD_EQ event clears DCBEVT2 latch.<br>11 Reserved.<br>Reset type: SYSRSn                                                      |
| 12    | EVT2LATSEL      | R/W  | 0h    | DCBEVT2 Latched signal select:<br>0 Does not select the DCBEVT2 latched signal (Refer figure 'Modifications to DCBEVT1.force/DCBEVT2.force generation.') as source of DCBEVT2.force.<br>1 Selects the DCBEVT2 latched signal as source of DCBEVT2.force.<br>Reset type: SYSRSn |
| 11-10 | RESERVED        | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                       |
| 9     | EVT2FRCSYN CSEL | R/W  | 0h    | DCBEVT2 Force Synchronization Signal Select<br>0: Source is synchronized with EPWMCLK<br>1: Source is passed through asynchronously<br>Reset type: SYSRSn                                                                                                                      |
| 8     | EVT2SRCSEL      | R/W  | 0h    | DCBEVT2 Source Signal Select<br>0: Source Is DCBEVT2 Signal<br>1: Source Is DCEVTFILT Signal<br>Reset type: SYSRSn                                                                                                                                                             |
| 7     | EVT1LAT         | R    | 0h    | Indicates the status of DCBEVT1LAT signal.<br>0 The DCBEVT1LAT latch is cleared.<br>1 The DCBEVT1LAT latch is set.<br>Reset type: SYSRSn                                                                                                                                       |
| 6-5   | EVT1LATCLRSEL   | R/W  | 0h    | DCBEVT1 Latched clear source select:<br>00 CNT_ZERO event clears DCBEVT1 latch.<br>01 PRD_EQ event clears DCBEVT1 latch.<br>10 CNT_ZERO event or PRD_EQ event clears DCBEVT1 latch.<br>11 Reserved.<br>Reset type: SYSRSn                                                      |

**Table 26-93. DCBCTL Register Field Descriptions (continued)**

| Bit | Field           | Type | Reset | Description                                                                                                                                                                                                                                                                    |
|-----|-----------------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4   | EVT1LATSEL      | R/W  | 0h    | DCBEVT1 Latched signal select:<br>0 Does not select the DCBEVT1 latched signal (Refer figure 'Modifications to DCBEVT1.force/DCBEVT2.force generation.') as source of DCBEVT1.force.<br>1 Selects the DCBEVT1 latched signal as source of DCBEVT1.force.<br>Reset type: SYSRSn |
| 3   | EVT1SYNCE       | R/W  | 0h    | DCBEVT1 SYNC, Enable/Disable<br>0: SYNC Generation Disabled<br>1: SYNC Generation Enabled<br>Reset type: SYSRSn                                                                                                                                                                |
| 2   | EVT1SOCE        | R/W  | 0h    | DCBEVT1 SOC, Enable/Disable<br>0: SOC Generation Disabled<br>1: SOC Generation Enabled<br>Reset type: SYSRSn                                                                                                                                                                   |
| 1   | EVT1FRCSYNCSSEL | R/W  | 0h    | DCBEVT1 Force Synchronization Signal Select<br>0: Source is synchronized with EPWMCLK<br>1: Source is passed through asynchronously<br>Reset type: SYSRSn                                                                                                                      |
| 0   | EVT1SRCSEL      | R/W  | 0h    | DCBEVT1 Source Signal Select<br>0: Source Is DCBEVT1 Signal<br>1: Source Is DCEVT1FILT Signal<br>Reset type: SYSRSn                                                                                                                                                            |

### 26.17.2.71 DCFCTL Register (Offset = C7h) [Reset = 0000h]

DCFCTL is shown in [Figure 26-163](#) and described in [Table 26-94](#).

Return to the [Summary Table](#).

Digital Compare Filter Control Register

**Figure 26-163. DCFCTL Register**

|            |             |          |           |          |        |          |   |
|------------|-------------|----------|-----------|----------|--------|----------|---|
| 15         | 14          | 13       | 12        | 11       | 10     | 9        | 8 |
| EDGESTATUS |             |          | EDGECOUNT |          |        | EDGEMODE |   |
| R-0h       |             |          | R/W-0h    |          |        | R/W-0h   |   |
| 7          | 6           | 5        | 4         | 3        | 2      | 1        | 0 |
| RESERVED   | EDGEFILTSEL | PULSESEL |           | BLANKINV | BLANKE | SRCSEL   |   |
| R-0-0h     | R/W-0h      | R/W-0h   |           | R/W-0h   | R/W-0h | R/W-0h   |   |

**Table 26-94. DCFCTL Register Field Descriptions**

| Bit   | Field       | Type | Reset | Description                                                                                                                                                                                                                                                                                                                              |
|-------|-------------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-13 | EDGESTATUS  | R    | 0h    | Edge Status:<br>These bits reflect the total number of edges currently captured. When the value matches the EDGECOUNT, the status bits are set to zero, and a TBCLK wide pulse is generated which can then be output on the DCEVTFILT signal. The edge counter can be reset by writing 000 to the EDGECOUNT value:<br>Reset type: SYSRSn |
| 12-10 | EDGECOUNT   | R/W  | 0h    | Edge Count: These bits select how many edges to count before generating a TBCLK wide pulse on the DCEVTFILT signal:<br>000: no edges, reset current EDGESTATUS bits to 0,0,0<br>001: 1 edge<br>010: 2 edges<br>011: 3 edges<br>100: 4 edges<br>101: 5 edges<br>110: 6 edges<br>111: 7 edges<br>Reset type: SYSRSn                        |
| 9-8   | EDGEMODE    | R/W  | 0h    | Edge Mode Select:<br>00: Low To High Edge<br>01: High To Low Edge<br>10: Both Edges<br>11: Reserved<br>Reset type: SYSRSn                                                                                                                                                                                                                |
| 7     | RESERVED    | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                 |
| 6     | EDGEFILTSEL | R/W  | 0h    | Edge Filter Select:<br>0: Edge Filter Not Selected<br>1: Edge Filter Selected<br>Reset type: SYSRSn                                                                                                                                                                                                                                      |
| 5-4   | PULSESEL    | R/W  | 0h    | Pulse Select For Blanking & Capture Alignment<br>00: Time-base counter equal to period (TBCTR = TBPRD)<br>01: Time-base counter equal to zero (TBCTR = 0x00)<br>10: Time-base counter equal to zero (TBCTR = 0x00) or period (TBCTR = TBPRD)<br>11: Reserved<br>Reset type: SYSRSn                                                       |
| 3     | BLANKINV    | R/W  | 0h    | Blanking Window Inversion<br>0: Blanking window not inverted<br>1: Blanking window inverted<br>Reset type: SYSRSn                                                                                                                                                                                                                        |

**Table 26-94. DCFCTL Register Field Descriptions (continued)**

| Bit | Field  | Type | Reset | Description                                                                                                                                                                             |
|-----|--------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2   | BLANKE | R/W  | 0h    | Blanking Window Enable/Disable<br>0: Blanking window is disabled<br>1: Blanking window is enabled<br>Reset type: SYSRSn                                                                 |
| 1-0 | SRCSEL | R/W  | 0h    | Filter Block Signal Source Select<br>00: Source Is DCAEVT1 Signal<br>01: Source Is DCAEVT2 Signal<br>10: Source Is DCBEVT1 Signal<br>11: Source Is DCBEVT2 Signal<br>Reset type: SYSRSn |

### 26.17.2.72 DCCAPCTL Register (Offset = C8h) [Reset = 0000h]

DCCAPCTL is shown in [Figure 26-164](#) and described in [Table 26-95](#).

Return to the [Summary Table](#).

Digital Compare Capture Control Register

**Figure 26-164. DCCAPCTL Register**

|          |            |        |          |    |    |          |        |
|----------|------------|--------|----------|----|----|----------|--------|
| 15       | 14         | 13     | 12       | 11 | 10 | 9        | 8      |
| CAPMODE  | CAPCLR     | CAPSTS | RESERVED |    |    |          |        |
| R/W-0h   | R-0/W1S-0h | R-0h   | R-0-0h   |    |    |          |        |
| 7        | 6          | 5      | 4        | 3  | 2  | 1        | 0      |
| RESERVED |            |        |          |    |    | SHDWMODE | CAPE   |
| R-0-0h   |            |        |          |    |    | R/W-0h   | R/W-0h |

**Table 26-95. DCCAPCTL Register Field Descriptions**

| Bit  | Field    | Type    | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------|----------|---------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | CAPMODE  | R/W     | 0h    | Counter Capture Mode<br>0: When a DCEVTFILT occurs and the counter capture is enabled, then the current TBCNT value is captured in the active register. When the respective trip event occurs, further trip (capture) events are ignored until the next PRD_eq or CNT_zero event (as selected by the PULSESEL bit in the DCFCTL register) re-triggers the capture mechanism.<br>If active mode is enabled, via SHDWMODE bit in DCCAPCTL register, CPU reads of this register will return the active register value.<br>If shadow mode is enabled, via SHDWMODE bit in DCCAPCTL register, the active register is copied to the shadow register on the PRD_eq or CNT_zero event (whichever is selected by PULSESEL bit in DCFCTL register). CPU reads of this register will return the shadow register value.<br>1: When a DCEVTFILT occurs and the counter capture is enabled, then the current TBCNT value is captured in the active register. When the respective trip event occurs - it will set the CAPSTS flag and further trip (capture) events are ignored until this bit is cleared. CAPSTS can be cleared by writing to CAPCLR bit in DCCAPCTL register and it re-triggers the capture mechanism.<br>If active mode is enabled, via SHDWMODE bit in DCCAPCTL register, CPU reads of this register will return the active register value.<br>If shadow mode is enabled, via SHDWMODE bit in DCCAPCTL register, the active register is copied to the shadow register on the PRD_eq or CNT_zero event (whichever is selected by PULSESEL bit in DCFCTL register). CPU reads of this register will return the shadow register value.<br>Reset type: SYSRSn |
| 14   | CAPCLR   | R-0/W1S | 0h    | DC Capture Latched Status Clear Flag<br>0: Writing a 0 has no effect.<br>1: Writing a 1 will clear this CAPSTS (set) condition.<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 13   | CAPSTS   | R       | 0h    | Latched Status Flag for Capture Event<br>0: No DC capture event occurred.<br>1: A DC capture event has occurred.<br>Reset type: SYSRSn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 12-2 | RESERVED | R-0     | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |



**Table 26-95. DCCAPCTL Register Field Descriptions (continued)**

| Bit | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----|----------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | SHDWMODE | R/W  | 0h    | <p>TBCTR Counter Capture Shadow Select Mode</p> <p>0: Enable shadow mode. The DCCAP active register is copied to shadow register on a TBCTR = TBPRD or TBCTR = zero event as defined by the DCFCTL[PULSESEL] bit. CPU reads of the DCCAP register will return the shadow register contents.</p> <p>1: Active Mode. In this mode the shadow register is disabled. CPU reads from the DCCAP register will always return the active register contents.</p> <p>Reset type: SYSRSn</p> |
| 0   | CAPE     | R/W  | 0h    | <p>TBCTR Counter Capture Enable/Disable</p> <p>0: Disable the time-base counter capture.</p> <p>1: Enable the time-base counter capture.</p> <p>Reset type: SYSRSn</p>                                                                                                                                                                                                                                                                                                            |

### 26.17.2.73 DCFOFFSET Register (Offset = C9h) [Reset = 0000h]

DCFOFFSET is shown in [Figure 26-165](#) and described in [Table 26-96](#).

Return to the [Summary Table](#).

Digital Compare Filter Offset Register

**Figure 26-165. DCFOFFSET Register**

|           |    |    |    |    |    |   |   |
|-----------|----|----|----|----|----|---|---|
| 15        | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| DCFOFFSET |    |    |    |    |    |   |   |
| R/W-0h    |    |    |    |    |    |   |   |
| 7         | 6  | 5  | 4  | 3  | 2  | 1 | 0 |
| DCFOFFSET |    |    |    |    |    |   |   |
| R/W-0h    |    |    |    |    |    |   |   |

**Table 26-96. DCFOFFSET Register Field Descriptions**

| Bit  | Field     | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|-----------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-0 | DCFOFFSET | R/W  | 0h    | <p>Blanking Window Offset</p> <p>These 16-bits specify the number of TBCLK cycles from the blanking window reference to the point when the blanking window is applied. The blanking window reference is either period or zero as defined by the DCFCTL[PULSESEL] bit. This offset register is shadowed and the active register is loaded at the reference point defined by DCFCTL[PULSESEL]. The offset counter is also initialized and begins to count down when the active register is loaded. When the counter expires, the blanking window is applied. If the blanking window is currently active, then the blanking window counter is restarted.</p> <p>Reset type: SYSRSn</p> |

### 26.17.2.74 DCFFOFFSETCNT Register (Offset = CAh) [Reset = 0000h]

DCFFOFFSETCNT is shown in [Figure 26-166](#) and described in [Table 26-97](#).

Return to the [Summary Table](#).

Digital Compare Filter Offset Counter Register

**Figure 26-166. DCFFOFFSETCNT Register**

|               |    |    |    |    |    |   |   |
|---------------|----|----|----|----|----|---|---|
| 15            | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| DCFFOFFSETCNT |    |    |    |    |    |   |   |
| R-0h          |    |    |    |    |    |   |   |
| 7             | 6  | 5  | 4  | 3  | 2  | 1 | 0 |
| DCFFOFFSETCNT |    |    |    |    |    |   |   |
| R-0h          |    |    |    |    |    |   |   |

**Table 26-97. DCFFOFFSETCNT Register Field Descriptions**

| Bit  | Field         | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------|---------------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-0 | DCFFOFFSETCNT | R    | 0h    | Blanking Offset Counter<br>These 16-bits are read only and indicate the current value of the offset counter. The counter counts down to zero and then stops until it is re-loaded on the next period or zero event as defined by the DCFCTL[PULSESEL] bit. The offset counter is not affected by the free/soft emulation bits. That is, it will always continue to count down if the device is halted by a emulation stop.<br>Reset type: SYSRSn |

### 26.17.2.75 DCFWINDOW Register (Offset = CBh) [Reset = 0000h]

DCFWINDOW is shown in [Figure 26-167](#) and described in [Table 26-98](#).

Return to the [Summary Table](#).

Digital Compare Filter Window Register

**Figure 26-167. DCFWINDOW Register**

|           |    |    |    |    |    |   |   |
|-----------|----|----|----|----|----|---|---|
| 15        | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| DCFWINDOW |    |    |    |    |    |   |   |
| R/W-0h    |    |    |    |    |    |   |   |
| 7         | 6  | 5  | 4  | 3  | 2  | 1 | 0 |
| DCFWINDOW |    |    |    |    |    |   |   |
| R/W-0h    |    |    |    |    |    |   |   |

**Table 26-98. DCFWINDOW Register Field Descriptions**

| Bit  | Field     | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------|-----------|------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-0 | DCFWINDOW | R/W  | 0h    | <p>Blanking Window Width</p> <p>00h: No blanking window is generated.</p> <p>01-FFFFh: Specifies the width of the blanking window in TBCLK cycles. The blanking window begins when the offset counter expires. When this occurs, the window counter is loaded and begins to count down. If the blanking window is currently active and the offset counter expires, the blanking window counter is not restarted and the blanking window is cut short prematurely. Care should be taken to avoid this situation. The blanking window can cross a PWM period boundary.</p> <p>Reset type: SYSRSn</p> |

### 26.17.2.76 DCFWINDOWCNT Register (Offset = CCh) [Reset = 0000h]

DCFWINDOWCNT is shown in [Figure 26-168](#) and described in [Table 26-99](#).

Return to the [Summary Table](#).

Digital Compare Filter Window Counter Register

**Figure 26-168. DCFWINDOWCNT Register**

|              |    |    |    |    |    |   |   |
|--------------|----|----|----|----|----|---|---|
| 15           | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| DCFWINDOWCNT |    |    |    |    |    |   |   |
| R-0h         |    |    |    |    |    |   |   |
| 7            | 6  | 5  | 4  | 3  | 2  | 1 | 0 |
| DCFWINDOWCNT |    |    |    |    |    |   |   |
| R-0h         |    |    |    |    |    |   |   |

**Table 26-99. DCFWINDOWCNT Register Field Descriptions**

| Bit  | Field        | Type | Reset | Description                                                                                                                                                                                                                                         |
|------|--------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-0 | DCFWINDOWCNT | R    | 0h    | Blanking Window Counter<br>These 16 bits are read only and indicate the current value of the window counter. The counter counts down to zero and then stops until it is re-loaded when the offset counter reaches zero again.<br>Reset type: SYSRSn |

### 26.17.2.77 DCCAP Register (Offset = CFh) [Reset = 0000h]

DCCAP is shown in [Figure 26-169](#) and described in [Table 26-100](#).

Return to the [Summary Table](#).

Digital Compare Counter Capture Register

**Figure 26-169. DCCAP Register**

|       |    |    |    |    |    |   |   |
|-------|----|----|----|----|----|---|---|
| 15    | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| DCCAP |    |    |    |    |    |   |   |
| R-0h  |    |    |    |    |    |   |   |
| 7     | 6  | 5  | 4  | 3  | 2  | 1 | 0 |
| DCCAP |    |    |    |    |    |   |   |
| R-0h  |    |    |    |    |    |   |   |

**Table 26-100. DCCAP Register Field Descriptions**

| Bit  | Field | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|-------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-0 | DCCAP | R    | 0h    | <p>Digital Compare Time-Base Counter Capture</p> <p>To enable time-base counter capture, set the DCCAPCTL[CAPE] bit to 1. If enabled, reflects the value of the time-base counter (TBCTR) on the low to high edge transition of a filtered (DCEVTFLT) event. Further capture events are ignored until the next period or zero as selected by the DCFCTL[PULSESEL] bit. Shadowing of DCCAP is enabled and disabled by the DCCAPCTL[SHDWMODE] bit. By default this register is shadowed.</p> <ul style="list-style-type: none"> <li>- If DCCAPCTL[SHDWMODE] = 0, then the shadow is enabled. In this mode, the active register is copied to the shadow register on the TBCTR = TBPRD or TBCTR = zero as defined by the DCFCTL[PULSESEL] bit. CPU reads of this register will return the shadow register value.</li> <li>- If DCCAPCTL[SHDWMODE] = 1, then the shadow register is disabled. In this mode, CPU reads will return the active register value. The active and shadow registers share the same memory map address.</li> </ul> <p>Reset type: SYSRSn</p> |

### 26.17.2.78 DCAHTRIPSEL Register (Offset = D2h) [Reset = 0000h]

DCAHTRIPSEL is shown in [Figure 26-170](#) and described in [Table 26-101](#).

Return to the [Summary Table](#).

Digital Compare AH Trip Select

**Figure 26-170. DCAHTRIPSEL Register**

| 15         | 14          | 13          | 12         | 11          | 10          | 9           | 8          |
|------------|-------------|-------------|------------|-------------|-------------|-------------|------------|
| RESERVED   | TRIPINPUT15 | TRIPINPUT14 | RESERVED   | TRIPINPUT12 | TRIPINPUT11 | TRIPINPUT10 | TRIPINPUT9 |
| R-0h       | R/W-0h      | R/W-0h      | R/W-0h     | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h     |
| 7          | 6           | 5           | 4          | 3           | 2           | 1           | 0          |
| TRIPINPUT8 | TRIPINPUT7  | TRIPINPUT6  | TRIPINPUT5 | TRIPINPUT4  | TRIPINPUT3  | TRIPINPUT2  | TRIPINPUT1 |
| R/W-0h     | R/W-0h      | R/W-0h      | R/W-0h     | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h     |

**Table 26-101. DCAHTRIPSEL Register Field Descriptions**

| Bit | Field       | Type | Reset | Description                                                                                                                                                           |
|-----|-------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15  | RESERVED    | R    | 0h    | Reserved                                                                                                                                                              |
| 14  | TRIPINPUT15 | R/W  | 0h    | TRIP Input 15<br>0: Trip Input 15 not selected as combinational ORed input<br>1: Trip Input 15 selected as combinational ORed input to DCAH mux<br>Reset type: SYSRSn |
| 13  | TRIPINPUT14 | R/W  | 0h    | TRIP Input 14<br>0: Trip Input 14 not selected as combinational ORed input<br>1: Trip Input 14 selected as combinational ORed input to DCAH mux<br>Reset type: SYSRSn |
| 12  | RESERVED    | R/W  | 0h    | Reserved                                                                                                                                                              |
| 11  | TRIPINPUT12 | R/W  | 0h    | TRIP Input 12<br>0: Trip Input 12 not selected as combinational ORed input<br>1: Trip Input 12 selected as combinational ORed input to DCAH mux<br>Reset type: SYSRSn |
| 10  | TRIPINPUT11 | R/W  | 0h    | TRIP Input 11<br>0: Trip Input 11 not selected as combinational ORed input<br>1: Trip Input 11 selected as combinational ORed input to DCAH mux<br>Reset type: SYSRSn |
| 9   | TRIPINPUT10 | R/W  | 0h    | TRIP Input 10<br>0: Trip Input 10 not selected as combinational ORed input<br>1: Trip Input 10 selected as combinational ORed input to DCAH mux<br>Reset type: SYSRSn |
| 8   | TRIPINPUT9  | R/W  | 0h    | TRIP Input 9<br>0: Trip Input 9 not selected as combinational ORed input<br>1: Trip Input 9 selected as combinational ORed input to DCAH mux<br>Reset type: SYSRSn    |
| 7   | TRIPINPUT8  | R/W  | 0h    | TRIP Input 8<br>0: Trip Input 8 not selected as combinational ORed input<br>1: Trip Input 8 selected as combinational ORed input to DCAH mux<br>Reset type: SYSRSn    |
| 6   | TRIPINPUT7  | R/W  | 0h    | TRIP Input 7<br>0: Trip Input 7 not selected as combinational ORed input<br>1: Trip Input 7 selected as combinational ORed input to DCAH mux<br>Reset type: SYSRSn    |
| 5   | TRIPINPUT6  | R/W  | 0h    | TRIP Input 6<br>0: Trip Input 6 not selected as combinational ORed input<br>1: Trip Input 6 selected as combinational ORed input to DCAH mux<br>Reset type: SYSRSn    |

**Table 26-101. DCAHTRIPSEL Register Field Descriptions (continued)**

| Bit | Field      | Type | Reset | Description                                                                                                                                                        |
|-----|------------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4   | TRIPINPUT5 | R/W  | 0h    | TRIP Input 5<br>0: Trip Input 5 not selected as combinational ORed input<br>1: Trip Input 5 selected as combinational ORed input to DCAH mux<br>Reset type: SYSRSn |
| 3   | TRIPINPUT4 | R/W  | 0h    | TRIP Input 4<br>0: Trip Input 4 not selected as combinational ORed input<br>1: Trip Input 4 selected as combinational ORed input to DCAH mux<br>Reset type: SYSRSn |
| 2   | TRIPINPUT3 | R/W  | 0h    | TRIP Input 3<br>0: Trip Input 3 not selected as combinational ORed input<br>1: Trip Input 3 selected as combinational ORed input to DCAH mux<br>Reset type: SYSRSn |
| 1   | TRIPINPUT2 | R/W  | 0h    | TRIP Input 2<br>0: Trip Input 2 not selected as combinational ORed input<br>1: Trip Input 2 selected as combinational ORed input to DCAH mux<br>Reset type: SYSRSn |
| 0   | TRIPINPUT1 | R/W  | 0h    | TRIP Input 1<br>0: Trip Input 1 not selected as combinational ORed input<br>1: Trip Input 1 selected as combinational ORed input to DCAH mux<br>Reset type: SYSRSn |



### 26.17.2.79 DCALTRIPSEL Register (Offset = D3h) [Reset = 0000h]

DCALTRIPSEL is shown in [Figure 26-171](#) and described in [Table 26-102](#).

Return to the [Summary Table](#).

Digital Compare AL Trip Select

**Figure 26-171. DCALTRIPSEL Register**

| 15         | 14          | 13          | 12         | 11          | 10          | 9           | 8          |
|------------|-------------|-------------|------------|-------------|-------------|-------------|------------|
| RESERVED   | TRIPINPUT15 | TRIPINPUT14 | RESERVED   | TRIPINPUT12 | TRIPINPUT11 | TRIPINPUT10 | TRIPINPUT9 |
| R-0h       | R/W-0h      | R/W-0h      | R/W-0h     | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h     |
| 7          | 6           | 5           | 4          | 3           | 2           | 1           | 0          |
| TRIPINPUT8 | TRIPINPUT7  | TRIPINPUT6  | TRIPINPUT5 | TRIPINPUT4  | TRIPINPUT3  | TRIPINPUT2  | TRIPINPUT1 |
| R/W-0h     | R/W-0h      | R/W-0h      | R/W-0h     | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h     |

**Table 26-102. DCALTRIPSEL Register Field Descriptions**

| Bit | Field       | Type | Reset | Description                                                                                                                                                           |
|-----|-------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15  | RESERVED    | R    | 0h    | Reserved                                                                                                                                                              |
| 14  | TRIPINPUT15 | R/W  | 0h    | TRIP Input 15<br>0: Trip Input 15 not selected as combinational ORed input<br>1: Trip Input 15 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 13  | TRIPINPUT14 | R/W  | 0h    | TRIP Input 14<br>0: Trip Input 14 not selected as combinational ORed input<br>1: Trip Input 14 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 12  | RESERVED    | R/W  | 0h    | Reserved                                                                                                                                                              |
| 11  | TRIPINPUT12 | R/W  | 0h    | TRIP Input 12<br>0: Trip Input 12 not selected as combinational ORed input<br>1: Trip Input 12 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 10  | TRIPINPUT11 | R/W  | 0h    | TRIP Input 11<br>0: Trip Input 11 not selected as combinational ORed input<br>1: Trip Input 11 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 9   | TRIPINPUT10 | R/W  | 0h    | TRIP Input 10<br>0: Trip Input 10 not selected as combinational ORed input<br>1: Trip Input 10 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 8   | TRIPINPUT9  | R/W  | 0h    | TRIP Input 9<br>0: Trip Input 9 not selected as combinational ORed input<br>1: Trip Input 9 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn    |
| 7   | TRIPINPUT8  | R/W  | 0h    | TRIP Input 8<br>0: Trip Input 8 not selected as combinational ORed input<br>1: Trip Input 8 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn    |
| 6   | TRIPINPUT7  | R/W  | 0h    | TRIP Input 7<br>0: Trip Input 7 not selected as combinational ORed input<br>1: Trip Input 7 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn    |
| 5   | TRIPINPUT6  | R/W  | 0h    | TRIP Input 6<br>0: Trip Input 6 not selected as combinational ORed input<br>1: Trip Input 6 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn    |

**Table 26-102. DCALTRIPSEL Register Field Descriptions (continued)**

| Bit | Field      | Type | Reset | Description                                                                                                                                                        |
|-----|------------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4   | TRIPINPUT5 | R/W  | 0h    | TRIP Input 5<br>0: Trip Input 5 not selected as combinational ORed input<br>1: Trip Input 5 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 3   | TRIPINPUT4 | R/W  | 0h    | TRIP Input 4<br>0: Trip Input 4 not selected as combinational ORed input<br>1: Trip Input 4 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 2   | TRIPINPUT3 | R/W  | 0h    | TRIP Input 3<br>0: Trip Input 3 not selected as combinational ORed input<br>1: Trip Input 3 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 1   | TRIPINPUT2 | R/W  | 0h    | TRIP Input 2<br>0: Trip Input 2 not selected as combinational ORed input<br>1: Trip Input 2 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 0   | TRIPINPUT1 | R/W  | 0h    | TRIP Input 1<br>0: Trip Input 1 not selected as combinational ORed input<br>1: Trip Input 1 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |

### 26.17.2.80 DCBHTRIPSEL Register (Offset = D4h) [Reset = 0000h]

DCBHTRIPSEL is shown in [Figure 26-172](#) and described in [Table 26-103](#).

Return to the [Summary Table](#).

Digital Compare BH Trip Select

**Figure 26-172. DCBHTRIPSEL Register**

| 15         | 14          | 13          | 12         | 11          | 10          | 9           | 8          |
|------------|-------------|-------------|------------|-------------|-------------|-------------|------------|
| RESERVED   | TRIPINPUT15 | TRIPINPUT14 | RESERVED   | TRIPINPUT12 | TRIPINPUT11 | TRIPINPUT10 | TRIPINPUT9 |
| R-0h       | R/W-0h      | R/W-0h      | R/W-0h     | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h     |
| 7          | 6           | 5           | 4          | 3           | 2           | 1           | 0          |
| TRIPINPUT8 | TRIPINPUT7  | TRIPINPUT6  | TRIPINPUT5 | TRIPINPUT4  | TRIPINPUT3  | TRIPINPUT2  | TRIPINPUT1 |
| R/W-0h     | R/W-0h      | R/W-0h      | R/W-0h     | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h     |

**Table 26-103. DCBHTRIPSEL Register Field Descriptions**

| Bit | Field       | Type | Reset | Description                                                                                                                                                           |
|-----|-------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15  | RESERVED    | R    | 0h    | Reserved                                                                                                                                                              |
| 14  | TRIPINPUT15 | R/W  | 0h    | TRIP Input 15<br>0: Trip Input 15 not selected as combinational ORed input<br>1: Trip Input 15 selected as combinational ORed input to DCBH mux<br>Reset type: SYSRSn |
| 13  | TRIPINPUT14 | R/W  | 0h    | TRIP Input 14<br>0: Trip Input 14 not selected as combinational ORed input<br>1: Trip Input 14 selected as combinational ORed input to DCBH mux<br>Reset type: SYSRSn |
| 12  | RESERVED    | R/W  | 0h    | Reserved                                                                                                                                                              |
| 11  | TRIPINPUT12 | R/W  | 0h    | TRIP Input 12<br>0: Trip Input 12 not selected as combinational ORed input<br>1: Trip Input 12 selected as combinational ORed input to DCBH mux<br>Reset type: SYSRSn |
| 10  | TRIPINPUT11 | R/W  | 0h    | TRIP Input 11<br>0: Trip Input 11 not selected as combinational ORed input<br>1: Trip Input 11 selected as combinational ORed input to DCBH mux<br>Reset type: SYSRSn |
| 9   | TRIPINPUT10 | R/W  | 0h    | TRIP Input 10<br>0: Trip Input 10 not selected as combinational ORed input<br>1: Trip Input 10 selected as combinational ORed input to DCBH mux<br>Reset type: SYSRSn |
| 8   | TRIPINPUT9  | R/W  | 0h    | TRIP Input 9<br>0: Trip Input 9 not selected as combinational ORed input<br>1: Trip Input 9 selected as combinational ORed input to DCBH mux<br>Reset type: SYSRSn    |
| 7   | TRIPINPUT8  | R/W  | 0h    | TRIP Input 8<br>0: Trip Input 8 not selected as combinational ORed input<br>1: Trip Input 8 selected as combinational ORed input to DCBH mux<br>Reset type: SYSRSn    |
| 6   | TRIPINPUT7  | R/W  | 0h    | TRIP Input 7<br>0: Trip Input 7 not selected as combinational ORed input<br>1: Trip Input 7 selected as combinational ORed input to DCBH mux<br>Reset type: SYSRSn    |
| 5   | TRIPINPUT6  | R/W  | 0h    | TRIP Input 6<br>0: Trip Input 6 not selected as combinational ORed input<br>1: Trip Input 6 selected as combinational ORed input to DCBH mux<br>Reset type: SYSRSn    |

**Table 26-103. DCBHTRIPSEL Register Field Descriptions (continued)**

| Bit | Field      | Type | Reset | Description                                                                                                                                                        |
|-----|------------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4   | TRIPINPUT5 | R/W  | 0h    | TRIP Input 5<br>0: Trip Input 5 not selected as combinational ORed input<br>1: Trip Input 5 selected as combinational ORed input to DCBH mux<br>Reset type: SYSRSn |
| 3   | TRIPINPUT4 | R/W  | 0h    | TRIP Input 4<br>0: Trip Input 4 not selected as combinational ORed input<br>1: Trip Input 4 selected as combinational ORed input to DCBH mux<br>Reset type: SYSRSn |
| 2   | TRIPINPUT3 | R/W  | 0h    | TRIP Input 3<br>0: Trip Input 3 not selected as combinational ORed input<br>1: Trip Input 3 selected as combinational ORed input to DCBH mux<br>Reset type: SYSRSn |
| 1   | TRIPINPUT2 | R/W  | 0h    | TRIP Input 2<br>0: Trip Input 2 not selected as combinational ORed input<br>1: Trip Input 2 selected as combinational ORed input to DCBH mux<br>Reset type: SYSRSn |
| 0   | TRIPINPUT1 | R/W  | 0h    | TRIP Input 1<br>0: Trip Input 1 not selected as combinational ORed input<br>1: Trip Input 1 selected as combinational ORed input to DCBH mux<br>Reset type: SYSRSn |

### 26.17.2.81 DCBLTRIPSEL Register (Offset = D5h) [Reset = 0000h]

DCBLTRIPSEL is shown in [Figure 26-173](#) and described in [Table 26-104](#).

Return to the [Summary Table](#).

Digital Compare BL Trip Select

**Figure 26-173. DCBLTRIPSEL Register**

| 15         | 14          | 13          | 12         | 11          | 10          | 9           | 8          |
|------------|-------------|-------------|------------|-------------|-------------|-------------|------------|
| RESERVED   | TRIPINPUT15 | TRIPINPUT14 | RESERVED   | TRIPINPUT12 | TRIPINPUT11 | TRIPINPUT10 | TRIPINPUT9 |
| R-0h       | R/W-0h      | R/W-0h      | R/W-0h     | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h     |
| 7          | 6           | 5           | 4          | 3           | 2           | 1           | 0          |
| TRIPINPUT8 | TRIPINPUT7  | TRIPINPUT6  | TRIPINPUT5 | TRIPINPUT4  | TRIPINPUT3  | TRIPINPUT2  | TRIPINPUT1 |
| R/W-0h     | R/W-0h      | R/W-0h      | R/W-0h     | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h     |

**Table 26-104. DCBLTRIPSEL Register Field Descriptions**

| Bit | Field       | Type | Reset | Description                                                                                                                                                           |
|-----|-------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15  | RESERVED    | R    | 0h    | Reserved                                                                                                                                                              |
| 14  | TRIPINPUT15 | R/W  | 0h    | TRIP Input 15<br>0: Trip Input 15 not selected as combinational ORed input<br>1: Trip Input 15 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 13  | TRIPINPUT14 | R/W  | 0h    | TRIP Input 14<br>0: Trip Input 14 not selected as combinational ORed input<br>1: Trip Input 14 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 12  | RESERVED    | R/W  | 0h    | Reserved                                                                                                                                                              |
| 11  | TRIPINPUT12 | R/W  | 0h    | TRIP Input 12<br>0: Trip Input 12 not selected as combinational ORed input<br>1: Trip Input 12 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 10  | TRIPINPUT11 | R/W  | 0h    | TRIP Input 11<br>0: Trip Input 11 not selected as combinational ORed input<br>1: Trip Input 11 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 9   | TRIPINPUT10 | R/W  | 0h    | TRIP Input 10<br>0: Trip Input 10 not selected as combinational ORed input<br>1: Trip Input 10 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 8   | TRIPINPUT9  | R/W  | 0h    | TRIP Input 9<br>0: Trip Input 9 not selected as combinational ORed input<br>1: Trip Input 9 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn    |
| 7   | TRIPINPUT8  | R/W  | 0h    | TRIP Input 8<br>0: Trip Input 8 not selected as combinational ORed input<br>1: Trip Input 8 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn    |
| 6   | TRIPINPUT7  | R/W  | 0h    | TRIP Input 7<br>0: Trip Input 7 not selected as combinational ORed input<br>1: Trip Input 7 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn    |
| 5   | TRIPINPUT6  | R/W  | 0h    | TRIP Input 6<br>0: Trip Input 6 not selected as combinational ORed input<br>1: Trip Input 6 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn    |

**Table 26-104. DCBLTRIPSEL Register Field Descriptions (continued)**

| Bit | Field      | Type | Reset | Description                                                                                                                                                        |
|-----|------------|------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4   | TRIPINPUT5 | R/W  | 0h    | TRIP Input 5<br>0: Trip Input 5 not selected as combinational ORed input<br>1: Trip Input 5 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 3   | TRIPINPUT4 | R/W  | 0h    | TRIP Input 4<br>0: Trip Input 4 not selected as combinational ORed input<br>1: Trip Input 4 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 2   | TRIPINPUT3 | R/W  | 0h    | TRIP Input 3<br>0: Trip Input 3 not selected as combinational ORed input<br>1: Trip Input 3 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 1   | TRIPINPUT2 | R/W  | 0h    | TRIP Input 2<br>0: Trip Input 2 not selected as combinational ORed input<br>1: Trip Input 2 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |
| 0   | TRIPINPUT1 | R/W  | 0h    | TRIP Input 1<br>0: Trip Input 1 not selected as combinational ORed input<br>1: Trip Input 1 selected as combinational ORed input to DCAL mux<br>Reset type: SYSRSn |

### 26.17.2.82 EPWMLOCK Register (Offset = FAh) [Reset = 00000000h]

EPWMLOCK is shown in [Figure 26-174](#) and described in [Table 26-105](#).

Return to the [Summary Table](#).

EPWM Lock Register

**Figure 26-174. EPWMLOCK Register**

|          |    |    |            |            |            |            |            |
|----------|----|----|------------|------------|------------|------------|------------|
| 31       | 30 | 29 | 28         | 27         | 26         | 25         | 24         |
| KEY      |    |    |            |            |            |            |            |
| R-0/W-0h |    |    |            |            |            |            |            |
| 23       | 22 | 21 | 20         | 19         | 18         | 17         | 16         |
| KEY      |    |    |            |            |            |            |            |
| R-0/W-0h |    |    |            |            |            |            |            |
| 15       | 14 | 13 | 12         | 11         | 10         | 9          | 8          |
| RESERVED |    |    |            |            |            |            |            |
| R-0h     |    |    |            |            |            |            |            |
| 7        | 6  | 5  | 4          | 3          | 2          | 1          | 0          |
| RESERVED |    |    | DCLOCK     | TZCLRLOCK  | TZCFGLOCK  | GLLOCK     | HRLOCK     |
| R-0h     |    |    | R/WOnce-0h | R/WOnce-0h | R/WOnce-0h | R/WOnce-0h | R/WOnce-0h |

**Table 26-105. EPWMLOCK Register Field Descriptions**

| Bit   | Field     | Type    | Reset | Description                                                                                                                                                                                                                                                                       |
|-------|-----------|---------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | KEY       | R-0/W   | 0h    | Write to this register succeeds only if this field is written with a value of 0xa5a5<br>Note:<br>[1] Due to this KEY, only 32-bit writes will succeed (provided the KEY matches). 16-bit writes to the upper or lower half of this register will be ignored<br>Reset type: SYSRSn |
| 15-5  | RESERVED  | R       | 0h    | Reserved                                                                                                                                                                                                                                                                          |
| 4     | DCLOCK    | R/WOnce | 0h    | 0: Digital Compare registers from 0xC0 to 0xD9 offsets are protected by EALLOW.<br>1: Digital Compare registers from 0xC0 to 0xD9 offsets are locked and not writable.<br>Reset type: SYSRSn                                                                                      |
| 3     | TZCLRLOCK | R/WOnce | 0h    | 0: Trip Zone registers from 0x97 to 0x9B offsets are protected by EALLOW.<br>1: Trip Zone registers from 0x97 to 0x9B offsets are locked and not writable.<br>Reset type: SYSRSn                                                                                                  |
| 2     | TZCFGLOCK | R/WOnce | 0h    | 0: TripZone registers from 0x80 to 0x8D and TZTRIPOUTSEL at 0x9D offsets are protected by EALLOW.<br>1: TripZone registers from 0x80 to 0x8D and TZTRIPOUTSEL at 0x9D offsets are locked and not writable.<br>Reset type: SYSRSn                                                  |
| 1     | GLLOCK    | R/WOnce | 0h    | 0: Global Load registers from 0x34 to 0x35 offsets are protected by EALLOW.<br>1: Global Load registers from 0x34 to 0x35 offsets are locked and not writable<br>Reset type: SYSRSn                                                                                               |

**Table 26-105. EPWMLOCK Register Field Descriptions (continued)**

| Bit | Field  | Type    | Reset | Description                                                                                                                                                              |
|-----|--------|---------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0   | HRLOCK | R/WOnce | 0h    | 0: HRPWM registers from 0x20 to 0x2D offsets are protected by EALLOW<br>1: HRPWM registers from 0x20 and 0x2D offsets are locked and not writable.<br>Reset type: SYSRSn |



### 26.17.2.83 HWVDELVAL Register (Offset = FDh) [Reset = 0000h]

HWVDELVAL is shown in [Figure 26-175](#) and described in [Table 26-106](#).

Return to the [Summary Table](#).

Hardware Valley Mode Delay Register

**Figure 26-175. HWVDELVAL Register**

|           |    |    |    |    |    |   |   |
|-----------|----|----|----|----|----|---|---|
| 15        | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| HWVDELVAL |    |    |    |    |    |   |   |
| R-0h      |    |    |    |    |    |   |   |
| 7         | 6  | 5  | 4  | 3  | 2  | 1 | 0 |
| HWVDELVAL |    |    |    |    |    |   |   |
| R-0h      |    |    |    |    |    |   |   |

**Table 26-106. HWVDELVAL Register Field Descriptions**

| Bit  | Field     | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                            |
|------|-----------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-0 | HWVDELVAL | R    | 0h    | <p>Hardware Valley Delay Value Register</p> <p>This read only register reflects the hardware delay value calculated by the equations defined in VCAPCTL[VDELAYDIV]. This reflects the latest value from the hardware calculations and can change every time valley capture sequence is triggered and VCAP1 and VCAP2 values are updated.</p> <p>Reset type: SYSRSn</p> |

### 26.17.2.84 VCNTVAL Register (Offset = FEh) [Reset = 0000h]

VCNTVAL is shown in [Figure 26-176](#) and described in [Table 26-107](#).

Return to the [Summary Table](#).

Hardware Valley Counter Register

**Figure 26-176. VCNTVAL Register**

|         |    |    |    |    |    |   |   |
|---------|----|----|----|----|----|---|---|
| 15      | 14 | 13 | 12 | 11 | 10 | 9 | 8 |
| VCNTVAL |    |    |    |    |    |   |   |
| R-0h    |    |    |    |    |    |   |   |
| 7       | 6  | 5  | 4  | 3  | 2  | 1 | 0 |
| VCNTVAL |    |    |    |    |    |   |   |
| R-0h    |    |    |    |    |    |   |   |

**Table 26-107. VCNTVAL Register Field Descriptions**

| Bit  | Field   | Type | Reset | Description                                                                                                                                                         |
|------|---------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-0 | VCNTVAL | R    | 0h    | Valley Time Base Counter Register<br>This register reflects the captured VCNT value upon occurrence of STOPEDGE selected in VCNTCFG register.<br>Reset type: SYSRSn |

### 26.17.3 SYNC\_SOC\_REGS Registers

Table 26-108 lists the memory-mapped registers for the SYNC\_SOC\_REGS registers. All register offset addresses not listed in Table 26-108 should be considered as reserved locations and the register contents should not be modified.

**Table 26-108. SYNC\_SOC\_REGS Registers**

| Offset | Acronym         | Register Name                               | Write Protection | Section            |
|--------|-----------------|---------------------------------------------|------------------|--------------------|
| 0h     | SYNCSELECT      | Sync Input and Output Select Register       | EALLOW           | <a href="#">Go</a> |
| 2h     | ADCSOCOUTSELECT | External ADC (Off Chip) SOC Select Register | EALLOW           | <a href="#">Go</a> |
| 4h     | SYNCSOCCLOCK    | SYNCSEL and EXTADC SOC Select Lock register | EALLOW           | <a href="#">Go</a> |

Complex bit access types are encoded to fit into small table cells. Table 26-109 shows the codes that are used for access types in this section.

**Table 26-109. SYNC\_SOC\_REGS Access Type Codes**

| Access Type              | Code       | Description                                                                                                                                                                                                                                                                          |
|--------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Read Type                |            |                                                                                                                                                                                                                                                                                      |
| R                        | R          | Read                                                                                                                                                                                                                                                                                 |
| R-0                      | R<br>-0    | Read<br>Returns 0s                                                                                                                                                                                                                                                                   |
| Write Type               |            |                                                                                                                                                                                                                                                                                      |
| W                        | W          | Write                                                                                                                                                                                                                                                                                |
| WSonce                   | W<br>Sonce | Write<br>Set once                                                                                                                                                                                                                                                                    |
| Reset or Default Value   |            |                                                                                                                                                                                                                                                                                      |
| -n                       |            | Value after reset or the default value                                                                                                                                                                                                                                               |
| Register Array Variables |            |                                                                                                                                                                                                                                                                                      |
| i,j,k,l,m,n              |            | When these variables are used in a register name, an offset, or an address, they refer to the value of a register array where the register is part of a group of repeating registers. The register groups form a hierarchical structure and the array is represented with a formula. |
| y                        |            | When this variable is used in a register name, an offset, or an address it refers to the value of a register array.                                                                                                                                                                  |

### 26.17.3.1 SYNCSELECT Register (Offset = 0h) [Reset = 00000000h]

SYNCSELECT is shown in [Figure 26-177](#) and described in [Table 26-110](#).

Return to the [Summary Table](#).

Sync Input and Output Select Register

**Figure 26-177. SYNCSELECT Register**

|          |          |          |    |          |          |    |          |
|----------|----------|----------|----|----------|----------|----|----------|
| 31       | 30       | 29       | 28 | 27       | 26       | 25 | 24       |
| RESERVED |          |          |    | SYNCOUT  |          |    |          |
| R/W-0h   |          |          |    | R/W-0h   |          |    |          |
| 23       | 22       | 21       | 20 | 19       | 18       | 17 | 16       |
| RESERVED |          |          |    | RESERVED |          |    |          |
| R/W-0h   |          |          |    | R/W-0h   |          |    |          |
| 15       | 14       | 13       | 12 | 11       | 10       | 9  | 8        |
| RESERVED | RESERVED |          |    | RESERVED |          |    | RESERVED |
| R-0-0h   | R/W-0h   |          |    | R/W-0h   |          |    | R/W-0h   |
| 7        | 6        | 5        | 4  | 3        | 2        | 1  | 0        |
| RESERVED |          | RESERVED |    |          | RESERVED |    |          |
| R/W-0h   |          | R/W-0h   |    |          | R/W-0h   |    |          |

**Table 26-110. SYNCSELECT Register Field Descriptions**

| Bit   | Field    | Type | Reset | Description |
|-------|----------|------|-------|-------------|
| 31-29 | RESERVED | R/W  | 0h    | Reserved    |

**Table 26-110. SYNCSELECT Register Field Descriptions (continued)**

| Bit   | Field    | Type | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 28-24 | SYNCOUT  | R/W  | 0h    | <p>Select Syncout Source:</p> <p>00000: EPWM1SYNCOUT selected to drive the SYNCOUT pin.<br/> 00001: EPWM2SYNCOUT selected to drive the SYNCOUT pin.<br/> 00010: EPWM3SYNCOUT selected to drive the SYNCOUT pin.<br/> 00011: EPWM4SYNCOUT selected to drive the SYNCOUT pin.<br/> 00100: EPWM5SYNCOUT selected to drive the SYNCOUT pin.<br/> 00101: EPWM6SYNCOUT selected to drive the SYNCOUT pin.<br/> 00110: EPWM7SYNCOUT selected to drive the SYNCOUT pin.<br/> 00111: EPWM8SYNCOUT selected to drive the SYNCOUT pin.<br/> 01000: EPWM9SYNCOUT selected to drive the SYNCOUT pin.<br/> 01001: EPWM10SYNCOUT selected to drive the SYNCOUT pin.<br/> 01010: EPWM11SYNCOUT selected to drive the SYNCOUT pin.<br/> 01011: EPWM12SYNCOUT selected to drive the SYNCOUT pin.<br/> 01100: EPWM13SYNCOUT selected to drive the SYNCOUT pin.<br/> 01101: EPWM14SYNCOUT selected to drive the SYNCOUT pin.<br/> 01110: EPWM15SYNCOUT selected to drive the SYNCOUT pin.<br/> 01111: EPWM16SYNCOUT selected to drive the SYNCOUT pin.<br/> 10000: Reserved<br/> 10001: Reserved<br/> 10010: Reserved<br/> 10011: Reserved<br/> 10100: Reserved<br/> 10101: Reserved<br/> 10110: Reserved<br/> 10111: Reserved<br/> 11000: ECAP1SYNCOUT selected to drive the SYNCOUT pin.<br/> 11001: ECAP2SYNCOUT selected to drive the SYNCOUT pin.<br/> 11010: ECAP3SYNCOUT selected to drive the SYNCOUT pin.<br/> 11011: ECAP4SYNCOUT selected to drive the SYNCOUT pin.<br/> 11100: ECAP5SYNCOUT selected to drive the SYNCOUT pin.<br/> 11101: ECAP6SYNCOUT selected to drive the SYNCOUT pin.<br/> 11110: ECAP7SYNCOUT selected to drive the SYNCOUT pin.<br/> 11111: Reserved</p> <p>Notes:<br/> [1] Reserved position defaults to 00 selection<br/> Reset type: CPU1.SYSRSn</p> |
| 23-20 | RESERVED | R/W  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 19-16 | RESERVED | R/W  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 15    | RESERVED | R-0  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 14-12 | RESERVED | R/W  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 11-9  | RESERVED | R/W  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 8-6   | RESERVED | R/W  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5-3   | RESERVED | R/W  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2-0   | RESERVED | R/W  | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

### 26.17.3.2 ADCSOCOUTSELECT Register (Offset = 2h) [Reset = 00000000h]

ADCSOCOUTSELECT is shown in [Figure 26-178](#) and described in [Table 26-111](#).

Return to the [Summary Table](#).

External ADC (Off Chip) SOC Select Register

**Figure 26-178. ADCSOCOUTSELECT Register**

|             |             |             |             |             |             |             |            |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------|
| 31          | 30          | 29          | 28          | 27          | 26          | 25          | 24         |
| PWM16SOCBEN | PWM15SOCBEN | PWM14SOCBEN | PWM13SOCBEN | PWM12SOCBEN | PWM11SOCBEN | PWM10SOCBEN | PWM9SOCBEN |
| R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h     |
| 23          | 22          | 21          | 20          | 19          | 18          | 17          | 16         |
| PWM8SOCBEN  | PWM7SOCBEN  | PWM6SOCBEN  | PWM5SOCBEN  | PWM4SOCBEN  | PWM3SOCBEN  | PWM2SOCBEN  | PWM1SOCBEN |
| R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h     |
| 15          | 14          | 13          | 12          | 11          | 10          | 9           | 8          |
| PWM16SOCAEN | PWM15SOCAEN | PWM14SOCAEN | PWM13SOCAEN | PWM12SOCAEN | PWM11SOCAEN | PWM10SOCAEN | PWM9SOCAEN |
| R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h     |
| 7           | 6           | 5           | 4           | 3           | 2           | 1           | 0          |
| PWM8SOCAEN  | PWM7SOCAEN  | PWM6SOCAEN  | PWM5SOCAEN  | PWM4SOCAEN  | PWM3SOCAEN  | PWM2SOCAEN  | PWM1SOCAEN |
| R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h      | R/W-0h     |

**Table 26-111. ADCSOCOUTSELECT Register Field Descriptions**

| Bit | Field       | Type | Reset | Description                                                                                                                                         |
|-----|-------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 31  | PWM16SOCBEN | R/W  | 0h    | ADCSOCBOn source select:<br>0: Respective EPWM SOCB output is not selected<br>1: Respective EPWM SOCB output is selected<br>Reset type: CPU1.SYSRSn |
| 30  | PWM15SOCBEN | R/W  | 0h    | ADCSOCBOn source select:<br>0: Respective EPWM SOCB output is not selected<br>1: Respective EPWM SOCB output is selected<br>Reset type: CPU1.SYSRSn |
| 29  | PWM14SOCBEN | R/W  | 0h    | ADCSOCBOn source select:<br>0: Respective EPWM SOCB output is not selected<br>1: Respective EPWM SOCB output is selected<br>Reset type: CPU1.SYSRSn |
| 28  | PWM13SOCBEN | R/W  | 0h    | ADCSOCBOn source select:<br>0: Respective EPWM SOCB output is not selected<br>1: Respective EPWM SOCB output is selected<br>Reset type: CPU1.SYSRSn |
| 27  | PWM12SOCBEN | R/W  | 0h    | ADCSOCBOn source select:<br>0: Respective EPWM SOCB output is not selected<br>1: Respective EPWM SOCB output is selected<br>Reset type: CPU1.SYSRSn |
| 26  | PWM11SOCBEN | R/W  | 0h    | ADCSOCBOn source select:<br>0: Respective EPWM SOCB output is not selected<br>1: Respective EPWM SOCB output is selected<br>Reset type: CPU1.SYSRSn |
| 25  | PWM10SOCBEN | R/W  | 0h    | ADCSOCBOn source select:<br>0: Respective EPWM SOCB output is not selected<br>1: Respective EPWM SOCB output is selected<br>Reset type: CPU1.SYSRSn |

**Table 26-111. ADCSOCOUTSELECT Register Field Descriptions (continued)**

| Bit | Field       | Type | Reset | Description                                                                                                                                         |
|-----|-------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 24  | PWM9SOCBEN  | R/W  | 0h    | ADCSOCBOn source select:<br>0: Respective EPWM SOCB output is not selected<br>1: Respective EPWM SOCB output is selected<br>Reset type: CPU1.SYSRSn |
| 23  | PWM8SOCBEN  | R/W  | 0h    | ADCSOCBOn source select:<br>0: Respective EPWM SOCB output is not selected<br>1: Respective EPWM SOCB output is selected<br>Reset type: CPU1.SYSRSn |
| 22  | PWM7SOCBEN  | R/W  | 0h    | ADCSOCBOn source select:<br>0: Respective EPWM SOCB output is not selected<br>1: Respective EPWM SOCB output is selected<br>Reset type: CPU1.SYSRSn |
| 21  | PWM6SOCBEN  | R/W  | 0h    | ADCSOCBOn source select:<br>0: Respective EPWM SOCB output is not selected<br>1: Respective EPWM SOCB output is selected<br>Reset type: CPU1.SYSRSn |
| 20  | PWM5SOCBEN  | R/W  | 0h    | ADCSOCBOn source select:<br>0: Respective EPWM SOCB output is not selected<br>1: Respective EPWM SOCB output is selected<br>Reset type: CPU1.SYSRSn |
| 19  | PWM4SOCBEN  | R/W  | 0h    | ADCSOCBOn source select:<br>0: Respective EPWM SOCB output is not selected<br>1: Respective EPWM SOCB output is selected<br>Reset type: CPU1.SYSRSn |
| 18  | PWM3SOCBEN  | R/W  | 0h    | ADCSOCBOn source select:<br>0: Respective EPWM SOCB output is not selected<br>1: Respective EPWM SOCB output is selected<br>Reset type: CPU1.SYSRSn |
| 17  | PWM2SOCBEN  | R/W  | 0h    | ADCSOCBOn source select:<br>0: Respective EPWM SOCB output is not selected<br>1: Respective EPWM SOCB output is selected<br>Reset type: CPU1.SYSRSn |
| 16  | PWM1SOCBEN  | R/W  | 0h    | ADCSOCBOn source select:<br>0: Respective EPWM SOCB output is not selected<br>1: Respective EPWM SOCB output is selected<br>Reset type: CPU1.SYSRSn |
| 15  | PWM16SOCAEN | R/W  | 0h    | ADCSOCAOn source select:<br>0: Respective EPWM SOCA output is not selected<br>1: Respective EPWM SOCA output is selected<br>Reset type: CPU1.SYSRSn |
| 14  | PWM15SOCAEN | R/W  | 0h    | ADCSOCAOn source select:<br>0: Respective EPWM SOCA output is not selected<br>1: Respective EPWM SOCA output is selected<br>Reset type: CPU1.SYSRSn |
| 13  | PWM14SOCAEN | R/W  | 0h    | ADCSOCAOn source select:<br>0: Respective EPWM SOCA output is not selected<br>1: Respective EPWM SOCA output is selected<br>Reset type: CPU1.SYSRSn |
| 12  | PWM13SOCAEN | R/W  | 0h    | ADCSOCAOn source select:<br>0: Respective EPWM SOCA output is not selected<br>1: Respective EPWM SOCA output is selected<br>Reset type: CPU1.SYSRSn |
| 11  | PWM12SOCAEN | R/W  | 0h    | ADCSOCAOn source select:<br>0: Respective EPWM SOCA output is not selected<br>1: Respective EPWM SOCA output is selected<br>Reset type: CPU1.SYSRSn |

**Table 26-111. ADCSOCOUTSELECT Register Field Descriptions (continued)**

| Bit | Field       | Type | Reset | Description                                                                                                                                         |
|-----|-------------|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 10  | PWM11SOCAEN | R/W  | 0h    | ADCSOCAOn source select:<br>0: Respective EPWM SOCA output is not selected<br>1: Respective EPWM SOCA output is selected<br>Reset type: CPU1.SYSRSn |
| 9   | PWM10SOCAEN | R/W  | 0h    | ADCSOCAOn source select:<br>0: Respective EPWM SOCA output is not selected<br>1: Respective EPWM SOCA output is selected<br>Reset type: CPU1.SYSRSn |
| 8   | PWM9SOCAEN  | R/W  | 0h    | ADCSOCAOn source select:<br>0: Respective EPWM SOCA output is not selected<br>1: Respective EPWM SOCA output is selected<br>Reset type: CPU1.SYSRSn |
| 7   | PWM8SOCAEN  | R/W  | 0h    | ADCSOCAOn source select:<br>0: Respective EPWM SOCA output is not selected<br>1: Respective EPWM SOCA output is selected<br>Reset type: CPU1.SYSRSn |
| 6   | PWM7SOCAEN  | R/W  | 0h    | ADCSOCAOn source select:<br>0: Respective EPWM SOCA output is not selected<br>1: Respective EPWM SOCA output is selected<br>Reset type: CPU1.SYSRSn |
| 5   | PWM6SOCAEN  | R/W  | 0h    | ADCSOCAOn source select:<br>0: Respective EPWM SOCA output is not selected<br>1: Respective EPWM SOCA output is selected<br>Reset type: CPU1.SYSRSn |
| 4   | PWM5SOCAEN  | R/W  | 0h    | ADCSOCAOn source select:<br>0: Respective EPWM SOCA output is not selected<br>1: Respective EPWM SOCA output is selected<br>Reset type: CPU1.SYSRSn |
| 3   | PWM4SOCAEN  | R/W  | 0h    | ADCSOCAOn source select:<br>0: Respective EPWM SOCA output is not selected<br>1: Respective EPWM SOCA output is selected<br>Reset type: CPU1.SYSRSn |
| 2   | PWM3SOCAEN  | R/W  | 0h    | ADCSOCAOn source select:<br>0: Respective EPWM SOCA output is not selected<br>1: Respective EPWM SOCA output is selected<br>Reset type: CPU1.SYSRSn |
| 1   | PWM2SOCAEN  | R/W  | 0h    | ADCSOCAOn source select:<br>0: Respective EPWM SOCA output is not selected<br>1: Respective EPWM SOCA output is selected<br>Reset type: CPU1.SYSRSn |
| 0   | PWM1SOCAEN  | R/W  | 0h    | ADCSOCAOn source select:<br>0: Respective EPWM SOCA output is not selected<br>1: Respective EPWM SOCA output is selected<br>Reset type: CPU1.SYSRSn |



### 26.17.3.3 SYNCSOCLOCK Register (Offset = 4h) [Reset = 00000000h]

SYNCSOCLOCK is shown in [Figure 26-179](#) and described in [Table 26-112](#).

Return to the [Summary Table](#).

SYNCSEL and EXTADCSOC Select Lock register

**Figure 26-179. SYNCSOCLOCK Register**

|          |    |    |    |    |    |                     |            |
|----------|----|----|----|----|----|---------------------|------------|
| 31       | 30 | 29 | 28 | 27 | 26 | 25                  | 24         |
| RESERVED |    |    |    |    |    |                     |            |
| R-0-0h   |    |    |    |    |    |                     |            |
| 23       | 22 | 21 | 20 | 19 | 18 | 17                  | 16         |
| RESERVED |    |    |    |    |    |                     |            |
| R-0-0h   |    |    |    |    |    |                     |            |
| 15       | 14 | 13 | 12 | 11 | 10 | 9                   | 8          |
| RESERVED |    |    |    |    |    |                     |            |
| R-0-0h   |    |    |    |    |    |                     |            |
| 7        | 6  | 5  | 4  | 3  | 2  | 1                   | 0          |
| RESERVED |    |    |    |    |    | ADCSOCOUTS<br>ELECT | SYNCSELECT |
| R-0-0h   |    |    |    |    |    | R/WOnce-0h          | R/WOnce-0h |

**Table 26-112. SYNCSOCLOCK Register Field Descriptions**

| Bit   | Field           | Type    | Reset | Description                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|-----------------|---------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31-16 | RESERVED        | R-0     | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 15-2  | RESERVED        | R-0     | 0h    | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 1     | ADCSOCOUTSELECT | R/WOnce | 0h    | ADCSOCOUTSELECT Register Lock bit:<br>0: Respective register is not locked<br>1: Respective register is locked.<br>Notes:<br>[1] Any bit in this register, once set can only be created through a CPU1.SYSRSn. Write of 0 to any bit of this register has no effect<br>[2] The locking mechanism applies to only writes. Reads to the registers which have LOCK protection are always allowed<br>Reset type: CPU1.SYSRSn |
| 0     | SYNCSELECT      | R/WOnce | 0h    | SYNCSELECT Register Lock bit:<br>0: Respective register is not locked<br>1: Respective register is locked.<br>Notes:<br>[1] Any bit in this register, once set can only be created through a CPU1.SYSRSn. Write of 0 to any bit of this register has no effect<br>[2] The locking mechanism applies to only writes. Reads to the registers which have LOCK protection are always allowed<br>Reset type: CPU1.SYSRSn      |

## 26.17.4 Register to Driverlib Function Mapping

### 26.17.4.1 EPWM Registers to Driverlib Functions

**Table 26-113. EPWM Registers to Driverlib Functions**

| File             | Driverlib Function                       |
|------------------|------------------------------------------|
| <b>TBCTL</b>     |                                          |
| epwm.c           | EPWM_setEmulationMode                    |
| epwm.h           | EPWM_setCountModeAfterSync               |
| epwm.h           | EPWM_setClockPrescaler                   |
| epwm.h           | EPWM_forceSyncPulse                      |
| epwm.h           | EPWM_setOneShotSyncOutTrigger            |
| epwm.h           | EPWM_setPeriodLoadMode                   |
| epwm.h           | EPWM_enablePhaseShiftLoad                |
| epwm.h           | EPWM_disablePhaseShiftLoad               |
| epwm.h           | EPWM_setTimeBaseCounterMode              |
| epwm.h           | EPWM_selectPeriodLoadEvent               |
| epwm.h           | EPWM_enableOneShotSync                   |
| epwm.h           | EPWM_disableOneShotSync                  |
| epwm.h           | EPWM_startOneShotSync                    |
| <b>TBCTL2</b>    |                                          |
| epwm.h           | EPWM_selectPeriodLoadEvent               |
| epwm.h           | EPWM_enableOneShotSync                   |
| epwm.h           | EPWM_disableOneShotSync                  |
| epwm.h           | EPWM_startOneShotSync                    |
| <b>SYNCINSEL</b> |                                          |
| epwm.h           | EPWM_setSyncInPulseSource                |
| <b>TBCTR</b>     |                                          |
| epwm.h           | EPWM_setTimeBaseCounter                  |
| epwm.h           | EPWM_getTimeBaseCounterValue             |
| <b>TBSTS</b>     |                                          |
| epwm.h           | EPWM_getTimeBaseCounterOverflowStatus    |
| epwm.h           | EPWM_clearTimeBaseCounterOverflowEvent   |
| epwm.h           | EPWM_getSyncStatus                       |
| epwm.h           | EPWM_clearSyncEvent                      |
| epwm.h           | EPWM_getTimeBaseCounterDirection         |
| <b>SYNCOUTEN</b> |                                          |
| epwm.h           | EPWM_enableSyncOutPulseSource            |
| epwm.h           | EPWM_disableSyncOutPulseSource           |
| <b>TBCTL3</b>    |                                          |
| epwm.h           | EPWM_setOneShotSyncOutTrigger            |
| <b>CMPCTL</b>    |                                          |
| epwm.h           | EPWM_setCounterCompareShadowLoadMode     |
| epwm.h           | EPWM_disableCounterCompareShadowLoadMode |
| epwm.h           | EPWM_getCounterCompareShadowStatus       |
| <b>CMPCTL2</b>   |                                          |
| epwm.h           | EPWM_setCounterCompareShadowLoadMode     |
| epwm.h           | EPWM_disableCounterCompareShadowLoadMode |

**Table 26-113. EPWM Registers to Driverlib Functions (continued)**

| File             | Driverlib Function                              |
|------------------|-------------------------------------------------|
| <b>DBCTL</b>     |                                                 |
| epwm.h           | EPWM_setDeadBandOutputSwapMode                  |
| epwm.h           | EPWM_setDeadBandDelayMode                       |
| epwm.h           | EPWM_setDeadBandDelayPolarity                   |
| epwm.h           | EPWM_setRisingEdgeDeadBandDelayInput            |
| epwm.h           | EPWM_setFallingEdgeDeadBandDelayInput           |
| epwm.h           | EPWM_setDeadBandControlShadowLoadMode           |
| epwm.h           | EPWM_disableDeadBandControlShadowLoadMode       |
| epwm.h           | EPWM_setRisingEdgeDelayCountShadowLoadMode      |
| epwm.h           | EPWM_disableRisingEdgeDelayCountShadowLoadMode  |
| epwm.h           | EPWM_setFallingEdgeDelayCountShadowLoadMode     |
| epwm.h           | EPWM_disableFallingEdgeDelayCountShadowLoadMode |
| epwm.h           | EPWM_setDeadBandCounterClock                    |
| <b>DBCTL2</b>    |                                                 |
| epwm.h           | EPWM_setDeadBandControlShadowLoadMode           |
| epwm.h           | EPWM_disableDeadBandControlShadowLoadMode       |
| <b>AQCTL</b>     |                                                 |
| epwm.h           | EPWM_setActionQualifierShadowLoadMode           |
| epwm.h           | EPWM_disableActionQualifierShadowLoadMode       |
| epwm.h           | EPWM_setActionQualifierAction                   |
| epwm.h           | EPWM_setActionQualifierActionComplete           |
| epwm.h           | EPWM_setAdditionalActionQualifierActionComplete |
| <b>AQTSRCSEL</b> |                                                 |
| epwm.h           | EPWM_setActionQualifierT1TriggerSource          |
| epwm.h           | EPWM_setActionQualifierT2TriggerSource          |
| <b>PCCTL</b>     |                                                 |
| epwm.h           | EPWM_enableChopper                              |
| epwm.h           | EPWM_disableChopper                             |
| epwm.h           | EPWM_setChopperDutyCycle                        |
| epwm.h           | EPWM_setChopperFreq                             |
| epwm.h           | EPWM_setChopperFirstPulseWidth                  |
| <b>VCAPCTL</b>   |                                                 |
| epwm.h           | EPWM_enableValleyCapture                        |
| epwm.h           | EPWM_disableValleyCapture                       |
| epwm.h           | EPWM_startValleyCapture                         |
| epwm.h           | EPWM_setValleyTriggerSource                     |
| epwm.h           | EPWM_enableValleyHWDelay                        |
| epwm.h           | EPWM_disableValleyHWDelay                       |
| epwm.h           | EPWM_setValleyDelayDivider                      |
| <b>VCNTCFG</b>   |                                                 |
| epwm.h           | EPWM_setValleyTriggerEdgeCounts                 |
| epwm.h           | EPWM_getValleyEdgeStatus                        |
| <b>HRCNFG</b>    |                                                 |
| -                |                                                 |
| <b>HRPWR</b>     |                                                 |

**Table 26-113. EPWM Registers to Driverlib Functions (continued)**

| File           | Driverlib Function                              |
|----------------|-------------------------------------------------|
| -              |                                                 |
| <b>HRMSTEP</b> |                                                 |
| -              |                                                 |
| <b>HRCNFG2</b> |                                                 |
| -              |                                                 |
| <b>HRPCTL</b>  |                                                 |
| -              |                                                 |
| <b>TRREM</b>   |                                                 |
| -              |                                                 |
| <b>GLDCTL</b>  |                                                 |
| epwm.h         | EPWM_enableGlobalLoad                           |
| epwm.h         | EPWM_disableGlobalLoad                          |
| epwm.h         | EPWM_setGlobalLoadTrigger                       |
| epwm.h         | EPWM_setGlobalLoadEventPrescale                 |
| epwm.h         | EPWM_getGlobalLoadEventCount                    |
| epwm.h         | EPWM_disableGlobalLoadOneShotMode               |
| epwm.h         | EPWM_enableGlobalLoadOneShotMode                |
| epwm.h         | EPWM_setGlobalLoadOneShotLatch                  |
| epwm.h         | EPWM_forceGlobalLoadOneShotEvent                |
| <b>GLDCFG</b>  |                                                 |
| epwm.h         | EPWM_enableGlobalLoadRegisters                  |
| epwm.h         | EPWM_disableGlobalLoadRegisters                 |
| <b>XLINK</b>   |                                                 |
| epwm.h         | EPWM_setupEPWMLinks                             |
| <b>AQCTLA</b>  |                                                 |
| epwm.h         | EPWM_setActionQualifierAction                   |
| epwm.h         | EPWM_setActionQualifierActionComplete           |
| epwm.h         | EPWM_setAdditionalActionQualifierActionComplete |
| <b>AQCTLA2</b> |                                                 |
| epwm.h         | EPWM_setActionQualifierAction                   |
| epwm.h         | EPWM_setAdditionalActionQualifierActionComplete |
| <b>AQCTLB</b>  |                                                 |
| -              | See AQCTLA                                      |
| <b>AQCTLB2</b> |                                                 |
| -              | See AQCTLA2                                     |
| <b>AQSFR</b>   |                                                 |
| epwm.h         | EPWM_setActionQualifierContSWForceShadowMode    |
| epwm.h         | EPWM_setActionQualifierSWAction                 |
| epwm.h         | EPWM_forceActionQualifierSWAction               |
| <b>AQCSFR</b>  |                                                 |
| epwm.h         | EPWM_setActionQualifierContSWForceAction        |
| <b>DBREDHR</b> |                                                 |
| -              |                                                 |
| <b>DBRED</b>   |                                                 |
| epwm.h         | EPWM_setRisingEdgeDelayCount                    |

**Table 26-113. EPWM Registers to Driverlib Functions (continued)**

| File             | Driverlib Function                           |
|------------------|----------------------------------------------|
| <b>DBFEDHR</b>   |                                              |
| -                |                                              |
| <b>DBFED</b>     |                                              |
| epwm.h           | EPWM_setFallingEdgeDelayCount                |
| <b>TBPHS</b>     |                                              |
| epwm.h           | EPWM_setPhaseShift                           |
| <b>TBPRDHR</b>   |                                              |
| -                |                                              |
| <b>TBPRD</b>     |                                              |
| epwm.h           | EPWM_setTimeBasePeriod                       |
| epwm.h           | EPWM_getTimeBasePeriod                       |
| <b>CMPA</b>      |                                              |
| epwm.h           | EPWM_setCounterCompareValue                  |
| epwm.h           | EPWM_getCounterCompareValue                  |
| <b>CMPB</b>      |                                              |
| -                | See CMPA                                     |
| <b>CMPC</b>      |                                              |
| epwm.h           | EPWM_setCounterCompareShadowLoadMode         |
| epwm.h           | EPWM_disableCounterCompareShadowLoadMode     |
| epwm.h           | EPWM_getCounterCompareShadowStatus           |
| <b>CPMD</b>      |                                              |
| -                | See CMPC                                     |
| <b>GLDCTL2</b>   |                                              |
| epwm.h           | EPWM_setGlobalLoadOneShotLatch               |
| epwm.h           | EPWM_forceGlobalLoadOneShotEvent             |
| <b>SWVDELVAL</b> |                                              |
| epwm.h           | EPWM_setValleySWDelayValue                   |
| <b>TZSEL</b>     |                                              |
| epwm.h           | EPWM_enableTripZoneSignals                   |
| epwm.h           | EPWM_disableTripZoneSignals                  |
| <b>TZDCSEL</b>   |                                              |
| epwm.h           | EPWM_setTripZoneDigitalCompareEventCondition |
| <b>TZCTL</b>     |                                              |
| epwm.h           | EPWM_enableTripZoneAdvAction                 |
| epwm.h           | EPWM_disableTripZoneAdvAction                |
| epwm.h           | EPWM_setTripZoneAction                       |
| epwm.h           | EPWM_setTripZoneAdvAction                    |
| epwm.h           | EPWM_setTripZoneAdvDigitalCompareActionA     |
| epwm.h           | EPWM_setTripZoneAdvDigitalCompareActionB     |
| <b>TZCTL2</b>    |                                              |
| epwm.h           | EPWM_enableTripZoneAdvAction                 |
| epwm.h           | EPWM_disableTripZoneAdvAction                |
| epwm.h           | EPWM_setTripZoneAdvAction                    |
| epwm.h           | EPWM_setTripZoneAdvDigitalCompareActionA     |
| epwm.h           | EPWM_setTripZoneAdvDigitalCompareActionB     |

**Table 26-113. EPWM Registers to Driverlib Functions (continued)**

| File            | Driverlib Function                        |
|-----------------|-------------------------------------------|
| <b>TZCTLDCA</b> |                                           |
| epwm.h          | EPWM_setTripZoneAdvDigitalCompareActionA  |
| <b>TZCTLDCB</b> |                                           |
| epwm.h          | EPWM_setTripZoneAdvDigitalCompareActionB  |
| <b>TZEINT</b>   |                                           |
| epwm.h          | EPWM_enableTripZoneInterrupt              |
| epwm.h          | EPWM_disableTripZoneInterrupt             |
| <b>TZFLG</b>    |                                           |
| epwm.h          | EPWM_getTripZoneFlagStatus                |
| <b>TZCBCFLG</b> |                                           |
| epwm.h          | EPWM_getCycleByCycleTripZoneFlagStatus    |
| <b>TZOSTFLG</b> |                                           |
| epwm.h          | EPWM_getOneShotTripZoneFlagStatus         |
| <b>TZCLR</b>    |                                           |
| epwm.h          | EPWM_selectCycleByCycleTripZoneClearEvent |
| epwm.h          | EPWM_clearTripZoneFlag                    |
| <b>TZCBCCLR</b> |                                           |
| epwm.h          | EPWM_clearCycleByCycleTripZoneFlag        |
| <b>TZOSTCLR</b> |                                           |
| epwm.h          | EPWM_clearOneShotTripZoneFlag             |
| <b>TZFRC</b>    |                                           |
| epwm.h          | EPWM_forceTripZoneEvent                   |
| <b>ETSEL</b>    |                                           |
| epwm.h          | EPWM_enableInterrupt                      |
| epwm.h          | EPWM_disableInterrupt                     |
| epwm.h          | EPWM_setInterruptSource                   |
| epwm.h          | EPWM_enableADCTrigger                     |
| epwm.h          | EPWM_disableADCTrigger                    |
| epwm.h          | EPWM_setADCTriggerSource                  |
| <b>ETPS</b>     |                                           |
| epwm.h          | EPWM_setInterruptEventCount               |
| epwm.h          | EPWM_setADCTriggerEventPrescale           |
| <b>ETFLG</b>    |                                           |
| epwm.h          | EPWM_getEventTriggerInterruptStatus       |
| epwm.h          | EPWM_getADCTriggerFlagStatus              |
| <b>ETCLR</b>    |                                           |
| epwm.h          | EPWM_clearEventTriggerInterruptFlag       |
| epwm.h          | EPWM_clearADCTriggerFlag                  |
| <b>ETFRC</b>    |                                           |
| epwm.h          | EPWM_forceEventTriggerInterrupt           |
| epwm.h          | EPWM_forceADCTrigger                      |
| <b>ETINTPS</b>  |                                           |
| epwm.h          | EPWM_setInterruptEventCount               |
| epwm.h          | EPWM_getInterruptEventCount               |
| <b>ETSOCPS</b>  |                                           |

**Table 26-113. EPWM Registers to Driverlib Functions (continued)**

| File                | Driverlib Function                            |
|---------------------|-----------------------------------------------|
| epwm.h              | EPWM_setADCTriggerEventPrescale               |
| epwm.h              | EPWM_getADCTriggerEventCount                  |
| <b>ETCNTINITCTL</b> |                                               |
| epwm.h              | EPWM_enableInterruptEventCountInit            |
| epwm.h              | EPWM_disableInterruptEventCountInit           |
| epwm.h              | EPWM_forceInterruptEventCountInit             |
| epwm.h              | EPWM_enableADCTriggerEventCountInit           |
| epwm.h              | EPWM_disableADCTriggerEventCountInit          |
| epwm.h              | EPWM_forceADCTriggerEventCountInit            |
| <b>ETCNTINIT</b>    |                                               |
| epwm.h              | EPWM_enableInterruptEventCountInit            |
| epwm.h              | EPWM_disableInterruptEventCountInit           |
| epwm.h              | EPWM_forceInterruptEventCountInit             |
| epwm.h              | EPWM_setInterruptEventCountInitValue          |
| epwm.h              | EPWM_enableADCTriggerEventCountInit           |
| epwm.h              | EPWM_disableADCTriggerEventCountInit          |
| epwm.h              | EPWM_forceADCTriggerEventCountInit            |
| epwm.h              | EPWM_setADCTriggerEventCountInitValue         |
| <b>DCTRIPSEL</b>    |                                               |
| epwm.h              | EPWM_selectDigitalCompareTripInput            |
| epwm.h              | EPWM_enableDigitalCompareTripCombinationInput |
| <b>DCACTL</b>       |                                               |
| epwm.h              | EPWM_setDigitalCompareEventSource             |
| epwm.h              | EPWM_setDigitalCompareEventSyncMode           |
| epwm.h              | EPWM_enableDigitalCompareADCTrigger           |
| epwm.h              | EPWM_disableDigitalCompareADCTrigger          |
| epwm.h              | EPWM_enableDigitalCompareSyncEvent            |
| epwm.h              | EPWM_disableDigitalCompareSyncEvent           |
| epwm.h              | EPWM_setDigitalCompareCBCLatchMode            |
| epwm.h              | EPWM_selectDigitalCompareCBCLatchClearEvent   |
| epwm.h              | EPWM_getDigitalCompareCBCLatchStatus          |
| <b>DCBCTL</b>       |                                               |
| -                   | See DCACTL                                    |
| <b>DCFCTL</b>       |                                               |
| epwm.h              | EPWM_enableDigitalCompareBlankingWindow       |
| epwm.h              | EPWM_disableDigitalCompareBlankingWindow      |
| epwm.h              | EPWM_enableDigitalCompareWindowInverseMode    |
| epwm.h              | EPWM_disableDigitalCompareWindowInverseMode   |
| epwm.h              | EPWM_setDigitalCompareBlankingEvent           |
| epwm.h              | EPWM_setDigitalCompareFilterInput             |
| epwm.h              | EPWM_enableDigitalCompareEdgeFilter           |
| epwm.h              | EPWM_disableDigitalCompareEdgeFilter          |
| epwm.h              | EPWM_setDigitalCompareEdgeFilterMode          |
| epwm.h              | EPWM_setDigitalCompareEdgeFilterEdgeCount     |
| epwm.h              | EPWM_getDigitalCompareEdgeFilterEdgeCount     |

**Table 26-113. EPWM Registers to Driverlib Functions (continued)**

| File                | Driverlib Function                              |
|---------------------|-------------------------------------------------|
| epwm.h              | EPWM_getDigitalCompareEdgeFilterEdgeStatus      |
| <b>DCCAPCTL</b>     |                                                 |
| epwm.h              | EPWM_enableDigitalCompareCounterCapture         |
| epwm.h              | EPWM_disableDigitalCompareCounterCapture        |
| epwm.h              | EPWM_setDigitalCompareCounterShadowMode         |
| epwm.h              | EPWM_getDigitalCompareCaptureStatus             |
| epwm.h              | EPWM_clearDigitalCompareCaptureStatusFlag       |
| epwm.h              | EPWM_configureDigitalCompareCounterCaptureMode  |
| <b>DCFOFFSET</b>    |                                                 |
| epwm.h              | EPWM_setDigitalCompareWindowOffset              |
| epwm.h              | EPWM_getDigitalCompareBlankingWindowOffsetCount |
| <b>DCFOFFSETCNT</b> |                                                 |
| epwm.h              | EPWM_getDigitalCompareBlankingWindowOffsetCount |
| <b>DCFWINDOW</b>    |                                                 |
| epwm.h              | EPWM_setDigitalCompareWindowLength              |
| epwm.h              | EPWM_getDigitalCompareBlankingWindowLengthCount |
| <b>DCFWINDOWCNT</b> |                                                 |
| epwm.h              | EPWM_getDigitalCompareBlankingWindowLengthCount |
| <b>DCCAP</b>        |                                                 |
| epwm.h              | EPWM_enableDigitalCompareCounterCapture         |
| epwm.h              | EPWM_disableDigitalCompareCounterCapture        |
| epwm.h              | EPWM_setDigitalCompareCounterShadowMode         |
| epwm.h              | EPWM_getDigitalCompareCaptureStatus             |
| epwm.h              | EPWM_clearDigitalCompareCaptureStatusFlag       |
| epwm.h              | EPWM_configureDigitalCompareCounterCaptureMode  |
| epwm.h              | EPWM_getDigitalCompareCaptureCount              |
| <b>DCAHTRIPSEL</b>  |                                                 |
| epwm.h              | EPWM_enableDigitalCompareTripCombinationInput   |
| epwm.h              | EPWM_disableDigitalCompareTripCombinationInput  |
| <b>DCALTRIPSEL</b>  |                                                 |
| -                   | See DCAHTRIPSEL                                 |
| <b>DCBHTRIPSEL</b>  |                                                 |
| -                   | See DCAHTRIPSEL                                 |
| <b>DCBLTRIPSEL</b>  |                                                 |
| -                   | See DCAHTRIPSEL                                 |
| <b>LOCK</b>         |                                                 |
| epwm.h              | EPWM_lockRegisters                              |
| <b>HWVDELVAL</b>    |                                                 |
| epwm.h              | EPWM_getValleyHWDelay                           |
| <b>VCNTVAL</b>      |                                                 |
| epwm.h              | EPWM_getValleyCount                             |



### 26.17.4.2 HRPWM Registers to Driverlib Functions

**Table 26-114. HRPWM Registers to Driverlib Functions**

| File                 | Driverlib Function                     |
|----------------------|----------------------------------------|
| <b>TBCTL</b>         |                                        |
| -                    |                                        |
| <b>TBCTL2</b>        |                                        |
| -                    |                                        |
| <b>EPWMSYNCINSEL</b> |                                        |
| -                    |                                        |
| <b>TBCTR</b>         |                                        |
| -                    |                                        |
| <b>TBSTS</b>         |                                        |
| -                    |                                        |
| <b>EPWMSYNCOUTEN</b> |                                        |
| -                    |                                        |
| <b>TBCTL3</b>        |                                        |
| -                    |                                        |
| <b>CMPCTL</b>        |                                        |
| -                    |                                        |
| <b>CMPCTL2</b>       |                                        |
| -                    |                                        |
| <b>DBCTL</b>         |                                        |
| -                    |                                        |
| <b>DBCTL2</b>        |                                        |
| -                    |                                        |
| <b>AQCTL</b>         |                                        |
| -                    |                                        |
| <b>AQTSRCSEL</b>     |                                        |
| -                    |                                        |
| <b>PCCTL</b>         |                                        |
| -                    |                                        |
| <b>VCAPCTL</b>       |                                        |
| -                    |                                        |
| <b>VCNTCFG</b>       |                                        |
| -                    |                                        |
| <b>HRCNFG</b>        |                                        |
| hrpwm.h              | HRPWM_setMEPEdgeSelect                 |
| hrpwm.h              | HRPWM_setMEPControlMode                |
| hrpwm.h              | HRPWM_setCounterCompareShadowLoadEvent |
| hrpwm.h              | HRPWM_setOutputSwapMode                |
| hrpwm.h              | HRPWM_setChannelBOutputPath            |
| hrpwm.h              | HRPWM_enableAutoConversion             |
| hrpwm.h              | HRPWM_disableAutoConversion            |
| hrpwm.h              | HRPWM_setDeadbandMEPEdgeSelect         |
| hrpwm.h              | HRPWM_setRisingEdgeDelayLoadMode       |
| hrpwm.h              | HRPWM_setFallingEdgeDelayLoadMode      |
| <b>HRPWR</b>         |                                        |

**Table 26-114. HRPWM Registers to Driverlib Functions (continued)**

| File             | Driverlib Function                 |
|------------------|------------------------------------|
| -                |                                    |
| <b>HRMSTEP</b>   |                                    |
| hrpwm.h          | HRPWM_setMEPStep                   |
| <b>HRCNFG2</b>   |                                    |
| hrpwm.h          | HRPWM_setDeadbandMEPEdgeSelect     |
| hrpwm.h          | HRPWM_setRisingEdgeDelayLoadMode   |
| hrpwm.h          | HRPWM_setFallingEdgeDelayLoadMode  |
| <b>HRPCTL</b>    |                                    |
| hrpwm.h          | HRPWM_enablePeriodControl          |
| hrpwm.h          | HRPWM_disablePeriodControl         |
| hrpwm.h          | HRPWM_enablePhaseShiftLoad         |
| hrpwm.h          | HRPWM_disablePhaseShiftLoad        |
| hrpwm.h          | HRPWM_setSyncPulseSource           |
| <b>TRREM</b>     |                                    |
| hrpwm.h          | HRPWM_setTranslatorRemainder       |
| <b>GLDCTL</b>    |                                    |
| -                |                                    |
| <b>GLDCFG</b>    |                                    |
| -                |                                    |
| <b>EPWMXLINK</b> |                                    |
| -                |                                    |
| <b>AQCTLA</b>    |                                    |
| -                |                                    |
| <b>AQCTLA2</b>   |                                    |
| -                |                                    |
| <b>AQCTLB</b>    |                                    |
| -                |                                    |
| <b>AQCTLB2</b>   |                                    |
| -                |                                    |
| <b>AQSFR</b>     |                                    |
| -                |                                    |
| <b>AQCSFR</b>    |                                    |
| -                |                                    |
| <b>DBREDHR</b>   |                                    |
| hrpwm.h          | HRPWM_setRisingEdgeDelay           |
| hrpwm.h          | HRPWM_setHiResRisingEdgeDelayOnly  |
| <b>DBRED</b>     |                                    |
| hrpwm.h          | HRPWM_setRisingEdgeDelay           |
| hrpwm.h          | HRPWM_setHiResRisingEdgeDelayOnly  |
| <b>DBFEDHR</b>   |                                    |
| hrpwm.h          | HRPWM_setFallingEdgeDelay          |
| hrpwm.h          | HRPWM_setHiResFallingEdgeDelayOnly |
| <b>DBFED</b>     |                                    |
| hrpwm.h          | HRPWM_setFallingEdgeDelay          |
| hrpwm.h          | HRPWM_setHiResFallingEdgeDelayOnly |

**Table 26-114. HRPWM Registers to Driverlib Functions (continued)**

| File             | Driverlib Function                    |
|------------------|---------------------------------------|
| <b>TBPHS</b>     |                                       |
| hrpwm.h          | HRPWM_setPhaseShift                   |
| hrpwm.h          | HRPWM_setHiResPhaseShiftOnly          |
| <b>TBPRDHR</b>   |                                       |
| hrpwm.h          | HRPWM_setTimeBasePeriod               |
| hrpwm.h          | HRPWM_setHiResTimeBasePeriodOnly      |
| hrpwm.h          | HRPWM_getTimeBasePeriod               |
| hrpwm.h          | HRPWM_getHiResTimeBasePeriodOnly      |
| <b>TBPRD</b>     |                                       |
| hrpwm.h          | HRPWM_setTimeBasePeriod               |
| hrpwm.h          | HRPWM_setHiResTimeBasePeriodOnly      |
| hrpwm.h          | HRPWM_getTimeBasePeriod               |
| hrpwm.h          | HRPWM_getHiResTimeBasePeriodOnly      |
| <b>CMPA</b>      |                                       |
| hrpwm.h          | HRPWM_setCounterCompareValue          |
| hrpwm.h          | HRPWM_setHiResCounterCompareValueOnly |
| hrpwm.h          | HRPWM_getCounterCompareValue          |
| hrpwm.h          | HRPWM_getHiResCounterCompareValueOnly |
| <b>CMPB</b>      |                                       |
| hrpwm.h          | HRPWM_setCounterCompareValue          |
| hrpwm.h          | HRPWM_setHiResCounterCompareValueOnly |
| hrpwm.h          | HRPWM_getCounterCompareValue          |
| hrpwm.h          | HRPWM_getHiResCounterCompareValueOnly |
| <b>CMPC</b>      |                                       |
| -                |                                       |
| <b>CMPD</b>      |                                       |
| -                |                                       |
| <b>GLDCTL2</b>   |                                       |
| -                |                                       |
| <b>SWVDELVAL</b> |                                       |
| -                |                                       |
| <b>TZSEL</b>     |                                       |
| -                |                                       |
| <b>TZDCSEL</b>   |                                       |
| -                |                                       |
| <b>TZCTL</b>     |                                       |
| -                |                                       |
| <b>TZCTL2</b>    |                                       |
| -                |                                       |
| <b>TZCTLDCA</b>  |                                       |
| -                |                                       |
| <b>TZCTLDCB</b>  |                                       |
| -                |                                       |
| <b>TZEINT</b>    |                                       |
| -                |                                       |

**Table 26-114. HRPWM Registers to Driverlib Functions (continued)**

| File                | Driverlib Function |
|---------------------|--------------------|
| <b>TZFLG</b>        |                    |
| -                   |                    |
| <b>TZCBCFLG</b>     |                    |
| -                   |                    |
| <b>TZOSTFLG</b>     |                    |
| -                   |                    |
| <b>TZCLR</b>        |                    |
| -                   |                    |
| <b>TZCBCCLR</b>     |                    |
| -                   |                    |
| <b>TZOSTCLR</b>     |                    |
| -                   |                    |
| <b>TZFRC</b>        |                    |
| -                   |                    |
| <b>ETSEL</b>        |                    |
| -                   |                    |
| <b>ETPS</b>         |                    |
| -                   |                    |
| <b>ETFLG</b>        |                    |
| -                   |                    |
| <b>ETCLR</b>        |                    |
| -                   |                    |
| <b>ETFRC</b>        |                    |
| -                   |                    |
| <b>ETINTPS</b>      |                    |
| -                   |                    |
| <b>ETSOCPS</b>      |                    |
| -                   |                    |
| <b>ETCNTINITCTL</b> |                    |
| -                   |                    |
| <b>ETCNTINIT</b>    |                    |
| -                   |                    |
| <b>DCTRIPSEL</b>    |                    |
| -                   |                    |
| <b>DCACTL</b>       |                    |
| -                   |                    |
| <b>DCBCTL</b>       |                    |
| -                   |                    |
| <b>DCFCTL</b>       |                    |
| -                   |                    |
| <b>DCCAPCTL</b>     |                    |
| -                   |                    |
| <b>DCFOFFSET</b>    |                    |
| -                   |                    |
| <b>DCFOFFSETCNT</b> |                    |

**Table 26-114. HRPWM Registers to Driverlib Functions (continued)**

| File                | Driverlib Function  |
|---------------------|---------------------|
| -                   |                     |
| <b>DCFWINDOW</b>    |                     |
| -                   |                     |
| <b>DCFWINDOWCNT</b> |                     |
| -                   |                     |
| <b>DCCAP</b>        |                     |
| -                   |                     |
| <b>DCAHTRIPSEL</b>  |                     |
| -                   |                     |
| <b>DCALTRIPSEL</b>  |                     |
| -                   |                     |
| <b>DCBHTRIPSEL</b>  |                     |
| -                   |                     |
| <b>DCBLTRIPSEL</b>  |                     |
| -                   |                     |
| <b>EPWMLOCK</b>     |                     |
| hrpwm.h             | HRPWM_lockRegisters |
| <b>HWVDELVAL</b>    |                     |
| -                   |                     |
| <b>VCNTVAL</b>      |                     |
| -                   |                     |